Solid Edge Draft Type Library

Add2dModelView Method

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Description

This method creates a 2D Model view as a DrawingView object.

Syntax

Visual Basic

```
Public Function Add2dModelView( _
    ByVal Scale As Double, _
    ByVal xl As Double, _
    ByVal yl As Double _
) As Drawingview
```

Parameters

Scale
Specifies the scale of the 2D Model view. The scale determines the size of the 2D Model view in that the view sizes automatically to display everything within the view definition box.

xl
Specifies the x coordinate of the 2D Model view origin.

yl
Specifies the y coordinate of the 2D Model view origin.

See Also

DrawingViews Collection | DrawingViews Members | Solid Edge V18 - What's New

Solid Edge Draft Type Library

AddAssemblyView Method

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Description

Adds an assembly drawing view based on the ViewType and an optional configuration name.

Syntax

Visual Basic

```
Public Function AddAssemblyView( _
    ByVai From As ModelLink, _
    ByVai Orientation As ViewOrientationConstants, _
    ByVai ScaLe As Double, _
    ByVai x As Double, _
    ByVai y As Double, _
    ByVai ViewType As AssemblyDrawingViewTypeConstants, _
    Optional ByVai ConfigurationName As String, _
    Optional ByVai Snapshot As Boolean = False, _
    Optional ByVai SnapShotQuaLity As Long = 0 _
) As Drawingview
```

Parameters

From

The ModelLink object from which the view is created.

Orientation

Value	Description
igBackView	View Orientation - Back View
igBottomBackLeftView	View Orientation - Bottom Back Left View
igBottomBackRightView	View Orientation - Bottom Back Right View
igBottomFrontLeftView	View Orientation - Bottom Front Left View
igBottomFrontRightView	View Orientation - Bottom Front Right View
igBottomView	View Orientation - Bottom View
igDimetricBottomBackLeftView	
igDimetricBottomBackRightView	
igDimetricBottomFrontLeftView	
igDimetricBottomFrontRightView	
igDimetricTopBackLeftView	
igDimetricTopBackRightView	
igDimetricTopFrontLeftView	
¡gDimetricTopFrontRightView	
igFrontView	View Orientation - Front View
igLeftView	View Orientation - Left View

igRightView	View Orientation - Right View
igTopBackLeftView	View Orientation - Top Back Left View
igTopBackRightView	View Orientation - Top Back Right View
igTopFrontLeftView	View Orientation - Top Front Left View
igTopFrontRightView	View Orientation - Top Front Right View
igTopView	View Orientation - Top View
igTrimetricBottomBackLeftView	
igTrimetricBottomBackRightView	
ígTrimetricBottomFrontLeftView	
igTrimetricBottomFrontRightView	
igTrimetricTopBackLeftView	
igTrimetricTopBackRightView	
igTrimetricTopFrontLeftView	
igTrimetricTopFrontRightView	

A member of the ViewOrientationConstants constant set that specifies the orientation of the view.

Scale

Specifies the scale of the view. The scale determines the size of the detail view in that the detail view sizes automatically to display everything encompassed within the detail envelope.

x

Specifies the coordinates of the assembly view's origin.

y

Specifies the coordinates of the assembly view's origin.

ViewType

Value	Description
seAssemblyConfigurationSimplifiedView	Assembly Configuration Simplified Drawing View Type - This is a view of an assembly with simplification. The display of the simplified parts will be based on the configuration.
seAssemblyDesignedView	Assembly Designed Drawing View Type - This is a view of an assembly with no simplification.
seAssemblySimplifiedView	Assembly Simplified Drawing View Type - This is a view of an assembly with simplification. All parts in the assembly that have simplification will be displayed as simplified.

A member of the AssemblyDrawingViewTypeConstants constant set that defines the type of view to be created.

ConfigurationName

Specifies a name for the view configuration.

Snapshot

Specifies whether a snapshot view is created.

SnapShotQuality

An integer from 1 to 5 that specifies the quality of the snapshot view. 1 is fastest, while 5 provides the best quality.

See Also

DrawingViews Collection | DrawingViews Members

Solid Edge Draft Type Library

AddByAuxiliaryFold Method

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Description

Creates an auxiliary DrawingView object and a ViewPlane object from an existing DrawingView object.

Syntax

Visual Basic

```
Public Function AddByAuxiliaryFold( _
    ByVai From As DrawingView, _
    ByVai xl As Double, _
    ByVai yl As Double, _
    ByVai x2 As Double, _
    ByVai y2 As Double, _
    ByVai x3 As Double, _
    ByVai y3 As Double _
) As DrawingView
```

Parameters

From
The DrawingView object from which the auxiliary drawing view is created.

xl
Specifies the coordinates for the first point of the fold line.

yi
Specifies the coordinates for the first point of the fold line.

x2
Specifies the coordinates for the second point of the fold line.

y2
Specifies the coordinates for the second point of the fold line.

x3
Specifies the coordinates for the origin point of the auxiliary view.

yi
Specifies the coordinates for the origin point of the auxiliary view.

Remarks

The AddByAuxiliaryFold method also creates a ViewPlane object for the auxiliary view. To create the auxiliary drawing view, the AddByAuxiliaryFold method accepts an existing DrawingView object, and three points for input. The first two points define the fold line, and the third point sets the origin point for the auxiliary view. The AddByAuxiallaryFold method cannot be used until a DrawingView object exists. The initial DrawingView object is created through the DrawingViews.Add method.

See Also

DrawingViews Collection | DrawingViews Members

Solid Edge Draft Type Library

AddByDetailEnvelope Method

1

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Description

Creates an detail DrawingView object and a DetailEnvelope object from an existing DrawingView object.

Syntax

Visual Basic

```
Public Function AddByDetailEnvelope( _
    ByVai From As DrawingView, _
    ByVai xl As Double, _
    ByVai yl As Double, _
    ByVai Radius As Double, _
    ByVai ScaLe As Double, _
    ByVai x2 As Double, _
    ByVai y2 As Double _
) As DrawingView
```

Parameters

From
The DrawingView object from which the detail drawing view is created.

xl
Specifies the coordinates for the center of the detail envelope used for generating the detail drawing view.

yi
Specifies the coordinates for the center of the detail envelope used for generating the detail drawing view.

Radius
Specifies the length of the detail envelope's radius.

Scale
Specifies the scale of the detail drawing view. The scale determines the size of the detail view in that the detail view sizes automatically to display everything encompassed within the detail envelope.

x2
Specifies the coordinates for the origin of the detail drawing view.

y¿
Specifies the coordinates for the origin of the detail drawing view.

Remarks

This method also creates a DetailEnvelope object for the detail drawing view. The AddByDetailEnvelope method requires detail envelope parameters as input. These parameters include the envelope's center point coordinates and its radius. This method also requires the coordinates for the detail drawing view and the scale of the view as inputs. The AddByDetailEnvelope method cannot be used until a DrawingView object exists. The initial DrawingView object is created through the DrawingViews.Add method.

See Also

DrawingViews Collection | DrawingViews Members

Solid Edge Draft Type Library

AddByDraftView Method

1

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Description

Adds a draft view that references the same user graphics as the input draft view.

Syntax

Visual Basic

```
Public Function AddByDraftView( _
    ByVai From As Drawingview, _
    ByVai ScaLe As Double, _
    ByVai xl As Double, _
    ByVai yl As Double _
) As Drawingview
```

Parameters

From
The DrawingView object from which the detail draft view is created.

Scale
Specifies the scale of the detail draft view. The scale determines the size of the draft view in that the draft view sizes automatically to display everything encompassed within the detail envelope.

xl
Specifies the coordinates for the origin of the detail drawing view.

y?
Specifies the coordinates for the origin of the detail drawing view.

See Also

DrawingViews Collection | DrawingViews Members

Solid Edge Draft Type Library

AddByFold Method

1

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Description

Creates a folded DrawingView object from an existing DrawingView object.

Syntax

Visual Basic

```
Public Function AddByFold( _
    ByVai From As DrawingView, _
    ByVai foLdDir As FoldTypeConstants, _
    ByVai x As Double, _
    ByVai y As Double _
) As DrawingView
```

Parameters

From

Specifies the DrawingView object from which the folded drawing view is created.

foLdDir

Value	Description
igFoldDown	Fold Type - Fold Down
igFoldDownLeft	Fold Type - Fold Down Left
igFoldDownRight	Fold Type - Fold Down Right
igFoldLeft	Fold Type - Fold Left
igFoldRight	Fold Type - Fold Right
igFoldllp	Fold Type - Fold Up
igFoldllpLeft	Fold Type - Fold Up Left
igFoldllpRight	Fold Type - Fold Up Right
igNullFold	Fold Type - Null Fold

A member of the FoldTypeConstants constant set that specifies the fold direction in which to create the folded drawing view.

x

Specifies the coordinates of the folded drawing view's origin.

y

Specifies the coordinates of the folded drawing view's origin.

Remarks

This method requires an existing drawing view, a fold direction in relation to the existing view, and an origin point for the auxiliary drawing view as inputs. The AddByFold method cannot be used until a DrawingView object exists. The initial DrawingView object is created through the DrawingViews.Add method.

See Also

DrawingViews Collection | DrawingViews Members

Solid Edge FrameworkSupport Type Library

AddByOrigin Method

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Description

Adds a SmartFrame2d object by specifying the origin and frame cropping distances.

Syntax

Visual Basic

```
Public Function AddByOrigin( _
    ByVai StyLeName As String, _
    ByVai xl As Double, _
    ByVai yl As Double, _
    ByVai top As Double, _
    ByVai bottom As Double, _
    ByVai Left As Double, _
    ByVai right As Double _
) As SmartFrame2d
```

Parameters

StyleName
Specifies the name of the style used a the source of default data for the SmartFrame2d.

xl
Specifies the coordinates of the origin of the contained object specified in the container's world coordinate system.

yi
Specifies the coordinates of the origin of the contained object specified in the container's world coordinate system.

top
Specifies the top cropping distance.

bottom
Specifies the bottom cropping distance.

left
Specifies the left cropping distance.

right
Specifies the right cropping distance.

See Also

SmartFrames2d Collection | SmartFrames2d Members

Solid Edge FrameworkSupport Type Library

AddCoordinate Method

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Description

Creates a Dimension object for the angular position of key points from a common origin.

Syntax

Visual Basic

```
Public Function AddCoordinate( _
    ByVal Obj1 As Object, _
    ByVal x1 As Double, _
    ByVal y1 As Double, _
    ByVal z1 As Double, _
    ByVal keyPoint1 As Boolean, _
    ByVal Obj2 As Object, _
    ByVal x2 As Double, _
    ByVal y2 As Double, _
    ByVal z2 As Double, _
    ByVal keyPoint2 As Boolean _
) As Dimension
```

Parameters

Obj1
Specifies the common origin object from where the coordinate dimension is to start.

x1
Specifies the coordinates of the common origin object's locate point. This point is used as a proximity point while calculating the key point. The z1 argument applies only to 3-D objects.

y?
Specifies the coordinates of the common origin object's locate point. This point is used as a proximity point while calculating the key point. The z1 argument applies only to 3-D objects.

z1
Specifies the coordinates of the common origin object's locate point. This point is used as a proximity point while calculating the key point. The z1 argument applies only to 3-D objects.

keyPoint1
Specifies whether the common origin object or a key point of the object is to be considered for measuring. If KeyPoint1 is True, the element is selected for the dimension's end point; if KeyPoint2 is False, the nearest key point is selected.

Obj2
Specifies the object where the coordinate dimension is to terminate.

x2
Specifies the coordinates of the second object's locate point. This point is used as a proximity point while calculating the key point. The z2 argument applies only to specified for 3-D objects.

y2
Specifies the coordinates of the second object's locate point. This point is used as a proximity point while calculating the key point. The z2 argument applies only to specified for 3-D objects.

z2
Specifies the coordinates of the second object's locate point. This point is used as a proximity point while calculating the key point. The z2 argument applies only to specified for 3-D objects.

keyPoint2
Specifies whether the object or a key point of the object is to be considered for measuring. If KeyPoint2 is True, the element is selected for the dimension's end point; if KeyPoint2 is False, the nearest key point is selected.

Remarks

A common origin dimension object must exist before dimensions can be added with this method. The common origin dimension is created through the AddCoordinateOrigin method. The dimension measures the distance between the common origin and the specified object along the dimension axis. The dimensions that reference the common origin are members of the coordinate dimension group.

Example

Example (Visual Basic)

```
Private Sub Form_Load()
    Dim objApp As SolidEdgeFramework.Application
    Dim objDoc As SolidEdgePart.PartDocument
    Dim objProfile As SolidEdgePart.Profile
    Dim objDimns As SolidEdgeFrameworkSupport.Dimensions
    Dim objDl As SolidEdgeFrameworkSupport.Dimension
    Dim objLines As SolidEdgeFrameworkSupport.Lines2d
    Dim objRelns As SolidEdgeFrameworkSupport.Relations2d
    Dim sDumpStatus As String     ' Used for temporary storage of datadump return string
```

```
    ' Report errors
    Const PI = 3.14159265358979
    ' Create/get the application with specific settings
    On Error Resume Next
    Set objApp = GetObject(, "SolidEdge.Application")
    If Err Then
        Err.Clear
        Set objApp = CreateObject("SolidEdge.Application")
        Set objDoc = objApp.Documents.Add("SolidEdge.PartDocument")
        objApp.Visible = True
    Else
        Set objDoc = objApp.ActiveDocument
    End If
    ' Create an empty Profile
    Set objProfile = objDoc.ProfileSets.Add.Profiles.Add(pRefPlaneDisp:= _
                                                         objDoc.RefPlanes(1))
    ' Create Line objects in the Profile
    Set objLines = objProfile.Lines2d
    Call objLines.AddBy2Points(x1:=0, y1:=0, x2:=0.01, y2:=0)
    Call objLines.AddBy2Points(x1:=0.01, y1:=0, x2:=0.01, y2:=0.01)
    Call objLines.AddBy2Points(x1:=0.01, y1:=0.01, x2:=0.04, y2:=0.01)
    Call objLines.AddBy2Points(x1:=0.04, y1:=0.01, x2:=0.04, y2:=0.04)
    Call objLines.AddBy2Points(x1:=0.04, y1:=0.04, x2:=0, y2:=0.04)
    Call objLines.AddBy2Points(x1:=0, y1:=0.04, x2:=0, y2:=0)
    ' Relate the Lines to make the Profile closed
    Set objRelns = objProfile.Relations2d
    Call objRelns.AddKeypoint(Object1:=objLines(1), Index1:=igLineEnd, object2:=objLines(2), Index2:=igLineStart)
    Call objRelns.AddKeypoint(Object1:=objLines(2), Index1:=igLineEnd, object2:=objLines(3), Index2:=igLineStart)
    Call objRelns.AddKeypoint(Object1:=objLines(3), Index1:=igLineEnd, object2:=objLines(4), Index2:=igLineStart)
    Call objRelns.AddKeypoint(Object1:=objLines(4), Index1:=igLineEnd, object2:=objLines(5), Index2:=igLineStart)
    Call objRelns.AddKeypoint(Object1:=objLines(5), Index1:=igLineEnd, object2:=objLines(6), Index2:=igLineStart)
    Call objRelns.AddKeypoint(Object1:=objLines(6), Index1:=igLineEnd, object2:=objLines(1), Index2:=igLineStart)
    If objProfile.End(ValidationCriteria:=igProfileClosed) <> 0 Then
        MsgBox ("Invalid Profile")
    End If
    ' Creating a Dimensions Collection object
    Set objDimns = objProfile.Dimensions
    'Add a dimension using AddCoordinate method
    Call objDimns.AddCoordinateOrigin(Object:=objLines(1), x:=0.01, y:=0, z:=0, _
                                      keyPoint:=True)
    Set objD1 = objDimns.AddCoordinate(Obj1:=objLines(3), x1:=0.01, y1:=0.01, z1:=0, keyPoint1:=True, _
                                       Obj2:=objLines(5), x2:=0.04, y2:=0.04, z2:=0, keyPoint2:=True)
    ' USER DISPLAY
    ' Release objects
    Set objApp = Nothing
    Set objDoc = Nothing
    Set objProfile = Nothing
    Set objDimns = Nothing
    Set objLines = Nothing
    Set objD1 = Nothing
    Set objRelns = Nothing
End Sub
```

See Also

Dimensions Collection | Dimensions Members

Solid Edge Draft Type Library

AddDetailView Method

1

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Description

Creates an detail DrawingView object and a DetailEnvelope object from an existing DrawingView object.

Syntax

Visual Basic

```
Public Function AddDetailView( _
    ByVai From As DrawingView, _
    ByVai xl As Double, _
    ByVai yl As Double, _
    ByVai Radius As Double, _
    ByVai ScaLe As Double, _
    ByVai x2 As Double, _
    ByVai y2 As Double, _
    ByVai Independent As Boolean _
) As DrawingView
```

Parameters

From
xl
y7
Radius
Scale
x2
y2
Independent

See Also

DrawingViews Collection | DrawingViews Members | Solid Edge V18 - What's New

Solid Edge FrameworkSupport Type Library

AddDimension Method

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Description

Creates a dimension. For creating PMI dimensions a plane must be set (XY plane is the default). The type of the dimension, the number of parents, and the collection object from which it is derived (sheet/profile2d/PMI) dictate the creation of dimensions.

Syntax

Visual Basic

```
Public Function AddDimension( _
    ByVai DimInitData As DimInitData _
) As Dimension
```

Parameters

DimInitData

See Also

Dimensions Collection | Dimensions Members | Solid Edge ST5 - What's New | Solid Edge ST4 MP1 - What's New | Solid Edge ST3 MP8 - What's New

Solid Edge FrameworkSupport Type Library

AddDistanceBetweenObjects Method

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Description

Creates a dimension to measure distance between two specified objects.

Syntax

Visual Basic

```
Public Function AddDistanceBetweenObjects( _
    ByVai Objectl As Object, _
    ByVai xl As Double, _
    ByVai yl As Double, _
    ByVai zl As Double, _
    ByVai keyPointl As Boolean, _
    ByVai 0bject2 As Object, _
    ByVai x2 As Double, _
    ByVai y2 As Double, _
    ByVai z2 As Double, _
    ByVai keyPoint2 As Boolean _
) As Dimension
```

Parameters

Objectl
Specifies the object from which the distance is to be measured. This can be another Dimension object. In that case, the newly created dimension will be added to the given dimension's group.

xl
Specifies the coordinates of the first object's locate point. This point is used as a proximity point while calculating the key point.

yi
Specifies the coordinates of the first object's locate point. This point is used as a proximity point while calculating the key point.

zl
Specifies the coordinates of the first object's locate point. This point is used as a proximity point while calculating the key point.

keyPointi
Specifies whether the object or a key point of the object is to be considered for measuring. If KeyPointl is True, the object is used only to extract the key point and this key point is used to dimension. For example, when a circle is treated as a key point, the dimension is measured from the center. If it is treated as an object (not as a key point) by setting the Boolean value to False, then the dimension is measured to the tangent point.

Object?
Specifies the object up to which the distance is to be measured.

x2
Specifies the coordinates of the second object's locate point. This point is used as a proximity point while calculating the key point.

y2
Specifies the coordinates of the second object's locate point. This point is used as a proximity point while calculating the key point.

z2
Specifies the coordinates of the second object's locate point. This point is used as a proximity point while calculating the key point.

keyPoint2

Specifies whether the object or a key point of the object is to be considered for measuring. If KeyPoint2 is True, the object is used only to extract the key point and this key point is used to dimension. For example, when a circle is treated as a key point, the dimension is measured from the center. If it is treated as an object (not as a key point) by setting the Boolean value to False, then the dimension is measured to the tangent point.

Remarks

This method takes the minimum required data and assumes other data from the defaults set in the collection object.

See Also

Dimensions Collection | Dimensions Members

Solid Edge FrameworkSupport Type Library

AddDistanceBetweenObjectsEX Method

collapseAll

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Description

This method creates a dimension to measure distance between two specified objects with tangency options.

Syntax

Visual Basic

```
Public Function AddDistanceBetweenObjectsEX( _
    ByVal Object1 As Object, _
    ByVal x1 As Double, _
    ByVal y1 As Double, _
    ByVal z1 As Double, _
    ByVal keyPoint1 As Boolean, _
    ByVal bTangent1 As Boolean, _
    ByVal Object2 As Object, _
    ByVal x2 As Double, _
    ByVal y2 As Double, _
    ByVal z2 As Double, _
    ByVal keyPoint2 As Boolean, _
    ByVal bTangent2 As Boolean _
) As Dimension
```

Parameters

Object1
: Specifies the object from which the distance is to be measured. This can be another Dimension object. In that case, the newly created dimension will be added to the given dimension's group.

x1
: Specifies the coordinates of the first object's locate point. This point is used as a proximity point while calculating the key point.

y1
: Specifies the coordinates of the first object's locate point. This point is used as a proximity point while calculating the key point.

z1
: Specifies the coordinates of the first object's locate point. This point is used as a proximity point while calculating the key point.

keyPoint1
: Specifies whether the object or a key point of the object is to be considered for measuring. If KeyPoint1 is True, the object is used only to extract the key point and this key point is used to dimension. For example, when a circle is treated as a key point, the dimension is measured from the center. If it is treated as an object (not as a key point) by setting the Boolean value to False, then the dimension is measured to the tangent point.

bTangent1
: Specifies whether Object1 is tangent.

Object2
: Specifies the object up to which the distance is to be measured.

x2
: Specifies the coordinates of the second object's locate point. This point is used as a proximity point while calculating the key point.

y2
: Specifies the coordinates of the second object's locate point. This point is used as a proximity point while calculating the key point.

z2

Specifies the coordinates of the second object's locate point. This point is used as a proximity point while calculating the key point.

keyPoint2

Specifies whether the object or a key point of the object is to be considered for measuring. If KeyPoint2 is True, the object is used only to extract the key point and this key point is used to dimension. For example, when a circle is treated as a key point, the dimension is measured from the center. If it is treated as an object (not as a key point) by setting the Boolean value to False, then the dimension is measured to the tangent point.

bTangent2

Specifies whether Object2 is tangent.

See Also

Dimensions Collection | Dimensions Members

Solid Edge FrameworkSupport Type Library

AddDistanceIntersectionToIntersection Method

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Description

Adds a dimension to measure distance between two specified intersections.

Syntax

Visual Basic

```
Public Function AddDistanceIntersectionToIntersection( _
    ByVai ObjectII As Object, _
    ByVai xII As Double, _
    ByVai yII As Double, _
    ByVai zII As Double, _
    ByVai ObjectI2 As Object, _
    ByVai xI2 As Double, _
    ByVai yI2 As Double, _
    ByVai zI2 As Double, _
    ByVai 0bject21 As Object, _
    ByVai x21 As Double, _
    ByVai y21 As Double, _
    ByVai z21 As Double, _
    ByVai 0bject22 As Object, _
    ByVai x22 As Double, _
    ByVai y22 As Double, _
    ByVai z22 As Double _
) As Dimension
```

Parameters

Object11
Specifies the first object of the first intersection on which the dimension is to be based.

xII
Specifies the coordinates of a point on ObjectI 1.

y7 7
Specifies the coordinates of a point on ObjectI 1.

z7 7
Specifies the coordinates of a point on ObjectI 1.

ObjectI2
Specifies the second object of the first intersection on which the dimension is to be based.

x12
Specifies the coordinates of a point on ObjectI 2.

y72
Specifies the coordinates of a point on ObjectI 2.

z12
Specifies the coordinates of a point on ObjectI 2.

Object2 7
Specifies the first object of the second intersection on which the dimension is to be based.

x21
Specifies the coordinates of a point on Object21.

y27

Specifies the coordinates of a point on Object21.

z27

Specifies the coordinates of a point on Object21.

Object22

Specifies the second object of the second intersection on which the dimension is to be based.

x22

Specifies the coordinates of a point on Object22.

y22

Specifies the coordinates of a point on Object22.

z22

Specifies the coordinates of a point on Object22.

See Also

Dimensions Collection | Dimensions Members

Solid Edge Draft Type Library

AddDraftView Method

1

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Description

Adds a scalable draft view to contain only user graphics.

Syntax

Visual Basic

```
Public Function AddDraftView( _
    ByVai ScaLe As Double, _
    ByVai xl As Double, _
    ByVai yl As Double _
) As Drawingview
```

Parameters

Scale
Specifies the scale of the detail drawing view. The scale determines the size of the detail view in that the detail view sizes automatically to display everything encompassed within the detail envelope.

xl
Specifies the coordinates for the origin of the detail drawing view.

yi
Specifies the coordinates for the origin of the detail drawing view.

See Also

DrawingViews Collection | DrawingViews Members

Solid Edge Draft Type Library

AddEmptyView Method

1

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Description

Adds an new empty, out-of-date drawing view. Specify parameters using view properties and then call Update.

Syntax

Visual Basic

```
Public Function AddEmptyView( _
    ByVai From As ModelLink, _
    Optional ByVai CreationDefauLts As DrawingViewDefaultsConstants = seDrawingViewDefaultsPrincipal _
) As DrawingView
```

Parameters

From
CreationDefaults

Value	Description
seDrawingViewDefaultsPictorial	Specifies the defaults for drawing view creation using AddEmptyView.
seDrawingViewDefaultsPrincipal	Specifies the defaults for drawing view creation using AddEmptyView.

See Also

DrawingViews Collection | DrawingViews Members | Solid Edge ST5 - What's New

Solid Edge Part Type Library

AddMeasureVariable Method

Collapse All

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Description

Returns the measure variable object.

Syntax

Visual Basic

```
Public Function AddMeasureVariable( _
    ByVal Type As MeasureVariableTypeConstants, _
    ByVal ValueType As MeasureVariableValueConstants, _
    ByVal Geom1 As Object, _
    ByVal Geom2 As Object, _
    Optional ByVal Geom3 As Variant _
) As Measurevariable
```

Parameters

Type

Value	Description
MeasureVariableTypeConstants_Angle	Measure variable type constants Angle
MeasureVariableTypeConstants_Distance	Measure variable type constants Distance
MeasureVariableTypeConstants_MinimumDistance	Measure variable type constants Minimum Distance
MeasureVariableTypeConstants_NormalDistance	Measure variable type constants Normal Distance

A member of the MeasureVariableTypeConstants constant set that specifies the measure variable type.

ValueType

Value	Description
MeasureVariableValueConstants_TrueMeasure	Measure variable value type constants True measure

A member of the MeasureVariableValueConstants constant set that specifies the measure variable value.

Geom1

Specifies the first geometric element that defines the measurement.

Geom2

Specifies the second geometric element that defines the measurement.

Geom3

Specifies the third geometric element that defines the measurement (if necessary).

See Also

PartDocument Object | PartDocument Members

Solid Edge Framework Type Library

AddMinimumDistanceSensor Method

Colla pseAI1

Solid EdcjO. Design better.

Description

Adds a minimum distance sensor to the sensors collection.

Syntax

Visual Basic

```
Public Function AddMinimumDistanceSensor( _
    ByVal ELementl As Object, _
    ByVal ELement2 As Object, _
    ByVal Name As String, _
    ByVal Description As String, _
    ByVal LowerRange As Double, _
    ByVal UpperRange As Double, _
    ByVal MinimumThreshoLd As Double, _
    ByVal MaximumThreshoLd As Double, _
    ByVal Operator As SensorOperatorConstants, _
    ByVal DispLayType As SensorDisplayTypeConstants, _
    ByVal UpdateMechanism As SensorUpdateMechanismConstants _
) As Object
```

Parameters

ElementI
Specifies the first object between which the minimum distance is to be reported.

Element2
Specifies the second object between which the minimum distance is to be reported.

Name
Specifies the name of the sensor.

Description
Contains a description for the sensor.

LowerRange
Specifies the lower value of the sensor's range, providing a mechanism for controlling the sensitivity of the sensor.

UpperRange
Specifies the upper value of the sensor's range.

MinimumThreshold
Specifies the minimum value of the threshold for the sensor.

MaximumThreshold
Specifies the maximum value of the threshold for the sensor.

Operator

Value	Description
seSensorOperatorBetween	Sensor operator - between
seSensorOperatorEqualTo	Sensor operator - equal to
seSensorOperatorGreaterThan	Sensor operator - greater than
seSensorOperatorInvalid	Invalid sensor operator
seSensorOperatorLessThan	Sensor operator - less than

seSensorOperatorNotBetween	Sensor operator - not between
seSensorOperatorNotEqualTo	Sensor operator - not equal to

Specifies the mathematical operator to be used for testing the sensor value against the threshold value. Valid values are members of the SensorOperatorConstants constant set.

DisplayType

Value	Description
seSensorDisplayTypeHorizontalRange	Sensor display - horizontal range
seSensorDisplayTypeInvalid	Invalid sensor display
seSensorDisplayTypeTrueFalse	Sensor display - true/false

Specifies the type of sensor display to use, either range type or true/false. Valid values are members of the SensorDisplayTypeConstants constant set.

UpdateMechanlsm

Value	Description
seSensorUpdateMechanismAutomatic	Automatic sensor update
seSensorUpdateMechanismInvalid	Invalid sensor update mechanism
seSensorUpdateMechanismManual	Manual sensor update

Specifies whether the sensor is to update manually or automatically. Valid values are members of the SensorUpdateMechanismConstants constant set.

See Also

Sensors Collection | Sensors Members

Solid Edge Draft Type Library

AddPartView Method

1

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Design better.

Description

Adds a part drawing view based on the ViewType.

Syntax

Visual Basic

```
Public Function AddPartView( _
    ByVai From As ModelLink, _
    ByVai Orientation As ViewOrientationConstants, _
    ByVai ScaLe As Double, _
    ByVai x As Double, _
    ByVai y As Double, _
    ByVai ViewType As PartDrawingViewTypeConstants _
) As Drawingview
```

Parameters

From

The ModelLink object from which the view is created.

Orientation

Value	Description
igBackView	View Orientation - Back View
igBottomBackLeftView	View Orientation - Bottom Back Left View
igBottomBackRightView	View Orientation - Bottom Back Right View
igBottomFrontLeftView	View Orientation - Bottom Front Left View
igBottomFrontRightView	View Orientation - Bottom Front Right View
igBottomView	View Orientation - Bottom View
igDimetricBottomBackLeftView	
igDimetricBottomBackRightView	
igDimetricBottomFrontLeftView	
igDimetricBottomFrontRightView	
igDimetricTopBackLeftView	
igDimetricTopBackRightView	
igDimetricTopFrontLeftView	
igDimetricTopFrontRightView	
igFrontView	View Orientation - Front View
igLeftView	View Orientation - Left View
igRightView	View Orientation - Right View
igTopBackLeftView	View Orientation - Top Back Left View

igTopBackRightView	View Orientation - Top Back Right View
igTopFrontLeftView	View Orientation - Top Front Left View
igTopFrontRightView	View Orientation - Top Front Right View
igTopView	View Orientation - Top View
igTrimetricBottomBackLeftView	
igTrimetricBottomBackRightView	
igTrimetricBottomFrontLeftView	
igTrimetricBottomFrontRightView	
igTrimetricTopBackLeftView	
igTrimetricTopBackRightView	
igTrimetricTopFrontLeftView	
igTrimetricTopFrontRightView	

Member of the ViewOrientationConstants constant set that specifies the orientation of the view.

Scale

Specifies the scale of the view. The scale determines the size of the detail view in that the detail view sizes automatically to display everything encompassed within the detail envelope.

x

Required Double that specify the coordinates of the view's origin.

y

Required Double that specify the coordinates of the view's origin.

ViewType

Value	Description
sePartDesignedView	Part Designed Drawing View Type - This is a view of a part with no simplification
sePartSimplifiedView	Part Simplified Drawing View Type - This is a view of a part with simplification

Member of the PartDrawingViewTypeConstants constant set that defines the type of view to be created.

See Also

DrawingViews Collection | DrawingViews Members

Solid Edge Draft Type Library

AddPMIModelView Method

1

Solid EdcjO.
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Description

Adds an drawing view based on the specified PMI model view.

Syntax

Visual Basic

```
Public Function AddPMIModelView( _
    ByVai From As ModelLink, _
    ByVai ModeLViewName As String, _
    ByVai ScaLe As Double, _
    ByVai x As Double, _
    ByVai y As Double, _
    Optional ByVai IncLudePMIDimensions As Boolean = True, _
    Optional ByVai UseDraftQuaLity As Boolean = False, _
    Optional ByVai DraftQuaLity As Long = 0 _
) As DrawingView
```

Parameters

From
Specifies the profile to be used to create the Contour Flange.

ModelViewName
Specifies the name of the model view.

Scale
Specifies the scale of the view. The scale determines the size of the detail view in that the detail view sizes automatically to display everything encompassed within the detail envelope.

x
Specifies the coordinates of the view's origin.

y
Specifies the coordinates of the view's origin.

IndudePM/Dimensions
Specifies whether PMI dimensions are included.

UseDraftQuality
Specifies whether a draft quality view is created.

DraftQuality
Specifies the quality of the draft view.

See Also

DrawingViews Collection | DrawingViews Members | Solid Edge V20 - What's New

Solid Edge Framework Type Library

AddSheetMetalCheckerSensor Method

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Design better.

Description

Adds a Sheet Metal checker sensor.

Syntax

Visual Basic

```
Public Function AddSheetMetalCheckerSensor( _
    ByVal LeftFeatureType As SheetMetalSensorFeatureTypeConstants, _
    ByVal RightFeatureType As SheetMetalSensorFeatureTypeConstants, _
    ByVal Name As String, _
    ByVal Description As String, _
    ByVal ThreshoLd As Double, _
    ByVal UpdateMechanism As SensorUpdateMechanismConstants, _
    Optional ByRef ELement As Variant _
) As Object
```

Parameters

LeftFeatureType

Value	Description
seSheetMetalSensorFeatureTypeBeads	
seSheetMetalSensorFeatureTypeCutouts	
seSheetMetalSensorFeatureTypeDimples	
seSheetMetalSensorFeatureTypeDrawnCutouts	
seSheetMetalSensorFeatureTypeExteriorEdges	
seSheetMetalSensorFeatureTypeHoles	
seSheetMetalSensorFeatureTypeInteriorEdges	
seSheetMetalSensorFeatureTypeLouvers	

Specifies the first edge set to be used to compare in evaluating the threshold distance. Valid values are members of the SheetMetalSensorFeatureTypeConstants constant set.

RightFeatureType

Value	Description
seSheetMetalSensorFeatureTypeBeads	
seSheetMetalSensorFeatureTypeCutouts	
seSheetMetalSensorFeatureTypeDimples	
seSheetMetalSensorFeatureTypeDrawnCutouts	
seSheetMetalSensorFeatureTypeExteriorEdges	
seSheetMetalSensorFeatureTypeHoles	
seSheetMetalSensorFeatureTypeInteriorEdges	

seSheetMetalSensorFeatureTypeLouvers	

Specifies the second edge set to be used to compare in evaluating the threshold distance. Valid values are members of the SheetMetalSensorFeatureTypeConstants constant set.

Name

Specifies the name of the sensor.

Description

Specifies a description for the sensor.

Threshold

Specifies the value at which the sensor status fails.

UpdateMechanism

Value	Description
seSensorUpdateMechanismAutomatic	Automatic sensor update
seSensorUpdateMechanismInvalid	Invalid sensor update mechanism
seSensorUpdateMechanismManual	Manual sensor update

Specifies whether the sensor is to update manually or automatically. Valid values are members of the SensorUpdateMechanismConstants constant set.

Element

Specifies a model or a collection of faces for which the sensor applies.

See Also

SheetMetalSensors Collection | SheetMetalSensors Members

Solid Edge Draft Type Library

AddSheetMetalView Method

1

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Description

Adds a sheet metal drawing view based on the ViewType.

Syntax

Visual Basic

```
Public Function AddSheetMetalView( _
    ByVai From As ModelLink, _
    ByVai Orientation As ViewOrientationConstants, _
    ByVai ScaLe As Double, _
    ByVai x As Double, _
    ByVai y As Double, _
    ByVai ViewType As SheetMetalDrawingViewTypeConstants _
) As Drawingview
```

Parameters

From

The ModelLink object from which the view is created.

Orientation

Value	Description
igBackView	View Orientation - Back View
igBottomBackLeftView	View Orientation - Bottom Back Left View
igBottomBackRightView	View Orientation - Bottom Back Right View
igBottomFrontLeftView	View Orientation - Bottom Front Left View
igBottomFrontRightView	View Orientation - Bottom Front Right View
igBottomView	View Orientation - Bottom View
igDimetricBottomBackLeftView	
igDimetricBottomBackRightView	
igDimetricBottomFrontLeftView	
igDimetricBottomFrontRightView	
igDimetricTopBackLeftView	
igDimetricTopBackRightView	
igDimetricTopFrontLeftView	
¡gDimetricTopFrontRightView	
igFrontView	View Orientation - Front View
igLeftView	View Orientation - Left View
igRightView	View Orientation - Right View
igTopBackLeftView	View Orientation - Top Back Left View

igTopBackRightView	View Orientation - Top Back Right View
igTopFrontLeftView	View Orientation - Top Front Left View
igTopFrontRightView	View Orientation - Top Front Right View
igTopView	View Orientation - Top View
igTrimetricBottomBackLeftView	
igTrimetricBottomBackRightView	
igTrimetricBottomFrontLeftView	
igTrimetricBottomFrontRightView	
igTrimetricTopBackLeftView	
igTrimetricTopBackRightView	
igTrimetricTopFrontLeftView	
igTrimetricTopFrontRightView	

Member of the ViewOrientationConstants constant set that specifies the orientation of the view.

Scale

Specifies the scale of the view. The scale determines the size of the detail view in that the detail view sizes automatically to display everything encompassed within the detail envelope.

x

Double that specify the coordinates of the view's origin.

y

Double that specify the coordinates of the view's origin.

ViewType

Value	Description
seSheetMetalDesignedView	Sheet Metal Designed Drawing View Type - This is a view of a sheet metal part with no simplification
seSheetMetalFlatView	Sheet Metal Flat Drawing View Type - This is a view of a flattened sheet metal part
seSheetMetalSimplifiedView	Sheet Metal Simplified Drawing View Type - This is a view of a sheet metal part with simplification

Member of the SheetMetalDrawingViewTypeConstants constant set that defines the type of view to be created.

See Also

Drawingviews Collection | Drawingviews Members

Solid Edge Draft Type Library

AddSheetMetalView Method

Collapse All

Solid Edge. Design better.

Description

Adds a sheet metal drawing view based on the ViewType.

Syntax

Visual Basic

```
Public Function AddSheetMetalView( _
    ByVal From As ModelLink, _
    ByVal Orientation As ViewOrientationConstants, _
    ByVal Scale As Double, _
    ByVal x As Double, _
    ByVal y As Double, _
    ByVal ViewType As SheetMetalDrawingViewTypeConstants _
) As Drawingview
```

Parameters

From

The ModelLink object from which the view is created.

Orientation

Value	Description
igBackView	View Orientation - Back View
igBottomBackLeftView	View Orientation - Bottom Back Left View
igBottomBackRightView	View Orientation - Bottom Back Right View
igBottomFrontLeftView	View Orientation - Bottom Front Left View
igBottomFrontRightView	View Orientation - Bottom Front Right View
igBottomView	View Orientation - Bottom View
igDimetricBottomBackLeftView	
igDimetricBottomBackRightView	
igDimetricBottomFrontLeftView	
igDimetricBottomFrontRightView	
igDimetricTopBackLeftView	
igDimetricTopBackRightView	
igDimetricTopFrontLeftView	
igDimetricTopFrontRightView	
igFrontView	View Orientation - Front View
igLeftView	View Orientation - Left View
igRightView	View Orientation - Right View
igTopBackLeftView	View Orientation - Top Back Left View

igTopBackRightView	View Orientation - Top Back Right View
igTopFrontLeftView	View Orientation - Top Front Left View
igTopFrontRightView	View Orientation - Top Front Right View
igTopView	View Orientation - Top View
igTrimetricBottomBackLeftView	
igTrimetricBottomBackRightView	
igTrimetricBottomFrontLeftView	
igTrimetricBottomFrontRightView	
igTrimetricTopBackLeftView	
igTrimetricTopBackRightView	
igTrimetricTopFrontLeftView	
igTrimetricTopFrontRightView	

Member of the ViewOrientationConstants constant set that specifies the orientation of the view.

Scale

Specifies the scale of the view. The scale determines the size of the detail view in that the detail view sizes automatically to display everything encompassed within the detail envelope.

x

Double that specify the coordinates of the view's origin.

y

Double that specify the coordinates of the view's origin.

ViewType

Value	Description
seSheetMetalDesignedView	Sheet Metal Designed Drawing View Type - This is a view of a sheet metal part with no simplification
seSheetMetalFlatView	Sheet Metal Flat Drawing View Type - This is a view of a flattened sheet metal part
seSheetMetalSimplifiedView	Sheet Metal Simplified Drawing View Type - This is a view of a sheet metal part with simplification

Member of the SheetMetalDrawingViewTypeConstants constant set that defines the type of view to be created.

See Also

Drawingviews Collection | Drawingviews Members

Solid Edge FrameworkSupport Type Library

AddSymmetricalDiameter Method

coHapseAn

Solid Edge. Design better.

Description

Adds a symmetrical diameter dimension to measure distance between two objects.

Syntax

Visual Basic

```
Public Function AddSymmetricalDiameter( _
    ByVai Object1 As Object, _
    ByVai x1 As Double, _
    ByVai y1 As Double, _
    ByVai z1 As Double, _
    ByVai keyPoint1 As Boolean, _
    ByVai Object2 As Object, _
    ByVai x2 As Double, _
    ByVai y2 As Double, _
    ByVai z2 As Double, _
    ByVai keyPoint2 As Boolean _
) As Dimension
```

Parameters

Object1
Specifies the object from which the distance is to be measured.

x1
Specifies the coordinates of the first object's locate point. This point is used as a proximity point while calculating the key point,

y?
Specifies the coordinates of the first object's locate point. This point is used as a proximity point while calculating the key point.

z1
Specifies the coordinates of the second object's locate point. This point is used as a proximity point while calculating the key point.

keyPoint1
Specifies whether the object or a key point of the object is to be considered for measuring. If True, the object is used only to extract the key point; and this key point is used to dimension. For example, when a circle is treated as a key point, the dimension is measured from the center. If it is treated as an object (not as a key point) by setting the Boolean value to False, then the dimension is measured to the tangent point.

Object2
Specifies the object up to which the distance is to be measured.

x2
Specifies the coordinates of the first object's locate point. This point is used as a proximity point while calculating the key point.

y2
Specifies the coordinates of the second object's locate point. This point is used as a proximity point while calculating the key point.

z2
Specifies the coordinates of the second object's locate point. This point is used as a proximity point while calculating the key point.

keyPoint2
Specifies whether the object or a key point of the object is to be considered for measuring. If the Boolean is set to True, the object is used only to extract the key point; and this key point is used to dimension. . For example, when a circle is treated as a key point, the dimension is measured from the center. If it is treated as an object (not as a key point) by setting the Boolean value to False, then the dimension is measured to the tangent point.

Example

Example (Visual Basic)

```
Private Sub Form_Load()
    Dim objApp As SolidEdgeFramework.Application
    Dim objDoc As SolidEdgePart.PartDocument
    Dim objProfile As SolidEdgePart.Profile
    Dim objC1 As SolidEdgeFrameworkSupport.Circle2d
    Dim objC2 As SolidEdgeFrameworkSupport.Circle2d
    Dim objDimns As SolidEdgeFrameworkSupport.Dimensions
    Dim objDim1 As SolidEdgeFrameworkSupport.Dimension
    Dim sDumpStatus As String     ' Used for temporary storage of datadump return string
```

```
    ' Report errors
    Const PI = 3.14159265358979
    ' Create/get the application with specific settings
    On Error Resume Next
    Set objApp - GetObject(j "SolidEdge.Application")
    If Err Then
        Err .Clear
        Set objApp = CreateObject("SolidEdge.Application")
        Set objDoc = objApp.Documents.Add("SolidEdge.PartDocument")
        objApp.Visible = True
    Else
        Set objDoc = objApp.ActiveDocument
    End If
    ' Create an empty Profile
    Set objProfile = objDoc.ProfileSets.Add.Profiles.Add(pRefPlaneDisp:= _
                                                         objDoc.RefPlanes(1) )
    ' Create 2 Circle2d objects in the Profile
    Set objCl = objProfile.Circles2d.AddByCenterRadius(x:=0, y:=0, Radius:=0.01)
    Set objC2 = objProfile.Circles2d.AddByCenterRadius(x:=0.02, y:=0.02j Radius:=0.01)
    ' Create a Dimension object
    Set objDimns - objProfile.Dimensions
    Set objDiml - objDimns.AddSymmetricalDiameter(Objectl:=objCl, xl:=0¿ yl:=0, _
                                                  zl:-0_, keyPointlz-True, object2:-objC2, x2:=0.02, _
                                                  y2:=0.02, z2:=0, keyPoint2:=True)
    ' USER DISPLAY
    ' Release objects
    Set objApp = Nothing
    Set objDoc = Nothing
    Set objProfile = Nothing
    Set objCl = Nothing
    Set objC2 = Nothing
    Set objDimns = Nothing
    Set objDiml = Nothing
End Sub
```

See Also

Dimensions Collection | Dimensions Members

Solid Edge Draft Type Library

AddWeldmentView Method

1

Solid EdcjO.
Design better.

Description

Adds a weldment drawing view based on the ViewType.

Syntax

Visual Basic

```
Public Function AddWeldmentView( _
    ByVai From As ModelLink, _
    ByVai Orientation As ViewOrientationConstants, _
    ByVai ScaLe As Double, _
    ByVai x As Double, _
    ByVai y As Double, _
    ByVai ViewType As WeldmentDrawingViewTypeConstants _
) As Drawingview
```

Parameters

From

The ModelLink object from which the view is created.

Orientation

Value	Description
igBackView	View Orientation - Back View
igBottomBackLeftView	View Orientation - Bottom Back Left View
igBottomBackRightView	View Orientation - Bottom Back Right View
igBottomFrontLeftView	View Orientation - Bottom Front Left View
igBottomFrontRightView	View Orientation - Bottom Front Right View
igBottomView	View Orientation - Bottom View
igDimetricBottomBackLeftView	
igDimetricBottomBackRightView	
igDimetricBottomFrontLeftView	
igDimetricBottomFrontRightView	
igDimetricTopBackLeftView	
igDimetricTopBackRightView	
igDimetricTopFrontLeftView	
¡gDimetricTopFrontRightView	
igFrontView	View Orientation - Front View
igLeftView	View Orientation - Left View
igRightView	View Orientation - Right View
igTopBackLeftView	View Orientation - Top Back Left View

igTopBackRightView	View Orientation - Top Back Right View
igTopFrontLeftView	View Orientation - Top Front Left View
igTopFrontRightView	View Orientation - Top Front Right View
igTopView	View Orientation - Top View
igTrimetricBottomBackLeftView	
igTrimetricBottomBackRightView	
igTrimetricBottomFrontLeftView	
igTrimetricBottomFrontRightView	
igTrimetricTopBackLeftView	
igTrimetricTopBackRightView	
igTrimetricTopFrontLeftView	
igTrimetricTopFrontRightView	

Member of the ViewOrientationConstants constant set that specifies the orientation of the view.

Scale

Specifies the scale of the view. The scale determines the size of the detail view in that the detail view sizes automatically to display everything encompassed within the detail envelope.

x

Double that specify the coordinates of the view's origin.

y

Double that specify the coordinates of the view's origin.

ViewType

Value	Description
seWeldmentAssembledView	Weldment View Type - Assembled Weldment Drawing View
seWeldmentMachinedView	Weldment View Type - Finish Machined Weldment Drawing View
seWeldmentWeldedView	Weldment View Type - Welded Pre-Machined Weldment Drawing View

Member of the WeldmentDrawingViewTypeConstants constant set that defines the type of view to be created.

See Also

Drawingviews Collection | Drawingviews Members

Solid Edge Draft Type Library

Align Method

1

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Description

Aligns selected principal and/or auxiliary DrawingView objects that are contained in a selection set.

Syntax

Visual Basic
`Public Sub Align()`

Remarks

As a result of alignment, views that are aligned move, rotate, and scale to maintain proper relationships with their related view(s). Detail and isometric drawing views cannot be aligned.

See Also

DrawingViews Collection | DrawingViews Members

Solid Edge Framework Type Library

Application Object Members

Overview Collapse All

Solid Edge Framework Type Library : Application Object

Public Methods

Activate	Makes the referenced object the active one.
ArrangeWindows	Arranges windows
CommandEnabled	
CreateCommand	Creates a Solid Edge command control.
DisableEventsForGivenAddIn	Disable firing the events for this Addin.
DoIdle	Gives idle time to the application. Call the DoIdleQ method to provide the application with the opportunity to perform background tasks that may not be up-to-date because of intense processing.
EnableEventsForGivenAddIn	Enable firing the events for this Addin.
FindFile	Displays the Find File dialog box.
GetCountOfOpenModelsInFemap	Get count of models currently open in FEMAP
GetDetailItTemplatePath	
GetDirectoryName	Displays a directory browser dialog box and returns the selected directory name.
GetDraftPrintUtility	An object that is used to print Draft files.
GetGlobalParameter	Returns the setting of a specified global parameter for the referenced Application or document object.
GetMaterialTable	
GetOpenFileName	Displays the File Open dialog box.
GetSaveAsFileName	Returns the new name of the file to save.
GetTemplateFileName	This method gets the template file name specified by the document type.
MailLogoff	Closes an established MAPI mail session.
MailLogon	Logs into MAPI mail and establishes a mail session.
Quit	Exits the referenced application.
ReplaceReference	Determines the documents with a specified referenced in user-defined directories and replaces those references with the new reference.
RunMacro	Executes a specified macro.
SEAdminIlpdate	
SearchDocuments	Searched documents at specified managed/unmanaged location based of defined criteria
SetAddInInterfaces	Preapre the map of Addins and their corresponding interfaces
SetDefaultTemplatePath	

SetGlobalParameter	Sets the value of a specified global parameter for the referenced Application or document object.
SetOLERequestPendingTimeout	Determines in milliseconds the amount of time after an automation or OLE request is made before a busy message is displayed. The default value is 8000 milliseconds (8 seconds)
SetOLEServerBusyTimeout	Determines in milliseconds how long Solid Edge will go on retrying your automation or OLE requests before displaying the Component Busy dialog. The default value is 10000 milliseconds (10 seconds).
ShowCommand	
StartCommand	Activates a specified Solid Edge command.
WriteDocumentFormulaIntoXML	

Public Properties

ActiveDocument	Returns the currently active document.
ActiveDocumentType	Returns the type of the currently active document.
ActiveEnvironment	Returns the environment that belongs to the current active window of the document.
ActiveObject	Returns an Object from the active document. The type of object returned is specified by Type.
ActivePrinter	Returns the name of the active printer.
ActiveSelectSet	Returns the SelectSet object for the currently active window.
ActiveStatusBarPart	Sets and returns the current active status bar part.
ActiveWindow	Returns the active Window object.
Addins	Returns the Addins object for the referenced Application object.
Application	Returns the active application object.
ApplicationEvents	Returns the ApplicationEvents object for the referenced Application object.
[*=] ApplicationV8AfterDocumentOpenEvent	Returns the ApplicationV8DocumentOpenEvent object for the referenced Application object.
ApplicationWindowEvents	Returns the ApplicationWindowEvents object for the referenced Application object.
ApprenticeMode	
Caption	Sets and returns the caption of the referenced object.
Customization	An object that is used to customize the user interface.
DefaultFilePath	Sets and returns the default path used when files are opened.
DelayCompute	Specifies whether or not to suspend the computing process.
0 DisplayAlerts	Enables or disables the display of alerts and messages.
DisplayFullScreen	Specifies whether or not the referenced Application object is maximized.

DisplayRecentFiles	Specifies whether or not the used file list is appended to the File menu of the referenced Application object.
Hsl DisplayRecentFilesCount	Sets and returns the number of files that are displayed in the most recently used (MRU) list.
Documents	Returns the collection of documents that are owned by the referenced Application object.
EdgeBarVisible	Specifies whether or not the Edge Bar is visible for the referenced Application object.
EnableStereo	Specifies whether or not the referenced Application object will support the creation of stereoscopic displays.
Environments	Returns a collection of Environment objects that are owned by the referenced Application object.
FeatureLibraryEvents	Returns the FeatureLibraryEvents object for the referenced Application object.
Hsl FileIllEvents	Returns the FileUIEvents object for the referenced Application object.
Hsl Height	Returns the height of the referenced object.
hWnd	Returns the window handle for the referenced object.
Insight	This read-only property returns the Insight object.
Interactive	Specifies whether or not the referenced Application object can be used interactively.
IsIdling	
Left	Specifies the distance (in twips) between the internal left edge of an object and the left edge of its container.
MailSession	Returns the MAPI mail session number.
Hsl Name	Returns the name of the referenced object.
NewFileUIEvents	Returns the NewFileUIEvents object for the referenced Application object.
Parent	Returns the parent object for the referenced object.
ProcessID	Returns the processID of the application.
Hsl ScreenUpdating	Determines if the screen updating is enabled or disabled.
0 SEECEvents	
ShortcutMenuEvents	Returns the ShortcutMenuEvents object for the referenced Application object.
Hsl ShowStartupScreen	
Hsl SolidEdgeInsightXT	
0 SolidEdgeTCE	
I±sl StatusBar	Sets and returns the caption text that is displayed in the referenced Application object.
Hsl StatusBarDelayUpdate	Specifies whether to immediately update the status bar or to defer the update to the idle loop.
Hsl StatusBarHeight	Returns the height of the status bar.

StatusBarPartCount	Sets and returns the number of parts or sections into which the status bar is divided.
StatusBarPartWidth	Returns the width of the current active part of the status bar.
StatusBarVisible	Specifies whether or not the status bar is visible.
0 Top	Sets and returns the distance from the top of the screen to the top of the referenced object or control.
UsableHeight	Returns the height of the usable vertical space in the application window.
UsableWidth	Returns the width of the usable horizontal space in the application window area.
UserName	Returns the name of the current user.
Value	Returns the name of the application, the value of a dimension, the value of a custom property, or the value of a variable.
Version	Returns the version number of the referenced Application object.
Visible	Specifies whether or not the referenced object is displayed to the screen.
Width	Returns the width of the referenced object.
Windows	Returns a collection of Window objects.
WindowState	Returns or sets the state of the referenced window.

See Also

Application Object

Solid Edge Framework Type Library

Application Object

CollapseAI1

Solid Edge.
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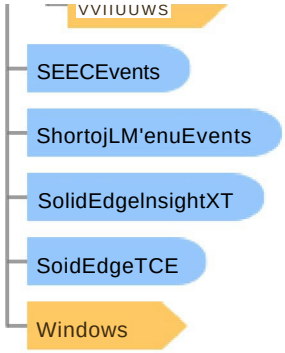
Description

Provides the properties and methods necessary for a Visual Basic user to drive Solid Edge programmatically.

Object Model

App¡cation
ActiveSelectSet (SelectSet)
Addins
AppicaLion
AdtiveSelectSet (SelectSet)
Addins
AppÍGdlHHl
AppicationEvents
App¡cationV8A iter DocumentOpenEvent (ApplicdtionV8DocumentOpenEvent)
App¡cationWindowEvents
CredteCommand (Command)
Customization
Documents
Erivronments
FedtureLibrdrYEvents
FitUIEvents
GetMaleiidITable (MdtTable)
Insight
NewFilelll Events
Pdient (Application)
SEECEvents
ShortcUtMenuEvents
SolidEdgeInsightXT
SolidEdgeTCE
*Wiidovis
AppicdLionEvents
AppicdLionV8ArLeiDocumentOpenEvent (ApplicdtionV8DocumentOpenEvent)
AppicdLionWindowEvents
CredteCommand (Command)
Mouse
Wiidow (CommandWindow)
Customization
Application

- Parent (Application)
- RadaMenu
- RbbonBarThemes
- SwtchWhdmvCust
- Documents
- Envronments
- FedLureLibraryEvents
- FileUIEvents
- GeLMatetiaFTable (MatTable)
- Insight
- NiewFileUI Events
- Parent (Application)
  - ActiveSelectSet (SelectSet)
  - Addins
  - Appicdtiun
  - AppicdLkmEvents
  - AppicdLieriVSAflerDocumentOpenEvent (ApplicatiünV8DocumentOpenEvent)
  - App icdlHoriWindowEvents
  - CreateComriiand (Command)
  - Customization
  - Documents
  - Environments
  - FeatureLbraryEvents
  - FileUIEvents
  - GetMatenalTable (MatTable)
  - Insight
  - NewFileUI Events
  - Parent (Application)
  - SEECEvents
  - ShortcuLMenuEvents
  - SolidEdgeInsightXT
  - SoidEdgeTCE
  - Windows



Remarks

All of the exposed objects for the application can be accessed once you have accessed the Application object. The Application object contains application-wide options and settings, including access to preference sets, version information, window states, status and toolbar controls, mail and routing slip access, and so forth. All top-level objects (ActiveDocument, ActiveEnvironment, and so forth) can be retrieved from the Application object.

Example

Example (VB.NET)

```
Imports SolidEdgeFramework
Imports System.Runtime.InteropServices

Module Example
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing

        Try
            OleMessageFilter.Register()

            ' Connect to a running instance of Solid Edge.
            objApplication = Marshal.GetActiveObject("SolidEdge.Application")

        Catch ex As Exception
            ' Write any errors to console.
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFliter. Revoke()
        End Try
    End Sub
End Module
```

See Also

Application Members

Solid Edge Draft Type Library

Application Property

Solid Edge.
Design better.

Description

Returns the active application object.

Property type

Read-only property

Syntax

Visual Basic
Public Property Application As Application

See Also

DrawingViews Collection | DrawingViews Members

Solid Edge Constants Type Library

ApplicationGlobalConstants Enumeration

Collapse All

Solid Edge. Design better.

Description

Application global constants

Members

Member	Value	Description
seApplicationGlobal3dInputdevice	195	3d Input Device
seApplicationGlobalActivateAllParts	66	
seApplicationGlobalActivateChangedPartsFileLargeAsm	357	Activate changed parts on file for Large asm
seApplicationGlobalActivateChangedPartsFileMedAsm	356	Activate changed parts on file for medium asm
seApplicationGlobalActivateChangePartFileSmallAsm	355	Activate changed parts on file for small asm
seApplicationGlobalActivePartColor	231	Active Part Color
seApplicationGlobalAdminFileLocation	53	AdminFileLocation
seApplicationGlobalAllowInterPart	253	Allow Inter-Part links
seApplicationGlobalAlwayallploadToServerCheekedOutToMe	274	Always upload to server but leave doc cheeked out to me
seApplicationGlobalAlwaysIploadToServerCheckDocIn	275	Always upload to server and check document in
seApplicationGlobalANSHnchToleranceFile	97	ANSI Inch Tolerance File
seApplicationGlobalANSIMetricToleranceFile	98	ANSI Metric Tolerance File
seApplicationGlobalApplicationColorScheme	312	Application color scheme
seApplicationGlobalApplicationDisplay	18	Application display mode for new windows
seApplicationGlobalApplyActivationOverridesToParts	65	
seApplicationGlobalApplySimplifyOverridesToParts	67	
seApplicationGlobalApplySimplifyOverridesToSubAssemblies	69	
seApplicationGlobalApplyStatusToLinks	36	Apply Status To Links
seApplicationGlobalArcSmoothness	203	Arc Smoothness
seApplicationGlobalAsmConfigurationChangesDrawingView	368	Asm Configuration changes make drawing view out of date
seApplicationGlobalAsmOpenas	334	Asm open as
seApplicationGlobalAssemblydrivenPartfeatures	258	Asm driven Part feature
seApplicationGlobalAssemblyReferencePlanefeature	257	Asm Reference Plane in feature
seApplicationGlobalAutoHideRelationshipPathfinder	333	Auto hide relationship pathfinder
seApplicationGlobalAutoMakeSiblingParentRevisionObsolute	277	Automatically make sibling and parent revision obsolute
seApplicationGlobalAutomaticallyReviseDrawingSelectedSDdoc	276	Automatically revise drawing that use selected 3D doc
seApplicationGlobalAutomaticSave	12	

seApplicationGlobalAutomaticSaveTime	13	
seApplicationGlobalAutomaticSelection	198	Automatic Selection
seApplicationGlobalAutoPreserveDoc	245	Auto preserve Document
seApplicationGlobalAutoPreserveDocAutoSave	246	Auto Preserve doc by Auto Save
seApplicationGlobalAutoPreserveDocBackup	206	Auto Preserve doc by Backup
seApplicationGlobalAutoSaveMinutes	247	Auto Save minutes
seApplicationGlobalAutoScrollASMPathfinder	351	Auto scroll assembly pathfinder
seApplicationGlobalAutoSetDraftStatus	278	Auto set draft status
seApplicationGlobalAutoSharpen	200	Auto sharpen
seApplicationGlobalAutoSharpenLevel	110	Automatic sharpen level
seApplicationGlobalAvailableRootFolder	39	Available Root Folder
seApplicationGlobalBackupDraftfiles	250	Backup Draft files
seApplicationGlobalBackupM¡ñutes	248	Auto save backup minutes
seApplicationGlobalBackupModelfiles	249	BAckup Model Files
seApplicationGlobalBaselinedRootFolder	44	Baselined Root Folder
seApplicationGlobalBendDownCenterlineStyle	422	Bend down centerline style
seApplicationGlobalBendupCenterlineStyle	421	Bend up centerline style
seApplicationGlobalBoundaryEdgesStyle	416	Boundary edges style
seApplicationGloba¡ChangedPartActivation	140	
seApplicationGlobalCheckInOnClose	20	Check In On Close
seApplicationGlobalCheckModelChanges	367	Check for model changes made by another user
seApplicationGlobalColorActive	2	The color for active elements
seApplicationGlobalColorBackground	3	The color for the background
seApplicationGlobalColorConstruction	4	The color for construction elements
seApplicationGlobalColorDisabled	5	The color for disabled items
seApplicationGlobaiColorFailed	6	The color for failed items
seApplicationGlobalColorHandle	7	The color for handles
seApplicationGlobalColorHighlight	8	The color for highlighted elements
seApplicationGlobalColorLiveSectionCenterline	92	
seApplicationGlobalColorLiveSectionEdge	91	
seApplicationGlobalColorLiveSectionOpacity	94	
seApplicationGlobalColorLiveSectionRegion	93	
seApplicationGlobalColorManagerCopyIndividualFaceColors	58	
seApplicationGlobaiColorManagerShowAssemblyStyleOverrides	57	
seApplicationGlobalColorManagerShowPartFaceColors	56	
seApplicationGlobalColorManagerIlseIndividualPartStyles	55	
seApplicationGlobalColorManagerIlseToolsOptionsColorSettings	54	
seApplicationGlobalColorProfile	9	The color for profiles, sketches and layouts
seApplicationGlobalColorRefPlane	31	

seApplicationGlobalColorSelected	10	The color for items in the select set
seApplicationGlobaIColorSheet	11	The color for the sheet
seApplicationGlobalColorSheetTabl	138	Specifies the first of two alternating colors for the sheet tabs.
seApplicationGlobalColorSheetTab2	139	Specifies the second of two alternating colors for the sheet tabs.
seApplicationGlobalCommandBarMode	105	Displays mode in command bar or quick bar.
seApplicationGlobalCommandUserlnterface	316	Command user interface
seApplicationGlobaiConfigNames	264	Config names
seApplicationGlobaIConfiguration	440	Use this constant and pass in the configuration name to be used during assembly file open.
seApplicationGlobaICullingBool	197	Culling
seApplicationGlobaiCullingValue	215	Culling Value
seApplicationGlobalCurvatureCombDensity	34	
seApplicationGlobaICurvatureCombMagnitude	35	
seApplicationGlobaiCustomSettingFile	63	
seApplicationGlobaiCutHardwareSectionViews	410	Cut hardware in section views
seApplicationGlobaIDefaultSharpness	19	Default sharpness for new windows
seApplicationGlobalDeriveDisplayAsReference	395	Derive Display as Reference from Assembly
seApplicationGlobaiDetailEnvelope	413	Detail Envelope
seApplicationGlobalDimKeylnValueAuto	371	Dimention Keyin values automatically
seApplicationGlobaIDimSurroundingComponents	208	Dimension Surrounding Components
seApplicationGlobaiDisabledEleColorDraft	385	Disabled Color
seApplicationGlobalDisperseAsmPlacePart	321	Disperse this assembly during Place Part
seApplicationGlobaIDisplayArcQuality	1	
seApplicationGlobalDisplayAsCircle	414	Display as Circle
seApplicationGlobaiDisplaydropshadows	210	Display Drop Shadows
seApplicationGlobaiDisplaylnterPartCopies	211	Recently used files Value
seApplicationGlobaIDisplayQuality	0	
seApplicationGlobaIDisplaySectionCaps	15	Enable the display of section caps with dynamic sections
seApplicationGlobaiDisplaySheet	381	Display sheet
seApplicationGlobaiDisplayStatistics	14	Enable display timing statistics in the status bar
seApplicationGlobaiDisplayllnitsMeasurement	364	Display units of measurement label
seApplicationGlobaiDocumentNameFormula	88	
seApplicationGlobaiDontCreateNewWndPlacePart	322	Do not create new window

seApplicationGlobalDoubleClickblock	365	Double Click a block
seApplicationGlobalDoubleclickEmbeddedObjects	366	Double click to open embedded objects
seApplicationGlobalDraftSaveAsPDFIncludeGridDisplay	169	Include grid display in PDF
seApplicationGlobalDraftSaveAsPDFPrintQualityDPI	171	Print quality DPI
seApplicationGlobalDraftSaveAsPDFSaveAllColorsBlack	168	Save all colors as black
seApplicationGlobalDraftSaveAsPDFSheetOptions	172	Sheet options
seApplicationGlobalDraftSaveAsPDFSheetsRange	173	Sheets range
seApplicationGlobalDraftSaveAsPDFTransparentDVBackgrounds	170	Transparent drawing view backgrounds
seApplicationGlobalDraftSaveAsPDFUseIndividualSheetSizes	174	Use individual sheet sizes
seApplicationGlobalDynamicClipping	213	Dynamic Clipping
seApplicationGlobalDynamicPreviewFeaturecreation	214	Dynamic Preview Feature creation
seApplicationGlobalDynamicTransition	17	Dynamic view transition smoothness
seApplicationGlobalECORootFolder	50	ECO Root Folder
seApplicationGlobalECRRootFolder	49	ECR Root Folder
seApplicationGlobalEdgeCondition	282	Edge Condition
seApplicationGlobalEdgeConditionDraft	408	Edge condition
seApplicationGlobalEnableDotNet2GC	441	Enable/Disable DotNet 2 Garbage Collection
seApplicationGlobalEnableDotNet4GC	442	Enable/Disable DotNet 4 Garbage Collection
seApplicationGlobalEnableDynamicEditProfiles	183	Enable Dynamic Edit of ProfilesSketches
seApplicationGlobalEnableDynamicTolerance	45	Enable dynamic tolerancing of facet data
seApplicationGlobalEnableSheetDistanceDimValue	370	Enable sheet distance dimension values
seApplicationGlobalEnableThreadedDisplay	30	Enable multithreaded displays
seApplicationGlobalEnableIndoAllProfilesketch	189	Enable undo All for profilesketch modifications
seApplicationGlobalEnableValuechangeUsingMouseWheel	185	Enable value changes using mouse wheel
seApplicationGlobalEnterProfileSketchCreateNewWindow	190	Entering Profile Sketch
seApplicationGlobalExpandWeldmentSubasms	347	Streering wheel Medium
seApplicationGlobalExplodedFlowLineStyle	415	Exploded flow line style
seApplicationGlobalExternalBom	272	External BOM
seApplicationGlobalFastlocateBoxDisplayAsm	324	Fast locate using box display for assemblies
seApplicationGlobalFastlocateboxDisplayParts	323	Fast locate using box display for parts
seApplicationGlobalFastlocateOverPathfinder	325	Fast locate when over path finder
seApplicationGlobalFeatureOriginSize	181	Feature origin size
seApplicationGlobalFramePipingIniquenessCutLength	348	Frame Piping Uniqueness Cut Length
seApplicationGlobalFramePipingUniquenessMass	349	Frame Piping Uniqueness Mass

seApplicationGlobalFramePipingllniquenessMiter	350	Frame Piping Uniqueness Miter
seApplicationGlobalGraphicsColorAngAccleration	295	Angular Accleration
seApplicationGlobalGraphicsColorAngVelocity	294	Angular Velocity Color
seApplicationGlobalGraphicsColorBeamCurve	291	Beam Color
seApplicationGlobalGraphicsColorBearing	290	Bearing Color
seApplicationGlobalGraphicsColorBodyTemp	297	Body Temperature
seApplicationGlobalGraphicsColorConnectorSource	300	Connector Suurce
seApplicationGlobalGraphicsColorConnectorTarget	299	Connector Target
seApplicationGlobalGraphicsColorConstraints	298	Constraints
seApplicationGlobalGraphicsColorConvection	303	Convection color
seApplicationGlobalGraphicsColorDisplacement	296	Displacement
seApplicationGlobalGraphicsColorForce	286	Force
seApplicationGlobalGraphicsColorGravity	288	Gravity Color
seApplicationGlobalGraphicsColorHeatFlux	301	Heat Flux
seApplicationGlobalGraphicsColorHeatGen	302	Material Table
seApplicationGlobalGraphicsColorNode	292	Node Color
seApplicationGlobalGraphicsColorPressure	287	Pressure
seApplicationGlobalGraphicsColorRadition	304	Radiation color
seApplicationGlobalGraphicsColorRigidLink	293	Rigid Link Color
seApplicationGlobalGraphicsColorTemperature	305	Temperature color
seApplicationGlobalGraphicsColorTorqueMomnent	289	Torque Color
seApplicationGlobalGRID_Draft_MajorLineColor	152	Specifies the major line color.
seApplicationGlobalGRID_Draft_MinorLineColor	153	Specifies the minor line color.
seApplicationGlobalGRID_Draft_ShowAlignmentLines	150	Specifies whether alignment lines are shown
seApplicationGlobalGRID_Draft_ShowGrid	147	Specifies whether the grid is shown.
seApplicationGlobalGRID_Draft_ShowKeylns	151	Specifies whether key-ins are shown.
seApplicationGlobalGRID_Draft_ShowReadout	149	Specifies whether position readouts are shown.
seApplicationGlobalGRID_Draft_SnapToGrid	148	Specifies whether to snap to the grid.
seApplicationGlobalGRID_Ordered_MajorLineColor	166	Specifies the major line color.
seApplicationGlobalGRID_Ordered_MinorLineColor	167	Specifies the minor line color.
seApplicationGlobalGRID_Ordered_ShowAlignmentLines	164	Specifies whether alignment lines are shown
seApplicationGlobalGRID_Ordered_ShowGrid	161	Specifies whether the grid is shown.
seApplicationGlobalGRID_Ordered_ShowKeylns	165	Specifies whether key-ins are shown.
seApplicationGlobalGRID_Ordered_ShowReadout	163	Specifies whether position readouts are shown.
seApplicationGlobalGRID_Ordered_SnapToGrid	162	Specifies whether to snap to the grid.
seApplicationGlobalGRID_Sync_MajorLineColor	159	Specifies the major line color.

seApplicationGlobalGRID_Sync_MinorLineColor	160	Specifies the minor line color.
seApplicationGlobalGRID_Sync_ShowAlignmentLines	157	Specifies whether alignment lines are shown
seApplicationGlobalGRID_Sync_ShowGrid	154	Specifies whether the grid is shown.
seApplicationGlobalGRID_Sync_ShowKeyIns	158	Specifies whether key-ins are shown.
seApplicationGlobalGRID_Sync_ShowReadout	156	Specifies whether position readouts are shown.
seApplicationGlobalGRID_Sync_SnapToGrid	155	Specifies whether to snap to the grid.
seApplicationGlobalHandlelColorDraft	386	Handlel Color
seApplicationGlobalHandle2ColorDraft	387	Handle3 Color
seApplicationGlobalHandle3ColorDraft	388	Handle3 Color
seApplicationGlobalHatchRibsSectionViews	411	Hatch Rib section Views
seApplicationGlobalHiddenEdgeStyle	400	Hidden edge style
seApplicationGlobalHiddenEdgeStyleFromAsm	397	Hidden edge style from asm
seApplicationGlobalHideAHComponents	64	
seApplicationGlobalHideAllComponentsLargeAsm	354	Hide all Components for large asm
seApplicationGlobalHideAllComponentsMedAsm	353	Hide all Components for medium asm
seApplicationGlobalHideAllComponentsSmallAsm	352	Hide all Components for small asm
seApplicationGlobalHighlightColorDraft	383	Highlight Color
seApplicationGlobalHoleSizeFile	61	
seApplicationGlobalHorizontalScrollBar	372	Horizontal scroll bar
seApplicationGlobalInactiveHiddenIlnusedComponents	329	Inactive hidden and unused components
seApplicationGlobalInclude2DModelSheets	361	Include 2D model sheets
seApplicationGlobalIncludeBackgroundSheets	362	Include background sheets
seApplicationGlobalIncludeCommandPartAssemblySketches	255	Include Command Part Asm Sketch
seApplicationGlobalIncludeDraftViewerData	358	Include Draft Viewer data in file
seApplicationGlobalIncludeWorkingSheets	360	Include working sheets
seApplicationGlobalIncreaseButtonsCommandRibbon2x	315	Increase buttons on Command Ribbons by 2x
seApplicationGlobalIndicateUnderConstraintProfilesPF	188	Indicate under constraint profiles in path finder
seApplicationGlobalInReviewRootFolder	43	In Review Root Folder
seApplicationGlobalInside	243	Inside Color
seApplicationGlobalInsideColorDraft	391	Inside Color in Draft
seApplicationGlobalInsideOutside	244	InsideOutside Color
seApplicationGlobalInsideOutsideColorDraft	392	Inside outside color in Draft
seApplicationGlobalInsightCacheLocation	22	Set InsightCacheLocation
seApplicationGlobalInsightFolderMappingFileLocation	23	Set InsightFolderMappingFileLocation

seApplicationGlobalInterPartCopyCommand	254	Inter Part Copy Command
seApplicationGlobalInWorkRootFolder	41	Obsolete Root Folder
seApplicationGlobalISOToleranceFile	99	ISO Tolerance File
seApplicationGlobalLargeAsmDynamicDisplay	431	Large Asm dynamic display
seApplicationGlobalLargeAsmOccurence	433	Large Asm occurance
seApplicationGlobaILimitedSave	435	Limited Save
seApplicationGlobalLimitedSaveEnable	438	Limited Save Enable
seApplicationGlobaiLimitedUpdate	434	Limited Update
seApplicationGlobalLimitedUpdateEnable	437	Limited Update Enable
seApplicationGlobaILimitsAndFits	283	Limits and fits
seApplicationGlobaILimitSearchToSearchServicesScope	280	Limit search to this search services
seApplicationGlobaiLimitsViewsDraft	412	Limits and Views
seApplicationGlobaiLinkMgmt	260	Link Management
seApplicationGlobalLogFilesLocation	21	Set LogFilesLocation
seApplicationGlobalLookAheadVersion	26	LookAheadVersion
seApplicationGlobaiMacros	261	MaAcros
seApplicationGlobaiMaintainItemNumbers	345	Maintain Item numbers
seApplicationGlobaiMaintainRelationships	330	Maintain relationships during place part
seApplicationGlobalMakeLiveRulePanelVertical	319	Make live rule panel as vertical
seApplicationGlobalMakeRevisionsObsolete	38	Make Revisions Obsolete
seApplicationGlobalManagedStdParts	271	Managed standard parts
seApplicationGlobaIMatTableFile	265	Material Table
seApplicationGlobalMatTableFolder	267	Material Table Folder
seApplicationGlobalMaximumPrintFileSize	180	Maximum print file size
seApplicationGlobalMediumAsmDynamicDisplay	430	Medium Asm dynamic display
seApplicationGlobalNumberSheetGroupSeparately	380	Number sheet group seprately
seApplicationGlobalNXNAsternScratch	273	NX Nastern Scratch
seApplicationGlobalObsoleteRootFolder	40	Obsolete Root Folder
seApplicationGlobaIObsolute	270	Obsolute
seApplicationGlobalOfflineMode	25	Set OfflineMode
seApplicationGlobaIOnLineHelpLocation	144	
seApplicationGlobaIOnlyGenerateEdgesInsideOverlappingCroppedBoundaries	402	Only generate edges inside and overlapping cropped boundaries
seApplicationGlobaIOpacityColorDraft	390	Opacity Color in Draft
seApplicationGlobaIOpacityRefPlane	32	
seApplicationGlobalOpenAsReadOnly3DFile	89	
seApplicationGlobaIOpenAsReadOnly3DFile_IndirectFiles	95	
seApplicationGlobaIOpenAsReadOnlyDftFile	90	
seApplicationGlobaIOpenAsReadOnlyDftFile_IndirectFiles	96	
seApplicationGlobaIOrientWindowSelectedPlane	191	Orient the window to the selected plane
seApplicationGlobaIQrientXpressSize	204	Orient Xpress Size

seApplicationGlobalOriginEdgeStyle	423	Origin edge style
seApplicationGlobalOverlayColor	28	The color for visible edge overlay
seApplicationGlobalOverlayColorMode	29	The color mode for visible edge overlay [1=Absolute, 2=Relative]
seApplicationGlobalPackagedCollaborationRootFolder	48	Packaged Collaboration Root Folder
seApplicationGlobalPartActivationLargeAsm	338	Part Activation for large asm
seApplicationGlobalPartActivationMediumAsm	337	Part Activation for medium asm
seApplicationGlobalPartActivationSmallAsm	336	Part Activation for small asm
seApplicationGlobalPartandAsmlIndoSteps	182	Part and asm undo steps
seApplicationGlobalPartDynamicDisplay	427	Part Dynamic Display
seApplicationGlobalPartIntersections	405	Part intersections
seApplicationGlobalPartSimplificationLargeAsm	341	Part simplification for large asm
seApplicationGlobalPartSimplificationMedAsm	340	Part simplification for medium asm
seApplicationGlobalPartSimplificationSmallAsm	339	Part simplification for small asm
seApplicationGlobalPaste2dBehaviourConstant	187	Paste2d Behaviour
seApplicationGlobalPasteLinkVariableTable	259	Paste Link to Variable Table
seApplicationGlobalPatternedPartsInherit	328	Patterned parts inherit
seApplicationGlobalPFAppereance	314	Path finder appearance
seApplicationGlobalPipeThreadfsFile	62	
seApplicationGlobalPrereleaseRootFolder	47	Prerelease Root Folder
seApplicationGlobalProcesshiddenedges	209	Process hidden edges
seApplicationGlobalProfileEdgeStyle	424	Profile edge style
seApplicationGlobalProfileUndoSteps	186	Profile undo steps
seApplicationGlobalProjectionAngleDraft	407	Projection Angle
seApplicationGlobalPromptForfileProp	251	Prompt for File Properties
seApplicationGlobalPromptMatmodelDoc	176	Prompt for material in new model document
seApplicationGlobalPropertyTextError	192	Property Text Error
seApplicationGlobalPropSeedFile	262	Property Seed File
seApplicationGlobalRecentlyUsedFilesBool	178	Recently used files Bool
seApplicationGlobalRecentlyllsedFilesValue	193	Recently used files Value
seApplicationGlobalRecomputeAssemblySketchEdit	184	Recompute assembly during sketch edits
seApplicationGlobalReferencePlaneSize	179	Reference plane size
seApplicationGlobalReferenceScale	201	Reference Scale
seApplicationGlobalReleasedRootFolder	42	Released Root Folder
seApplicationGlobalReports	263	Reports
seApplicationGlobalRevisionDelimiter	33	Sets Revision Delimiter
seApplicationGlobalSaveColorsBlackWhite	359	Save colors as black and white
seApplicationGlobalSaveMultiCADDatasettoTC	135	
seApplicationGlobalSearchScope	24	Set SearchScope
seApplicationGlobalSectionCurvatureCombDensity	141	

seApplicationGlobalSectionCurvatureCombMagnitude	142	
seApplicationGlobalSelElementColorDraft	384	Select Element Color
seApplicationGlobalSessionDraftOpenInactive	60	Draft inactive mode on open (boolean)
seApplicationGlobalSet2DSelectionFenceOpacity	242	SecRegionOpacity
seApplicationGlobalSetBackGroundColor	216	Background color
seApplicationGlobalSetChildFeatureColor	225	Child Feature Color
seApplicationGlobalSetConstructionColor	233	Construction Color
seApplicationGlobalSetDrivenColor	229	Driven Color
seApplicationGlobalSetFailedColor	230	Failed Color
seApplicationGlobalSetFullyDefinedColor	235	Fully Defined Color
seApplicationGlobalSetGeneralBodyColor	234	General Body Color
seApplicationGlobalSetGuidePathColor	220	Guide Path color
seApplicationGlobalSetHandlel Color	226	Handlel Color
seApplicationGlobalSetHandle2Color	227	Handle2 Color
seApplicationGlobalSetHandleSColor	228	Handle3 Color
seApplicationGlobalSetHighlightColor	217	Highlight color
seApplicationGlobalSetInactiveColor	232	Inactive Color
seApplicationGlobalSetInconsistantColor	238	Inconsistant Color
seApplicationGlobalSetOverDefinedColor	237	Over Defined Color
seApplicationGlobalSetPlaneColor	222	Plane color
seApplicationGlobalSetPlaneEdgesColor	223	Plane Edge color
seApplicationGlobalSetPlaneFeatureColor	224	Plane Feature Color
seApplicationGlobalSetProfileSelectionColor	219	Profile Selection color
seApplicationGlobalSetRegionColor	239	Region Color
seApplicationGlobalSetRegionOpacity	240	PriRegionOpacity
seApplicationGlobalSetSelectedColor	218	Selected color
seApplicationGlobalSetSketchColor	221	Sketch color
seApplicationGlobalSetIlnderDefinedColor	236	User Defined Color
seApplicationGlobalSheetColorDraft	382	Sheet Color
seApplicationGlobalSheetMetalDynamicDisplay	428	Sheet Metal Dynamic Display
seApplicationGlobalSheetMetalGage	266	Sheet Metal Gage
seApplicationGlobalSheetNoNameSeprator	379	Sheet Number and name seprator
seApplicationGlobalSheetTabIColorDraft	393	Sheet Tab1 Color in Draft
seApplicationGlobalSheetTab2ColorDraft	394	Sheet Tab2 Color in Draft
seApplicationGlobalSheetTabDisplayInfo	136	Specifies what information is displayed in the sheet tab : 0 = sheet number; 1 - sheet name; 2 = sheet number and sheet name
seApplicationGlobalSheetTabDisplayInfoSeparator	137	Specifies the separator string used between the sheet number and sheet name when both are displayed in the sheet tabs.
seApplicationGlobalShowCommandTips	310	Show command tips

seApplicationGlobalShowDragRectTimeValue	374	Show drag rectangle when full display time exceeds
seApplicationGlobalShowDrawingViewScale	418	Show drawing view scale
seApplicationGlobalShowEdgesByCuttingPlane	403	Show Edges by cutting plane
seApplicationGlobalShowLiveRules	318	Show Live rules
seApplicationGlobalShowOrientationTriad	194	Show Orientaion Triad
seApplicationGlobalShowPartReference	331	Show Part reference
seApplicationGlobalShowPFDocView	313	Show Path finder in doc view
seApplicationGlobalShowRotationAngle	419	Show rotation angle
seApplicationGlobalShowSensorIndicator	311	Show sensor indicator
seApplicationGlobalShowSheetNumberParentAnnotation	417	Show sheet number if parent annotation
seApplicationGlobalShowStartUpScreen	306	Show Start up screen
seApplicationGlobalShowtriangulationbendlines	212	Show triangulation bend lines
seApplicationGlobalShowlInitsInValueFields	104	Shows units of measure in value fields.
seApplicationGlobalShowUnitsValueField	175	Show units in value field
seApplicationGlobalSimplifyBSplineEdges	404	Simplify BSpline edges
seApplicationGlobalSketchRelationshipPeerEdges	256	Sketch Relationship to peer Edges
seApplicationGlobalSmallAsmDynamicDisplay	429	Small Asm dynamic display
seApplicationGlobalSmallAsmOccurence	432	Small Asm Occurance
seApplicationGlobalSoftwareVHL	16	Enable software VHL/Disable hardware VHL
seApplicationGlobalStartEnvironmentSyncOrOrdered	309	Start part or sheet metal
seApplicationGlobalStartLastSaveDoc	308	Start with my last saved document
seApplicationGlobalStartUsingTemplate	307	Start using template
seApplicationGlobalStatusBarCmdFinder	84	Status Bar Command Finder
seApplicationGlobalStatusBarCommonViews	78	Status Bar Common Views
seApplicationGlobalStatusBarFit	73	Status Bar Fit
seApplicationGlobalStatusBarLookAtFace	77	Status Bar Look At Face
seApplicationGlobalStatusBarNamedViews	80	Status Bar Named Views
seApplicationGlobalStatusBarPan	74	Status Bar Pan
seApplicationGlobalStatusBarPreviousView	79	Status Bar Previous View
seApplicationGlobalStatusBarQuickViewCube	436	Quick View Cube
seApplicationGlobalStatusBarRecordVideo	145	
seApplicationGlobalStatusBarRotateView	75	Status Bar Rotate View
seApplicationGlobalStatusBarSketchView	86	Status Bar Sketch View
seApplicationGlobalStatusBarSpinAbout	76	Status Bar Spin About
seApplicationGlobalStatusBarIlploadToYouTube	146	
seApplicationGlobalStatusBarViewStyles	81	Status Bar View Styles
seApplicationGlobalStatusBarZoom	71	
seApplicationGlobalStatusBarZoomArea	72	Status Bar Zoom Area
seApplicationGlobalStatusBarZoomSlider	82	Status Bar Zoom Slider

seApplicationGlobalStatusBarZoomTool	83	Status Bar Zoom Tool
seApplicationGlobalSteeringwheelSize	202	Steerinf Wheel Size
seApplicationGlobalStopLifeCycleProcess	279	Stop life cycle process
seApplicationGlobalStoreGeomPart	177	Store geometry in part file
seApplicationGlobalStudyType	285	Study Type
seApplicationGlobalSubAsmSimplificationLargeAsm	344	Sub asm simplification for large asm
seApplicationGlobalSubAsmSimplificationMedAsm	343	Sub asm simplification for medium asm
seApplicationGlobalSubAsmSimplificationSmallAsm	342	Sub asm simplification for small asm
seApplicationGlobalSystemInfo	46	Retrieve system information [readonly]
seApplicationGlobalTangentEdgeStyle	401	Tangent edge style
seApplicationGlobalTangentEdgeStyleFromAsm	398	Tangent edge style from asm
seApplicationGlobalTeamcenterFormula	87	
seApplicationGlobalTeamCenterMode	59	
seApplicationGlobalTemplatePathFolder	443	
seApplicationGlobalTemplateRootFolder	51	
seApplicationGlobalThreadDisplayMode	406	Thread Display mode
seApplicationGlobalUpdateAII	252	Inter-Part_Genral Update All
seApplicationGlobalUpdateDraft	37	Update draft files also
seApplicationGlobalUpdateLinkAutomatically	363	Update links automatically
seApplicationGlobalIIploadOnClose	27	Upload On Close
seApplicationGlobalUseAIISimplifiedParts	68	
seApplicationGlobalUseAHSimplifiedSubAssemblies	70	
seApplicationGlobalUseConfigOrModelViewShow	369	Use configuration or model view show/hide states
seApplicationGlobalUseDefaultPlacementName	327	Use default placement name during Replace Part
seApplicationGlobalUseDetailEnvelopeDisplayAsCircle	102	Show Detail Envelope as Circle when TRUE.
seApplicationGlobalUseDetailEnvelopeStandard	101	Use to set the Detail Envelope Standard Constants. Valid values are defined by enum DetailEnvelopeStandardConstants.
seApplicationGlobalUseDimensionStyleElementMapAngularCoordinateDim	118	
seApplicationGlobalUseDimensionStyleElementMapAngularDim	117	
seApplicationGlobalUseDimensionStyleElementMapBalloon	123	
seApplicationGlobalIIseDimensionStyleElementMapBHC	122	
seApplicationGlobalUseDimensionStyleElementMapBlockLabels	134	
seApplicationGlobalUseDimensionStyleElementMapCallouts	124	
seApplicationGlobalUseDimensionStyleElementMapCenterline	120	
seApplicationGlobalUseDimensionStyleElementMapCenterMark	121	
seApplicationGlobalUseDimensionStyleElementMapChamferDim	119	
seApplicationGlobalUseDimensionStyleElementMapCircularDiameterDim	114	

seApplicationGlobalUseDimensionStyleElementMapConnectors	126	
seApplicationGlobalUseDimensionStyleElementMapCoordinateDim	116	
seApplicationGlobalUseDimensionStyleElementMapDatumFrames	130	
seApplicationGlobalUseDimensionStyleElementMapDatumPoints	131	
seApplicationGlobalUseDimensionStyleElementMapDatumTargets	132	
seApplicationGlobalUseDimensionStyleElementMapEdgeCondition	129	
seApplicationGlobalUseDimensionStyleElementMapFCF	133	
seApplicationGlobalUseDimensionStyleElementMapLeaders	125	
seApplicationGlobalUseDimensionStyleElementMapLinDim	111	
seApplicationGlobalUseDimensionStyleElementMapRadialDiameterDim	113	
seApplicationGlobalUseDimensionStyleElementMapRadialDim	112	
seApplicationGlobalUseDimensionStyleElementMapSTSymbols	127	
seApplicationGlobalUseDimensionStyleElementMapSymmetricDiameterDim	115	
seApplicationGlobalUseDimensionStyleElementMapWeldSymbols	128	
seApplicationGlobalUseDimensionStyleMapping	52	
seApplicationGlobalUseDrawingViewCommandBarDRWcommand	426	Use Drawing View Command Bar when the Drawing View Wizard command is run
seApplicationGlobalUseDrawingViewShowCroppingEdges	103	Show cropping edges when TRUE.
seApplicationGlobalUseDrawingViewShowEdgesHiddenTangentEdgesHiddenByOtherParts	107	Show tangent edges that are hidden by other parts when TRUE.
seApplicationGlobalUseDrawingViewShowEdgesHiddenTangentEdgesSelfHidden	106	Show the hidden tangent edges when TRUE.
seApplicationGlobalUseDrawingViewWizardModelsDragged	425	Use drawing view wizard when models are dragged onto the drawing sheet
seApplicationGlobalUseformulaPlacementName	326	Use formula for placement name
seApplicationGlobalUseGesters	317	Use gesters
seApplicationGlobalUselSOToleranceTable	100	Use ISO Tolerance Table when TRUE. Use ANSI Tolerance Table when FALSE.
seApplicationGlobalUselevelbaseNumbers	346	Use level bases numbers
seApplicationGlobalUseOnLineHelp	143	
seApplicationGlobalUserType	85	Global User Type
seApplicationGloball)seshading2DFence	241	Use Shading 2D Fence
seApplicationGlobalUseShadingDraft	389	Use shading in Draft
seApplicationGlobalUseShadingRefplanes	207	Use Shading Reference Plane
seApplicationGlobalUseShadingSelection	205	Use Shading on Selection
seApplicationGlobalUseSimplifiedModels	332	Use simplified models
seApplicationGlobalUseWebBrowser	320	Use my web browser
seApplicationGlobalUsewhenPlacingAsmPartslib	335	Use when placing asm from parts lib
seApplicationGlobalVerticalScrollBar	373	Vertical scroll bar
seApplicationGlobalViewTransitionValue	196	View Translation
seApplicationGlobalVisibleEdgesStyleFromAsm	396	Visible edge style from asm

seApplicationGlobalVisibleEdgeStyle	399	Visible edge style
seApplicationGlobalWeldSymbols	284	Weld Symbols
seApplicationGlobalWeldSymbolsDraft	409	Weld symbols
seApplicationGlobalZebraFolderMapping	109	Set ZebraFolderMapping
seApplicationGlobalZebraFormula	108	
seApplicationGlobalZone	439	Use this constant and pass in the zone name to be used during assembly file open.
seApplicationGlobalZoomToolLeftClick	375	Zoom Tool left click
seApplicationGlobalZoomToolLeftDrag	377	Zoom Tool left drag
seApplicationGlobalZoomToolRightclick	376	Zoom Tool right click
seApplicationGlobalZoomToolRightDrag	378	Zoom Tool right drag

See Also

Solid Edge ST5 - What's New | Solid Edge ST4 - What's New | Solid Edge ST4 MP8 - What's New | Solid Edge ST3 - What's New | Solid Edge ST2 - What's New | Solid Edge ST - What's New | Solid Edge V20 - What's New | Solid Edge V19 - What's New | Solid Edge V18 - What's New | Solid Edge V18 SP4 - What's New | Solid Edge V17 - What's New

Solid Edge FrameworkSupport Type Library

Arcs2d Collection Members

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Sonu cuye rrameworkSupport Type Library : Arcs2d Collection

Public Methods

AddAsFillet	Creates an Arc2d object that connects to two intersecting graphics objects.
AddAsFilletNoTrim	Creates an Arc2d object that connects to two intersecting graphics objects.
AddByCenterStartEnd	Creates an Arc2d object by three points: the center point, the start point, and the end point.
AddByStartAlongEnd	Creates an Arc2d object by three points: the start point, a point along the arc, and the end point.
Item	The items in a collection.

Public Properties

Application	Returns the active application object.
Count	Returns the number of objects in the referenced collection.
Parent	Returns the parent object for the referenced object.

See Also

Arcs2d Collection

Solid Edge FrameworkSupport Type Library

Arcs2d Collection

Collapse All

Soiiu nuye rrameworkSupport Type Library : Arcs2d Collection

Description

A collection of Arc2d objects.

See Also

Arc$2d Members

Solid Edge Draft Type Library

ArrangeDimensionsInSelectSet Method

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Description

Arranges the dimensions in the select set. API SolidEdgeDraft.DrawingView.AddConnectedDimensionsToSelectSet() can be used to add dimensions to select set which are connected to DV

Syntax

Visual Basic

```
Public Sub ArrangeDimensionsInSelectSet( _
    ByVal iStackPitchMuLtipLier As Double, _
    ByVal bAssociative As Boolean _
)
```

Parameters

iStackPitchMultiplier
Sets the stack pitch for dimension.

bAssociative
If "True" then function will create associative groups of selected dimensions.

Example

Example (VB.NET)

```
Imports System.Runtime.InteropServices
Public Class DimensionAutoArrange

    Private Sub btnDimensionAutoArrange_Click(sender As System.Object, e As System.EventArgs) Handles btnDimensionAutoArrange.Cli<
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim ObjDraftDocument As SolidEdgeDraft.DraftDocument = Nothing
        Dim objSheet As SolidEdgeDraft.Sheet = Nothing
        Dim objDrawingViews As SolidEdgeDraft.Drawingviews = Nothing
        Dim objDrawingview As SolidEdgeDraft.Drawingview = Nothing
        Dim objType As Type = Nothing

        Try

            objApplication = Marshal.GetActiveObject("SolidEdge.Application")

            If objApplication Is Nothing Then
                ' Get the type from the Solid Edge ProgID
                objType = Type.GetTypeFromProgID("SolidEdge.Application")
                ' Start Solid Edge
                objApplication = Activator.CreateInstance(objType)
                ' Make Solid Edge visible
                objApplication.Visible = True
            End If

            ObjDraftDocument = objApplication.ActiveDocument

            'getting the Active Sheet object
            objSheet = ObjDraftDocument.ActiveSheet

            objDrawingViews = objSheet.Drawingviews
            objDrawingView = objDrawingViews(l)

            'Add dimensions to select set which are connected to Drawing View.
            objDrawingView.AddConnectedAnnotationsToSelectSet()

            'Arrange the dimensions in select set.
            objSheet.ArrangeDimensionsInSelectSet(1, False)

            If Not ObjDraftDocument Is Nothing Then
                ObjDraftDocument.Close(False)
                ObjDraftDocument = Nothing
            End If
        Catch ex As Exception
            MsgBox(ex.ToString)
        End Try
    End Sub

End Class
```

See Also

Sheet Object | Sheet Members

Solid Edge Constants Type Library

AutoConstrainDimPlacementOptionConstants Enumeration

Collapse All

Solid Edge Constants Type Library : AutoConstrainDimPlacementOptionConstants Enumeration

Description

AutoConstrain Dimension Placement Constants

Members

Member	Value	Description
seAutoConstrainDimChain	0	AutoConstrain Dimension Placement - Chain
seAutoConstrainDimCoordinate	1	AutoConstrain Dimension Placement - Coordinate
seAutoConstrainDimStack	2	AutoConstrain Dimension Placement - Stack

See Also

Solid Edge V20 - What's New | Solid Edge V19 SP4 - What's New

Solid Edge Constants Type Library

AutoConstrainLinearDimOptionConstants Enumeration

Collapse All

Solid Edge Constants Type Library : AutoConstrainLinearDimOptionConstants Enumeration

Description

AutoConstrain Linear Dimension Constants

Members

Member	Value	Description
seAutoConstrainLinearDimAll	2	AutoConstrain Linear Dimension - All
seAutoConstrainLinearDimDistanceBetween	0	AutoConstrain Linear Dimension - Distance Between
seAutoConstrainLinearDimLineLength	1	AutoConstrain Linear Dimension - Line Length

See Also

Solid Edge V20 - What's New | Solid Edge V19 SP4 - What's New

Solid Edge Draft Type Library

Block Object Members

Overview Collapse All

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Solid Edge Draft Type Library : Block Object

Public Methods

Delete	Deletes the referenced object.

Public Properties

Application	Returns the active application object.
AttributeSets	Returns the AttributeSets collection object for the referenced object.
Defaultview	Gets or sets the default view associated with a block.
IsAttributeSetPresent	Returns True if a specified Attribute Set exists on the referenced object.
Name	Returns the name of the referenced object.
OccurrenceCount	Returns the number of objects in the referenced collection.
Parent	Returns the parent object for the referenced object.
ScaleDimAndAnnot	Whether or not dimensions and annotations should use the block scale.
Views	Block views collection

See Also

Block Object | Solid Edge V19 - What's New

Solid Edge Draft Type Library

Block Object

Members Collapse All

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Description

Represents a block.

Object Model

- Block
  - AppicaLion
    - AdbveSelectSel (SelectSet)
    - Addins
    - AppicaLion
    - AppicationEvents
    - AppIcationVaAfterDooumentOpenEvent (AppÍcat¡onV8DooumentOpenEventt)
    - App¡cationWindowEvents
    - CreateCommand (Command)
    - Customization
    - Documents
    - Envronments
    - FeatureLibrarvEvents
    - FifeUI Events
    - GetMatenaiTable (MatTable)
    - Insight
    - NewFilelll Events
    - Parent (Application)
    - SEECEvents
    - ShortcutMenuEvents
    - SolidEdgeInsightXT
    - SolidEdgeTCE
    - Windows
  - DefaultView (BlockView)
    - App ¡cation
    - ArcsZd
    - Balloons
    - BlocicLabels
    - BlockOccurrences
    - BotHoleCrcles
    - Boundaries2d
    - BspineCuives2d (BSplineCurves2d)
    - ¡Centerlines

- CenterMarks
- Crclesid
- ComplexStrhgs2d
- Connectors
- CornerAnnoldLions
- CurVes2d
- DatumFrames
- DatumPurits
- DatumTargets
- Dimensions
- DrawingViews
- Elípses2d
- EllpticalArcs2d
- FeatureCoritrolFrames
- Leaders
- Lines2d
- LineStrings2d
- Points2d
- Relations2d
- SmartFrames2d
- SurfdceFiriishSymbuls;
- TexlBuxes
- WeldSymbols
- Views (BlockViews)

See Also

Block Members | Solid Edge V19 - What's New

Solid Edge Draft Type Library

BlockView Object Members

Overview Collapse All

Solid Edge Draft Type Library : BlockView Object

Public Methods

Delete	Deletes the referenced object.
GetOrigin	Set the origin position of the block view.
ReplaceGraphics	Replace block view contents with the contents in the select set.
SetOrigin	Get the origin position of the block view.
UpdatePreview	Updates the block preview after edit of a block.

Public Properties

0 Application	Returns the active application object.
Arcs2d	Returns the Arcs2d collection object for the referenced object.
AttributeSets	Returns the AttributeSets collection object for the referenced object.
Balloons	Returns the Balloons collection object for the referenced object.
BlockLabels	Returns the collection of block labels for the block view
BlockOccurrences	Returns the BlockOccurrences collection.
BoltHoleCircles	Returns the BoltHoleCircles collection object for the referenced object.
Boundaries2d	Returns the Boundaries2d collection object for the referenced object.
BsplineCurves2d	Returns the BSplineCurves2d collection object for the referenced object.
CenterLines	Returns the CenterLines collection object for the referenced object.
CenterMarks	Returns the CenterMarks collection object for the referenced object.
Circles2d	Returns the Circles2d collection object for the referenced object.
ComplexStrings2d	Returns the ComplexStrings2d collection object for the referenced object.
Connectors	This read-only Connectors property returns the connectors collection for the referenced object.
S CornerAnnotations	Returns the CornerAnnotations collection object for the referenced object.
Curves2d	Returns the Curves2d collection object for the referenced object.
DatumFrames	Returns the DatumFrames collection object for the referenced object.
DatumPoints	Returns the DatumPoints collection object for the referenced object.
DatumTargets	Returns the DatumTargets collection object for the referenced object.
Dimensions	Returns the Dimensions collection object for the referenced object.
Document	Returns the document to which the referenced object belongs.
DrawingViews	Returns the DrawingViews collection.

Ellipses2d	Returns the Ellipses2d collection object for the referenced object.
EllipticalArcs2d	Returns the EllipticalArcs2d collection object for the referenced object.
0 FeatureControlFrames	Returns the FeatureControlFrames collection object for the referenced object.
Index	Returns the numeric index for the active object.
IsAttributeSetPresent	Returns True if a specified Attribute Set exists on the referenced object.
Leaders	Returns the Leaders collection object for the referenced object.
Lines2d	Returns the Lines2d collection object for the referenced object.
LineStrings2d	Returns the LineStrings2d collection object for the referenced object.
Name	Returns the name of the referenced object.
Parent	Returns the parent object for the referenced object.
Points2d	Returns the Points2d collection object for the referenced object.
Relations2d	Returns the Relations2d collection.
SmartFrames2d	Returns the SmartFrames2d collection.
SurfaceFinishSymbols	Returns the SurfaceFinishSymbols collection object for the referenced object.
TextBoxes	Returns the TextBoxes collection object for the referenced object.
H2] WeldSymbols	Returns the WeldSymbols collection object for the referenced object.

See Also

BlockView Object | Solid Edge V19 - What's New

Solid Edge FrameworkSupport Type Library

BreakDistance Property

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Description

Sets and returns the break distance for the referenced Dimension object.

Property type

Read-write property

Syntax

Visual Basic
`Public Property BreakDistance As Double`

Remarks

The break distance defines where the text is positioned. If the BreakPosition property is igDimBreakPositonCenter, the break distance is a ratio of dimension value to distance required. For example, a ratio of 0.5 places the text at the center point between the Origin and Measure terminators. In all other cases, this distance is the absolute distance from the respective terminator ends.

Example

Example (Visual Basic)

```
Private Sub Form_Load()
    Dim objApp As SolidEdgeFrameWork.Application
    Dim objDoc As SolidEdgeDraft.DraftDocument
    Dim objSheet As SolidEdgeDraft.Sheet
    Dim objline As SolidEdgeFrameworkSupport.Line2d
    Dim objDimension As SolidEdgeFrameworkSupport.Dimension
    Dim dblBrkDistance As Double
    ' Report errors
    Const PI = 3.14159265358979
    ' Create/get the application with specific settings
    On Error Resume Next
    Set objApp = GetObjectG "SolidEdge.Application")
    If Err Then
        Err.Clear
        Set objApp = CreateObject("SolidEdge.Application")
        Set objDoc = objApp.Documents.Add("SolidEdge.DraftDocument")
        objApp.Visible = True
    Else
        Set objDoc = objApp.ActiveDocument
    End If
    'getting the Active Sheet object
    Set objSheet = objDoc.ActiveSheet
    'create a line2d
    Set objline = objSheet.Lines2d.AddBy2Points(0.1, 0.1, 0.3, 0.3)
    'create a dimension object
    Set objDimension = objSheet.Dimensions.AddLength(Object:=objline)
    'get break distance
    dblBrkDistance = objDimension.BreakDistance
```

```
        'set break distance
        objDimension.BreakDistance = 0.9
        ' USER DISPLAY
        ' Release objects
        Set objApp = Nothing
        Set objDoc = Nothing
        Set objSheet = Nothing
        Set objDimension = Nothing
        Set objline = Nothing
    End Sub
```

See Also

Dimension Object | Dimension Members

Solid Edge FrameworkSupport Type Library

BSplineCurvesZd Collection Members

Collapse All

Soiiu cuye rrameworkSupport Type Library : BSplineCurves2d Collection

Public Methods

Add	
AddByPoints	Creates the referenced object by specifying a set of points.
AddByPointsWithCloseOption	Same as AddByPoints with a new option to create tangentially closed curve.
Item	The items in a collection.

Public Properties

Application	Returns the active application object.
Count	Returns the number of objects in the referenced collection.
Parent	Returns the parent object for the referenced object.

See Also

BSplineCurves2d Collection

Solid Edge FrameworkSupport Type Library

BSplineCurves2d Collection

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Description

A collection of BSplineCurve2d objects.

See Also

BSplineCurves2d Members

Solid Edge FrameworkSupport Type Library

CaptionPrimary Property

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Description

Sets and returns the primary caption source string to be used by the specified view type.

Property type

Read-write property

Syntax

Visual Basic

```
Public Property CaptionPrimary( _
    ByVal dvtype As DrawingViewCaptionTypeConstants _
) As String
```

Parameters

dvtype

Value	Description
se2DModelView	constant identifier for 2D Model Views.
seAuxiliaryView	constant identifier for Auxiliary Views.
seDetailView	constant identifier for Detail Views.
seNailboardView	constant identifier for Nailboard Views.
sePrincipalView	constant identifier for Principal Views.
seSectionView	constant identifier for Section Views.

See Also

DrawingViewStyle Object | DrawingViewStyle Members | Solid Edge ST4 - What's New

Solid Edge FrameworkSupport Type Library

Circles2d Collection Members

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Sonu cuye rrameworkSupport Type Library : Circles2d Collection

Public Methods

AddBy3Points	Creates a Circle2d object from three nonlinear points that define the circle's circumference.
AddByCenterRadius	Uses a center point and radius to create a Circle2d object.
Item	The items in a collection.

Public Properties

Application	Returns the active application object.
H2] Count	Returns the number of objects in the referenced collection.
Parent	Returns the parent object for the referenced object.

See Also

Circles2d Collection

Solid Edge FrameworkSupport Type Library

Circles2d Collection

Collapse All

Suiiu cuye rrameworkSupport Type Library : Circles2d Collection

Description

A collection of Circle2d objects.

See Also

Circles2d Members

Solid Edge Framework Type Library

Command Object

Members Collapse All

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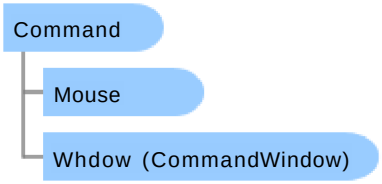
Description

Provides interfaces for Visual Basic users to capture user input to process Solid Edge events.

Object Model



Interfaces

Implemented Interface	Description
ISECommand	Provides interfaces for Visual Basic users to capture user input to process Solid Edge events.

See Also

Command Members

Solid Edge Framework Type Library

Command Object

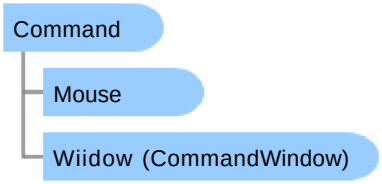
Colla pseAH

Sonu cuye rramework Type Library : Command Object

Description

Provides interfaces for Visual Basic users to capture user input to process Solid Edge events.

Object Model



Interfaces

Implemented Interface	Description
ISECommand	Provides interfaces for Visual Basic users to capture user input to process Solid Edge events.

See Also

Command Members

Solid Edge Framework Type Library

Command Object Members

Overview Collapse All

Solid Edge Framework Type Library : Command Object

Public Methods

-V Start	Runs the referenced Command object.

Public Properties

0 Done	Sets and returns the command finished property.
Mouse	Returns the MouseEvents object.
OnEditOwnerChange	Sets and returns the termination property for edit owner changes.
OnEnvironmentChange	Sets and returns the termination property for environment changes.
Window	Returns the Window object.

Events

Activate	
Deactivate	
f Idle	
KeyDown	
KeyPress	
KeyUp	
Terminate	

See Also

Command Object

Solid Edge ST7 SDK

Connecting To Solid Edge

Collapse All

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You may still use the familiar GetObjectQ method from the Visual Basic 6 days if you'd like. In Visual Basic .NET, GetObjectQ simply calls Marshal.GetActiveObject. This is important to understand because GetObjectQ is a language specific feature of Visual Basic .NET. No other .NET language has a comparable feature. The other .NET languages must rely on Marshal.GetActiveObject.

See Handling 'Application is Busy' and 'Call was Rejected By Callee' errors for information regarding the use of OleMessageFilter.

Connecting To Solid Edge from Visual Basic .NET Copy Code

```
Imports System.Runtime.InteropServices

Module Program
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing

        Try
            'Old VB6 Syntax
            'objApp = GetObjectG "SolidEdge.Application")

            OleMessageFilter.Register()

            ' Connect to a running instance of Solid Edge
            objApplication = Marshal.GetActiveObject("SolidEdge.Application")

        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter.Revoke()
        End Try
    End Sub
End Module
```

Connecting To Solid Edge from C# Copy Code

```
using Examples;
using System;
using System.Runtime.InteropServices;

namespace SolidEdge.SDK
{
    class Program
    {
        [STAThread]
        static void Main(string[] args)
        {
            SolidEdgeFramework.Application application = null;

            try
            {
                OleMessageFilter.Register();

                // Connect to a running instance of Solid Edge
                application = (SolidEdgeFramework.Application)
                    Marshal.GetActiveObject("SolidEdge.Application");
            }
            catch (System.Exception ex)
            {
                Console.WriteLine(ex.Message);
            }
            finally
            {
                OleMessageFilter.Revoke();
            }
        }
    }
}
```

See Also

Starting Solid Edge
Handling 'Application is Busy' and 'Call was Rejected By Callee' errors

Solid Edge Draft Type Library

CoordinateSystems2d Collection

Collapse All

Sonu cuye Draft Type Library : CoordinateSystems2d Collection

Description

Represents a collection of CoordinateSystem2d objects.

See Also

CoordinateSystems2d Members

Solid Edge Draft Type Library

Count Property

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Description

Returns the number of objects in the referenced collection.

Property type

Read-only property

Syntax

Visual Basic
`Public Property Count As Long`

See Also

DrawingViews Collection | DrawingViews Members

Solid Edge ST7 SDK

Creating Documents

Collapse All

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All Solid Edge document objects implement the SolidEdgeFramework.SolidEdgeDocument interface. This generic interface allows document ambiguity when you don't know what type of document that you're working with.

The following table lists the classes (also called ProgIDs) associated with Solid Edge and each type of document environment. These class names can be used in the Visual Basic function CreateObject and can also be used in the Add method of the Documents object.

ProgID	Type	Description
SolidEdge.AssemblyDocument	AssemblyDocument Object	Solid Edge Assembly Document
SolidEdge.DraftDocument	DraftDocument Object	Solid Edge Draft Document
SolidEdge.PartDocument	PartDocument Object	Solid Edge Part Document
SolidEdge.SheetMetalDocument	SheetMetalDocument Object	Solid Edge Sheet Metal Document

See Handling 'Application is Busy' and 'Call was Rejected By Callee' errors for information regarding the use of OleMessageFilter.

Creating Documents from Visual Basic.NET Copy Code

```
Imports System.10
Imports System.Runtime.InteropServices

Module Program
    <DllImport("ole32.dll", CharSet:=CharSet.Unicode, PreserveSig:=False)> _
    Function ProgIDFromCLSID(ByRef clsid As Guid) As String
    End Function

    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objAssemblyDocument As SolidEdgeAssembly.AssemblyDocument = Nothing
        Dim objDraftDocument As SolidEdgeDraft.DraftDocument = Nothing
        Dim objPartDocument As SolidEdgePart.PartDocument = Nothing
        Dim objSheetMetalDocument As SolidEdgePart.SheetMetalDocument = Nothing

        Try
            OleMessageFilter.Register/)

            ' Connect to Solid Edge
            objApplication = Marshal.GetActiveObject("SolidEdge.Application")

            ' Create new SolidEdgeAssembly.AssemblyDocument.
            objAssemblyDocument = CreateSolid EdgeDocument(objApplication, GetType(SolidEdgeAssembly.AssemblyDocument))

            ' Create new SolidEdgeDraft.DraftDocument.
            objDraftDocument = CreateSolid EdgeDocument(objApplication, GetType(SolidEdgeDraft.DraftDocument))

            ' Create new SolidEdgePart.PartDocument.
            objPartDocument = CreateSolidEdgeDocument(objApplication, GetType(SolidEdgePart.PartDocument))

            ' Create new SolidEdgePart.SheetMetalDocument.
            objSheetMetalDocument = CreateSolid EdgeDocument(objApplication, GetType(Solid EdgePart.SheetMetalDocument))
        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter.Revoke()
        End Try
    End Sub

    Function CreateSolidEdgeDocument(objApplication As SolidEdgeFramework.Application, objType As Type) As Object
        Dim objGuid As Guid
        Dim strProgId As String
        Dim objDocuments As SolidEdgeFramework.Documents

        ' Get the Type GUID.
        objGuid = objType.GUID

        ' Get the Progid for the specified Type.
        strProgId = ProgIDFromCLSID(objGuid)

        ' Get a reference to the Documents object.
        objDocuments = objApplication.Documents

        ' Add specified document.
        Return objDocuments.Add(strProgId)
    End Function
```

```
End Module
```

Creating Documents from C# Copy Code

```
using System;
using System.Reflection;
using System.Runtime.InteropServices;

namespace SolidEdge.SDK
{
    class Program
    {
        [DllImport("ole32.dll", CharSet = CharSet.Unicode, PreserveSig = false)]
        static extern string ProgIDFromCLSID([In()]ref Guid clsid);

        [STAThread]
        static void Main(string[] args)
        {
            SolidEdgeFramework.Application application = null;
            SolidEdgeAssembly.AssemblyDocument assemblyDocument = null;
            SolidEdgeDraft.DraftDocument draftDocument = null;
            SolidEdgePart.PartDocument partDocument = null;
            SolidEdgePart.SheetMetalDocument sheetMetalDocument = null;

            try
            {
                OleMessageFilter.Register();

                // Connect to a running instance of Solid Edge.
                application = (SolidEdgeFramework.Application)
                    Marshal.GetActiveObject("Solid Edge.Application");

                // Create new SolidEdgeAssembly.AssemblyDocument.
                assemblyDocument = (SolidEdgeAssembly.AssemblyDocument)
                    CreateSolidEdgeDocument(application, typeof(SolidEdgeAssembly.AssemblyDocument));

                // Create new SolidEdgeDraft.DraftDocument.
                draftDocument = (SolidEdgeDraft.DraftDocument)
                    CreateSolidEdgeDocument(application, typeof(SolidEdgeDraft.DraftDocument));

                // Create new SolidEdgePart.PartDocument.
                partDocument = (SolidEdgePart.PartDocument)
                    CreateSolidEdgeDocument(application, typeof(SolidEdgePart.PartDocument));

                // Create new SolidEdgePart.SheetMetalDocument.
                sheetMetalDocument = (SolidEdgePart.SheetMetalDocument)
                    CreateSolidEdgeDocument(application, typeof(SolidEdgePart.SheetMetalDocument));
            }
            catch (System.Exception ex)
            {
                Console.WriteLine(ex.Message);
            }
            finally
            {
                OleMessageFilter.Revoke();
            }
        }

        static object CreateSolidEdgeDocument(SolidEdgeFramework.Application application, Type t)
        {
            // Get the Type GUID.
            Guid guid = t.GUID;

            // Get the Progid for the specified Type.
            string progid = ProgIDFromCLSID(ref guid);

            // Get a reference to the Documents object.
            SolidEdgeFramework.Documents documents = application.Documents;

            // Add specified document.
            return documents.Add(progId, Missing.Value);
        }
    }
}
```

See Also

Handling 'Application is Busy' and 'Call was Rejected By Callee' errors

Solid Edge ST7 SDK

Creating Macros

Collapse All

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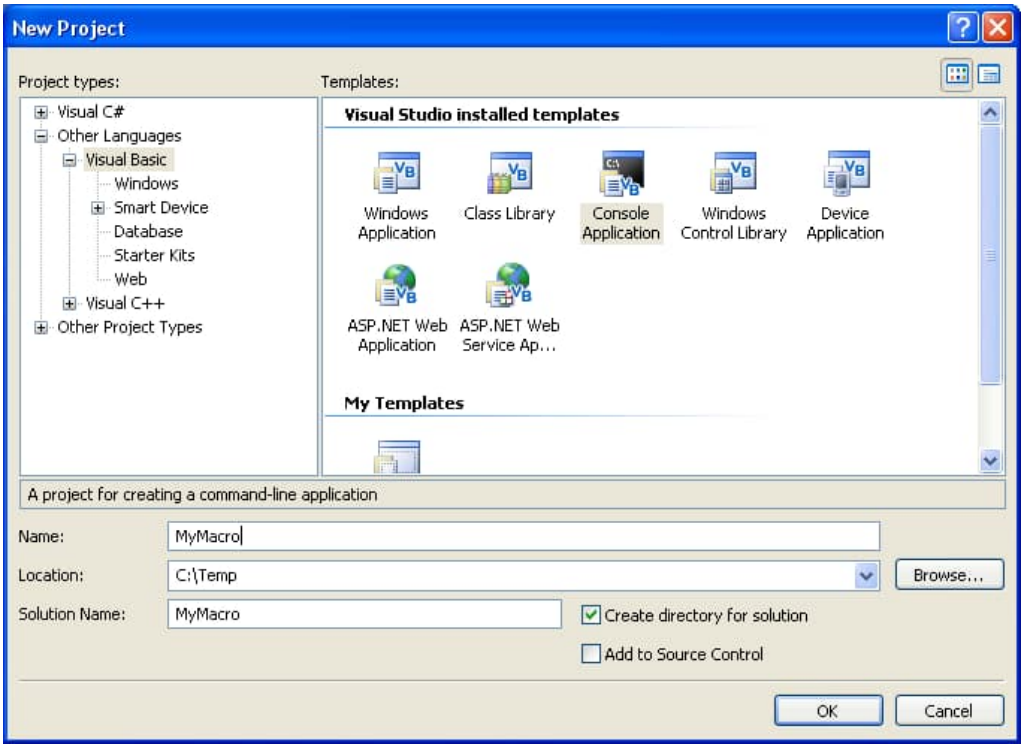
Create a new Visual Basic .NET project

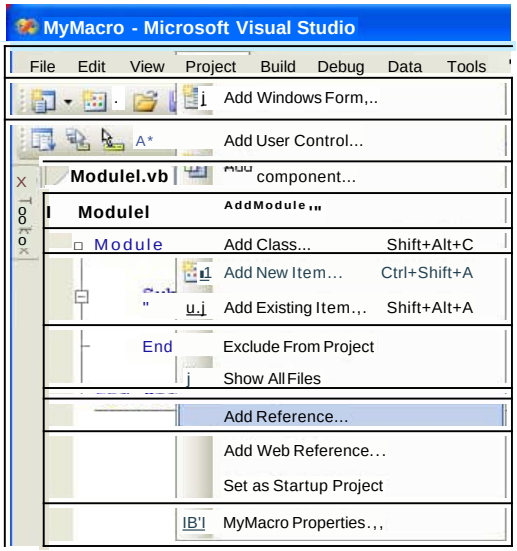
Microsoft Visual Studio offers many different templates for different project types. Depending on your needs, the template that you choose will vary. If you simply want to automate Solid Edge without any user interaction, the Console Application template generally works well.



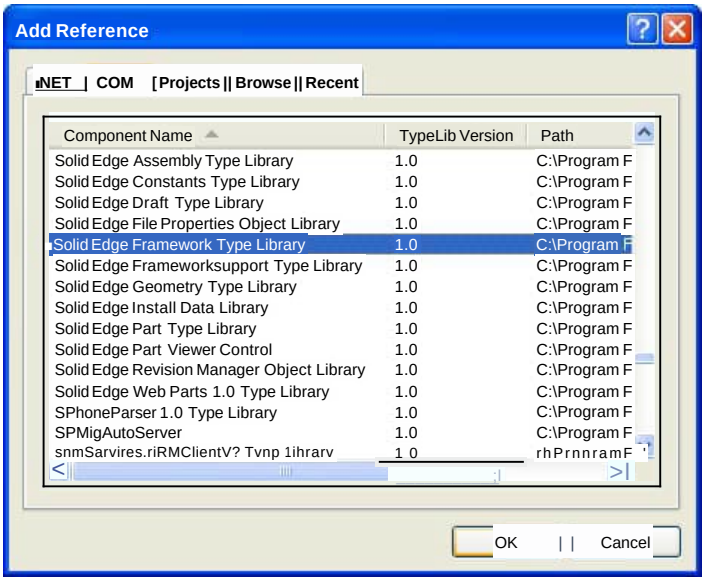
This creates an empty console application project that you can use to begin your work.

Adding a reference to the Solid Edge API

Before you can start working with the Solid Edge API, you'll need to add a reference to the type libraries that you'll be using. You can begin by selecting Project -> Add Reference from the main menu in Visual Studio.



The Add Reference dialog window will appear. You will first need to click the COM tab. Scroll down until you get to the Solid Edge Type Libraries. Select the Solid Edge Framework Type Library as shown below.

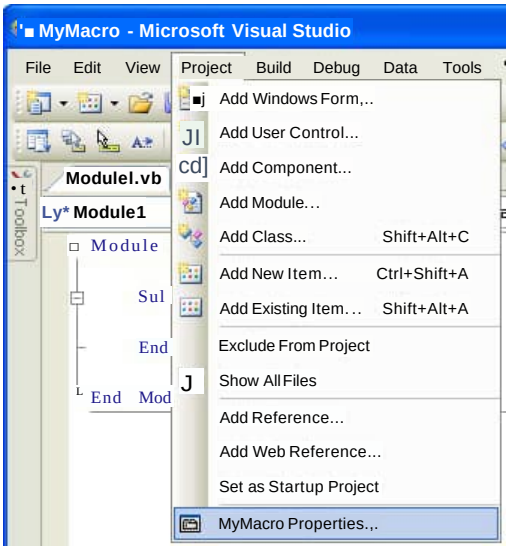


Visual Basic 6.0 users will immediately recognize this as a step they've had to do in the past. While the steps appear similar, the actual details of what's going on under the hood are quite different. Because Visual Basic 6.0 was written to be a COM programming language, it could reference existing COM Type Libraries directly.

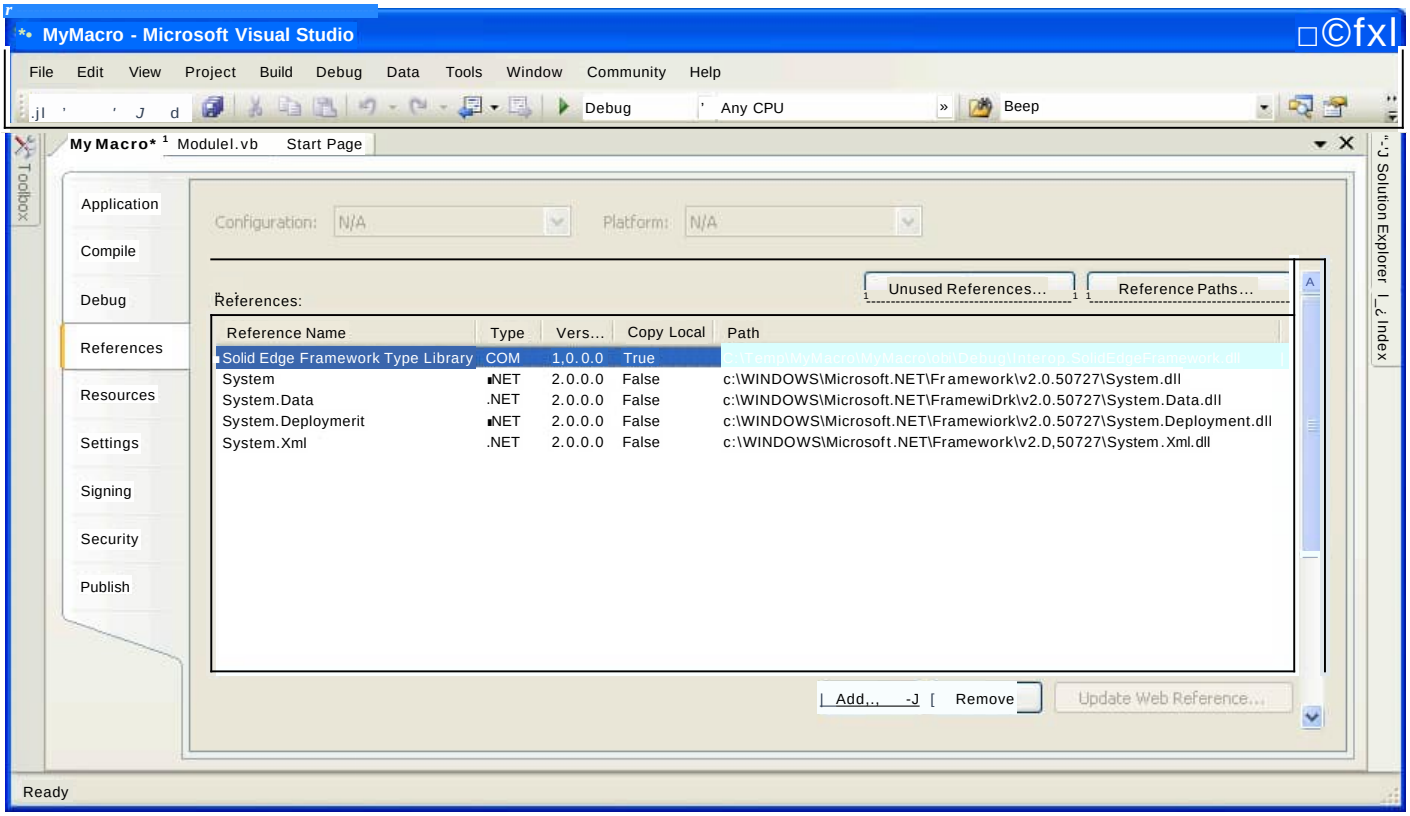
When you add a COM reference in .NET, Visual Studio actually calls a Type Library Importer program that generates an "Interop Assembly" for your project to use. This also means that your executable will be dependent upon the generated Interop Assembly. If you plan on deploying your application to users, you'll need to be sure to include any Interop Assemblies that you've generated for your project.

Viewing Interop Assembly References

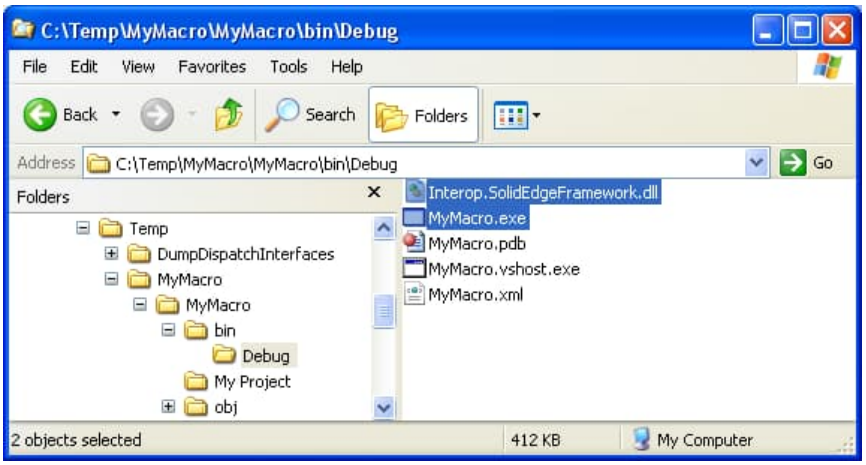
Now that you know how to add Solid Edge API support to your project, let's see how you can view your project's dependencies, specifically Interop Assemblies. Select Project -> (Project Name) Properties from the main menu in Visual Studio.



Once the project properties appear, select the References tab. Here you can see the Interop Assembly, Interop.SolidEdgeFramework.dll, that the Type Library Importer created for your project. Your executable is dependent upon the references that you see in this window.



For generated Interop Assemblies like the Interop.SolidEdgeFramework.dll, they will have the "Copy Local" flag set to True. This means that these .dll's will be placed in the projects output folder when you build the project.



See Also

Connecting To Solid Edge
Starting Solid Edge
Handling 'Application is Busy' and 'Call was Rejected By Callee' errors

Solid Edge FrameworkSupport Type Library

CuttingPlaneLineArrowOffset Property

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Description

Sets and returns offset distance (0.0 - 1.0) for the arrow terminator.

Property type

Read-write property

Syntax

Visual Basic
`Public Property CuttingPlaneLineArrowOffset As Double`

See Also

DrawingViewStyle Object | DrawingViewStyle Members | Solid Edge ST4 - What's New

Solid Edge ST7 SDK

Determining Document Type

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Since all Solid Edge Documents implement the SolidEdgeFramework.SolidEdgeDocument interface, we can use the interface to query generic information about the document before performing specific document environment routines.

See Handling 'Application is Busy' and 'Call was Rejected By Callee' errors for information regarding the use of OleMessageFilter.

Determining Document Type from Visual Basic.NET — Copy Code

```
Imports System.Runtime.InteropServices

Module Program
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objDocument As SolidEdgeFramework.SolidEdgeDocument = Nothing

        Try
            OleMessageFilter.Register()

            ' Connect to a running instance of Solid Edge.
            objApplication = Marshal.GetActiveObject("SolidEdge.Application")

            ' Get a reference to the active document.
            objDocument = objApplication.ActiveDocument

            ' Using Type property, determine document type.
            Select Case objDocument.Type
                Case SolidEdgeFramework.DocumentTypeConstants.igAssemblyDocument
                    Console.WriteLine("Assembly Document")
                Case SolidEdgeFramework.DocumentTypeConstants.igDraftDocument
                    Console.WriteLine("Draft Document")
                Case SolidEdgeFramework.DocumentTypeConstants.igPartDocument
                    Console.WriteLine("Part Document")
                Case SolidEdgeFramework.DocumentTypeConstants.igSheetMetalDocument
                    Console.WriteLine("SheetMetal Document")
                Case SolidEdgeFramework.DocumentTypeConstants.igUnknownDocument
                    Console.WriteLine("Unknown Document")
            End Select
        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter.Revoke()
        End Try
    End Sub
End Module
```

Determining Document Type from C# — Copy Code

```
using Examples;
using System;
using System.Runtime.InteropServices;

namespace SolidEdge.SDK
{
    class Program
    {
        [STAThread]
        static void Main(string[] args)
        {
            SolidEdgeFramework.Application application = null;
            SolidEdgeFramework.SolidEdgeDocument document = null;

            try
            {
                OleMessageFilter.Register();

                // Connect to a running instance of Solid Edge,
                application = (SolidEdgeFramework.Application)
                    Marshal.GetActiveObject("SolidEdge.Application");
```

```
                // Get a reference to the active document.
                document = (SolidEdgeFramework.SolidEdgeDocument)
                    application.ActiveDocument;

                // Using Type property, determine document type,
                switch (document.Type)
                {
                    case SolidEdgeFramework.DocumentTypeConstants.igAssemblyDocument:
                        Console.WriteLine("Assembly Document");
                        break;
                    case SolidEdgeFramework.DocumentTypeConstants.igDraftDocument:
                        Console.WriteLine("Draft Document");
                        break;
                    case SolidEdgeFramework.DocumentTypeConstants.igPartDocument:
                        Console.WriteLine("Part Document");
                        break;
                    case SolidEdgeFramework.DocumentTypeConstants.igSheetMetalDocument:
                        Console.WriteLine("SheetMetal Document");
                        break;
                    case SolidEdgeFramework.DocumentTypeConstants.igUnknownDocument:
                        Console.WriteLine("Unknown Document");
                        break;
                    case SolidEdgeFramework.DocumentTypeConstants.igWeldmentAssemblyDocument:
                        Console.WriteLine("Weldment Assembly Document");
                        break;
                    case SolidEdgeFramework.DocumentTypeConstants.igWeldmentDocument:
                        Console.WriteLine("Weldment Document");
                        break;
                }
            }
            catch (System.Exception ex)
            {
                Console.WriteLine(ex.Message);
            }
            finally
            {
                OleMessageFilter.Revoke();
            }
        }
    }
}
```

Solid Edge Constants Type Library

DimDispTypeConstants Enumeration

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Description

Dimension Display Types

Members

Member	Value	Description
igDimDisplayTypeBasic	5	Dimension Basic
igDimDisplayTypeBlank	8	Dimension Blank
igDimDisplayTypeClassfit	3	Dimension Class Fit
igDimDisplayTypeClassPlusMinus	9	Dimension Class Plus Minus
igDimDisplayTypeFitWithLimits	11	Display the tolerance class along with the tolerance values in parentheses.
igDimDisplayTypeHoleCallout	12	Dimension Hole Callout.
igDimDisplayTypeHoleShaft	10	Display Hole and Shaft tolerance classes as Hole/Shaft.
igDimDisplayTypeHoleShaftTolOnly	13	Display Hole and Shaft tolerance classes as tolerance values.
igDimDisplayTypeHoleShaftWithTol	14	Display Hole and Shaft tolerance classes along with tolerance values in parentheses.
igDimDisplayTypeLimits	4	Dimension Limits
igDimDisplayTypeNominal	1	Dimension Nominal
igDimDisplayTypeReference	6	Dimension Reference
igDimDisplayTypeTolerance	2	Dimension Tolerance
igDimDisplayTypeUnitTolerance	15	Dimension Unit Tolerance.

See Also

Solid Edge ST5 - What's New | Solid Edge ST4 - What's New | Solid Edge ST2 - What's New

Solid Edge FrameworkSupport Type Library

Dimension Object

Members Collapse All

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Solid Edge FrameworkSupport Type Library : Dimension Object

Description

Stores characteristics such as size, distance, and angle.

Object Model

Remarks

The dimensioning objects are associative to the geometry they are measuring. The Dimension object is created only through the Dimensions collection object.

See Also

Dimension Members

Solid Edge FrameworkSupport Type Library

Dimension Object Members

Overview Collapse All

Solid Edge FrameworkSupport Type Library : Dimension Object

Public Methods

AddProjectionLineBreaks	
BreakAlignmentSet	Breaks the alignment set and allows the dimensions to move independently.
BringForward	Changes the display order of the referenced object by moving it in front of the first object that obscures it.
BringToFront	Changes the display order of the referenced object by moving it in front of all other overlapping objects.
ClearProjectionLineBreaks	This method clears the list of breaking dimensions.
Copy	Places a copy of the referenced object on the clipboard.
Cut	Moves the referenced object.
Delete	Deletes the referenced object.
GetComment	Returns the variable table comment.
GetDiscreteValues	Returns discrete values and number of discrete values of the referenced dimension object.
GetDisplayData	This method returns available display data for the referenced Dimension object.
GetJogPointl Distance	Returns the JogPointl distance for the jogged line in a Dimension.
GetJogPoint2	Returns the JogPoint2 distance for the jogged line in a Dimension.
GetJogPointSDistance	Sets the JogPointB distance for the jogged line in a Dimension.
GetKeyPoint	Returns information about a specified key point.
GetMultipleJogPoints	Sets and returns number and offsets of jog points of the referenced dimension object.
GetRelated	Returns the related object at the specified index.
GetRelatedCount	Returns the number of objects to which the referenced Dimension is related.
GetTextOffsets	Returns the offsets for the dimension text of the referenced Dimension object.
Range	Returns the high and low range values for the referenced object.
RemoveFromAlignmentSet	Removes the dimension from alignment set
Select	Selects the referenced object in the active window.
SendBackward	Changes the display order of visible objects by moving the referenced object so that it is in back of the first object behind it.
SendToBack	Changes the display order of visible objects by moving the referenced object behind all overlapping objects.
SetComment	Sets the variable table comment.
SetDiscreteValues	Sets discrete values to the referenced dimension object.

SetJogPointl Distance	Sets the JogPointl distance for the jogged line in a Dimension.
SetJogPoint?	Sets the JogPoint2 distance for the jogged line in a Dimension.
SetJogPointBDistance	Sets the JogPointB distance for the jogged line in a Dimension.
SetKeyPoint	Sets the position for a specified key point.
SetMultipleJogPoints	Sets and returns number and offsets of jog points of the referenced dimension object.
SetPMIEditDirection	Set the edit direction of the PMI dimension. The input constant determines which direction to move on edit value.
SetTextOffsets	Sets the offsets for the dimension text of the referenced Dimension object.
UpdateStatus	Updates the reference dimension object and returns its status after update.

Public Properties

0 AbsoluteTrackDistance	
AngleClockwise	Returns True if the referenced Dimension object measures the angle clockwise.
AngleCompliment	Specifies whether or not the referenced dimension measures the compliment value.
Application	Returns the active application object.
AttributeSets	Returns the AttributeSets collection object for the referenced object.
Axis	Sets and returns the object used as the axis for the referenced object. It must be a line.
BreakDistance	Sets and returns the break distance for the referenced Dimension object.
BreakPosition	Sets and returns the break position for the referenced Dimension object.
ChamferDimensionMode	Returns the mode of chamfer dimension (along axis, callout perpendicular, or callout parallel).
Constraint	Sets and returns the Constraint property for the referenced dimension object.
DimensionType	Sets and returns the type of dimension for the referenced object.
DisplayHalf	Sets and returns the Display Half flag for symmetrical and radial diameter dimensions.
DisplayName	This read-only String property returns the display name for the referenced object.
DisplayType	Sets and returns the display type for the referenced object.
Document	Returns the document to which the referenced object belongs.
Expose	
ExposeName	This read-write String property sets and returns the name of the referenced object when the object is exposed as a custom property.
Formula	Sets and returns the formula of the referenced Variable object.
GroupMemberType	Returns the type of group(Stack, Chain, Coordinate or Not a Group) to which referenced dimension object belongs.

0 HidePMI	Show or Hide the referenced PMI Dimension object.
HoleClassString	Hole Class String. This string must match a class in the standards table, e.g. H7.
Index	Returns the numeric index for the active object.
S InhibitZeroToleranceDisplay	Inhibit display of zero tolerance value
Inspection	Determines whether or not an inspection bubble is applied to the referenced Dimension object.
IsAttributeSetPresent	Returns True if a specified Attribute Set exists on the referenced object.
Jogged	Specifies whether or not the referenced Dimension object is jogged.
Key	Returns a key (a string) to find the object relative to its owning parents.
KeyPointCount	Returns the number of keypoints/handles that are supported by the referenced object.
Layer	Sets and returns the named layer on which the referenced object resides.
0 Leader	Specifies whether or not the referenced object has a leader line.
Hsl LeaderDistance	Sets and returns the leader distance of the referenced Dimension object.
Hsl MeasureLineDirection	Determines the direction of measurement for coordinate dimensions.
MeasurementAxisDirection	Specifies the direction in which the linear distance is measured for the referenced Dimension object.
MeasurementAxisEx	Sets and returns the axis mode of a Dimension.
Name	Returns the name of the referenced object.
OffsetLeader	Specifies whether or not the referenced Dimension object has the offset leader line.
0 OriginLineDirection	Specifies the orientation of the origin line for coordinate dimensions.
OverrideChamferAngleString	Returns the chamfer dimension angle override string.
OverrideString	Sets and returns the override value set on the referenced Dimension object.
Parent	Returns the parent object for the referenced object.
0 PMIPlane	
PMITrackDistance	
PrefixDisplayedText	Returns the evaluated Prefix text.
Prefixinside	Specifies whether or not prefix text for the referenced Dimension object is placed inside the associated basic dimension or inspection box.
PrefixString	Sets and returns the prefix string to be displayed along with the dimension text.
Hsl PrimaryLowerTolerance	Sets and returns the text for the primary lower tolerance value.
Hsl PrimaryToleranceSymbol	Sets and returns the text for the primary tolerance symbol of the referenced Dimension object.
Hsl PrimaryUpperTolerance	Sets and returns the text for the primary upper tolerance value.
Hsl ProjectionArc	Sets and returns the projection arc for the referenced Dimension object.

ProjectionLineAnglel	They return double values that represent the angle between the projection line and the x-axis of the sheet.
ProjectionLineAngle2	They return double values that represent the angle between the projection line and the x-axis of the sheet.
1*21 ProjectionLineDirection	Specifies the projection line direction.
1*21 RadialExtensionDistance	Sets and returns the extension distance of radial and half displayed diameter dimensions. Negative is away from the terminator.
1*21 SecondaryLowerTolerance	Sets and returns the text for the secondary lower tolerance value.
1*21 SecondaryUpperTolerance	Sets and returns the text for the secondary upper tolerance value.
1*21 SecondValue	Returns the chamfer dimension angle.
1*2, ShaftClassString	Shaft Class String. This string must match a class in the standards table, e.g. h7.
1*21 StatusOfDimension	Returns the status(Detached, Error, Driving, Driven) of the referenced dimension object.
1*21 Style	Sets and returns the active style on dimension
1*21 Subfix2DisplayedText	Returns the evaluated Subfix2 text.
1*2, SubfixAlignment	
1*21 SubfixDisplayedText	Returns the evaluated Subfix text.
1*21 SubfixString	Sets and returns the subfix text to be displayed along with the dimension text.
1*21 SubfixString2	Sets and returns the subfix2 text to be displayed along with the dimension text.
1*21 SuffixDisplayedText	Returns the evaluated Suffix text.
1*21 SuffixString	Sets and returns the suffix string to be displayed along with the dimension text.
1*21 SuperfixDisplayedText	Returns the evaluated Superfix text.
1*21 SuperfixString	Sets and returns the superfix text to be displayed along with the dimension text.
1*21 SystemName	This read-only String property returns the system name for the referenced object.
1*21 TerminatorPosition	Determines the position (inside or outside of the projection lines) of the terminator for the referenced Dimension object.
1*2, TextScale	Sets and returns the text scale of the object.
1*21 TrackAngle	Sets and returns the angle between the referenced Dimension object and the object it measures.
1*21 TrackDistance	Sets and returns the track distance of the referenced Dimension object.
1*21 TrueSDSketchDimension	Sets and returns the True3DSketchDimension property for the referenced dimension object.
1*2, Type	Specifies the type of the object being referenced.
1*21 UseStandardLimitAndFitTolerance	Extract tolerances from standards tables when TRUE. Use user-defined tolerances when FALSE. When TRUE, use appropriate DisplayType and use

	HoleClass and/or ShaftClass properties.
Value	Sets or returns the value of the referenced Dimension object.
VariableTableName	Returns the name used in the variable table.
S ZOrder	Specifies the front-to-back order in which an object is displayed relative to other objects.

See Also

Dimension Object

Solid Edge Assembly Type Library

Dimension Property

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Description

Property type

Read-only property

Syntax

Visual Basic
`Public Property Dimension As Object`

See Also

SegmentAngularRelationSd Object | SegmentAngularRelation3d Members | Solid Edge V19 - What's New | Solid Edge V18 SP2 - What's New

Solid Edge FrameworkSupport Type Library

Dimensions Collection Members

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Sunu cuye rrameworkSupport Type Library : Dimensions Collection

Public Methods

AddAngle	Creates a dimension to measure the angle of a line or an arc.
AddAngleBetweenSObjects	Creates an angular dimension between three objects.
AddAngleBetweenObjects	Creates an angular dimension between two given elements.
AddAngularCoordinate	Creates a Dimension object for the position of key points or elements from a common origin.
AddAngularCoordinateOrigin	Creates a Dimension object for the position of key points or elements from a common origin.
AddChamfer	Creates a dimension to measure the angle and the setback of a chamfer.
AddCircularDiameter	Creates a dimension to measure the circular diameter of a circle or arc object.
AddCoordinate	Creates a Dimension object for the angular position of key points from a common origin.
AddCoordinateEx	Creates a Dimension object for the angular position of key points from a common origin.
AddCoordinateOrigin	Creates a Dimension object for the angular position of key points from a common origin.
AddCoordinateOriginEx	Creates a Dimension object for the angular position of key points from a common origin.
AddDimension	Creates a dimension. For creating PMI dimensions a plane must be set (XY plane is the default). The type of the dimension, the number of parents, and the collection object from which it is derived (sheet/profile2d/PMI) dictate the creation of dimensions.
AddDistanceBetweenObjects	Creates a dimension to measure distance between two specified objects.
AddDistanceBetweenObjectsEX	This method creates a dimension to measure distance between two specified objects with tangency options.
AddDistanceIntersectionToIntersection	Adds a dimension to measure distance between two specified intersections.
AddDistanceIntersectionToObject	Adds a dimension from an intersection to a key point on an object.
AddDistanceObjectToIntersection	Adds a dimension from a key point on an object to an intersection.
AddLength	Creates a dimension to measure the length of the input object, which may be a line or an arc or a curve.
AddRadialDiameter	Creates a dimension to measure the radial diameter of a Circle2d or Arc2d object.
AddRadius	Creates a dimension to measure the radius of a Circle2d or an Arc2d object.

AddSymmetricalDiameter	Adds a symmetrical diameter dimension to measure distance between two objects.
ClearAxis	Clears any previously set axis.
ClearPlane	Clears any previously set plane object.
ClearStyle	Clears any previously set style object.
Item	The items in a collection.

Public Properties

AngleCompliment	Specifies whether or not the referenced dimension measures the compliment value.
Application	Returns the active application object.
Axis	Sets and returns the object used as the axis for the referenced object. It must be a line.
AxisMode	Sets and returns the axis mode from which the Dimension measurement axis is found.
Constraint	Sets and returns the Constraint property for the referenced dimension object.
Count	Returns the number of objects in the referenced collection.
DimInitData	Initialization data object used for creating a new dimension using the AddDimensionO API.
Parent	Returns the parent object for the referenced object.
Style	For collection objects, Body objects, RefPlane objects, Sketch objects, CoordinateSystem objects, CommandBarButton objects, SmartFrame2d objects, View objects, and TextCharStyle objects, Style is read-write. For all other objects, Style is read-only.

See Also

Dimensions Collection

Solid Edge FrameworkSupport Type Library

Dimensions Collection

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Sonu Luyt rrameworkSupport Type Library : Dimensions Collection

Description

A collection of Dimension objects.

Remarks

The collection object provides properties to set default values for creating Dimension objects.

See Also

Dimensions Members

Solid Edge ST7 SDK

Working with Dimensions

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Working with Dimensions

This topic contains these sections:

Solid Edge allows you to place and edit dimensions on elements. In the Profile environment, dimension objects control the size and orientation of geometry.

Dimensions can be linear, radial, or angular. Dimensions supply information about the measurements of elements, such as the angle of a line or the distance between two points. A dimension is related to the element on which it is placed. In the Profile environment, the dimensions control the geometry; if the dimension changes, the geometry updates.

In a Part document, the Dimensions collection is accessed through the Profile object. The Dimensions collection provides the methods for placing dimensions and for iterating through all the dimensions on the entire sheet or profile.

Linear Dimensions
A linear dimension measures the distance between two or more elements, the length of a line, or an arc's length. For a complete description of the properties that define how a linear dimension is placed, see the Programming with Solid Edge on-line Help.

Radial Dimensions
Radial dimensions measure the radius or diameter at a point on the element. These dimensions are similar except that they show the radius or diameter value depending on the type. With the ProjectionArc and TrackAngle properties, you can define the measurement point on the element.

Angular Dimensions
Angular dimensions measure the angle between two lines or three points. An angular dimension defines two intersecting vectors and four minor sectors. These sectors are distinguished according to whether the angle is measured in the sector where the vector direction goes outward from the intersection point or comes inward, and whether the angle is measured in the clockwise or counterclockwise direction.

The angles are always measured in the counterclockwise direction with both vector directions going outward from the intersection point (sector one condition). To measure in any other angle, certain properties are set so that the dimension object modifies the vector direction and computes the angle.

Placing Dimensions

In the Profile environment, driving dimensions control the elements to which they refer. When you edit a driving dimension, the geometry of the element that is related to that dimension is modified.

You can place dimensions only on existing elements. A set of Add methods is provided on the Dimensions collection, one for each type of dimension. The element to which the dimension is attached determines the type of dimension (driving or driven) that will be placed. The Add methods on the Dimensions collection object take minimal input and place the dimensions with specific default values.

When you place dimensions between two elements interactively, the dimensions are measured at a specific location on an element. For example, when you place a dimension between the end points of two lines, you select one end of each line. When you place dimensions through automation, you specify a point on the element and a key point flag to define the dimension.

In the following program, four lines are drawn and connected with key point relationships. The lengths of two of the lines and the distance between two lines are dimensioned. The dimension is set to be a driving dimension so it will control the length and position of the geometry. The sample also shows how to modify a dimension style by changing the units of measurement of one of the dimensions to meters.

| See Handling 'Application is Busy' and 'Call was Rejected By Callee' errors for information regarding the use of OleMessageFilter.

Placing Dimensions in Visual Basic.NET Copy Code

```
Imports SolidEdgeConstants
Imports System.Runtime.InteropServices

Module Program
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objDocuments As SolidEdgeFramework.Documents = Nothing
        Dim objPart As SolidEdgePart.PartDocument = Nothing
        Dim objProfileSets As SolidEdgePart.ProfileSets = Nothing
        Dim objProfileSet As SolidEdgePart.ProfileSet = Nothing
        Dim objProfiles As SolidEdgePart.Profiles = Nothing
        Dim objProfile As SolidEdgePart.Profile = Nothing
        Dim objRefPlanes As SolidEdgePart.RefPlanes = Nothing
        Dim objRefPlane As SolidEdgePart.RefPlane = Nothing
        Dim objLines2d As SolidEdgeFrameworkSupport.Lines2d = Nothing
        Dim aLine2d(0 To 3) As SolidEdgeFrameworkSupport.Line2d
        Dim objRelations2d As SolidEdgeFrameworkSupport.Relations2d = Nothing
        Dim objDimensions As SolidEdgeFrameworkSupport.Dimensions = Nothing
        Dim objDimension As SolidEdgeFrameworkSupport.Dimension = Nothing
        Dim objDimStyle As SolidEdgeFrameworkSupport.DimStyle = Nothing
        Dim i As Integer

        Try
            OleMessageFilter.Register()

            ' Connect to a running instance of Solid Edge
            objApplication = Marshal.GetActiveObject("SolidEdge.Application")

            ' Get a reference to the documents collection
            objDocuments = objApplication.Documents

            ' Add a Part document
```

```
objPart = objDocuments.Add("SolidEdge.PartDocument")

' Get a reference to the ref planes collection
objRefPlanes = objPart.RefPlanes

' Get a reference to the first reference plane
objRefPlane = objRefPlanes.Item(l)

' Get a reference to the profile sets collection
objProfileSets = objPart.ProfileSets

' Add a new profile set
objProfileSet = objProfileSets.Add()

' Get a reference to the profiles collection
objProfiles = objProfileSet.Profiles

[1] Add a new profile on the selected reference plane
objProfile = objProfiles .Add(objRefPlane)

' Get a reference to the lines 2d collection
objLines2d = objProfile .Lines2d

' Get a reference to the relations 2d collection
objRelations2d = objProfile.Relations2d

' Get a reference to the dimensions collection
objDimensions = objProfile .Dimensions

' Draw the geometry.
aLine2d(0) = objLines2d.AddBy2Points(0, 0, 0.1, 0)
aLine2d(l) = objLines2d.AddBy2Points(0.1, 0, 0.1, 0.1)
aLine2d(2) = objLines2d.AddBy2Points(0.1, 0.1, 0, 0.05)
aLine2d(3) = objLines2d.AddBy2Points(0, 0.05, 0, 0)

' Add endpoint relationships between the lines
objRelations2d.AddKeypoint( _
  aLine2d(0), _
  KeypointIndexConstants.igLineEnd, _
  aLine2d(l), _
  KeypointIndexconstants.igLineStart)

objRelations2d.AddKeypoint( _
  aLine2d(l), _
  KeypointIndexconstants.igLineEnd, _
  aLine2d(2), _
  KeypointIndexconstants.igLineStart)

objRelations2d.AddKeypoint( _
  aLine2d(2), _
  KeypointIndexconstants.igLineEnd, _
  aLine2d(3), _
  KeypointIndexconstants.igLineStart)

objRelations2d.AddKeypoint( _
              aLine2d(3), _
  KeypointIndexconstants.igLineEnd, _
  aLine2d(0), _
  KeypointIndexConstants.igLineStart)

' Add dimensions, and change the dimension units to meters
objDimension = objDimensions.AddLength(aLine2d(l))

objDimension.Constraint = True
objDimStyle = objDimension.Style
objDimStyle.Primaryunits = _
  DimLinearUnitConstants.IgDimStyleLinearMeters

objDimension = objDimensions.AddLength(aLine2d(3))

objDimension.Constraint = True
objDimStyle = objDimension.Style
objDimStyle.Primaryunits = _
  DimLinearUnitConstants.IgDimStyleLinearMeters

objDimension = objDimensions.AddDistanceBetweenObjects( _
  aLine2d(l), 0.1, 0.1, 0, False, _
  aLine2d(2), 0, 0.05, 0, False)
```

```
            objDimension.Constraint = True
            objDimStyle = objDimension.Style
            objDimStyle.Primaryunits = _
              DimLinearUnitConstants.igDimStyleLinearMeters

        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter.Revoke()
        End Try
    End Sub
End Module
```

Placing Dimensions in C# Copy Code

```
using SolidEdgeConstants;
using System;
using System.Reflection;
using System.Runtime.InteropServices;

namespace SolidEdge.SDK
{
    class Program
    {
        [STAThread]
        static void Main(string[] args)
        {
            SolidEdgeFramework.Application application = null;
            SolidEdgeFramework.Documents documents = null;
            SolidEdgePart.PartDocument part = null;
            SolidEdgePart.ProfileSets profileSets = null;
            SolidEdgePart.ProfileSet profileSet = null;
            SolidEdgePart.Profiles profiles = null;
            SolidEdgePart.Profile profile = null;
            SolidEdgePart.RefPlanes refPlanes = null;
            SolidEdgePart.RefPlane refPlane = null;
            SolidEdgeFrameworkSupport.Lines2d lines2d = null;
            SolidEdgeFrameworkSupport.Line2d[] aLine2d =
              new SolidEdgeFrameworkSupport.Line2d[4];
            SolidEdgeFrameworkSupport.Relations2d relations2d = null;
            SolidEdgeFrameworkSupport.Dimensions dimensions = null;
            SolidEdgeFrameworkSupport.Dimension dimension = null;
            SolidEdgeFrameworkSupport.DimStyle dimStyle = null;

            try
            {
                OleMessageFilter.Register();

                // Connect to a running instance of Solid Edge
                application = (SolidEdgeFramework.Application)
                  Marshal.GetActiveObject("SolidEdge.Application");

                // Get a reference to the documents collection
                documents = application.Documents;

                // Add a Part document
                part = (SolidEdgePart.PartDocument)
                  documents.Add("SolidEdge.PartDocument", Missing.Value);

                // Get a reference to the ref planes collection
                refPlanes = part.RefPlanes;

                // Get a reference to the first reference plane
                refPlane = refPlanes.Item(1);

                // Get a reference to the profile sets collection
                profileSets = part.ProfileSets;

                // Add a new profile set
                profileSet = profileSets.Add();

                // Get a reference to the profiles collection
                profiles = profileSet.Profiles;

                // Add a new profile on the selected reference plane
                profile = profiles.Add(refPlane);

                // Get a reference to the lines 2d collection
                lines2d = profile.Lines2d;
```

```
                // Get a reference to the relations 2d collection
                relations2d = (SolidEdgeFrameworkSupport.Relations2d)
                  profile.Relations2d;

                // Get a reference to the dimensions collection
                dimensions = (SolidEdgeFrameworkSupport.Dimensions)
                  profile.Dimensions;

                // Draw the geometry.
                aLine2d[0] = lines2d.AddBy2Points(0, 0, 0.1, 0);
                aLine2d[1] = lines2d.AddBy2Points(0.1, 0, 0.1, 0.1);
                aLine2d[2] = lines2d.AddBy2Points(0.1, 0.1, 0, 0.05);
                aLine2d[3] = lines2d.AddBy2Points(0, 0.05, 0, 0);

                // Add endpoint relationships between the lines
                relations2d.AddKeypoint(
                  aLine2d[0],
                  (int) KeypointIndexConstants.igLineEnd,
                  aLine2d[1],
                  (int)KeypointIndexConstants.igLineStart,
                  Missing.Value);

                relations2d.AddKeypoint(
                  aLine2d[1],
                  (int)KeypointIndexConstants.igLineEnd,
                  aLine2d[2],
                  (int) KeypointIndexConstants.igLineStart,
                  Missing.Value);

                relations2d.AddKeypoint(
                  aLine2d[2],
                  (int)KeypointIndexConstants.igLineEnd,
                  aLine2d[3],
                  (int) KeypointIndexConstants.igLineStart,
                  Missing.Value);

                relations2d.AddKeypoint(
                  aLine2d[3],
                  (int) KeypointIndexConstants.igLineEnd,
                  aLine2d[0],
                  (int)KeypointIndexConstants.igLineStart,
                  Missing.Value);

                // Add dimensions, and change the dimension units to meters
                dimension = dimensions.AddLength(aLine2d[1]);
                dimension.Constraint = true;
                dimStyle = dimension.Style;
                dimStyle.PrimaryUnits = SolidEdgeFrameworkSupport.DimLinearUnitConstants.igDimStyleLinearMeters;

                dimension = dimensions.AddLength(aLine2d[3]);
                dimension.Constraint = true;
                dimStyle = dimension.Style;
                dimStyle.Primaryunits = SolidEdgeFrameworkSupport.DimLinearUnitConstants.igDimStyleLinearMeters;

                dimension = dimensions.AddDistanceBetweenObjects(
                  aLine2d[1], 0.1, 0.1, 0, false, aLine2d[2], 0, 0.05, 0, false);
                dimension.Constraint = true;
                dimStyle = dimension.Style;
                dimStyle.PrimaryUnits = SolidEdgeFrameworkSupport.DimLinearUnitConstants.igDimStyleLinearMeters;

            }
            catch (System.Exception ex)
            {
                Console.WriteLine(ex.Message);
            }
            finally
            {
                OleMessageFilter.Revoke();
            }
        }
    }
}
```

Dimensions via Variables

When working interactively with the Variable Table, both variables and dimensions are displayed in the table. This enables you to create formulas using the values of dimensions and also have formulas that drive the values of dimensions. In this workflow, there is no apparent difference between variables and dimensions. Internally, however, variables and dimensions are two distinct types of objects that have their own unique collections, properties, and methods.

The Variables collection allows you to work with dimensions in the context of variables through several methods on the collection. These methods include Edit, GetFormula, GetName, PutName, Query, and Translate. The following program uses dimensions through the Variables collection. The following programs assume that Solid Edge is running and in the Profile environment.

See Handling 'Application is Busy' and 'Call was Rejected By Callee' errors for information regarding the use of OleMessageFilter.

Accessing Dimensions through the Variable Table in Visual Basic .NET Copy

```
Imports System.Runtime.InteropServices

Module Program
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objPart As SolidEdgePart.PartDocument = Nothing
        Dim objProfileSets As SolidEdgePart.ProfileSets = Nothing
        Dim objProfileSet As SolidEdgePart.ProfileSet = Nothing
        Dim objProfiles As SolidEdgePart.Profiles = Nothing
        Dim objProfile As SolidEdgePart.Profile = Nothing
        Dim objVariables As SolidEdgeFramework.Variables = Nothing
        Dim objVariable As Object = Nothing
        Dim objLines2d As SolidEdgeFrameworkSupport.Lines2d = Nothing
        Dim objLine2d As SolidEdgeFrameworkSupport.Line2d = Nothing
        Dim objDimensions As SolidEdgeFrameworkSupport.Dimensions = Nothing
        Dim objDimension1 As SolidEdgeFrameworkSupport.Dimension = Nothing
        Dim objDimension2 As SolidEdgeFrameworkSupport.Dimension = Nothing
        Dim objVariableList As SolidEdgeFramework.VariableList = Nothing
        Dim sName As String
        Dim sFormula As String

        Try
            OleMessageFilter.Register()

            ' Connect to a running instance of Solid Edge
            objApplication = Marshal.GetActiveObject("SolidEdge.Application")

            ' Get a reference to the active document
            objPart = objApplication.ActiveDocument

            ' Get a reference to the active document
            objProfileSets = objPart.ProfileSets

            ' Get a reference to the profile set
            objProfileSet = objProfileSets.Item(objProfileSets.Count)

            ' Get a reference to the profiles collection
            objProfiles = objProfileSet.Profiles

            ' Get a reference to the profile
            objProfile = objProfiles.Item(1)

            ' Get a reference to the variables collection
            objVariables = objPart.Variables

            ' Get a reference to the lines2d collection
            objLines2d = objProfile.Lines2d

            ' Get a reference to the dimensions collection
            objDimensions = objProfile.Dimensions

            ' Create a new line
            objLine2d = objLines2d.AddBy2Points(0, 0, 0.1, 0.1)

            ' Place a dimension on the line to control its length
            objDimension1 = objDimensions.AddLength(objLine2d)

            ' Make the dimension a driving dimension
            objDimension1.Constraint = True

            ' Create a second line
            objLine2d = objLines2d.AddBy2Points(0, 0.1, 0.1, 0.2)

            ' Place a dimension on the line to control its length
            objDimension2 = objDimensions.AddLength(objLine2d)

            ' Make the dimension a driving dimension
            objDimension2.Constraint = True

            ' Assign a name to the dimension placed on the first line
            objVariables.PutName(objDimension1, "Dimension1")
```

```
            ' Retrieve the system name of the second dimension, and display
            ' it in the debug window


            sName = objVariables.GetName(objDimension2)
            Console.WriteLine(String.Format("Dimension = {0}", sName))

            ' Edit the formula of the second dimension to be half
            ' the value of the first dimension
            objVariables.Edit(sName, "Dimensionl/2.0")

            ' Retrieve the formula from the dimension, and print it to the
            ' debug window to verify
            sformula = objVariables.GetFormula(sName)
            Console.WriteLine(String.Format("Formula = {0}", sFormula))

            ' This demonstrates the ability to reference a dimension object
            ' by its name
            objDimensionl = objVariables.Translate("Dimensionl")

            ' To verify a dimension object was returned, change its
            ' TrackDistance property to cause the dimension to change
            objDimensionl.TrackDistance = objDimensionl.TrackDistance * 2

            ' Use the Query method to list all all user-defined
            ' variables and user-named Dimension objects and
            ' display in the debug window
            objVariableList = objVariables.Query("*")
            For Each objVariable In objVariableList
                Console.WriteLine(objVariables.GetName(objVariable))
            Next

        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter.Revoke()
        End Try
    End Sub
End Module
```

Accessing Dimensions through the Variable Table in C# Copy Code

```
using SolidEdgeFramework;
using System;
using System.Runtime.InteropServices;

namespace SolidEdge.SDK
{
    class Program
    {
        [STAThread]
        static void Main(string[] args)
        {
            SolidEdgeFramework.Application application = null;
            SolidEdgePart.PartDocument part = null;
            SolidEdgePart.ProfileSets profilesets = null;
            SolidEdgePart.ProfileSet profileSet = null;
            SolidEdgePart.Profiles profiles = null;
            SolidEdgePart.Profile profile = null;
            SolidEdgeFramework.Variables variables = null;
            Object oVariable = null;
            SolidEdgeFrameworkSupport.Lines2d lines2d = null;
            SolidEdgeFrameworkSupport.Line2d line2d = null;
            SolidEdgeFrameworkSupport.Dimensions dimensions = null;
            SolidEdgeFrameworkSupport.Dimension dimensionl = null;
            SolidEdgeFrameworkSupport.Dimension dimension2 = null;
            SolidEdgeFramework.VariableList variableList = null;
            String sName;
            String sFormula;

            try
            {
                OleMessageFilter.Register();

                // Connect to a running instance of Solid Edge
                application = (SolidEdgeFramework.Application)
                  Marshal.GetActiveObject("SolidEdge.Application");

                // Get a reference to the active document
```

```
part = (SolidEdgePart.PartDocument)
  application.ActiveDocument;


// Get a reference to the profile sets collection
profileSets = part.ProfileSets;

// Get a reference to the profile set
profileset = profileSets.Item/profileSets.Count);

// Get a reference to the profiles collection
profiles = profileset.Profiles;

// Get a reference to the profile
profile = profiles.Item(l);

// Get a reference to the variables collection
variables = (SolidEdgeFramework.Variables)part.Variables;

// Get a reference to the lines2d collection
lines2d = profile.Lines2d;

// Get a reference to the dimensions collection
dimensions = (SolidEdgeFrameworkSupport.Dimensions)
  profile.Dimensions;

// Create a line
line2d = lines2d.AddBy2Points(0, 0, 0.1, 0.1);

// Place a dimension on the line to control its length
dimensionl = dimensions.AddLength(line2d);

// Make the dimension a driving dimension
dimensionl.Constraint = true;

// Create a second line
line2d = lines2d.AddBy2Points(0, 0.1, 0.1, 0.2);

// Place a dimension on the line to control its length
dimension2 = dimensions.AddLength(line2d);

// Make the dimension a driving dimension
dimension2.Constraint = true;

// Assign a name to the dimension placed on the first line
variables.PutName(dimensionl, "Dimensionl");

// Retrieve the system name of the second dimension, and display
// it in the debug window
sName= variables.GetName(dlmension2);
Console.WriteLine(String.Format("Dimension = {0}", sName));

// Edit the formula of the second dimension to be half
// the value of the first dimension
variables.Edit (sName, "Dimensionl/2.0");

// Retrieve the formula from the dimension, and print it to the
// debug window to verify
sFormula = variables.GetFormula(sName);
Console.WriteLine(String.Format("Formula = {0}", sFormula));

// This demonstrates the ability to reference a dimension object
// by its name
dimensionl = (SolidEdgeFrameworkSupport.Dimension)
  variables.Translate/"Dimensionl");

//To verify a dimension object was returned, change its
// TrackDistance property to cause the dimension to change
dimensionl.TrackDistance = dimensionl.TrackDistance * 2;

// Use the Query method to list all all user-defined
// variables and user-named Dimension objects and
// display in the debug window
variableList = (SolidEdgeFramework.VariableList)
  variables.Query/"*", null, null, false);

for (int i = 1; i <= variableList.Count; i++)
{
    oVariable = variableList.Item(i);
```

```
                    Console.WriteLine(variables.GetName(oVariable));
                }
            }
            catch (System.Exception ex)
            {
                Console.WriteLine (ex.Message);
            }
            finally
            {
                OleMessageFilter.Revoke();
            }
        }
    }
}
```

Solid Edge FrameworkSupport Type Library

Dimensionstyle Object Members

Overview Collapse All

Solid Edge FrameworkSupport Type Library : DimensionStyle Object

Public Properties

S AboveGap	Sets and returns the gap between the dimension text and the dimension line when the text position is set to above.
AllAroundMultiplier	
AllowNegativeValueForCoordinate	Sets and returns the option to allow to set the negative values on coordinate dimensions
AllowOriginValueChangeForCoordinate	Sets and returns the option to allow to change the coordinate origin dimension value
AngularDecimalRoundOff	Sets and returns the angular round-off when angular units is set to degrees or radians and round off type is set to decimal.
AngularDMSRoundOff	Sets and return the angular round off when angular units is set to Deg-Min-Sec.
AngularFractionRoundOff	Sets and returns the angular round off when angular units is set to degrees or radians and round off type is set to fraction.
AngularRoundOffType	Sets and returns the angular round off type (decimal or fraction) for the referenced dimension style object.
AngularToleranceDecimalRoundoff	Sets and returns the angular tolerance decimal roundoff
AngularToleranceDMSRoundOff	Sets and returns the angular tolerance Degree Minute Second roundoff
AngularToleranceFractionRoundoff	Sets and returns the angular tolerance fractional roundoff
AngularToleranceRoundOffType	Sets and returns the angular tolerance round off type
AngularToleranceUnits	Sets and returns the angular tolerance units
AngularUnits	Sets and returns the angular dimensioning unit separate from the working units of the file.
ApplyBreakLineGap	
AttributeSets	Returns the AttributeSets collection object for the referenced object.
AutoCenterProjectionLine	Sets and returns the center projection line.
AutoPhase	If True, specifies that the negative dash gaps for the referenced style object are automatically extended based on the size of the edge.
AutoSuppressDiameter	Specifies whether or not to automatically suppress the diameter symbol when a circle is measured by a radial or circular diameter dimension.
CalloutTextAspectRatio	Sets the callout text aspect ratio for the style.
CenterLineType	Sets and returns the centerline type name for the referenced object.

CenterMarkExtension	Specifies whether keypoint extensions are used for the center mark.
CenterMarkSize	Sets and returns the size of the center mark for the referenced style object.
ChamferUse45DegreeCharacter	Specifies chamfer 45 degree character use.
CommonOrigin	Sets and returns the type of the terminator displayed at the common origin of a coordinate dimension group.
ConnectorColor	
ConnectorLineType	
ConnectorWidth	
0 CoordinateAutoJog	Determines if the referenced coordinate dimension is jogged automatically when the dimension value is less than the coordinate stack pitch.
0 CoordinateStackPitch	Sets and returns the minimum coordinate dimension value after which the coordinate dimension will be jogged when the CoordAutoJog property is set to True.
CoordinateTextOrientation	Sets and returns the text orientation of Coordinate Dimension
CoordinateTextPosition	Sets and returns the position of the coordinate text in relation to the projection line.
DatumFrameDashes	Sets and returns the datum frame dashes for the referenced DatumFrame object.
DatumFrameLineTermThickness	Sets and returns the datum frame line terminator thickness.
DatumFrameShape	Sets and returns the datum frame shape.
DatumTargetTerminatorType	Sets and returns the datum target terminator type.
DatumTerminatorGap	Sets and returns the datum terminator gap.
DatumTerminatorType	Sets and returns the datum terminator type.
Delimiter	Sets and returns the delimiter for the referenced object.
Description	Returns a user-specified description of the referenced object.
DimensionScale	Sets and returns the scale factor of the referenced object.
0 DimensionScaleMode	Sets and returns the scale mode of the referenced object.
DisplayCoordinateOriginText	Sets and returns the display of coordinate origin text.
DisplayDegreeSymbolOnNumericTolerance	Display degree symbol on numeric angular tolerance values.
DrivenColor	Sets and returns the color in which driven dimensions are displayed.
DrivingColor	Sets and returns the color in which driving dimensions are displayed.
DualDisplay	Turns on and off the dual display for the referenced object.
DualModeVerticalGap	Sets and returns the vertical gap distance between Dimension Value text when the DualModeDisplay property is True.

0 DualUnitJustification	Sets and returns the dual unit justification,Only applicable for dual unit position below primary
DualUnitPosition	Sets and returns the dual unit position
EnableAutoJogForCoordinate	Sets and returns the option to allow enable automatic jogging on coordinate dimensions
ErrorColor	Sets and returns the color in which dimensions in error are displayed.
Font	Sets and returns the name of the font.
0 FontStyle	Sets and returns the style of the font for the text of the referenced object.
FractionSeparator	
FreeSpaceTerminatorType	Sets and returns the type of terminator for an annotation whose terminator end is placed on a free-space point.
Height	Returns the height of the referenced object.
HeightPMI	
HoleCalloutCounterbore	This read-write String property sets and returns the hole callout counterbore settings for the referenced object.
HoleCalloutCounterboreThreaded	This read-write String property sets and returns the hole callout counterbore threaded settings for the referenced object.
HoleCalloutCountersink	This read-write String property sets and returns the hole callout countersink settings for the referenced object.
HoleCalloutCountersinkThreaded	This read-write String property sets and returns the hole callout countersink threaded settings for the referenced object.
HoleCalloutSimple	This read-write String property sets and returns the hole callout simple settings for the referenced object.
0 HoleCalloutSimpleThreaded	This read-write String property sets and returns the hole callout simple threaded settings for the referenced object.
0 HoleShaftToleranceSeparatorType	Sets and returns class fit hole/shaft Separator type.
HorizontalBoxGap	Sets and returns the horizontal gap for the dimension text box.
HorizontalToleranceGap	Sets and returns the horizontal tolerance gap between the dimension value and tolerance text.
S InhibitZeroToleranceDisplay	
IsAttributeSetPresent	Returns True if a specified Attribute Set exists on the referenced object.
LeaderLineLength	Sets and returns the default leader distance.
LimitTextArrangement	Sets and returns the arrangement of the dimension text for a limit dimension.
0 LineConnect	Specifies whether or not a line is drawn between the terminators of a dimension when text and terminators are to be outside of the measured value.
LineSpacing	Sets and returns the amount of space between two lines in a paragraph for the referenced object.

0 LineWidth	Sets and returns the line width (in paper space units) for the referenced object.
LowerCaseMultiplicationSymbol	This read-write Boolean property specifies whether the referenced object uses the lowercase multiplication symbol (x) for chamfer dimensions.
MatchPrimaryDelimiter	
MaximumSubunits	Sets and returns the maximum number of subunits displayed by the associated dimension before truncating and adding to the unit.
MaximumViewAngle	Sets and returns the maximum view angle after which the associated dimension will not be displayed in a 3-D environment.
Name	Returns the name of the referenced object.
0 NTSSymbol	Sets and returns the display characteristics of a not-to-scale dimension for the referenced object.
OffsetLeaderType	Sets and returns the type of offset leader to be drawn.
OriginTerminatorType	Sets and returns the terminator type for the origin terminator of the referenced object.
OrigTerminatorSize	
OverridePulledOutText	
OverridePulledOutText?	Override dimension text position and orientation when dimension dragged along the dimension line.
Parent	Returns the parent object for the referenced object.
PlacePrefixInsideBasicDimensionBox	Sets and returns the placement of prefix inside basic dimension box.
0 PrefixSuffixGap	Sets and returns the gap between prefix/suffix string and the dimension value text.
PrimaryDecimalRoundOff	Sets and returns the accuracy for the primary decimal value.
PrimaryFractionRoundOff	Sets and returns the accuracy for the primary fraction value.
PrimaryLeadingZeros	Specifies whether or not the primary leading zeros are on.
PrimaryLinearToleranceDecimalRoundOff	Sets and returns the primary linear tolerance decimal roundoff
PrimaryLinearToleranceFractionRoundOff	Sets and returns the primary linear tolerance fractional roundoff
PrimaryLinearToleranceFractionSeparator	Sets and returns the primary linear tolerance fraction separator
PrimaryLinearToleranceMaximumSubunits	Sets and returns the primary linear tolerance maximum subunits
PrimaryLinearToleranceRoundOffType	Sets and returns the primary linear tolerance round off type
PrimaryLinearToleranceSubunitsLabel	Sets and returns the primary linear tolerance subunit label
PrimaryLinearToleranceUnitLabel	Sets and returns the primary linear tolerance unit label
PrimaryLinearToleranceUnits	Sets and returns the primary linear tolerance units
PrimarySubunitsLabel	Sets and returns the text string for the primary subunits label.
PrimaryTrailingZeros	Specifies whether or not primary trailing zeros are on.
PrimaryUnitLabel	Sets and returns the text string of the primary unit's label.

PrimaryUnits	Sets and returns the dimensioning unit separate from the working units of the file.
ProjectionAngle	Sets and returns the angle of the projection lines with respect to the dimension text.
ProjectionBreakGap	This property creates an override style on the dimension and updates the break gap.
ProjectionExtensionGap	Sets and returns the distance by which the projection lines are extended beyond the dimension line.
ProjectionLineDisplay	Specifies the projection line direction.
ProjectionOffsetGap	Sets and returns the gap between the object being measured and the projection line.
ProjectionOffsetGapAlignment	Specifies whether or not the projection offset is aligned.
ProjectionToleranceZonePosition	Sets and returns the projection tolerance position for the dimension.
PulledOutText2Orientation	Sets and returns the text orientation of OverridePulledOutText2.
[±=1 PulledOutText2Position	Sets and returns the text orientation of OverridePulledOutText2.
1*21 PulledOutTextOrientation	
1*21 PulledOutTextPosition	
H2I Ratio	Sets and returns the text aspect ratio.
1*21 RoundOffType	Sets and returns the round off type.
1*21 RoundUp	Sets and returns round up type (round up all or round up odd) for the referenced dimension style object.
1*21 SecondaryDecimalRoundOff	Sets and returns the accuracy for the secondary decimal value.
1*21 SecondaryDelimiter	
1*21 SecondaryFractionRoundOff	Sets and returns the accuracy for secondary fraction value.
1*21 SecondaryLeadingZeros	Specifies whether or not secondary leading zeros are on.
1*21 SecondaryLinearFractionSeparator	Sets and returns the secondary linear fraction separator
1*21 SecondaryLinearRoundOffType	Sets and returns the secondary linear roundoff type.
1*21 SecondaryLinearToleranceDecimalRoundoff	Sets and returns the secondary linear tolerance decimal roundoff
1*21 SecondaryLinearToleranceFractionRoundOff	Sets and returns the secondary linear tolerance fractional roundoff
1*21 SecondaryLinearToleranceFractionSeparator	Sets and returns the secondary linear tolerance fraction separator
1*21 SecondaryLinearToleranceMaximumSubunits	Sets and returns the secondary linear tolerance maximum subunits
1*21 SecondaryLinearToleranceRoundOffType	Sets and returns the secondary linear tolerance round off type
1*21 SecondaryLinearToleranceSubunitsLabel	Sets and returns the secondary linear tolerance subunit label
1*21 SecondaryLinearToleranceUnitLabel	Sets and returns the secondary linear tolerance unit label
1*21 SecondaryLinearToleranceUnits	Sets and returns the secondary linear tolerance units
1*21 SecondarySubunitsLabel	Sets and returns the text string for the secondary subunits label.

Hsl SecondaryToleranceEquivalent	Specifies whether or not the secondary tolerance is equivalent.
SecondaryTrailingZeros	Specifies whether or not the secondary trailing zeros are on.
O SecondaryUnitLabel	Sets and returns the text string for the secondary units label.
Hsl SecondaryUnitRoundOff	Sets and returns the round off type for secondary units.
SecondaryUnits	Sets and returns the secondary dimensioning unit.
Hsl SecondaryUnitSeparator	Sets and returns the secondary unit separator.
Hsl ShowOriginLineForCoordinate	Sets and returns the option to allow to display the coordinate origin dimension line
Hsl SlotCalloutCounterbore	Sets and returns text for Counterbore slots.
Hsl SlotCalloutSimple	Sets and returns text for Simple slots.
Hsl SmartDepthHoleFinite	Sets and returns SmartDepth for finite holes.
Hsl SmartDepthHoleThru	Sets and returns SmartDepth for through holes.
Hsl SmartDepthSlotFinite	Sets and returns SmartDepth text for finite slots.
Hsl SmartDepthSlotThru	Sets and returns SmartDepth text for through slots.
Hsl SmartDepthThreadFinite	Sets and returns SmartDepth for finite threads.
Hsl SmartDepthThreadThru	Sets and returns SmartDepth for through threads.
Hsl StackPitch	Sets and returns the minimum distance track distance maintained between individual dimensions within a group.
Hsl SymbolFont	Sets and returns the Symbol Font used by this style.
Hsl SymbolGap	Sets and returns the gap between the dimension symbol and the dimension value text.
Hsl SymbolPosition	Sets and returns the symbol position with respect to the dimension text value.
Hsl TerminatorDisplay	Sets and returns the terminator display.
Hsl TerminatorInsideLimit	Toggles the associated dimension to the outside of the projection when the dimension value is less than the terminator inside limit.
Hsl TerminatorSize	Sets and returns the size of the terminator for the referenced DimensionStyle object.
Hsl TerminatorType	Sets and returns the type of terminator for both origin and measure.
Hsl TextAutoFillBackground	
Hsl TextClearanceGap	Sets and returns the gap distance between the dimension text and the dimension line.
Hsl TextOrientation	Sets and returns the orientation of the dimension text with respect to the dimension plane axis.
Hsl TextPosition	Sets and returns the text position.
Hsl TextWeight	Sets and returns the text line width.
Hsl ThreeSidedSymbolOffsetGap	Sets and returns the offset gap value of Three Sided Weld Symbol.

0 ToleranceHeightRatio	Sets and returns the tolerance height.
ToleranceTextHorizontalAlignment	Sets and returns horizontal alignment for upper and lower tolerance.
ToleranceTextVerticalAlignment	Sets and returns vertical alignment for tolerance text.
S ToleranceWeight	Sets and returns the tolerance text weight.
UseTextSizeForCombinedTolValueDisplay	If True, specifies that the dimension text size will be used for the tolerance when upper and lower tolerance have the same absolute value but opposite signs.
VerticalBoxGap	Sets and returns the vertical gap for the text box.
VerticalLimitsGap	Sets and returns the vertical limits gap between the limit value texts.
VerticalToleranceGap	Sets and returns the vertical tolerance gap between the dimension value and the tolerance text.
0 ViewCaptionLocation	Sets and returns the caption location of the drawing view.
ViewCPLCaptionLocation	Sets and returns the caption location of the cutting plane line display.
ViewCuttingPlaneLineDisplay	Sets and returns the direction of the terminator.
ViewLineArrowOffset	
ViewLineType	Sets and returns the name of the view line type to be displayed for the referenced object.
ViewLineWidth	Sets and returns the line width of the viewing and cutting plane lines.
ViewPlaneLineDisplay	Sets and returns the number of terminator lines displayed.
ViewSuffixAsPrefix	Sets and returns the suffix as prefix for drawing view caption.
ViewTerminatorRatio	Sets and returns the size of the terminator on the viewing and cutting plane lines.
ViewTextRatio	Sets and returns the size of caption text displayed on the drawing view.
ViewThickLineLength	
ViewThinTerminator	Sets and returns the thin/thick direction lines of cutting plane line or view plane lines.
ViewVPLCaptionAtEnd	Sets and returns the caption location of the view plane.
WeldSymbolLineWidth	Sets and returns the line width of Weld Symbol. This value is always in meters.
ZeroInchesForFTIN	Display leading zero for inches when value includes fraction of inch for unit type Feet-Inch.

See Also

DimensionStyle Object

Solid Edge Constants Type Library

DimGroupMemberTypeConstants Enumeration

Collapse All

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Description

Dimension Group Type

Members

Member	Value	Description
seDimChainGroupMember	3	Dimension Chain Group Member
seDimCoordinateGroupMember	4	Dimension Coordinate Group Member
seDimNotAGroupMember	1	Dimension Not A Group Member
seDimStackGroupMember	2	Dimension Stack Group Member

Solid Edge FrameworkSupport Type Library

DimlnitData Object

Members Collapse All

Solid Edge.
DPSICJH bettCT

Description

Stores the data required to create a dimension. Use with AddDimension() method. This is the only way to create PMI dimensions.

Object Model

See Also

DimInitData Members | Solid Edge ST5 - What's New | Solid Edge ST4 MP1 - What's New | Solid Edge ST3 MP8 - What's New

Solid Edge Constants Type Library

DimLimitTextArrangmentConstants Enumeration

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Solid Edge Constants Type Library : DimLimitTextArrangmentConstants Enumeration

Description

Dimension Limit Text Arrangment

Members

Member	Value	Description
igDimStyleLimitTextHorizontal	1	Limit Text Horizontal
igDimStyleLimitTextVertical	2	Limit Text Vertical

Solid Edge FrameworkSupport Type Library

DimStyle Object Members

Overview Collapse All

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Solid Edge FrameworkSupport Type Library : DimStyle Object

Public Methods

UseActiveStyle	Assigns the current active linear style to the referenced graphic object.

Public Properties

S AboveGap	Sets and returns the gap between the dimension text and the dimension line when the text position is set to above.
AllAroundMultiplier	
AllowNegativeValueForCoordinate	Sets and returns the option to allow the negative values for coordinate dimensions
AllowOriginValueChangeForCoordinate	Sets and returns the option to allow changing the coordinate origin dimension value
AngularDecimalRoundOff	Sets and returns the angular round-off when angular units is set to degrees or radians and round off type is set to decimal.
AngularDMSRoundOff	Sets and return the angular round off when angular units is set to Deg-Min-Sec.
AngularFractionRoundOff	Sets and returns the angular round off when angular units is set to degrees or radians and round off type is set to fraction.
AngularRoundOffType	Sets and returns the angular round off type (decimal or fraction) for the referenced dimension style object.
AngularToleranceDecimalRoundoff	Sets and returns the angular tolerance decimal roundoff
AngularToleranceDMSRoundOff	Sets and returns the angular tolerance Degree Minute Second roundoff
AngularToleranceFractionRoundoff	Sets and returns the angular tolerance fractional roundoff
AngularToleranceRoundOffType	Sets and returns the angular tolerance round off type
AngularToleranceUnits	Sets and returns the angular tolerance units
AngularUnits	Sets and returns the angular dimensioning unit separate from the working units of the file.
ApplyBreakLineGap	
AutoCenterProjectionLine	Sets and returns the center projection line.
AutoPhase	If True, specifies that the negative dash gaps for the referenced style object are automatically extended based on the size of the edge.
AutoSuppressDiameter	Specifies whether or not to automatically suppress the diameter symbol when a circle is measured by a radial or circular diameter dimension.
CenterLineType	Sets and returns the centerline type name for the referenced object.

CenterMarkExtension	Specifies whether keypoint extensions are used for the center mark.
CenterMarkSize	Sets and returns the size of the center mark for the referenced style object.
ChamferUse45DegreeCharacter	Specifies chamfer 45 degree character use.
CommonOrigin	Sets and returns the type of the terminator displayed at the common origin of a coordinate dimension group.
ConnectorColor	
ConnectorLineType	
ConnectorWidth	
0 CoordinateAutoJog	Determines if the referenced coordinate dimension is jogged automatically when the dimension value is less than the coordinate stack pitch.
0 CoordinateStackPitch	Sets and returns the minimum coordinate dimension value after which the coordinate dimension will be jogged when the CoordAutoJog property is set to True.
CoordinateTextOrientation	Sets and returns the text orientation of Coordinate Dimension
CoordinateTextPosition	Sets and returns the position of the coordinate text in relation to the projection line.
DatumFrameDashes	Sets and returns the datum frame dashes for the referenced DatumFrame object.
DatumFrameLineTermThickness	Sets and returns the datum frame line terminator thickness.
DatumFrameShape	Sets and returns the datum Frame Shape.
DatumTargetTerminatorType	Sets and returns the datum target terminator type.
DatumTerminatorGap	Sets and returns the datum terminator gap.
DatumTerminatorType	Sets and returns the datum terminator type.
Delimiter	Sets and returns the delimiter for the referenced object.
DimensionScale	Sets and returns the scale factor of the referenced object.
DimensionScaleMode	Sets and returns the scale mode of the referenced object.
DisplayCoordinateOriginText	Sets and returns the display of coordinate origin text.
DisplayDegreeSymbolOnNumericTolerance	Display degree symbol on numeric angular tolerance values.
DrivenColor	Sets and returns the color in which driven dimensions are displayed.
DrivingColor	Sets and returns the color in which driving dimensions are displayed.
DualDisplay	Turns on and off the dual display for the referenced object.
DualModeVerticalGap	Sets and returns the vertical gap distance between Dimension Value text when the DualModeDisplay property is True.
DualUnitJustification	Sets and returns the dual unit justification,Only applicable for dual unit position below primary

0 DualUnitPosition	Sets and returns the dual unit position
EnableAutoJogForCoordinate	Sets and returns the enable automatic jogging option for coordinate dimensions
ErrorColor	Sets and returns the color in which dimensions in error are displayed.
Font	Sets and returns the name of the font.
Fontstyle	Sets and returns the style of the font for the text of the referenced object.
FractionSeparator	
FreeSpaceTerminatorType	Sets and returns the type of terminator for an annotation whose terminator end is placed on a free-space point.
Height	Returns the height of the referenced object.
S HeightPMI	
HoleCalloutCounterbore	This read-write String property sets and returns the hole callout counterbore settings for the referenced object.
HoleCalloutCounterboreThreaded	This read-write String property sets and returns the hole callout counterbore threaded settings for the referenced object.
HoleCalloutCountersink	This read-write String property sets and returns the hole callout countersink settings for the referenced object.
HoleCalloutCountersinkThreaded	This read-write String property sets and returns the hole callout countersink threaded settings for the referenced object.
HoleCalloutSimple	This read-write String property sets and returns the hole callout simple settings for the referenced object.
HoleCalloutSimpleThreaded	This read-write String property sets and returns the hole callout simple threaded settings for the referenced object.
HoleShaftToleranceSeparatorType	Sets and returns class fit hole/shaft Separator type.
HorizontalBoxGap	Sets and returns the horizontal gap for the dimension text box.
HorizontalToleranceGap	Sets and returns the horizontal tolerance gap between the dimension value and tolerance text.
InhibitZeroToleranceDisplay	
LeaderLineLength	Sets and returns the default leader distance.
LimitTextArrangement	Sets and returns the arrangement of the dimension text for a limit dimension.
LineConnect	Specifies whether or not a line is drawn between the terminators of a dimension when text and terminators are to be outside of the measured value.
LineSpacing	Sets and returns the amount of space between two lines in a paragraph for the referenced object.
LineWidth	Sets and returns the line width (in paper space units) for the referenced object.
LowerCaseMultiplicationSymbol	This read-write Boolean property specifies whether the referenced object uses the lowercase multiplication symbol (x) for chamfer dimensions.

0 MatchPrimaryDelimiter	
MaximumSubunits	Sets and returns the maximum number of subunits displayed by the associated dimension before truncating and adding to the unit.
MaximumViewAngle	Sets and returns the maximum view angle after which the associated dimension will not be displayed in a 3-D environment.
Name	Returns the name of the referenced object.
0 NTSSymbol	Sets and returns the display characteristics of a not-to-scale dimension for the referenced object.
OffsetLeaderType	Sets and returns the type of offset leader to be drawn.
OriginTerminatorType	Sets and returns the terminator display.
OrigTerminatorSize	
OverridePulledOutText	
OverridePulledOutText?	Override dimension text position and orientation when dimension dragged along the dimension line.
PlacePrefixInsideBasicDimensionBox	Sets and returns the placement of prefix inside basic dimension box.
PrefixSuffixGap	Sets and returns the gap between prefix/suffix string and the dimension value text.
PrimaryDecimalRoundOff	Sets and returns the accuracy for the primary decimal value.
PrimaryFractionRoundOff	Sets and returns the accuracy for the primary fraction value.
PrimaryLeadingZeros	Specifies whether or not the primary leading zeros are on.
PrimaryLinearToleranceDecimalRoundOff	Sets and returns the primary linear tolerance decimal roundoff
PrimaryLinearToleranceFractionRoundOff	Sets and returns the primary linear tolerance fractional roundoff
PrimaryLinearToleranceFractionSeparator	Sets and returns the primary linear tolerance fraction separator
PrimaryLinearToleranceMaximumSubunits	Sets and returns the primary linear tolerance maximum subunits
PrimaryLinearToleranceRoundOffType	Sets and returns the primary linear tolerance round off type
PrimaryLinearToleranceSubunitsLabel	Sets and returns the primary linear tolerance subunit label
PrimaryLinearToleranceUnitLabel	Sets and returns the primary linear tolerance unit label
PrimaryLinearToleranceUnits	Sets and returns the primary linear tolerance units
PrimarySubunitsLabel	Sets and returns the text string for the primary subunits label.
PrimaryTrailingZeros	Specifies whether or not primary trailing zeros are on.
PrimaryUnitLabel	Sets and returns the text string of the primary unit's label.
PrimaryUnits	Sets and returns the dimensioning unit separate from the working units of the file.
ProjectionAngle	Sets and returns the angle of the projection lines with respect to the dimension text.
ProjectionBreakGap	This property creates an override style on the dimension and updates the break gap.

0 ProjectionExtensionGap	Sets and returns the distance by which the projection lines are extended beyond the dimension line.
Hsl ProjectionLineDisplay	Specifies the projection line direction.
ProjectionOffsetGap	Sets and returns the gap between the object being measured and the projection line.
ProjectionOffsetGapAlignment	Specifies whether or not the projection offset is aligned.
ProjectionToleranceZonePosition	Sets and returns the projection tolerance position for the dimension.
PulledOutText2Orientation	Sets and returns the text orientation of OverridePulledOutText2.
PulledOutText2Position	Sets and returns the text orientation of OverridePulledOutText2.
0 PulledOutTextOrientation	
PulledOutTextPosition	
H2] Ratio	Sets and returns the text aspect ratio.
Hsl RoundOffType	Sets and returns the round off type.
Hsl RoundUp	Sets and returns round up type (round up all or round up odd) for the referenced dimension style object.
Hsl SecondaryDecimalRoundOff	Sets and returns the accuracy for the secondary decimal value.
H2] SecondaryDelimiter	
Hsl SecondaryFractionRoundOff	Sets and returns the accuracy for secondary fraction value.
Hsl SecondaryLeadingZeros	Specifies whether or not secondary leading zeros are on.
H2] SecondaryLinearFractionSeparator	Sets and returns the secondary linear fraction separator
H2l SecondaryLinearRoundOffType	Sets and returns the secondary linear roundoff type.
Us] SecondaryLinearToleranceDecimalRoundoff	Sets and returns the secondary linear tolerance decimal roundoff
Hsl SecondaryLinearToleranceFractionRoundOff	Sets and returns the secondary linear tolerance fractional roundoff
Hsl SecondaryLinearToleranceFractionSeparator	Sets and returns the secondary linear tolerance fraction separator
Hsl SecondaryLinearToleranceMaximumSubunits	Sets and returns the secondary linear tolerance maximum subunits
Hsl SecondaryLinearToleranceRoundOffType	Sets and returns the secondary linear tolerance round off type
Hsl SecondaryLinearToleranceSubunitsLabel	Sets and returns the secondary linear tolerance subunit label
Hsl SecondaryLinearTolerancellnitLabel	Sets and returns the secondary linear tolerance unit label
Hsl SecondaryLinearTolerancellnits	Sets and returns the secondary linear tolerance units
@ SecondarySubunitsLabel	Sets and returns the text string for the secondary subunits label.
l±sl SecondaryToleranceEquivalent	Specifies whether or not the secondary tolerance is equivalent.
Hsl SecondaryTrailingZeros	Specifies whether or not the secondary trailing zeros are on.
Hsl SecondaryllnitLabel	Sets and returns the text string for the secondary units label.
Hsl SecondaryUnitRoundOff	Sets and returns the round off type for secondary units.

SecondaryUnits	Sets and returns the secondary dimensioning unit.
SecondaryUnitSeparator	Sets and returns the secondary unit separator.
0 ShowOriginLineForCoordinate	Sets and returns the display of coordinate dimension origin line
SlotCalloutCounterbore	Sets and returns text for Counterbore slots.
SlotCalloutSimple	Sets and returns text for Simple slots.
SmartDepthHoleFinite	Sets and returns SmartDepth for finite holes.
SmartDepthHoleThru	Sets and returns SmartDepth for through holes.
SmartDepthSlotFinite	Sets and returns SmartDepth text for finite slots.
SmartDepthSlotThru	Sets and returns SmartDepth text for through slots.
SmartDepthThreadFinite	Sets and returns SmartDepth for finite threads.
SmartDepthThreadThru	Sets and returns SmartDepth for through threads.
0 StackPitch	Sets and returns the minimum distance track distance maintained between individual dimensions within a group.
SymbolFont	Sets and returns the Symbol Font used by this style.
SymbolGap	Sets and returns the gap between the dimension symbol and the dimension value text.
SymbolPosition	Sets and returns the symbol position with respect to the dimension text value.
TerminatorDisplay	Gets/Sets the Terminator Display.
TerminatorInsideLimit	Toggles the associated dimension to the outside of the projection when the dimension value is less than the terminator inside limit.
TerminatorSize	Sets and returns the size of the terminator for the referenced Dimensionstyle object.
TerminatorType	Sets and returns the type of terminator for both origin and measure.
TextAutoFillBackground	
TextClearanceGap	Sets and returns the gap distance between the dimension text and the dimension line.
TextOrientation	Sets and returns the orientation of the dimension text with respect to the dimension plane axis.
TextPosition	Sets and returns the text position.
TextWeight	Sets and returns the text line width.
ThreeSidedSymbolOffsetGap	Sets and returns the offset gap value of Three Sided Weld Symbol.
ToleranceHeightRatio	Sets and returns the tolerance height.
ToleranceTextHorizontalAlignment	Sets and returns horizontal alignment for upper and lower tolerance.
ToleranceTextVerticalAlignment	Sets and returns vertical alignment for tolerance text.
ToleranceWeight	Sets and returns the tolerance text weight.

0 UseTextSizeForCombinedTolValueDisplay	If True, specifies that the dimension text size will be used for the tolerance when upper and lower tolerance have the same absolute value but opposite signs.
0 VerticalBoxGap	Sets and returns the vertical gap for the text box.
VerticalLimitsGap	Sets and returns the vertical limits gap between the limit value texts.
VerticalToleranceGap	Sets and returns the vertical tolerance gap between the dimension value and the tolerance text.
ViewCuttingPlaneLineDisplay	Sets and returns the direction of the terminator.
ViewLineArrowOffset	
ViewLineType	Sets and returns the name of the view line type to be displayed for the referenced object.
ViewLineWidth	Sets and returns the line width of the viewing and cutting plane lines.
ViewPlaneLineDisplay	Sets and returns the number of terminator lines displayed.
ViewTerminatorRatio	Sets and returns the size of the terminator on the viewing and cutting plane lines.
ViewTextRatio	Sets and returns the size of caption text displayed on the drawing view.
ViewThickLineLength	
WeldSymbolLineWidth	Sets and returns the line width of Weld Symbol. This value is always in meters.
ZeroInchesForFTIN	Display leading zero for inches when value includes fraction of inch for unit type Feet-Inch.

See Also

DimStyle Object

Solid Edge Constants Type Library

DimTermDisplayTypeConstants Enumeration

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Description

Dimension Terminators Display Type

Members

Member	Value	Description
igDimStyleTermBoth	3	Terminators Display Both
igDimStyleTermMeas	2	Terminators Display Measure
igDimStyleTermNone	0	Terminators Display None
igDimStyleTermOrig	1	Terminators Display Origin

Solid Edge Constants Type Library

DimTextOrientationConstants Enumeration

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Description

Dimension Text Orientation

Members

Member	Value	Description
igDimStyleTextHorizontal	1	Text Horizontal
igDimStyleTextParallel	3	Text Parallel
igDimStyleTextPerpendicular	4	Text Perpendicular
igDimStyleTextVertical	2	Text Vertical

Solid Edge Constants Type Library

DimTypeConstants Enumeration

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Description

Dimension Types

Members

Member	Value	Description
igDimTypeAngular	3	Angular Dimension
igDimTypeAngularCoordinate	11	Angular Coordinate Dimension
igDimTypeArcAngle	7	Arc Angle Dimension
igDimTypeArcLength	6	Arc Length Dimension
igDimTypeCDiameter	5	Circular Diameter Dimension
igDimTypeChamfer	10	Symmetrical Diameter Dimension
igDimTypeCoordinate	8	Coordinate Dimension
igDimTypeCurveLength	12	Curve Length Dimension
igDimTypeLinear	1	Linear Dimension
igDimTypeRadial	2	Radial Dimension
igDimTypeRDiameter	4	Radial Diameter Dimension
igDimTypeSDiameter	9	Symmetrical Diameter Dimension

Solid Edge Constants Type Library

DimViewCuttingPlaneDisplayTypeConstants Enumeration

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Description

Dimension View Cutting Plane Display Type

Members

Member	Value	Description
igDimViewCuttingPlaneLineDisplayFrom	2	View Cutting Plane Display From
igDimViewCuttingPlaneLineDisplayTo	1	View Cutting Plane Display To

Solid Edge Constants Type Library

DimViewPlaneDisplayTypeConstants Enumeration

Collapse All

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Description

Dimension View Plane Display Type

Members

Member	Value	Description
igDimViewPlaneLineDisplayDouble	2	Dimension View Plane Display Double
igDimViewPlaneLineDisplaySingle	1	Dimension View Plane Display Single

Solid Edge Assembly Type Library

Distance Property

Description

Sets and returns the constraint distance for the referenced SegmentDistanceRelationSd object.

Property type

Read-write property

Syntax

Visual Basic
`Public Property Distance As Double`

See Also

SegmentDistanceRelation3d Object | SegmentDistanceRelation3d Members

Solid Edge Draft Type Library

DraftDocument Object

Members JI

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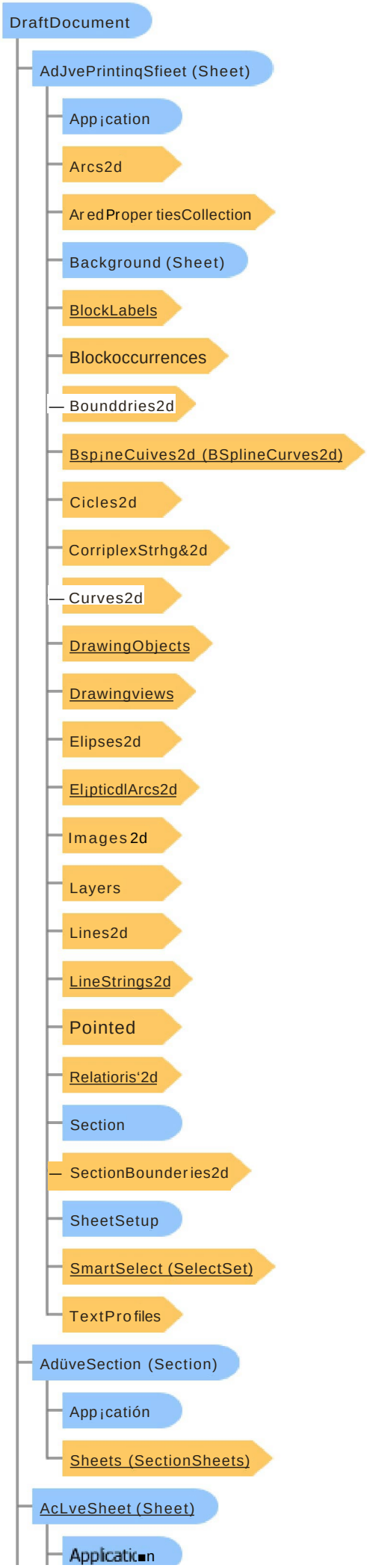
Solid Edge Dian iype Library : DraftDocument Object

Description

Represents the data file that typically contains drafting information, such as part and assembly views, dimensions, annotation, and so forth.

Object Model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App¡cation
Arcs2d
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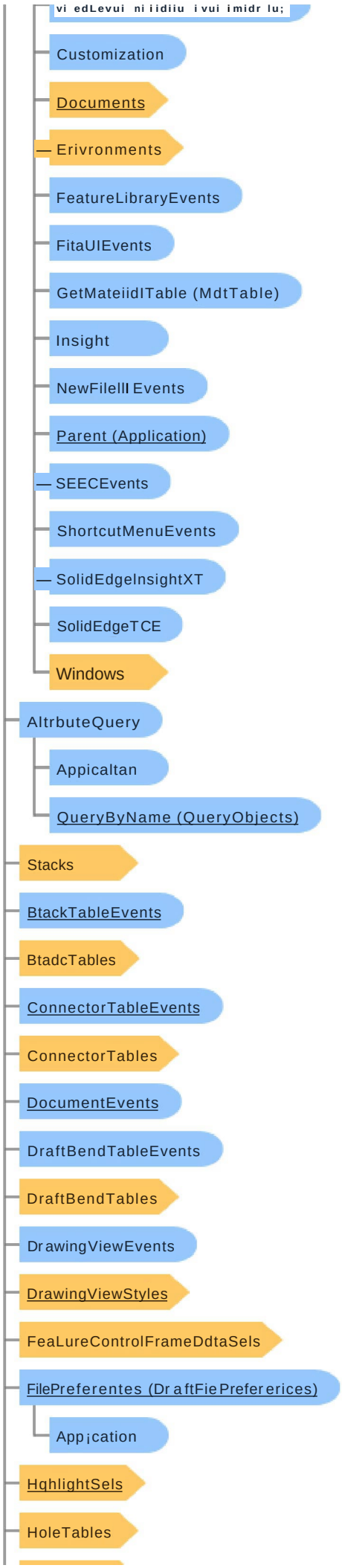
CorriplexStrhg&2d
Curves2d
DrawingObjects
Drawingviews
Elipses2d
El¡pticdlArcs2d
Images 2d
Layers
Lines2d
LineStrings2d
Pointed
Relatioris'2d
Section
SectionBounderies2d
SheetSetup
SmartSelect (SelectSet)
TextPro files
AdüveSection (Section)
App¡catión
Sheets (SectionSheets)
AcLveSheet (Sheet)
Applicatiɑn

- Arcs2d
- AreaPropertiesColleclion
- Background (Sheet)
- BlockLabels
- BlockOccurrences
- Boundaries2d
- BsplneCuives2d (BSplineCurves2d)
- Cicles2d
- CornplexStrhgs2d
- Curves2d
- DrawhgObjects
- DrawingViev/s
- Eipses2d
- ElípticalAics2d
- Images2d
- Layers
- Lines2d
- LineStrings2d
- Points2d
- Relations2d
- Section
- SectionBoundaries2d
- SheetSetup
- SmartSelect (SeleclSet)
- "JjexlPro files
- App¡cation
  - AdbveSelectSet (SelectSet)
  - Addins
  - Appication
  - AppicdtiunEvents
  - App¡cationV8AÍLerDocurnentOpenEvent (ApplicdtiunV8DocurTientOpenEvent)
  - App¡cationWindowEvents
  - CreateCommand (Command)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ShortcutMenuEvents
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SolidEdgeTCE
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BtadcTables
ConnectorTableEvents
ConnectorTables
DocumentEvents
DraftBendTableEvents
DraftBendTables
DrawingViewEvents
DrawingViewStyles
FeaLureControlFrameDdtaSels
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- ModelLriks
- Parent (Application)
  - AdtrveSelectSet (Selectset)
  - Addins
  - Appication
  - AppScationEvents
  - AppicatioriVSAiterDocumentOpenEvent (ApplicdtionV8DocurnentOpenEvent)
  - AppicationWindowEvents
  - CreateCommand (Command)
  - Customization
  - Documents
  - Environments
  - FeatureLibr aryEvents
  - FileUIEvents
  - GetMatenaUable (MatTable)
  - Insight
  - NewFilelI Events
  - Parent (Application)
  - SEECEvents
  - ShortoutMenuEvents
  - SolidEdgeInsightXT
  - SolidEdgeTCE
  - Windows
- PartsListEvents
- PartsLists
- Sections
- SelectSet
- Sheets
- Sui faceFinishSymbolDataSets
- Tables
- TableSLyles
- UnitsOFMeasure
- WeldSymbolDataSets

Whdaws

See Also

DraftDocument Members

Solid Edge Draft Type Library

DraftFilePreferences Object Members

Overview Collapse All

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Solid Edge Draft Type Library : DraftFilePreferences Object

Public Properties

S Application	Returns the active application object.
CenterOfMassBlockName	Gets or sets the block used for center of mass. Use empty string to clear this property.
CoordinateSystemBlockName	Gets or sets the block used for coordinate system. Use empty string to clear this property.
DVConfigChangesMakeViewsOutOfDate	
0 DVCroppedEdgeProcessing	
DVDeriveBendDirectionFromView	
DVDeriveFillStyleFromPart	
DVDeriveReferenceFromAssembly	
DVDetailEnvelopeDisplayAsCircle	
DVEdgeStyleName	Gets or sets the drawing view edge styles for the specified edge type.
DVEnableAlternatingHatchAngle	
DVEnableAlternatingHatchSpacing	
DVHatchRibsInSectionViews	
0 DVIncludePMIAnnotations	
DVIncludePMIDimensions	
DVNailboardSetDisplayWhiteWiresAs	Specifies the color for displaying white wires in nailboard drawing views.
DVProjectionAngle	
DVReferenceComponentsByConfigurations	Specifies whether show/hide settings are applied to reference components, e.g. ref planes, in drawing views using a configuration.
DVRetrieveDimensionsOnViewCreation	
DVShowBoundaryEdges	
DVShowCenterlines	
DVShowDeformationFeatureOrigins	
DVShowDeformationFeatureProfiles	
DVShowDetailViewBorders	
DVShowEdgesOfCuttingPlaneVertices	
DVShowFillStyleInSectionViews	

Hsl DVShowFillStyleInSectionWeldBeads	
Hsl DVShowFlowlines	
Hsl DVShowHiddenEdgesInConnectorViews	Specifies whether hidden edges in nailboard connector drawing views are shown by default.
S DVShowHiddenEdgesInOrthographicViews	
Hsl DVShowHiddenEdgesInPictorialViews	
Hsl DVShowHiddenEdgesInSectionViews	
Hsl DVShowHiddenOtherEdgesInOrthographicViews	
Hsl DVShowHiddenOtherEdgesInPictorialViews	
Hsl DVShowHiddenOtherTangentEdges	
Hsl DVShowHiddenTangentEdges	
Hsl DVShowTangentEdgesInOrthographicViews	
Hsl DVShowTangentEdgesInPictorialViews	
Hsl DVShowThreadsInSectionOnlyViews	
Hsl DVShowTubeCenterlines	
Hsl DVSolidFillSectionWeldBeads	
Hsl DVStyleNameFromTable	Gets/Sets the drawing view style name for a given element type in the drawing view style mapping table.
Hsl DVThreadDisplayMode	
0 GRIDAngle	Specifies the grid angle.
Hsl GRIDDisplayGridMode	Specifies whether grid is displayed as points or lines.
Hsl GRIDMajorLine	Specifies the grid major line spacing.
Hsl GRIDMinorSpaces	Specifies the number of minor spaces per major grid space.
Hsl GRIDSnapToGridMode	Specifies whether to snap to points or lines.
Hsl NumberSheetGroupsSeparately	Specifies whether sheet numbering is relative to all working sheets or to each sheet group.
Hsl Parent	Returns the parent object for the referenced object.
Hsl ViewerDataInclude2DModelSheet	
Hsl ViewerDataIncludeBackgroundSheets	
Hsl ViewerDataIncludeWorkingSheets	

See Also

DraftFilePreferences Object | Solid Edge ST4 - What's New

Solid Edge Draft Type Library

DraftGlobalConstants Enumeration

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Description

Draft Global Constants

Members

Member	Value	Description
seDraftSelectToolActiveLayerFilter	6	Select Tool - Active Layer filter
seDraftSelectToolDimensionAnnotationFilter	3	Select Tool - Dimension Annotation filter
seDraftSelectToolDrawingViewFilter	5	Select Tool - Drawing View filter
seDraftSelectToolRelationHandleFilter	2	Select Tool - Relation Handle filter
seDraftSelectToolTextFilter	4	Select Tool - Text filter
seDraftSelectToolWireFrameFilter	1	Select Tool - Wireframe filter

Solid Edge Draft Type Library

DraftPrintUtility Object

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Description

An object that is sued to print one or more Draft files.

Object Model

- DraftPrintUtifcy
  - App¡cation
    - AdbveSelectSel (SelectSet)
    - Addins
    - AppÍGdlHJ'FI
    - AppicalioriEvents
    - App¡cationV8AfLerDocumentOpenEvent (Appl¡LdtionV8DocurTientOpenEvent)
    - App icalioriWmdowEvents
    - CreateCommand (Command)
    - Customization
    - Documents
    - Envionments
    - FeatureLibraryEvents
    - FitUIEvents
    - GetMateiidiTable (MdtTable)
    - Insight
    - IMewFileUI Events
    - Parent (Application)
    - SEECEvents
    - ShortcutMenuEvents
    - SolidEdgeInsightXT
    - SolidEdgeTCE
    - Windows
  - Paierit (Application)
    - AütÑeSetedtSat (SeledtSet)
    - IAddins
    - Appicaliori
    - AppicalionEvents
    - AppicalioriV8AlterDocurTientOpenEvent (ApplicdtwnV8DocurTientOpenEvent)
    - App icaliuriWindowEvents
    - CreateCommand (Command)
    - Customization
    - Documents

- Erivronments
- FeatureLibraryEvenis
- FieUIEvents
- GetMateiialTable (MatTable)
- Insight
- NewFileUI Events
- Paient (Application)
- SEECEvents
- ShortcxiLMenuEvents
- SolidEdgeInsightXT
- SoidEdgeTCE
- Windows

Example

Example (VB.NET)

```
Imports System.10
Imports System.Runtime.InteropServices

Module Example
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objDraftPrintUtility As SolidEdgeDraft.DraftPrintUtility = Nothing

        Try
            OleMessageFliter .Register()

            objApplication = Marshal.GetActiveObject("SolidEdge.Application")
            objDraftPrintUtility = objApplication.GetDraftPrintUtility()

        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter.Revoke()
        End Try
    End Sub
End Module
```

See Also

DraftPrintUtility Members | Solid Edge ST3 - What's New

Solid Edge Draft Type Library

DraftProfile Object

Collapse All

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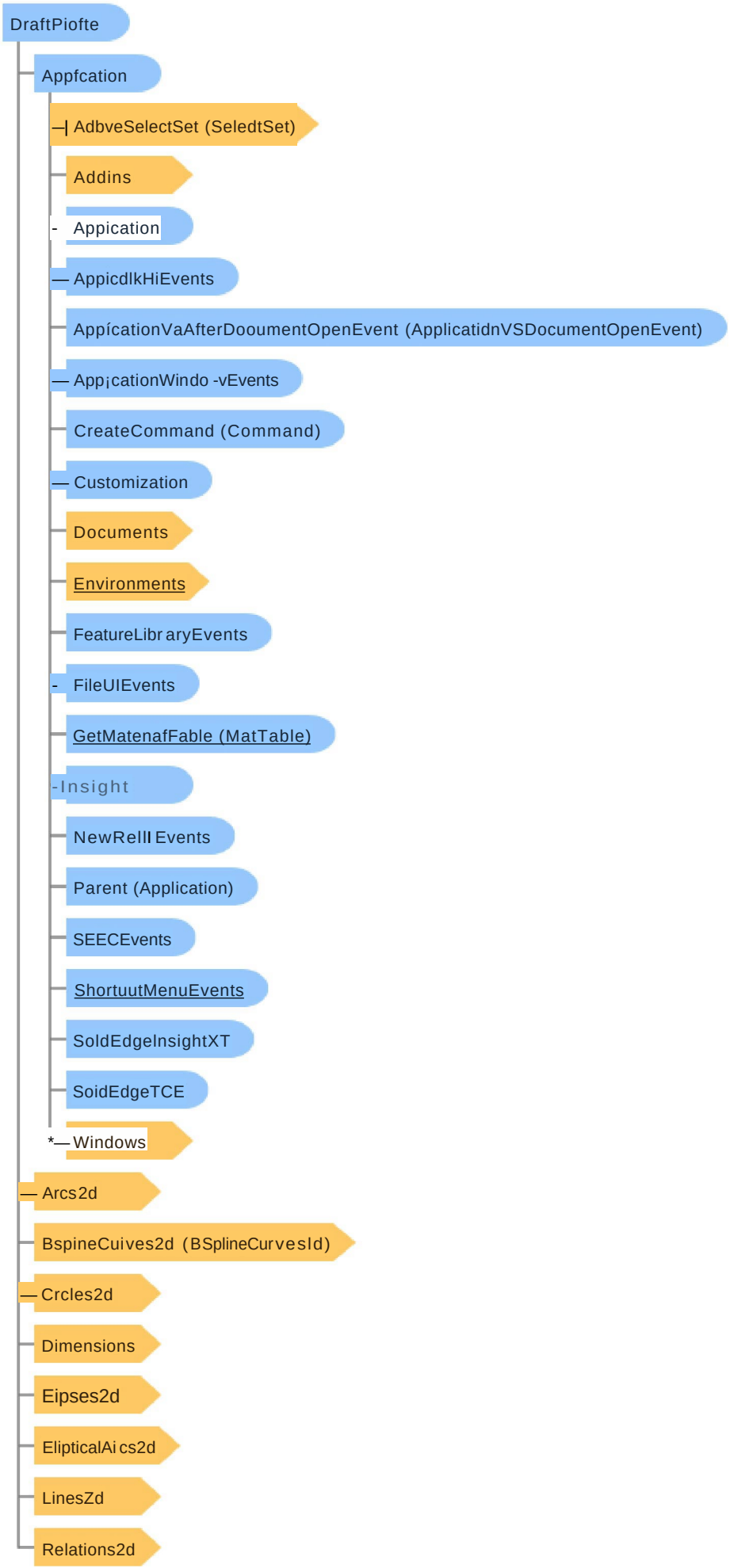
Description

Draft Profile Object

Object Model


DraftPiofte
Appfcation
AdbveSelectSet (SeledtSet)
Addins
Appication
AppicdlkHiEvents
AppícationVaAfterDooumentOpenEvent (ApplicatidnVSDocumentOpenEvent)
App¡cationWindo -vEvents
CreateCommand (Command)
Customization
Documents
Environments
FeatureLibr aryEvents
FileUIEvents
GetMatenafFable (MatTable)
-Insight
NewRelII Events
Parent (Application)
SEECEvents
ShortuutMenuEvents
SoldEdgeInsightXT
SoidEdgeTCE
Windows
Arcs2d
BspineCuives2d (BSplineCurvesId)
Crcles2d
Dimensions
Eipses2d
ElipticalAi cs2d
LinesZd
Relations2d

See Also

DraftProfile Members

Solid Edge Draft Type Library

DrawingView Object Members

Overview Collapse All

Solid Edge Draft Type Library : DrawingView Object

Public Methods

AddAlternateMember	Adds the specified alternate member. A drawing view update will be required to see the new member in the view.
AddConnectedAnnotationsToSelectSet	Adds annotations to select set which are connected to drawing view.
AddConnectedDimensionsToSelectSet	Adds dimensions to select set which are connected to drawing view.
BringForward	Changes the display order of the referenced object by moving it in front of the first object that obscures it.
BringToFront	Changes the display order of the referenced object by moving it in front of all other overlapping objects.
Checkconfiguration	Check to see if the Drawing View is up to date with the current configuration.
ClearEdgeOverrides	Clears all user edge overrides on the drawing view.
ConvertToIndependentDetailView	Convert the current detail view to an independent detail view. Only applies to detail views.
Copy	Places a copy of the referenced object on the clipboard.
Cut	Moves the referenced object to the system clipboard.
Delete	Deletes the referenced object.
Drop	Converts the current DrawingView to an igUserView type containing simple geometry and disassociates the drawing view from the source 3d Model.
ExecuteQuery	Executes query and returns ModelMembers that match the query criteria.
ForceUpdate	Updates the drawing view even if it is not out-of-date.
GetActiveCoordinateSystem	Returns the active coordinate system for the referenced DrawingView object.
GetAlternateMembers	Returns the number and names of alternate members displayed in the alternate position assembly drawing view.
GetAlternatePositionMemberCandidates	Returns the number and names of alternate position members available in the associated assembly model.
GetCaptionPosition	Returns the x and y co-ordinates of the drawing view's caption in sheet space.
GetListOfHiddenLayers	Gets a list of hidden layers for the specified object.
GetOrigin	Returns the origin for the referenced object.
GetReferenceToGraphicMember	Generates a Reference object to user-drawn 2D graphics and creates a dimension to them by passing the Reference to the method that creates the dimension. Use this method with Microsoft Visual Basic version 6.

GetRotationAngle	Returns the rotation angle for the specified object.
HideLayer	Hides the specified layer.
IsLayerHidden	Returns True if the specified layer is hidden.
ModelToView	Converts a specified position from Model coordinates to View coordinates.
Range	Returns the high and low range values for the referenced object.
RemoveAlternateMember	Removes the specified alternate member. A drawing view update will be required to completely remove the member from the view.
RetrieveCenterLinesCenterMarks	This method retrieves or removes the center lines and center marks as specified for the referenced DrawingView object.
RetrieveDimensions	Retrieves dimensions for the referenced object.
Select	Selects the referenced object in the active window.
SendBackward	Changes the display order of visible objects by moving the referenced object so that it is in back of the first object behind it.
SendToBack	Changes the display order of visible objects by moving the referenced object behind all overlapping objects.
SetActiveCoordinateSystem	Defines an active coordinate system for the referenced DrawingView object.
SetCaptionPosition	Sets the x and y co-ordinates of the drawing view's caption in sheet space.
SetOrigin	Redefines the offset between the lower left corner of the window and the object's origin.
SetPerspectiveFromNamedView	Specifies the view to use for a perspective drawing view. The specified view must have perspective defined.
SetPerspectiveOff	Turns off the perspective on a drawing view.
SetPerspectiveOn	Specifies the focal length for a perspective drawing view.
SetRotationAngle	Sets the rotation angle for the specified object.
SetViewOrientation	Sets orientation of the drawing view.
SetViewOrientationFromNamedView	Sets the view orientation from the specified named view.
SetViewOrientationStandard	Sets the specified standard orientation of the drawing view.
SheetToView	Converts a specified position from Sheet coordinates to Drawing View coordinates.
ShowLayer	Shows the specified layer.
Update	Updates an out-of-date drawing view.
ViewOrientation	Returns orientation of the drawing view.
ViewToSheet	Converts a specified position from View coordinates to Sheet coordinates.

Public Properties

0 Annotation	Returns the DetailEnvelope object, the ViewPlane object, or the CuttingPlane object for the referenced DrawingView, if one exists.
AnnotationSheetNumberDefinitionText	Sets and returns the definition text for the AnnotationSheetNumber (%LN) property.
Application	Returns the active application object.
AttributeSets	Returns the AttributeSets collection object for the referenced object.
BrokenOutSectionProfiles	Returns the collection of broken out section profiles defined on this view.
BsplineSimplification	Gets or sets the level of simplification to use when representing 3D bspline edges in drawing views.
CaptionDefinitionTextPrimary	Sets and returns the definition text for the primary caption. Definition text can contain codes for Property Text and Symbols.
CaptionDefinitionTextSecondary	Sets and returns the definition text for the secondary caption. Definition text can contain codes for Property Text and Symbols.
CaptionDisplayedTextPrimary	Returns the text for the primary caption as displayed after processing Property Text and Symbols.
CaptionDisplayedTextSecondary	Returns the text for the secondary caption as displayed after processing Property Text and Symbols.
CaptionLocation	Sets and returns the caption location of the drawing view.
Configuration	Specifies the configuration for an assembly Drawing View.
CoordinateSystems2d	Returns the CoordinateSystems2d object for the referenced DrawingView object.
Crop	Specifies whether or not the contents of the referenced object are to be cropped.
CropBottom	Sets and returns the distance from the origin of the referenced object to the bottom of the frame or view.
CropLeft	Sets and returns the distance from the origin of the referenced object to the left side of the frame or view.
CroppedBoundarystyle	Specifies the default style name for the drawing view's cropped boundary. The cropped boundary shows only the portion of the border that is over visible geometry.
CroppedVHL	Specifies whether to generate only edges inside or overlapping the cropping boundary of the drawing view.

O CroppingBoundaryProfile	Returns the profile associated with the parent object.
Hsl CropRight	Sets and returns the distance from the origin of the referenced object to the right side of the frame or view.
CropTop	Sets and returns the distance from the origin of the referenced object to the top of the frame or view.
Us] CuttingPlanes	Returns the collection of cutting plane lines defined on this view.
Defaults_CoordinateSystemEdgeStyleName	Gets or sets the default Center of Mass and Coordinate System edge style name for a new part added to an assembly for the next drawing view update.
Hsl Defaults_DeriveSectionHatchStyleNameFromPart	Specifies whether to derive the cross hatching style name from the part model for a new part added to an assembly for the next drawing view update.
Us] Defaults_EnableAlternatingHatchAngle	Specifies whether to show alternating hatch angle display by default for a new part added to an assembly for the next drawing view update.
Defaults_EnableAlternatingHatchSpacing	Specifies whether to show alternating hatch spacing display by default for a new part added to an assembly for the next drawing view update.
Defaults_HiddenEdgeStyleName	Specifies the default hidden edge style name for a new part added to an assembly for the next drawing view update.
Defaults_SectionHatchStyleName	Specifies the default cross hatching style name for a new part added to an assembly for the next drawing view update.
Us] Defaults_ShowEdgesHiddenByOtherParts	Specifies whether to display edges hidden by other parts by default for a new part added to an assembly for the next drawing view update.
Us] Defaults_ShowEdgesHiddenTangentEdgesHiddenByOtherParts	Specifies whether tangent edges that are hidden by other parts should be displayed by default for a new part added to an assembly for the next drawing view update. This setting can be set to TRUE only if ShowEdgesHiddenByOtherParts is set to TRUE.
S Defaults_ShowEdgesHiddenTangentEdgesSelfHidden	Specifies whether tangent edges that are hidden should be displayed by default for a new part added to an assembly for the next drawing view update. This setting can be set to TRUE only if ShowHiddenEdges is set to TRUE.
Us] Defaults_ShowHiddenEdges	Specifies whether to display hidden edges by default for a new part added to an assembly for the next drawing view update.

Defaults_ShowSectionHatch	Specifies whether to display cross hatching by default for a new part added to an assembly for the next drawing view update.
Defaults_ShowTangentEdges	Specifies whether to display tangent edges by default for a new part added to an assembly for the next drawing view update.
Defaults_ShowTubeCenterlines	Specifies whether to display tube centerlines by default for a new part added to an assembly for the next drawing view update.
0 Defaults_SolidFillSectionWeldBeads	Specifies whether to display solid fill style by default for a new sectioned weld bead added to an assembly for the next drawing view update.
Defaults_TangentEdgeStyleName	Specifies the default tangent edge style name for a new part added to an assembly for the next drawing view update.
Defaults_TubeCenterlineStyleName	Specifies the default tube centerline style name for a new part added to an assembly for the next drawing view update.
Defaults_VisibleEdgeStyleName	Specifies the default visible edge style name for a new part added to an assembly for the next drawing view update.
0 Description	Returns the description of the refererenced object.
DetailEnvelopes	Returns the collection of detail envelopes defined on this view.
DisplayAnnotation	Specifies whether or not the annotation geometry is to be displayed. The annotation is typically either the DetailEnvelope object or the ViewPlane object.
DisplayAnnotationSheetNumber	Specifies whether or not the referenced drawing view's AnnotationSheetNumber (%LN) property is displayed in the caption.
DisplayBorder	Specifies whether to display the drawing view's border. Set to TRUE to show the border. Set to FALSE to hide the border.
DisplayBreakLinesInBrokenState	Specifies whether or not break lines are displayed when the drawing view is in the broken state.
DisplayCaption	Specifies whether or not the referenced drawing view's primary and secondary captions are displayed. Returns True if primary caption or secondary caption is shown.
DisplayCaptionPrimary	Specifies whether or not the referenced drawing view's primary caption is displayed.
DisplayCaptionSecondary	Specifies whether or not the referenced drawing view's secondary caption is displayed.
0 DisplayCroppedBoundary	Specifies whether to display the drawing view's cropped boundary. The cropped boundary

	shows only the portion of the border that is over visible geometry. Set to TRUE to show cropped boundary. Set to FALSE to hide cropped boundary.
0 DisplayRotationAngle	Specifies whether or not the referenced drawing view's RotationAngle (%VR) property is displayed in the caption.
Hsl DisplayScale	Specifies whether or not the scale is displayed.
Hsl DisplaySuffix	Specifies whether or not the suffix is added to the display of the caption.
Hsl Displaywindow	Specifies whether or not the edit window is displayed.
Hsl Document	Returns the document to which the referenced object belongs.
Hsl DrawingViewType	Returns the view type for the referenced DrawingView object.
DVArcs2d	Returns the DVArc2d collection of the Drawing View.
Hsl DVBSplineCurves2d	Returns the DVBSplineCurve2d collection of the Drawing View.
Hsl DVCircles2d	Returns the DVCircle2d collection of the Drawing View.
Hsl DVEIlipses2d	Returns the DVEIlipse2d collection of the Drawing View.
Hsl DVEIlipticalArcs2d	Returns the DVEIlipticalArc2d collection of the Drawing View.
Hsl DVLines2d	Returns the DVLine2d collection of the Drawing View.
Hsl DVLineStrings2d	Returns the DVLineString2d collection of the Drawing View.
Hsl DVPoints2d	Returns the DVPoint2d collection of the Drawing View.
Hsl EdgesUseAssemblyOverrideColor	Specifies whether or not edge and hatch style use assembly override colors. 'EdgesUseModelColor' must be true before this option is honored.
Hsl EdgesUseModelColor	Specifies whether or not edge and hatch style use model colors
Hsl EdgesUsePartBaseColor	Specifies whether or not edge and hatch style use part base colors. 'EdgesUseModelColor' must be true before this option is honored.
I±sl GetReferenceToGraphicMember2	Generates a Reference object to user-drawn 2D graphics and creates a dimension to them by passing the Reference to the method that creates the dimension. Use this method with Microsoft Visual Basic.NET.

GraphicMembers	Returns the GraphicMember collection of the Drawing View.
HatchRibsInSectionViews	Specifies whether sectioned ribs should be hatched.
HighlightDrawingViewMembers	Returns the HighlightDrawingViewMembers which can be used to highlight members.
IncludePMIAnnotations	Specifies whether or not PMI Annotations are included when a PMI model view is specified
IncludePMIDimensions	Specifies whether or not PMI dimensions are included when a PMI model view is specified
Index	Returns the numeric index for the active object.
Intersectionprocessing	Gets or sets the level of interference edge processing computed for the drawing view.
Hsl IsAttributeSetPresent	Returns True if a specified Attribute Set exists on the referenced object.
Hsl IsBroken	Specifies whether the referenced view is a broken view.
Hsl IsBrokenOutSectionTarget	Returns true if the view is the target view of a broken out section
IsCut	Returns true if the view is a section or revolved section view, target view of a broken out section, or has a model section applied
IsPerspective	Specifies whether the referenced view is a perspective view.
IsSnapShot	Specifies whether the referenced view is a Draft Quality view.
IsIpToDate	Specifies whether or not the drawing view is up to date with respect to the 3d model it references.
Key	Returns a key (a string) to find the object relative to its owning parents.
LastUpdateDate	Returns the date the last update was performed.
LastUpdateSEVersion	Returns the Solid Edge version of the last update.
@ LastUpdateTimeToUpdate	Returns the elapsed time of the last update.
LastUpdateUsername	Returns the username of the user who performed the last update.
Layer	Sets and returns the named layer on which the referenced object resides.
LockView	Specifies whether to Lock/Unlock the drawing view.Set to TRUE for Lock.Set to FALSE to Unlock the Drawing View.
Hsl Matchconfiguration	Specifies whether show/hide settings in the drawing view match the configuration.

ModelLink	Returns the ModelLink object for the referenced object.
ModelMembers	Returns the ModelMember collection of the Drawing View.
Name	Returns the name of the referenced object.
Parent	Returns the parent object for the referenced object.
PMIModelView	Gets or sets the model view associated with this drawing view.
[±3 PrimaryCaptionColor	Sets and returns the color of the text for the caption of the referenced object.
PrimaryCaptionFont	Sets and returns the name of the font.
PrimaryCaptionFontStyle	Sets and returns the style of the font for the text of the referenced object.
PrimaryCaptionTextAlignment	Sets and returns the horizontal and vertical justification of the text for the referenced TextBox object.
Hsl PrimaryCaptionTextSize	Sets and returns the text height of the caption of the referenced object.
PrimaryMember	Specifies the name of the primary member of the alternate position assembly drawing view.
ProcessPartIntersections	Specifies whether part intersections are to be processed.
ReferenceComponentsByConfigurations	Specifies whether show/hide settings are applied to reference components, e.g. ref planes, when using a configuration on this drawing view.
RotationAngleDefinitionText	Sets and returns the definition text for the RotationAngle (%VR) property.
ScaleDefinitionText	Sets and returns the definition text for the Scale.
ScaleDimensionsAndAnnotations	Enables display of dimensions and annotations at drawing view scale.
ScaleFactor	Sets and returns the scale factor between the referenced object and its source document.
0 SecondaryCaptionColor	Sets and returns the color of the text for the caption of the referenced object.
SecondaryCaptionFont	Sets and returns the name of the font.
SecondaryCaptionFontStyle	Sets and returns the style of the font for the text of the referenced object.
SecondaryCaptionTextAlignment	Sets and returns the horizontal and vertical justification of the text for the referenced TextBox object.
SecondaryCaptionTextSize	Sets and returns the text height of the caption of the referenced object.

O SectionHatchUseModelColor	Specifies whether or not hatch styles use model colors. 'EdgesIIseModelColor' must be true before this option is honored.
Hsl SectionOnly	Specifies whether or not the view should be a section only view.
Shading	Specifies whether view shading is enabled for the referenced drawing view.
Hsl ShadingAllowAssemblyOverrides	Specifies whether assembly override colors are shown in the referenced drawing view.
Hsl ShadingAllowPartFaceColors	Specifies whether part face colors are shown in the referenced drawing view.
Hsl ShadingFlat	Specifies whether flat colors are used in the referenced drawing view.
Hsl ShadingGrayscale	Specifies whether grayscale is shown in the referenced drawing view.
@ ShadingQuality	Specifies the shading quality for the referenced drawing view.
Hsl ShadingReflections	Specifies whether reflections colors are shown in the referenced drawing view.
Hsl ShadingShowVisibleEdges	Specifies whether visible edges are shown in the referenced drawing view.
Hsl ShadingTextures	Specifies whether textures are shown in the referenced drawing view.
Hsl Sheet	Returns the drawing view sheet. All drawing view geometry objects (e.g. DVxxx objects) are contained by the drawing view sheet.
Hsl ShowEdgesHiddenByOtherParts	Specifies whether edges hidden by other parts are to be displayed.
Hsl ShowPrimarySecondayCaptionSeparator	Sets and returns the display status of the separator line between the primary and secondary captions.
Hsl SimplifiedAssemblyOption	Sets and returns the simplified assembly option associated with the view.
Hsl SimplifiedPartOption	Sets and returns the simplified part option associated with the view.
Hsl Snapshot	Gets whether or not a view is a Draft Quality view. If set to FALSE, then the Draft Quality view will be converted to a High Quality View.
Hsl SnapShotQuality	Specifies the snapshot quality.
Hsl SourceDrawingView	Returns the DrawingView object from which the detail drawing view is derived.
Hsl StyleName	Sets and returns the name of the style that is applied to the referenced object.
Hsl SuffixDefinitionText	Sets and returns the definition text for the Suffix.

0 TangencyTolerance	Gets or sets the tangency tolerance, in degrees. 0.0 means use the default; otherwise the tolerance in degrees is > 0.0 but <= 5.0
0 TextColor	Sets and returns the color of the text for the caption of the referenced object.
TextFont	Sets and returns the text font of the caption of the referenced object.
TextFontStyle	Sets and returns the text font style of the caption of the referenced object.
TextHeight	Sets and returns the text height of the caption of the referenced object.
TextRatio	Sets and returns the text ratio of the caption of the referenced object.
ThreadAxisTolerance	Gets or sets the thread axis tolerance, in degrees. 0.0 means use the default; otherwise the tolerance in degrees is > 0.0 but <= 5.0
Type	Specifies the type of the object being referenced.
VHL_ToleranceOverrideQuality	Sets and returns the VHL tolerance override value for the referenced drawing view.
ViewType	Returns an integer to indicate the model options on the DV: 0 = Design body, 1 = Simplified body, 2 - Flat sheet metal or Simplified by config.
Visible	Specifies whether or not the referenced object is displayed to the screen.
Zone	Sets and returns the assembly zone associated with the view.
ZOrder	Specifies the front-to-back order in which an object is displayed relative to other objects.

See Also

DrawingView Object

Solid Edge Draft Type Library

DrawingView Object

Members dl

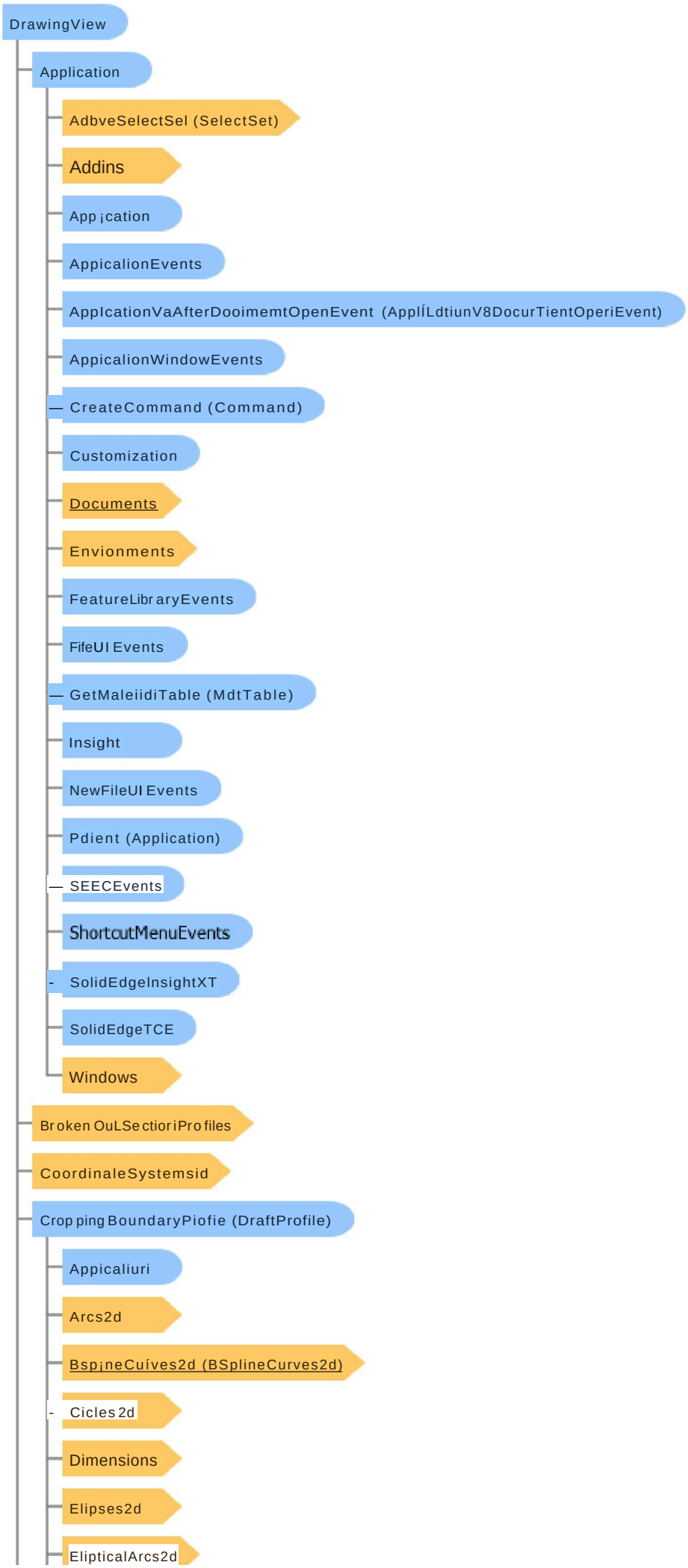
Solid Edge. Design better.

Solid Edge Dran lype Library : DrawingView Object

Description

A 2-D representation of a 3-D part or assembly model.

Object Model


DrawingView
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NewFileUI Events
Pdient (Application)
SEECEvents
ShortcutMenuEvents
SolidEdgeInsightXT
SolidEdgeTCE
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Bsp¡neCuíves2d (BSplineCurves2d)
Cicles2d
Dimensions
Elipses2d
ElipticalArcs2d

- Lines2d
- Relations2d
- CuttingPlanes
- DeldilEnvelopes
- DVArcs2d
- DVBSpfcneCurves2d
- DVCirdes2d
- DVElípses2d
- DVEllptJcalArcs2d
- DVLines2d
- DVLineStrinqs2d
- DVPorits2d
- ExecuLeQuery (ModelMembers)
- GelReferenceToGrdphicMember2 (Reference)
- GraphicMembers
- HqhlighlDrdWingViewMembers
- ModelMembers
- Sour ceDrawiriqView (Di awingView)
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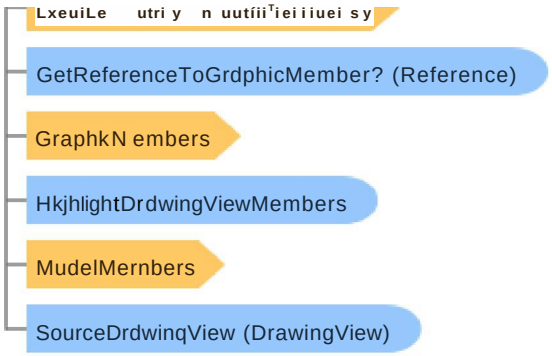
  - Br okeri OuLSe ctior iPro files
  - Coo rdinateSystems 2d
  - CroppingBounddryPiuile (Drd flProfile)
  - CuttingPlanes
  - DeldilEnvelopes
  - DVArcs2d
  - DVBSpirieCurves2d
  - DVCircles2d
  - DVElípses2d
  - DVElliplicalArcs2d
  - DVLiries2d
  - DVLneStrings2d
  - DVPohts2d



Remarks

A drawing view is used to display design space geometry in document space. A view of design space is enclosed by the drawing view border (a handle that allows manipulation of the drawing view). Only one part or assembly document can be used as the basis for drawing views in a draft document.

Example

Example (VB.NET)

```
Imports System.10
Imports System.Runtime.InteropServices

Module Example
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objDraftDocument As SolidEdgeDraft.DraftDocument
        Dim objSheet As SolidEdgeDraft.Sheet = Nothing
        Dim objDrawingViews As SolidEdgeDraft.Drawingviews = Nothing
        Dim objDrawingView As SolidEdgeDraft.DrawingView = Nothing

        Try
            OleMessageFliter .Register()

            objApplication = Marshal.GetActiveObject("SolidEdge.Application")
            objDraftDocument = objApplication.ActiveDocument
            objSheet = objDraftDocument.ActiveSheet
            objDrawingViews = objSheet.Drawingviews
            objDrawingView = objDrawingViews.Item(l)

        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter.Revoke()
        End Try
    End Sub
End Module
```

See Also

DrawingView Members

Solid Edge Constants Type Library

DrawingViewCaptionTextAlignment Enumeration

Collapse All

Solid Edge.
Design better.

Description

Drawing View Caption Text Alignment constants

Members

Member	Value	Description
seTextAlignmentCenter	2	Align center.
seTextAlignmentLeft	1	Align left.
seTextAlignmentRight	3	Align right.

See Also

Solid Edge ST4 - What's New

Solid Edge Constants Type Library

DrawingViewCaptionTypeConstants Enumeration

Collapse All

Solid Edge. Design better.

Description

Drawing View Type constants for defining captions.

Members

Member	Value	Description
se2DModelView	5	constant identifier for 2D Model Views.
seAuxiliaryView	3	constant identifier for Auxiliary Views.
seDetailView	4	constant identifier for Detail Views.
seNailboardView	6	constant identifier for Nailboard Views.
sePrincipalView	1	constant identifier for Principal Views.
seSectionView	2	constant identifier for Section Views.

See Also

Solid Edge ST5 - What's New | Solid Edge ST4 - What's New

Solid Edge Draft Type Library

DrawingViews Collection Members

Overview Collapse All

Solid Edge Draft Type Library : DrawingViews Collection

Public Methods

Add2dModelView	This method creates a 2D Model view as a DrawingView object.
AddAssemblyView	Adds an assembly drawing view based on the ViewType and an optional configuration name.
AddByAuxiliaryFold	Creates an auxiliary DrawingView object and a ViewPlane object from an existing DrawingView object.
AddByDetailEnvelope	Creates an detail DrawingView object and a DetailEnvelope object from an existing DrawingView object.
AddByDraftView	Adds a draft view that references the same user graphics as the input draft view.
AddByFold	Creates a folded DrawingView object from an existing DrawingView object.
AddDetailView	Creates an detail DrawingView object and a DetailEnvelope object from an existing DrawingView object.
AddDraftView	Adds a scalable draft view to contain only user graphics.
AddEmptyView	Adds an new empty, out-of-date drawing view. Specify parameters using view properties and then call Update.
AddPartView	Adds a part drawing view based on the ViewType.
AddPMIModelView	Adds an drawing view based on the specified PMI model view.
AddSheetMetalView	Adds a sheet metal drawing view based on the ViewType.
AddWeldmentView	Adds a weldment drawing view based on the ViewType.
Align	Aligns selected principal and/or auxiliary DrawingView objects that are contained in a selection set.
Item	The items in a collection.
Unalign	Removes the alignment relationship between selected principal and/or auxiliary DrawingView objects that are contained in a selection set.

Public Properties

H2] Application	Returns the active application object.
Count	Returns the number of objects in the referenced collection.
Parent	Returns the parent object for the referenced object.

See Also

DrawingViews Collection

Solid Edge Draft Type Library

DrawingViews Collection

Members Collapse All

Solid Edge Draft Type Library : DrawingViews Collection

Description

Represents a collecion of DrawingView objects.

Remarks

The DrawingViews collection object has four methods for creating a drawing view. These methods produce different types of drawing views (such as a detail view). The Add method places the initial drawing view (primary view). Because the other add methods create views that are derived from an existing drawing view, they are disabled until the first view has been created. The AddByFold method creates a folded drawing view (either principal or isometric) from an existing drawing view. The AddByAuxiliaryFold method creates an auxiliary view and a view plane from an existing view. The AddByDetailEnvelope method creates a detail view and a detail envelope from an existing view.

Example

Example (VB.NET)

```
Imports System.10
Imports System.Runtime.InteropServices

Module Example
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objDraftDocument As SolidEdgeDraft.DraftDocument
        Dim objSheet As SolidEdgeDraft.Sheet = Nothing
        Dim objDrawingViews As SolidEdgeDraft.DrawingViews = Nothing

        Try
            OleMessageFilter.Register()

            objApplication = Marshal.GetActiveObject("SolidEdge.Application")
            objDraftDocument = objApplication.ActiveDocument
            objSheet = objDraftDocument.ActiveSheet
            objDrawingViews = objSheet.DrawingViews

        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter.Revoke()
        End Try
    End Sub
End Module
```

See Also

DrawingViews Members

Solid Edge Draft Type Library

DrawingViews Collection

Members Collapse All

Solid Edge Draft Type Library : DrawingViews Collection

Description

Represents a collecion of DrawingView objects.

Remarks

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Example

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Imports System.Runtime.InteropServices

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    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objDraftDocument As SolidEdgeDraft.DraftDocument
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        Dim objDrawingViews As SolidEdgeDraft.DrawingViews = Nothing

        Try
            OleMessageFilter.Register()

            objApplication = Marshal.GetActiveObject("SolidEdge.Application")
            objDraftDocument = objApplication.ActiveDocument
            objSheet = objDraftDocument.ActiveSheet
            objDrawingViews = objSheet.DrawingViews

        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter.Revoke()
        End Try
    End Sub
End Module
```

See Also

DrawingViews Members

Solid Edge FrameworkSupport Type Library

DrawingViewStyle Object Members

Overview Collapse All

Solid Edge FrameworkSupport Type Library : DrawingViewStyle Object

Public Properties

S AttributeSets	Returns the AttributeSets collection object for the referenced object.
CaptionAnnotation	Sets and returns the caption string to be used by the specified view annotation type.
CaptionPrimary	Sets and returns the primary caption source string to be used by the specified view type.
CaptionSecondary	Sets and returns the secondary caption source string to be used by the specified view type.
CodeASSuffix	Sets and returns the Suffix (%AS) source string for the specified view type. This is intended for use in a form like VA-VA and may include other formatting options, e.g. "()".
CodeLNAnnotationSheetNumber	Sets and returns the Annotation Sheet Number (%LN) source string. This is intended for use with %{Annotation Sheet Number|DV} property text and may include other formatting options, e.g. "()".
CodeVNViewSheetNumber	Sets and returns the View Sheet Number (%VN) source string. This is intended for use with %{View Sheet Number|DV} property text and may include other formatting options.
CodeVRViewRotationAngle	Sets and returns the View Rotation Angle (%VR) source string. This is intended for use with %{View Angle|DV} property text and may include other formatting options.
CodeVSViewScale	Sets and returns the View Scale (%VS) source string. This is intended for use with %{Primary Scale|DV} and/or %{Secondary Scale|DV} property text and may include other formatting options.
CPLCaptionLocation	Sets and returns the caption location of the cutting plane line display.
CuttingPlaneLineArrowOffset	Sets and returns offset distance (0.0 - 1.0) for the arrow terminator.
CuttingPlaneLineDisplay	Sets and returns the direction of the terminator.
CuttingPlaneLineDisplayStyle	Sets and returns the display style for the cutting plane.
Description	Returns a user-specified description of the referenced object.
Dimensionstyle	Sets and returns the dimension style referenced by the drawing view style for settings not defined within the drawing view style itself. Used by property text in captions.
IsAttributeSetPresent	Returns True if a specified Attribute Set exists on the referenced object.
Name	Returns the name of the referenced object.
Parent	Sets and returns the name of the parent style.
PrimaryCaptionColor	Sets and returns the color in which PrimaryCaption is displayed.
PrimaryCaptionFont	Sets and returns the name of the PrimaryCaption's font.

PrimaryCaptionFontStyle	Sets and returns the style of the font for the PrimaryCaption.
PrimaryCaptionTextAlignment	Sets and returns text alignment used for the PrimaryCaption.
0 PrimaryCaptionTextAutoFillBackground	Sets and returns whether background color is used for the PrimaryCaption.
PrimaryCaptionTextSize	Sets and returns the text size for the PrimaryCaption.
SecondaryCaptionColor	Sets and returns the color in which SecondaryCaption is displayed.
SecondaryCaptionFont	Sets and returns the name of the SecondaryCaption's font.
SecondaryCaptionFontStyle	Sets and returns the style of the font for the SecondaryCaption.
SecondaryCaptionTextAlignment	Sets and returns text alignment used for the SecondaryCaption.
SecondaryCaptionTextAutoFillBackground	Sets and returns whether background color is used for the SecondaryCaption.
SecondaryCaptionTextSize	Sets and returns the text size for the SecondaryCaption.
ShowAnnotationCaption	Sets and returns whether the caption string is shown for the specified view annotation type.
ShowCodeASSuffix	Sets and returns the display directive for the Suffix (%AS) source string for the specified view type.
ShowCodeLNAnnotationSheetNumber	Sets and returns the display directive for the AnnotationSheetNumber (%LN) source string.
ShowCodeVNViewSheetNumber	Sets and returns the display directive for the ViewSheetNumber source string.
ShowCodeVRViewRotationAngle	Sets and returns the display directive for the View Rotation Angle source string.
ShowCodeVSViewScale	Sets and returns the display directive for the View Scale source string.
ShowPrimaryCaption	Sets and returns whether the primary caption is shown for the specified view type.
0 ShowPrimarySecondayCaptionSeparator	Sets and returns the display status of the separator line between the primary and secondary captions.
ShowSecondaryCaption	Sets and returns whether the secondary caption is shown for the specified view type.
TerminatorDisplay	Sets and returns the terminator display.
TerminatorSize	Sets and returns the size of the terminator for the referenced DrawingViewStyle object.
TerminatorType	Sets and returns the type of terminator for both origin and measure.
ThickLineLengthRatio	Sets and returns multiplication factor for the line length. Ex. line length = ThickLineLengthRatio x font size.
ThinTerminatorLines	Sets and returns the thin/thick direction lines of cutting plane line or view plane lines.
ViewAnnotationLineType	Sets and returns the name of the line type of the viewing and cutting plane lines.
ViewAnnotationLineWidth	Sets and returns the line width of the viewing and cutting plane lines.

0 ViewBorderLineType	Sets and returns the name of the line type of the drawing view border.
ViewBorderLineWidth	Sets and returns the line width of the drawing view border.
ViewCaptionLocation	Sets and returns the caption location of the drawing view.
ViewPlaneLineDisplay	Sets and returns the number of terminator lines displayed for view plane.
ViewSheetNumberLocation	Sets and returns the Sheet Number Location of the view plane.
VPLCaptionAtEnd	Sets and returns the caption location of the view plane.

See Also

DrawingViewStyle Object | Solid Edge ST4 - What's New

Solid Edge FrameworkSupport Type Library

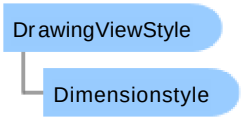
DrawingViewStyle Object

Members Collapse All

Description

Consists of a group of parameters that control the format and presentation characteristics of captions and lines of drawing views and other view annotations such as, cutting plane line, view plane line and detail envelope.

Object Model



See Also

DrawingViewStyle Members | Solid Edge ST4 - What's New

Solid Edge Constants Type Library

DrawingViewStyleMappingElementType Enumeration

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Description

Drawing view type used to get or set the style name to use when creating drawing views of the specified type.

Members

Member	Value	Description
seDVStyleMapping_2DmodelViews	1	
seDVStyleMapping_AuxiliaryViews	3	
seDVStyleMapping_ConnectorView	9	
seDVStyleMapping_CuttingPlane	5	
seDVStyleMapping_DetailEnvelope	7	
seDVStyleMapping_DetailViews	4	
seDVStyleMapping_NailboardView	8	
seDVStyleMapping_PrincipalAndPictorialViews	0	
seDVStyleMapping_SectionViews	2	
seDVStyleMapping_ViewingPlane	6	

See Also

Solid Edge ST5 - What's New

Solid Edge Draft Type Library

DrawingViews Collection

Members Collapse All

Solid Edge Draft Type Library : DrawingViews Collection

Description

Represents a collecion of DrawingView objects.

Remarks

The DrawingViews collection object has four methods for creating a drawing view. These methods produce different types of drawing views (such as a detail view). The Add method places the initial drawing view (primary view). Because the other add methods create views that are derived from an existing drawing view, they are disabled until the first view has been created. The AddByFold method creates a folded drawing view (either principal or isometric) from an existing drawing view. The AddByAuxiliaryFold method creates an auxiliary view and a view plane from an existing view. The AddByDetailEnvelope method creates a detail view and a detail envelope from an existing view.

Example

Example (VB.NET)

```
Imports System.10
Imports System.Runtime.InteropServices

Module Example
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objDraftDocument As SolidEdgeDraft.DraftDocument
        Dim objSheet As SolidEdgeDraft.Sheet = Nothing
        Dim objDrawingViews As SolidEdgeDraft.DrawingViews = Nothing

        Try
            OleMessageFilter.Register()

            objApplication = Marshal.GetActiveObject("SolidEdge.Application")
            objDraftDocument = objApplication.ActiveDocument
            objSheet = objDraftDocument.ActiveSheet
            objDrawingViews = objSheet.DrawingViews

        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter.Revoke()
        End Try
    End Sub
End Module
```

See Also

DrawingViews Members

Solid Edge FrameworkSupport Type Library

DualModeVerticalGap Property

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Description

Sets and returns the vertical gap distance between Dimension Value text when the DualModeDisplay property is True.

Property type

Read-write property

Syntax

Visual Basic
`Public Property DualModeVerticalGap As Double`

Remarks

The value is expressed as a ratio of dimension text height.

See Also

DimStyle Object | DimStyle Members

Solid Edge FrameworkSupport Type Library

DualModeVerticalGap Property

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Description

Sets and returns the vertical gap distance between Dimension Value text when the DualModeDisplay property is True.

Property type

Read-write property

Syntax

Visual Basic
`Public Property DualModeVerticalGap As Double`

Remarks

The value is expressed as a ratio of dimension text height.

See Also

DimensionStyle Object | DimensionStyle Members

Solid Edge FrameworkSupport Type Library

Duplicate Method

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Description

Copies the referenced object to a specified location, creating a new instance of the object with the same property values as the original.

Syntax

Visual Basic

```
Public Function Duplicatef _
    Optional ByVal XDistance As Variant, _
    Optional ByVal YDistance As Variant _
) As Object
```

Parameters

XDistance
Specifies the distance in the x direction from the original that the copy of the referenced object is to be placed.

YDistance
Specifies the distance in the y direction from the original that the copy of the referenced object is to be placed.

Remarks

The referenced object is copied to the new location relative to the original (source) location. The distance is the difference from the object's original location to the XDistance and YDistance values. The new object is returned.

Example

Example (Visual Basic)

```
Private Sub Form_Load()
    Dim objApp As SolidEdgeFramework.Application
    Dim objDoc As SolidEdgePart.PartDocument
    Dim objEllipses As SolidEdgeFrameworkSupport.Ellipses2d
    Dim objEllipse As SolidEdgeFrameworkSupport.Ellipse2d
    Dim sDumpStatus As String     ' Used for temporary storage of datadump return string
    ' Report errors
    Const PI = 3.14159265358979
    ' Create/get the application with specific settings
    On Error Resume Next
    Set objApp = GetObject(, "SolidEdge.Application")
    If Err Then
        Err.Clear
        Set objApp = CreateObject("SolidEdge.Application")
        Set objDoc = objApp.Documents.Add("SolidEdge.PartDocument")
        objApp.Visible = True
    Else
        Set objDoc = objApp.ActiveDocument
    End If
    'Create an Ellipses collection object
    Set objEllipses = objDoc.ProfileSets.Add.Profiles.Add(pRefPlaneDisp:= _
                                                          objDoc.RefPlanes(l)).Ellipses2d
    'Create an Ellipse object
    Set objEllipse = objEllipses.AddByCenter(xCenter:=0.04, yCenter:=0.03, xMajor:= _
                                              0.01, yMajor:=0.01, Ratio:=0.5, Orientation:=igGeom2dOrientClockwise)
    ' Make Duplicates of the Ellipse object at a specified distance
    Call objEllipse.Duplicate(xDistance:=0.1)
    Call objEllipse.Duplicate(yDistanee:=0.1)
    Call objEllipse.Duplicate(xDistance:=0.1, yDistance:=0.1)
    ' USER DISPLAY
    ' Release objects
    Set objApp = Nothing
    Set objDoc = Nothing
    Set objEllipses = Nothing
    Set objEllipse = Nothing
End Sub
```

See Also

Ellipse2d Object | Ellipse2d Members

Solid Edge Draft Type Library

DVArcs2d Collection Members

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Public Methods

Item	The items in a collection.

Public Properties

Application	Returns the active application object.
Count	Returns the number of objects in the referenced collection.
Parent	Returns the parent object for the referenced object.

See Also

DVArcs2d Collection

Solid Edge Draft Type Library

DVBSplineCurves2d Collection Members

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Sonu Luye Draft Type Library : DVBSplineCurves2d Collection

Public Methods

Item	The items in a collection.

Public Properties

Application	Returns the active application object.
Count	Returns the number of objects in the referenced collection.
Parent	Returns the parent object for the referenced object.

See Also

DVBSpl¡neCurves2d Collection

Solid Edge Draft Type Library

DVCircle2d Object Members

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Solid Edge Draft Type Library : DVCircle2d Object

Public Methods

GetCenterPoint	Returns the point that is the center of the referenced object.
GetKeyPoint	Returns information about a specified key point.
GetMajorAxis	Returns the coordinates of the major axis vector for the referenced object.
GetRefereneeKey	Returns a reference key for the edge in the model.
GetSegmentedStyle	Returns the start point, end point, and linear style applied to a specified segment of a 2-D geometry object.
Range	Returns the high and low range values for the referenced object.

Public Properties

Application	Returns the active application object.
Area	Returns the area of the referenced object.
AttributeSets	Returns the AttributeSets collection object for the referenced object.
BendAngle	Returns the bend angle from the underlying sheet metal model.
BendRadius	Returns the bend radius from the underlying sheet metal model.
Circumference	Returns the dimension along the line of a circle or ellipse.
Diameter	Sets and returns the diameter of the referenced Circle2d object.
Document	Returns the document to which the referenced object belongs.
DrawingView	Returns the DrawingView for this object.
EdgeType	Returns the EdgeType for this object.
IsAttributeSetPresent	Returns True if a specified Attribute Set exists on the referenced object.
Key	Returns a key (a string) to find the object relative to its owning parents.
KeyPointCount	Returns the number of keypoints/handles that are supported by the referenced object.
ModelMember	Returns the ModelMember for this object.
ModelWelds	Returns the ModelWelds collection object.
Name	Returns the name of the referenced object.
Radius	Returns the radius of the referenced object.
Reference	Returns a Reference object for this object.
Relationships	Returns a collection of objects that are related to the referenced graphic object.
SegmentedStyleCount	Returns the number of segmented styles defined for the referenced graphic object.
ShowHideEdgeOverride	Gets or sets a show/hide override on an edge or segment.

0 Type	Specifies the type of the object being referenced.

See Also

DVCircle2d Object

Solid Edge Draft Type Library

DVCircles2d Collection Members

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Solid Edge Draft Type Library : DVCircles2d Collection

Public Methods

Item	The items in a collection.

Public Properties

Application	Returns the active application object.
Count	Returns the number of objects in the referenced collection.
Parent	Returns the parent object for the referenced object.

See Also

DVCircles2d Collection

Solid Edge Draft Type Library

DVEllipses2d Collection Members

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Soiu Luye Draft Type Library : DVEllipses2d Collection

Public Methods

Item	The items in a collection.

Public Properties

0 Application	Returns the active application object.
Count	Returns the number of objects in the referenced collection.
Parent	Returns the parent object for the referenced object.

See Also

DVEllipses2d Collection

Solid Edge Draft Type Library

DVEllipticalArcsZd Collection Members

Collapse All

Sonu Luye Draft Type Library : DVEllipticalArcs2d Collection

Public Methods

Item	The items in a collection.

Public Properties

Application	Returns the active application object.
Count	Returns the number of objects in the referenced collection.
Parent	Returns the parent object for the referenced object.

See Also

DVEllipticalArcs2d Collection

Solid Edge Draft Type Library

DVLines2d Collection Members

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Solid Edge Draft Type Library : DVLines2d Collection

Public Methods

Item	The items in a collection.

Public Properties

Application	Returns the active application object.
Count	Returns the number of objects in the referenced collection.
Parent	Returns the parent object for the referenced object.

See Also

DVLines2d Collection

Solid Edge Draft Type Library

DVLineStrings2d Collection Members

Collapse All

Sonu Luye Draft Type Library : DVLineStrings2d Collection

Public Methods

Item	The items in a collection.

Public Properties

Application	Returns the active application object.
Count	Returns the number of objects in the referenced collection.
Parent	Returns the parent object for the referenced object.

See Also

DVLineStrings2d Collection

Solid Edge Draft Type Library

DVPoints2d Collection Members

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Public Methods

Item	The items in a collection.

Public Properties

Application	Returns the active application object.
Count	Returns the number of objects in the referenced collection.
Parent	Returns the parent object for the referenced object.

See Also

DVPoints2d Collection

Solid Edge FrameworkSupport Type Library

Ellipses2d Collection Members

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Sonu Luyt rrameworkSupport Type Library : Ellipses2d Collection

Public Methods

AddByCenter	Uses a center point to create a CenterMark, Ellipse2d, or EllipticalArc2d object.
Item	The items in a collection.

Public Properties

Application	Returns the active application object.
Count	Returns the number of objects in the referenced collection.
Parent	Returns the parent object for the referenced object.

See Also

EllipsesZd Collection

Solid Edge FrameworkSupport Type Library

EllipticalArcs2d Collection Members

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Sonu Luyt rrameworkSupport Type Library : EllipticalArcs2d Collection

Public Methods

AddByCenter	Uses a center point to create a CenterMark, Ellipse2d, or EllipticalArc2d object.
AddByStartEndEllipse	Creates an EllipticalArc2d object based on an existing Ellipse object and bounded by a start angle and end angle.
Item	The items in a collection.

Public Properties

Application	Returns the active application object.
H2] Count	Returns the number of objects in the referenced collection.
Parent	Returns the parent object for the referenced object.

See Also

EllipticalArcs2d Collection

Solid Edge ST7 SDK

File Properties

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Solid Edge supports five property sets: summary information, extended summary information, project information, document information, and mechanical modeling. In addition, there is a Custom property set which gives access to all user-created properties defined through the Custom pane on the Properties dialog box.

Solid Edge file properties can be accessed via the SolidEdgeFramework.PropertySets, SolidEdgeFileProperties.PropertySets and RevisionManager.PropertySets API classes.

Although each Solid Edge file type is different, the following table lists common file properties.

Property Set	Property Name	Maintained	Variant Type	.NET Type
Custom	(User Defined)	X	Any	Any
Documentsummaryinformation	Byte Count		**VTJ4**	Int32
Documentsummaryinformation	Category	X	**VTLPSTR**	String
Documentsummaryinformation	Category	X	**VTJ.PSTR**	String
DocumentSummaryInformation	Company		**VTJ.PSTR**	String
Documentsummaryinformation	Hidden Objects		**VTJ4**	Int32
DocumentSummaryInformation	Lines		**VTJ4**	Int32
DocumentSummaryInformation	Manager	X	**VTLPSTR**	String
DocumentSummaryInformation	Multimedia Clips		**VTJ4**	Int32
DocumentSummaryInformation	Notes		**VTJ4**	Int32
DocumentSummaryInformation	Paragraphs		**VTJ4**	Int32
DocumentSummaryInformation	Presentation Format		**VTJ.PSTR**	String
DocumentSummaryInformation	Slides		**VTJ4**	Int32
ExtendedSummaryInformation	CreationLocale	X	**VTJ4**	Int32
ExtendedSummaryInformation	DocumentID	X	**VT_CLSID**	Guid
ExtendedSummaryInformation	Hardware	X	**VT_BOOL**	Boolean
ExtendedSummaryInformation	Name of Saving Application	X	**VTJ.PWSTR**	String
ExtendedSummaryInformation	Status	X	**VTJ4**	Int32
ExtendedSummaryInformation	Username	X	**VTJ.PWSTR**	String
MechanicalModeling	Material	X	**VTJ.PWSTR**	String
Projectinformation	Document Number	X	**VTJ.PWSTR**	String
Projectinformation	Project Name	X	**VTJ.PWSTR**	String
Projectinformation	Revision	X	**VTJ.PWSTR**	String
Summaryinformation	Application Name	X	**VTJ.PSTR**	String
Summaryinformation	Author	X	**VTLPSTR**	String
Summaryinformation	Comments	X	**VTJ.PSTR**	String
Summaryinformation	Creation Date	X	**VT_FILETIME**	DateTime

Summaryinformation	Keywords	X	VT.LPSTR	String
Summaryinformation	Last Author	X	VT_LPSTR	String
Summaryinformation	Last Print Date		VT_FILETIME	DateTime
Summaryinformation	Last Save Date	X	VT.FILETIME	DateTime
Summaryinformation	Number of characters		VTJ4	Int32
Summaryinformation	Number of pages		VTJ4	Int32
Summaryinformation	Number of words		VTJ4	Int32
Summaryinformation	Origination Date		VT_FILETIME	DateTime
Summaryinformation	Revision Number		VTJ.PSTR	String
Summaryinformation	Security		VTJ4	Int32
Summaryinformation	Subject	X	VTJ.PSTR	String
Summaryinformation	Template	X	VTLPSTR	String
Summaryinformation	Title	X	VT_LPSTR	String
Summaryinformation	Total Editing Time		VTFILETIME	DateTime

See Also

SolidEdgeFramework.PropertySets
SolidEdgeFileProperties.PropertySets
RevisionManager.PropertySets

Solid Edge FrameworkSupport Type Library

FirstLineIndent Property

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Description

Sets and returns the distance from the margin to the start of the line.

Property type

Read-write property

Syntax

Visual Basic
Public Property FirstLineIndent As Double

Remarks

The TextEdit object always has an insert index or an active selection set; setting any property on the TextEdit object applies to the current selection set, if one is present, or to the text at the current insert index.

Example

Example (VB.NET)

```
Imports System.10
Imports System.Runtime.InteropServices

Module Example
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objDraftDocument As SolidEdgeDraft.DraftDocument = Nothing
        Dim objSheet As SolidEdgeDraft.Sheet = Nothing
        Dim objTextBoxs As SolidEdgeFrameworkSupport.TextBoxes = Nothing
        Dim objTextBox As SolidEdgeFrameworkSupport.TextBox = Nothing
        Dim objTextEdit As SolidEdgeFrameworkSupport.TextEdit = Nothing

        Try
            OleMessageFilter.Register()

            objApplication = Marshal.GetActiveObject("SolidEdge.Application")
            objDraftDocument = objApplication.ActiveDocument
            objSheet = objDraftDocument.ActiveSheet
            objTextBoxs = objSheet.TextBoxes
            objTextBox = objTextBoxs.Add(xl:=0.1, yl:=0.3, zl:=0)
            objTextBox.Text = "Some text"
            objTextEdit = objTextBox.Edit
            ' Set first line indent
            objTextEdit.FirstLineIndent = 0.021

        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter.Revoke()
```

```
        End Try
    End Sub

End Module
```

See Also

TextEdit Object | TextEdit Members

Solid Edge ST7 SDK

Frequently Used Programs

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Get Solid Edge Version

Following function will return Solid Edge Version.

Get Solid Edge Version Copy Code

```
Public Function oGetSolidEdgeVersion() As String
    Try
        Dim install As New SEInstallDataLib.SEInstallData
        Dim strSEVersion As String = String.Empty

        strSEVersion = install.GetVersion.ToString
        oReleaseObject(insta11)
        oForceGarbageCollection()

        Return strSEVersion
    Catch ex As Exception
        Return Nothing

    End Try

End Function
```

Get Solid Edge Status

Following function will return Solid Edge Status.

Get Solid Edge Version Copy Code

```
Public Function oGetSEStatus(ByVal strFName As String) As SolidEdgeFramework.Documentstatus
    Dim objPropertySets As SolidEdgeFileProperties.PropertySets = Nothing

    Try
        objPropertySets = New SolidEdgeFileProperties.PropertySets
        Call objPropertySets.Open(strFName, True)
        Return objPropertySets.Item("ExtendedSummaryinformation").item("Status").value
        objPropertySets.Close()
        oReleaseObject(objPropertySets)
        oForceGarbageCollection()
        Exit Function
    Catch ex As Exception
        Return SolidEdgeFramework.DocumentStatus.igStatusUnknown
        oReleaseObject(objPropertySets)
        oForceGarbageCollection()
        Exit Function
    End Try


End Function
```

Get Solid Edge Path

Following function will return Solid Edge Path.

Get Solid Edge Path Copy Code

```
Public Function () As String

    Dim strSEPath As String = String.Empty
    Dim install As SEInstallDataLib.SEInstallData = Nothing
    Try
        install = New SEInstallDataLib.SEInstallData
        strSEPath = install.GetInstalledPath
    Catch ex As Exception
        Return Nothing
    Finally
        oReleaseObject(install)
        oForceGarbageCollection()
    End Try

    Return strSEPath

End Function
```

Release Solid Edge Object

Following function will Release Solid Edge Object.

Release Solid Edge Object Copy Code

```
Sub (ByVal obj As Object)
    Try
        If Not (obj Is Nothing) Then
            'this should only be used when programming applications that run in their own process space
            Marshal.FinalReleaseComObject(obj)
        End If
    Catch ex As Exception
        obj = Nothing
    End Try
End Sub
```

Kill Solid Edge Process

Following function will kill Solid Edge process.

Kill Solid Edge Copy Code

```
Public Function (ByVal Name As String) As Long

    Dim LocalProcs As Process()
    Dim Proc As Process
    Dim i As Integer

    LocalProcs = System.Diagnostics.Process.GetProcesses
    For Each Proc In LocalProcs
        If UCase(Proc.ProcessName) = UCase(Name) Then
            Try
                Proc.Kill()
                oReleaseObject(Proc)
                oReleaseObject(LocalProcs)
                oForceGarbageCollection()
                Return 0
            Catch ex As System.Exception
                oReleaseObject(Proc)
                oReleaseObject(LocalProcs)
                oForceGarbageCollection()
                Return 1
                Exit Function
            End Try
        End If
        i += 1
    Next
    oReleaseObject(Proc)
    oReleaseObject(LocalProcs)
    oForceGarbageCollection()
    Return 1


End Function
```

Check File Attribute

Following function will check file attributes.

Cehck File Attribute Copy Code

```
Public Function (ByVal filename As String., ByVal attribute As 10.FileAttributes) As Boolean
    If 10.File.Exists(filename) Then
        If (IO.File.GetAttributes(filename) And attribute) > 0 Then
            Return True
        Else
            Return False
        End If
    Else
        Return False
    End If
End Function
```

Connecting to Revision Manager

Following function will help you to connect to Revision Manager.

Connecting to Revision Manager Copy Code

```
Public Function (ByVal blnAppVisibility As Boolean, ByVal blnDisplayAlerts As Boolean) As Boolean

    'to connect to a running instance of Revision Manager
    Try
        '("Word.Application")
        '("Excel.Application")
        '("RevisionManager.Application")
        objRevManApp = Marshal.GetActiveObject("RevisionManager.Application")
        objRevManApp.DisplayAlerts = blnDisplayAlerts
        objRevManApp.Visible = blnAppVisibility
        Return True
```

```
    Catch ex As System.Exception

        'SE not running then start it
        Try
            objRevManApp = Activator.CreateInstance(objRevManType)
            objRevManApp.DisplayAlerts = blnDisplayAlerts
            objRevManApp.Visible = blnAppVisibility
            Return True
        Catch exl As Exception
            Return False
        End Try
    End Try

    Return False
End Function
```

Is Valid File Name?

Following function will determine whether File Name is valid or not.

Is Valid File Name? Copy Code

```
Public Function (ByVai fn As String) As Boolean
    Try
        If System.10.File.Exists(fn) Then
            Return True
        Else
            Return False
        End If
    Catch ex As Exception
        Return False
    End Try
End Function
```

Is Solid Edge File?

Following function will determine if File belongs to Solid Edge or not.

Is Solid Edge File? Copy Code

```
'pass in the full path including filename or just the filename
Public Function (ByVai Filename As String) As Boolean
    Try
        Dim validFileTypes As StringO = {"par"., "psm", "asm", "dft", "pwd"}
        Dim strFileNameOnly As String = String.Empty

        strFileNameOnly = System.10.Path.GetFileName(Filename)
        If strFileNameOnly = String.Empty Then
            Return False
        End If
        If System.10.Path.HasExtension(strFileNameOnly) Then
            'it has an extension now check to see if it is a valid SE one
            Dim stnExtension As String = System.10.Path.GetExtension(strFileNameOnly)
            For i As Integer = 0 To validFileTypes.Length - 1
                If strExtension.ToLower = "." + validFileTypes(i).ToLower Then
                    Return True
                End If
            Next
            Return False
        Else
            Return False
        End If
    Catch ex As Exception
        Return False
    End Try
End Function
```

Is Solid Edge Part File?

Following function will determine if File is Solid Edge Part file or not.

Is Solid Edge Part File? Copy Code

```
'pass in the full path including filename or just the filename
Public Function (ByVai Filename As String) As Boolean
    Dim sExtension As String = Nothing
    Dim bPart As Boolean = False

    Try
        sExtension = System.10.Path.GetExtension(Filename)
        If (sExtension.ToLower = ".par") Then
            Return True
        Else
            Return False
```

```
        End If
    Catch ex As Exception
        Return False
    End Try

End Function
```

Is Solid Edge Sheet Metal File?

Following function will determine if File is Solid Edge Sheet Metal file or not.

Is Solid Edge Sheet Metal File? Copy Code

```
'pass in the full path including filename or just the filename
Public Function oIsSolidEdgeSheetMetalFile(ByVal Filename As String) As Boolean
    Dim sExtension As String = Nothing
    Dim bPart As Boolean = False

    Try
        sExtension = System.10.Path.GetExtension(Filename)
        If (sExtension.ToLower = ".psm") Then
            Return True
        Else
            Return False
        End If
    Catch ex As Exception
        Return False
    End Try
End Function
```

Is Solid Edge Draft File?

Following function will determine if File is Solid Edge Draft file or not.

Is Solid Edge Draft File? Copy Code

```
'pass in the full path including filename or just the filename
Public Function (ByVai Filename As String) As Boolean
    Dim sExtension As String = Nothing
    Dim bPart As Boolean = False

    Try
        sExtension = System.10.Path.GetExtension(Filename)
        If (sExtension.ToLower = ".dft") Then
            Return True
        Else
            Return False
        End If
    Catch ex As Exception
        Return False
    End Try
End Function
```

Is Solid Edge Assembly File?

Following function will determine if File is Solid Edge Assembly file or not.

Is Solid Edge Assembly File? Copy Code

```
'pass in the full path including filename or just the filename
Public Function (ByVai Filename As String) As Boolean
    Dim sExtension As String = Nothing
    Dim bPart As Boolean = False

    Try
        sExtension = System.10.Path.GetExtension(Filename)
        If (sExtension.ToLower = ".asm") Then
            Return True
        Else
            Return False
        End If
    Catch ex As Exception
        Return False
    End Try
End Function
```

Is Solid Edge Weldment File?

Following function will determine if File is Solid Edge Weldment file or not.

Is Solid Edge Weldment File? Copy Code

```
'pass in the full path including filename or just the filename
Public Function (ByVai Filename As String) As Boolean
    Dim sExtension As String = Nothing
    Dim bPart As Boolean = False
```

```
    Try
        sExtension = System.10.Path.GetExtension(Filename)
        If (sExtension.ToLower = ".pwd") Then
            Return True
        Else
            Return False
        End If
    Catch ex As Exception
        Return False
    End Try
End Function
```

Solid Edge Part Type Library

GetCurvelData Method

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Description

Returns data parameters for the select curve.

Syntax

Visual Basic

```
Public Sub GetCurvelData( _
    ByRef Anchorside As PatternCurveAnchorSideConstants, _
    ByRef AnchorAtDistance As Double, _
    ByRef PatternType As PatternOffsetTypeConstants, _
    ByRef NumberOfOccurrences As Long, _
    ByRef Occurrencespacing As Double _
)
```

Parameters

AnchorSide

Value	Description
sePatternCurveLeftSide	
sePatternCurveRightSide	

Member of the PatternCurveAnchorSideConstants constant set that specifies the side of the curve on which the anchor point resides.

AnchorAtDistance

Specifies the distance from the anchor point at which the pattern begins.

PatternType

Value	Description
sePatternFillOffset	Fill Offset
sePatternFitOffset	Fit Offset
sePatternFixedOffset	Fixed Offset

Member of the PatternOffsetTypeConstants constant set that specifies the pattern offset type.

NumberOfOccurrences

Specifies the number of occurrences in the pattern.

Occurrencespacing

Specifies the space between occurrences in the pattern.

See Also

Pattern Object | Pattern Members

Solid Edge Part Type Library

GetCurve2Data Method

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Description

Returns data parameters for the path curve.

Syntax

Visual Basic

```
Public Sub GetCurve2Data( _
    ByRef Anchorside As PatternCurveAnchorSideConstants, _
    ByRef AnchorAtDistance As Double, _
    ByRef PatternType As PatternOffsetTypeConstants, _
    ByRef NumberOfOccurrences As Long, _
    ByRef Occurrencespacing As Double _
)
```

Parameters

AnchorSide

Value	Description
sePatternCurveLeftSide	
sePatternCurveRightSide	

Member of the PatternCurveAnchorSideConstants constant set that specifies the side of the curve on which the anchor point resides.

AnchorAtDistance

Specifies the distance from the anchor point at which the pattern begins.

PatternType

Value	Description
sePatternFillOffset	Fill Offset
sePatternFitOffset	Fit Offset
sePatternFixedOffset	Fixed Offset

Member of the PatternOffsetTypeConstants constant set that specifies the pattern offset type.

NumberOfOccurrences

Specifies the number of occurrences in the pattern.

Occurrencespacing

Specifies the space between occurrences in the pattern.

See Also

Pattern Object | Pattern Members

Solid Edge Assembly Type Library

GetElementl Method

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Description

Returns the first of the two elements constrained by the referenced SegmentDirectionRelation3d object.

Syntax

Visual Basic

```
Public Sub GetElementl( _
    ByRef ELementl As Object, _
    ByRef Type As ObjectType, _
    ByRef GeometryType As SegmentRelation3dGeometryConstants _
)
```

Parameters

Element1

Returns the first of two constrained elements.

Type

Value	Description
igAngularRelationSd	Assembly Angular Relation3d
igArc2d	2d Circular Arc
igArc3D	
igAsmRefPlane	Assembly Reference Plane
igAxialRelation3d	Assembly Axial Relation3d
igBalloon	Balloon
igBoltHoleCircle	IgBoltHoleCircle
igBoundary2d	2d Boundary
igBsplineCurve2d	2d Bspline Curve
igBSplineCurve3D	
igCamFollowerRelationSd	Assembly Cam-follower Relation3d
igCenterLine	Center Line
igCenterMark	Center Mark
igCenterPlaneRelation3d	
igChamferRelation2d	2d Chamfer Relation
igCircle2d	2d Circle
igCircleSD	
igCircularPattern2d	2d Circular Pattern
igColinearRelation2d	2d Colinear Relation

igComplexString2d	2d Complex String
igComponentImage2d	IgComponentImage2d
igComponentSketch	IgComponentSketch
igConcentricRelation2d	2d Concentric Relation
igConnector	Connector
igCuttingPlane	CuttingPlane
igDatumFrame	Datum Frame
igDatumPoint	Datum Point
igDatumTarget	Datum Target
¡gDetailEnvelope	DetailEnvelope
igDimension	Dimension
igDividedPart	Divided Part object
igDrawingView	DrawingView
igEllipse2d	2d Ellipse
igEllipticalArc2d	2d Elliptical Arc
igEqualRelation2d	2d Equal Relation
igFamilyMember	Family Member object
igFeatureContolFrame	Feature Control Frame
¡gFeatureControlFrame	Feature Control Frame
igFilletRelation2d	2d Fillet Relation
¡gFittedCurve2d	2d Fitted Curve
igFixRelation2d	2d Fix Relation
igGearRelation3d	Assembly Gear Relation3d
igGostWeldSymbol	GOST Weld Symbol
igGroundRelation3d	Assembly Ground Relation3d
igGroup	Group
igHole2d	2d Hole
igHorizontalAlignRelation2d	2d Horizontal Align Relation
igHorizontalRelation2d	2d Horizontal Relation
igImage2d	2d Image
igIntersectRelation2d	2d Intersect Relation
igKeyPointRelation2d	2d Key Point Relation
igLeader	Leader
igLine2d	2d Line
igLine3D	
igLineString2d	2d Line String
igLinkRelation2d	2d Link Relation

igOccurrence	
igParallelRelation2d	2d Parallel Relation
igPart	Assembly Part Object
igPathRelationSd	Assembly Path Relation3d
igPerpendicularRelation2d	2d Perpendicular Relation
igPlanarRelation3d	Assembly Planar Relation3d
igPoint2d	2d Point
igPointOnRelation2d	2d Point On Relation
igPointRelation3d	Assembly Point Relation3d
igProfile	Profile
igProfileSet	ProfileSet
igRectangularPattern2d	2d Rectangular Pattern
igRefAxis	Reference Axis
igReference	Membassy
igRefPlane	Reference Plane
igRigidSetRelation3d	Assembly Rigid Set Relation3d
igSetRelation2d	2d Set Relation
igSheetView	Sheetview
igSketch	Sketch
igSketch3D	
igSmartFrame2d	SmartFrame2d
igSubAssembly	Assembly Sub-Assembly Object
igSubOccurrence	Assembly SubOccurrence object
igSurfaceFinishTexture	Surface Finish Texture
igSymbol2d	2d Symbol
igSymmetricRelation2d	2d Symmetric Relation
igTangentRelation2d	2d Tangent Relation
igTangentRelationSd	Assembly Tangent Relation3d
igTextBox	Text Box
¡gTextProfile	Text Profile
igTopologyReference	Topology Reference object
igTube	Tube object
igVariable	Variable
igVerticalAl¡gnRelat¡on2d	2d Vertical Align Relation
igVerticalRelation2d	2d Vertical Relation
igViewPlane	ViewPlane
igWeldSymbol	Weld Symbol

seArcSegment	Arc Segment object
seAssemblyBodies	
seAssemblyBody	
seAssemblyGroup	
seAssemblyGroups	
seComponentLayout	
seConfiguration	
seCurveSegment	Curve Segment object
seDVArc2d	DrawingView Derived Arc
seDVBSplineCurve2d	DrawingView Derived BSplineCurve
seDVCircle2d	DrawingView Derived Circle
seDVEllipse2d	DrawingView Derived Ellipse
seDVEIlipticalArc2d	DrawingView Derived EllipticalArc
seDVLine2d	DrawingView Derived Line
seDVLineString2d	DrawingView Derived LineString
seDVPoint2d	DrawingView Derived Point
seLayout	
seLineSegment	Line Segment object
seOccurrences	
seSegmentAngularRelationBd	Segment Angular Relation
seSegmentDirectionRelationBd	Segment Direction Relation
seSegmentDistanceRelation3d	Segment Distance Relation
seSegmentPointRelation3d	Segment 3d Point Relation
seSegmentRadiusRelation3d	Segment 3d Radius Relation
seSegmentTangentRelation3d	Segment Tangent Relation
seSelectSet	
seSimplifiedAssembly	Simplified Assembly
seSubassemblyBodies	
seSubassemblyBody	
seVirtualComponent	A virtual component object
seVirtualComponentOccurrence	A virtual component occurrence object

Member of the ObjectType constant set that specifies what type of object ElementI is.

GeometryType

Value	Description
seSegmentRelation3dArc	
seSegmentRelation3dArcCenter	
seSegmentRelation3dEllipseCenter	
seSegmentRelation3dEndPoint	

seSegmentRelation3dLineEndPoint	
seSegmentRelation3dLineStartPoint	
seSegmentRelation3dRefPlane	
seSegmentRelation3dStartPoint	
seSegmentRelation3dUnbounded	

Member of the SegmentRelation3dGeometryConstants constant set that specifies the type of geometry that is associated with ElementI.

See Also

SegmentAngularRelation3d Object | SegmentAngularRelation3d Members | Solid Edge V19 - What's New | Solid Edge V18 SP2 - What's New

Solid Edge Assembly Type Library

GetElement2 Method

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Description

Returns the second of the two elements constrained by the referenced SegmentDirectionRelation3d object.

Syntax

Visual Basic

```
Public Sub GetElement2( _
    ByRef ELement2 As Object, _
    ByRef Type As ObjectType, _
    ByRef GeometryType As SegmentRelation3dGeometryConstants _
)
```

Parameters

Element2

Returns the second of two constrained elements.

Type

Value	Description
igAngularRelationSd	Assembly Angular Relation3d
igArc2d	2d Circular Arc
igArc3D	
igAsmRefPlane	Assembly Reference Plane
igAxialRelation3d	Assembly Axial Relation3d
igBalloon	Balloon
igBoltHoleCircle	IgBoltHoleCircle
igBoundary2d	2d Boundary
igBsplineCurve2d	2d Bspline Curve
igBSplineCurve3D	
igCamFollowerRelationSd	Assembly Cam-follower Relation3d
igCenterLine	Center Line
igCenterMark	Center Mark
igCenterPlaneRelation3d	
igChamferRelation2d	2d Chamfer Relation
igCircle2d	2d Circle
igCircleSD	
igCircularPattern2d	2d Circular Pattern
igColinearRelation2d	2d Colinear Relation

igComplexString2d	2d Complex String
igComponentImage2d	IgComponentImage2d
igComponentSketch	IgComponentSketch
igConcentricRelation2d	2d Concentric Relation
igConnector	Connector
igCuttingPlane	CuttingPlane
igDatumFrame	Datum Frame
igDatumPoint	Datum Point
igDatumTarget	Datum Target
¡gDetailEnvelope	DetailEnvelope
igDimension	Dimension
igDividedPart	Divided Part object
igDrawingView	DrawingView
igEllipse2d	2d Ellipse
igEllipticalArc2d	2d Elliptical Arc
igEqualRelation2d	2d Equal Relation
igFamilyMember	Family Member object
igFeatureContolFrame	Feature Control Frame
¡gFeatureControlFrame	Feature Control Frame
igFilletRelation2d	2d Fillet Relation
¡gFittedCurve2d	2d Fitted Curve
igFixRelation2d	2d Fix Relation
igGearRelation3d	Assembly Gear Relation3d
igGostWeldSymbol	GOST Weld Symbol
igGroundRelation3d	Assembly Ground Relation3d
igGroup	Group
igHole2d	2d Hole
igHorizontalAlignRelation2d	2d Horizontal Align Relation
igHorizontalRelation2d	2d Horizontal Relation
igImage2d	2d Image
igIntersectRelation2d	2d Intersect Relation
igKeyPointRelation2d	2d Key Point Relation
igLeader	Leader
igLine2d	2d Line
igLine3D	
igLineString2d	2d Line String
igLinkRelation2d	2d Link Relation

igOccurrence	
igParallelRelation2d	2d Parallel Relation
igPart	Assembly Part Object
igPathRelationSd	Assembly Path Relation3d
igPerpendicularRelation2d	2d Perpendicular Relation
igPlanarRelation3d	Assembly Planar Relation3d
igPoint2d	2d Point
igPointOnRelation2d	2d Point On Relation
igPointRelation3d	Assembly Point Relation3d
igProfile	Profile
igProfileSet	ProfileSet
igRectangularPattern2d	2d Rectangular Pattern
igRefAxis	Reference Axis
igReference	Membassy
igRefPlane	Reference Plane
igRigidSetRelation3d	Assembly Rigid Set Relation3d
igSetRelation2d	2d Set Relation
igSheetView	Sheetview
igSketch	Sketch
igSketch3D	
igSmartFrame2d	SmartFrame2d
igSubAssembly	Assembly Sub-Assembly Object
igSubOccurrence	Assembly SubOccurrence object
igSurfaceFinishTexture	Surface Finish Texture
igSymbol2d	2d Symbol
igSymmetricRelation2d	2d Symmetric Relation
igTangentRelation2d	2d Tangent Relation
igTangentRelationSd	Assembly Tangent Relation3d
igTextBox	Text Box
¡gTextProfile	Text Profile
igTopologyReference	Topology Reference object
igTube	Tube object
igVariable	Variable
igVerticalAl¡gnRelat¡on2d	2d Vertical Align Relation
igVerticalRelation2d	2d Vertical Relation
igViewPlane	ViewPlane
igWeldSymbol	Weld Symbol

seArcSegment	Arc Segment object
seAssemblyBodies	
seAssemblyBody	
seAssemblyGroup	
seAssemblyGroups	
seComponentLayout	
seConfiguration	
seCurveSegment	Curve Segment object
seDVArc2d	DrawingView Derived Arc
seDVBSplineCurve2d	DrawingView Derived BSplineCurve
seDVCircle2d	DrawingView Derived Circle
seDVEllipse2d	DrawingView Derived Ellipse
seDVEIlipticalArc2d	DrawingView Derived EllipticalArc
seDVLine2d	DrawingView Derived Line
seDVLineString2d	DrawingView Derived LineString
seDVPoint2d	DrawingView Derived Point
seLayout	
seLineSegment	Line Segment object
seOccurrences	
seSegmentAngularRelationBd	Segment Angular Relation
seSegmentDirectionRelationBd	Segment Direction Relation
seSegmentDistanceRelation3d	Segment Distance Relation
seSegmentPointRelation3d	Segment 3d Point Relation
seSegmentRadiusRelation3d	Segment 3d Radius Relation
seSegmentTangentRelation3d	Segment Tangent Relation
seSelectSet	
seSimplifiedAssembly	Simplified Assembly
seSubassemblyBodies	
seSubassemblyBody	
seVirtualComponent	A virtual component object
seVirtualComponentOccurrence	A virtual component occurrence object

Member of the ObjectType constant set that specifies what type of object Element2 is.

GeometryType

Value	Description
seSegmentRelation3dArc	
seSegmentRelation3dArcCenter	
seSegmentRelation3dEllipseCenter	
seSegmentRelation3dEndPoint	

seSegmentRelation3dLineEndPoint	
seSegmentRelation3dLineStartPoint	
seSegmentRelation3dRefPlane	
seSegmentRelation3dStartPoint	
seSegmentRelation3dUnbounded	

Returns a member of the SegmentRelation3dGeometryConstants constant set that specifies the type of geometry that is associated with Element2.

See Also

SegmentAngularRelation3d Object | SegmentAngularRelation3d Members | Solid Edge V19 - What's New | Solid Edge V18 SP2 - What's New

Solid Edge Framework Type Library

GetGlobalParameter Method

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Description

Returns the setting of a specified global parameter for the referenced Application or document object.

Syntax

Visual Basic

```
Public Sub GetGlobalParameten( _
    ByVai Parameter As ApplicationGlobalConstants, _
    ByRef VaLue As Variant _
)
```

Parameters

Parameter

Value	Description
seApplicationGlobal3dInputdevice	3d Input Device
seApplicationGlobalActivateAIIParts	Activate All Parts
seApplicationGlobalActivateChangedPartsFileLargeAsm	Activate changed parts on file for Large asm
seApplicationGlobalActivateChangedPartsFileMedAsm	Activate changed parts on file for medium asm
seApplicationGlobalActivateChangePartFileSmallAsm	Activate changed parts on file for small asm
seApplicationGlobalActivePartColor	Active Part Color
seApplicationGlobalAdminFileLocation	AdminFileLocation
seApplicationGlobalAllowInterPart	Allow Inter-Part links
seApplicationGlobalAlwayaUploadToServerCheekedOutToMe	Always upload to server but leave doc cheeked out to me
seApplicationGlobalAlwaysUploadToServerCheckDocIn	Always upload to server and check document in
seApplicationGlobalANSHnchToleranceFile	ANSI Inch Tolerance File
seApplicationGlobalANSIMetricToleranceFile	ANSI Metric Tolerance File
seApplicationGlobalApplicationColorScheme	Application color scheme
seApplicationGlobalApplicationDisplay	Application display mode for new windows
seApplicationGlobalApplyActivationOverridesToParts	Apply Activation Overrides To Parts
seApplicationGlobalApplySimplifyOverridesToParts	Apply Simplify Overrides To Parts
seApplicationGlobalApplySimplifyOverridesToSubAssemblies	Apply Simplify Overrides To SubAssemblies
seApplicationGlobalApplyStatusToLinks	Apply Status To Links
seApplicationGlobalArcSmoothness	Arc Smoothness
seApplicationGlobalAsmConfigurationChangesDrawingView	Asm Configuration changes make drawing view out of date
seApplicationGlobalAsmOpenas	Asm open as
seApplicationGlobalAssemblydrivenPartfeatures	Asm driven Part feature
seApplicationGlobalAssemblyReferencePlanefeature	Asm Reference Plane in feature
seApplicationGlobalAutoHideRelationshipPathfinder	Auto hide relationship pathfinder
seApplicationGlobalAutoMakeSiblingParentRevisionObsolute	Automatically make sibling and parent revision obsolute
seApplicationGlobalAutomaticallyReviseDrawingSelectedSDdoc	Automatically revise drawing that use selected 3D doc
seApplicationGlobalAutomaticSave	Automatic Save
seApplicationGlobalAutomaticSaveTime	Automatic Save Time
seApplicationGlobalAutomaticSelection	Automatic Selection
seApplicationGlobalAutoPreserveDoc	Auto preserve Document
seApplicationGlobalAutoPreserveDocAutoSave	Auto Preserve doc by Auto Save
seApplicationGlobalAutoPreserveDocBackup	Auto Preserve doc by Backup
seApplicationGlobalAutoSaveMinutes	Auto Save minutes
seApplicationGlobalAutoScrollASMPathfinder	Auto scroll assembly pathfinder
seApplicationGlobalAutoSetDraftStatus	Auto set draft status
seApplicationGlobalAutoSharpen	Auto sharpen
seApplicationGlobalAutoSharpenLevel	Automatic sharpen level
seApplicationGlobalAvailableRootFolder	Available Root Folder
seApplicationGlobalBackupDraftfiles	Backup Draft files
seApplicationGlobalBackupM¡ñutes	Auto save backup minutes
seApplicationGlobalBackupModelfiles	BAckup Model Files

seApplicationGlobalBaselinedRootFolder	Baselined Root Folder
seApplicationGlobalBendDownCenterlineStyle	Bend down centerline style
seApplicationGlobalBendupCenterlineStyle	Bend up centerline style
seApplicationGlobalBoundaryEdgesStyle	Boundary edges style
seApplicationGlobalChangedPartActivation	
seApplicationGlobalChecklnOnClose	Check In On Close
seApplicationGlobalCheckModelChanges	Check for model changes made by another user
seApplicationGlobalColorActive	The color for active elements
seApplicationGlobalColorBackground	The color for the background
seApplicationGlobalColorConstruction	The color for construction elements
seApplicationGlobalColorDisabled	The color for disabled items
seApplicationGlobalColorFailed	The color for failed items
seApplicationGlobalColorHandle	The color for handles
seApplicationGlobalColorHighlight	The color for highlighted elements
seApplicationGlobalColorLiveSectionCenterline	
seApplicationGlobalColorLiveSectionEdge	
seApplicationGlobalColorLiveSectionOpacity	
seApplicationGlobalColorLiveSectionRegion	
seApplicationGlobalColorManagerCopylndividualFaceColors	Color Manager Copy Individual Face Colors
seApplicationGlobalColorManagerShowAssemblyStyleOverrides	Color Manager Show Assembly Style Overrides
seApplicationGlobalColorManagerShowPartFaceColors	Color Manager Show Part Face Colors
seApplicationGlobalColorManagerUselndividualPartStyles	ColorManager Use Individual Part Styles
seApplicationGlobalColorManagerUseToolsOptionsColorSettings	Color Manager Use Tools Options Color Settings
seApplicationGlobalColorProfile	The color for profiles, sketches and layouts
seApplicationGlobalColorRefPlane	Color RefPlane
seApplicationGlobalColorSelected	The color for items in the select set
seApplicationGlobalColorSheet	The color for the sheet
seApplicationGlobalColorSheetTabl	Specifies the first of two alternating colors for the sheet tabs.
seApplicationGlobalColorSheetTab2	Specifies the second of two alternating colors for the sheet tabs.
seApplicationGlobalCommandBarMode	Displays mode in command bar or quick bar.
seApplicationGlobalCommandUserInterface	Command user interface
seApplicationGlobalConfigNames	Config names
seApplicationGlobalConfiguration	
seApplicationGlobalCullingBool	Culling
seApplicationGlobalCullingValue	Culling Value
seApplicationGlobalCurvatureCombDensity	Curvature Comb Density
seApplicationGlobalCurvatureCombMagnitude	Curvature Comb Magnitude
seApplicationGlobalCustomSettingFile	Custom Setting File
seApplicationGlobalCutHardwareSectionViews	Cut hardware in section views
seApplicationGlobalDefaultSharpness	Default sharpness for new windows
seApplicationGlobalDeriveDisplayAsReference	Derive Display as Reference from Assembly
seApplicationGlobalDetailEnvelope	Detail Envelope
seApplicationGlobalDimKeylnValueAuto	Dimention Keyin values automatically
seApplicationGlobalDimSurroundingComponents	Dimension Surrounding Components
seApplicationGlobalDisabledEleColorDraft	Disabled Color
seApplicationGlobalDisperseAsmPlacePart	Disperse this assembly during Place Part
seApplicationGlobalDisplayArcQuality	Display Arc Quality
seApplicationGlobalDisplayAsCircle	Display as Circle
seApplicationGlobalDisplaydropshadows	Display Drop Shadows
seApplicationGlobalDisplaylnterPartCopies	Recently used files Value
seApplicationGlobalDisplayQuality	Display Quality
seApplicationGlobalDisplaySectionCaps	Enable the display of section caps with dynamic sections
seApplicationGlobalDisplaySheet	Display sheet
seApplicationGlobalDisplayStatistics	Enable display timing statistics in the status bar
seApplicationGlobalDisplayUnitsMeasurement	Display units of measurement label
seApplicationGlobalDocumentNameFormula	

seApplicationGlobalDontCreateNewWndPlacePart	Do not create new window
seApplicationGlobalDoubleClickblock	Double Click a block
seApplicationGlobalDoubleclickEmbeddedObjects	Double click to open embedded objects
seApplicationGlobalDraftSaveAsPDFIncludeGridDisplay	Include grid display in PDF
seApplicationGlobalDraftSaveAsPDFPrintQualityDPI	Print quality DPI
seApplicationGlobalDraftSaveAsPDFSaveAllColorsBlack	Save all colors as black
seApplicationGlobalDraftSaveAsPDFSheetOptions	Sheet options
seApplicationGlobalDraftSaveAsPDFSheetsRange	Sheets range
seApplicationGlobalDraftSaveAsPDFTransparentDVBackgrounds	Transparent drawing view backgrounds
seApplicationGlobalDraftSaveAsPDFUseIndividualSheetSizes	Use individual sheet sizes
seApplicationGlobalDynamicClipping	Dynamic Clipping
seApplicationGlobalDynamicPreviewFeaturecreation	Dynamic Preview Feature creation
seApplicationGlobalDynamicTransition	Dynamic view transition smoothness
seApplicationGlobalECORootFolder	ECO Root Folder
seApplicationGlobalECRRootFolder	ECR Root Folder
seApplicationGlobalEdgeCondition	Edge Condition
seApplicationGlobalEdgeConditionDraft	Edge condition
seApplicationGlobalEnableDotNet2GC	Enable/Disable DotNet 2 Garbage Collection
seApplicationGlobalEnableDotNet4GC	Enable/Disable DotNet 4 Garbage Collection
seApplicationGlobalEnableDynamicEditProfiles	Enable Dynamic Edit of ProfilesSketches
seApplicationGlobalEnableDynamicTolerance	Enable dynamic tolerancing of facet data
seApplicationGlobalEnableSheetDistanceDimValue	Enable sheet distance dimension values
seApplicationGlobalEnableThreadedDisplay	Enable multithreaded displays
seApplicationGlobalEnableUndoAllProfileSketch	Enable undo All for profilesketch modifications
seApplicationGlobalEnableValuechangeUsingMouseWheel	Enable value changes using mouse wheel
seApplicationGlobalEnterProfileSketchCreateNewWindow	Entering Profile Sketch
seApplicationGlobalExpandWeldmentSubasms	Streering wheel Medium
seApplicationGlobalExplodedFlowLineStyle	Exploded flow line style
seApplicationGlobalExternalBom	External BOM
seApplicationGlobalFastlocateBoxDisplayAsm	Fast locate using box display for assemblies
seApplicationGlobalFastlocateboxDisplayParts	Fast locate using box display for parts
seApplicationGlobalFastlocateOverPathfinder	Fast locate when over path finder
seApplicationGlobalFeatureOriginSize	Feature origin size
seApplicationGlobalFramePipingUniquenessCutLength	Frame Piping Uniqueness Cut Length
seApplicationGlobalFramePipingIlniquenessMass	Frame Piping Uniqueness Mass
seApplicationGlobalFramePipingUniquenessMiter	Frame Piping Uniqueness Miter
seApplicationGlobalGraphicsColorAngAccleration	Angular Accleration
seApplicationGlobalGraphicsColorAngVelocity	Angular Velocity Color
seApplicationGlobalGraphicsColorBeamCurve	Beam Color
seApplicationGlobalGraphicsColorBearing	Bearing Color
seApplicationGlobalGraphicsColorBodyTemp	Body Temperature
seApplicationGlobalGraphicsColorConnectorSource	Connector Suurce
seApplicationGlobalGraphicsColorConnectorTarget	Connector Target
seApplicationGlobalGraphicsColorConstraints	Constraints
seApplicationGlobalGraphicsColorConvection	Convection color
seApplicationGlobalGraphicsColorDisplacement	Displacement
seApplicationGlobalGraphicsColorForce	Force
seApplicationGlobalGraphicsColorGravity	Gravity Color
seApplicationGlobalGraphicsColorHeatFlux	Heat Flux
seApplicationGlobalGraphicsColorHeatGen	Material Table
seApplicationGlobalGraphicsColorNode	Node Color
seApplicationGlobalGraphicsColorPressure	Pressure
seApplicationGlobalGraphicsColorRadition	Radiation color
seApplicationGlobalGraphicsColorRigidLink	Rigid Link Color
seApplicationGlobalGraphicsColorTemperature	Temperature color
seApplicationGlobalGraphicsColorTorqueMomnent	Torque Color

seApplicationGlobalGRID_Draft_MajorLineColor	Specifies the major line color.
seApplicationGlobalGRID_Draft_MinorLineColor	Specifies the minor line color.
seApplicationGlobalGRID_Draft_ShowAlignmentLines	Specifies whether alignment lines are shown
seApplicationGlobalGRID_Draft_ShowGrid	Specifies whether the grid is shown.
seApplicationGlobalGRID_Draft_ShowKeyIns	Specifies whether key-ins are shown.
seApplicationGlobalGRID_Draft_ShowReadout	Specifies whether position readouts are shown.
seApplicationGlobalGRID_Draft_SnapToGrid	Specifies whether to snap to the grid.
seApp|icationGlobalGR|D_Ordered_MajorLineColor	Specifies the major line color.
seApplicationGlobalGRID_Ordered_MinorLineColor	Specifies the minor line color.
seApplicationGlobalGRID_Ordered_ShowAlignmentLines	Specifies whether alignment lines are shown
seApplicationGlobalGRID_Ordered_ShowGrid	Specifies whether the grid is shown.
seApplicationGlobalGRID_Ordered_ShowKeyIns	Specifies whether key-ins are shown.
seApplicationGlobalGRID_Ordered_ShowReadout	Specifies whether position readouts are shown.
seApplicationGlobalGRID_Ordered_SnapToGrid	Specifies whether to snap to the grid.
seApplicationGlobalGRID_Sync_MajorLineColor	Specifies the major line color.
seApplicationGlobalGRID_Sync_Minorl_ineColor	Specifies the minor line color.
seApplicationGlobalGRID_Sync_ShowAlignmentLines	Specifies whether alignment lines are shown
seApplicationGlobalGRID_Sync_ShowGrid	Specifies whether the grid is shown.
seApplicationGlobalGRID_Sync_ShowKeyIns	Specifies whether key-ins are shown.
seApplicationGlobalGRID_Sync_ShowReadout	Specifies whether position readouts are shown.
seApplicationGlobalGRID_Sync_SnapToGrid	Specifies whether to snap to the grid.
seApplicationGlobalHandleIColorDraft	Handlei Color
seApplicationGlobalHandle2ColorDraft	Handles Color
seApplicationGlobalHandle3ColorDraft	Handles Color
seApplicationGlobalHatchRibsSectionViews	Hatch Rib section Views
seApplicationGlobalHiddenEdgeStyle	Hidden edge style
seApplicationGlobalHiddenEdgeStyleFromAsm	Hidden edge style from asm
seApplicationGlobalHideAIIComponents	Hide All Components
seApplicationGlobalHideAIIComponentsLargeAsm	Hide all Components for large asm
seApplicationGlobalHideAHComponentsMedAsm	Hide all Components for medium asm
seApplicationGlobalHideAIIComponentsSmallAsm	Hide all Components for small asm
seApplicationGlobalHighlightColorDraft	Highlight Color
seApplicationGlobalHoleSizeFile	Hole Size File
seApplicationGlobalHorizontalScrollBar	Horizontal scroll bar
seApplicationGlobalInactiveHiddenIInusedComponents	Inactive hidden and unused components
seApplicationGlobalInclude2DModelSheets	Include 2D model sheets
seApplicationGlobalIncludeBackgroundSheets	Include background sheets
seApplicationGlobalIncludeCommandPartAssemblySketches	Include Command Part Asm Sketch
seApplicationGlobalIncludeDraftViewerData	Include Draft Viewer data in file
seApplicationGlobalIncludeWorkingSheets	Include working sheets
seApplicationGlobalIncreaseButtonsCommandRibbon2x	Increase buttons on Command Ribbons by 2x
seApplicationGlobalIndicateUnderConstraintProfilesPF	Indicate under constraint profiles in path finder
seApplicationGlobalInReviewRootFolder	In Review Root Folder
seApplicationGlobalInside	Inside Color
seApplicationGlobalInsideColorDraft	Inside Color in Draft
seApplicationGlobalInsideOutside	InsideOutside Color
seApplicationGlobalInsideOutsideColorDraft	Inside outside color in Draft
seApplicationGlobalInsightCacheLocation	Set InsightCacheLocation
seApplicationGlobalInsightFolderMappingFileLocation	Set InsightFolderMappingFileLocation
seApplicationGlobalInterPartCopyCommand	Inter Part Copy Command
seApplicationGlobalInWorkRootFolder	Obsolete Root Folder
seApplicationGlobalISOToleranceFile	ISO Tolerance File
seApplicationGlobalLargeAsmDynamicDisplay	Large Asm dynamic display
seApplicationGlobalLargeAsmOccurence	Large Asm occurance
seApplicationGlobalLimitedSave	Limited Save
seApplicationGlobalLimitedSaveEnable	Limited Save Enable

seApplicationGlobalLimitedUpdate	Limited Update
seApplicationGlobalLimitedUpdateEnable	Limited Update Enable
seApplicationGlobalLimitsAndFits	Limits and fits
seApplicationGlobalLimitSearchToSearchServicesScope	Limit search to this search services
seApplicationGlobalLimitsViewsDraft	Limits and Views
seApplicationGlobalLinkMgmt	Link Management
seApplicationGlobalLogFilesLocation	Set LogFilesLocation
seApplicationGlobalLookAheadVersion	LookAheadVersion
seApplicationGlobalMacros	MaAcros
seApplicationGlobalMaintainItemNumbers	Maintain Item numbers
seApplicationGlobalMaintainRelationships	Maintain relationships during place part
seApplicationGlobalMakeLiveRulePanelVertical	Make live rule panel as vertical
seApplicationGlobalMakeRevisionsObsolete	Make Revisions Obsolete
seApplicationGlobalManagedStdParts	Managed standard parts
seApplicationGlobalMatTableFile	Material Table
seApplicationGlobalMatTableFolder	Material Table Folder
seApplicationGlobalMaximumPrintFileSize	Maximum print file size
seApplicationGlobalMediumAsmDynamicDisplay	Medium Asm dynamic display
seApplicationGlobalNumberSheetGroupSeparately	Number sheet group seprately
seApplicationGlobalNXNAsternScratch	NX Nastern Scratch
seApplicationGlobalObsoleteRootFolder	Obsolete Root Folder
seApplicationGlobalObsolute	Obsolute
seApplicationGlobalOfflineMode	Set OfflineMode
seApplicationGlobalOnLineHelpLocation	
seApplicationGlobalOnlyGenerateEdgesInsideOverlappingCroppedBoundaries	Only generate edges inside and overlapping cropped boundaries
seApplicationGlobalOpacityColorDraft	Opacity Color in Draft
seApplicationGlobalOpacityRefPlane	Opacity RefPlane
seApplicationGlobalOpenAsReadOnly3DFile	
seApplicationGlobalOpenAsReadOnlyBDFileJndirectFiles	
seApplicationGlobalOpenAsReadOnlyDftFile	
seApplicationGlobalOpenAsReadOnlyDftFileJndirectFiles	
seApplicationGlobalOrientWindowSelectedPlane	Orient the window to the selected plane
seApplicationGlobalOrientXpressSize	Orient Xpress Size
seApplicationGlobalOriginEdgeStyle	Origin edge style
seApplicationGlobalOverlayColor	The color for visible edge overlay
seApplicationGlobalOverlayColorMode	The color mode for visible edge overlay [1=Absolute, 2=Relative]
seApplicationGlobalPackagedCollaborationRootFolder	Packaged Collaboration Root Folder
seApplicationGlobalPartActivationLargeAsm	Part Activation for large asm
seApplicationGlobalPartActivationMediumAsm	Part Activation for medium asm
seApplicationGlobalPartActivationSmallAsm	Part Activation for small asm
seApplicationGlobalPartandAsmUndoSteps	Part and asm undo steps
seApplicationGlobalPartDynamicDisplay	Part Dynamic Display
seApplicationGlobalPartIntersections	Part intersections
seApplicationGlobalPartSimplificationLargeAsm	Part simplification for large asm
seApplicationGlobalPartSimplificationMedAsm	Part simplification for medium asm
seApplicationGlobalPartSimplificationSmallAsm	Part simplification for small asm
seApp|icationGlobalPaste2dBehaviourConstant	Paste2d Behaviour
seApplicationGlobalPasteLinkVariableTable	Paste Link to Variable Table
seApplicationGlobalPatternedPartsInherit	Patterned parts inherit
seApplicationGlobalPFAppereance	Path finder appearance
seApplicationGlobalPipeThreadfsFile	PipeThreadfsFile
seApplicationGlobalPrereleaseRootFolder	Prerelease Root Folder
seApplicationGlobalProcesshiddenedges	Process hidden edges
seApplicationGlobalProfileEdgeStyle	Profile edge style
seApplicationGlobalProfileIndoSteps	Profile undo steps
seApplicationGlobalProjectionAngleDraft	Projection Angle

seApplicationGlobalPromptForfileProp	Prompt for File Properties
seApplicationGlobalPromptMatmodelDoc	Prompt for material in new model document
seApplicationGlobalPropertyTextError	Property Text Error
seApplicationGlobalPropSeedFile	Property Seed File
seApplicationGlobalRecentlyUsedFilesBool	Recently used files Bool
seApplicationGlobalRecentlyUsedFilesValue	Recently used files Value
seApplicationGlobalRecomputeAssemblySketchEdit	Recompute assembly during sketch edits
seApplicationGlobalReferencePlaneSize	Reference plane size
seApplicationGlobalReferenceScale	Reference Scale
seApplicationGlobalReleasedRootFolder	Released Root Folder
seApplicationGlobalReports	Reports
seApplicationGlobalRevisionDelimiter	Sets Revision Delimiter
seApplicationGlobalSaveColorsBlackWhite	Save colors as black and white
seApplicationGlobalSaveMultiCADDatasettoTC	
seApplicationGlobalSearchScope	Set SearchScope
seApplicationGlobalSectionCurvatureCombDensity	
seApplicationGlobalSectionCurvatureCombMagnitude	
seApplicationGlobalSelElementColorDraft	Select Element Color
seApplicationGlobalSessionDraftOpenInactive	Draft inactive mode on open (boolean)
seApplicationGlobalSet2DSelectionFenceOpacity	SecRegionOpacity
seApplicationGlobalSetBackGroundColor	Background color
seApplicationGlobalSetChildFeatureColor	Child Feature Color
seApplicationGlobalSetConstructionColor	Construction Color
seApplicationGlobalSetDrivenColor	Driven Color
seApplicationGlobalSetFailedColor	Failed Color
seApplicationGlobalSetFullyDefinedColor	Fully Defined Color
seApplicationGlobalSetGeneralBodyColor	General Body Color
seApplicationGlobalSetGuidePathColor	Guide Path color
seApplicationGlobalSetHandleIColor	HandleI Color
seApplicationGlobalSetHandle2Color	Handle2 Color
seApplicationGlobalSetHandle3Color	Handle3 Color
seApplicationGlobalSetHighlightColor	Highlight color
seApplicationGlobalSetInactiveColor	Inactive Color
seApplicationGlobalSetInconsistantColor	Inconsistant Color
seApplicationGlobalSetOverDefinedColor	Over Defined Color
seApplicationGlobalSetPlaneColor	Plane color
seApplicationGlobalSetPlaneEdgesColor	Plane Edge color
seApplicationGlobalSetPlaneFeatureColor	Plane Feature Color
seApplicationGlobalSetProfileSelectionColor	Profile Selection color
seApplicationGlobalSetRegionColor	Region Color
seApplicationGlobalSetRegionOpacity	PriRegionOpacity
seApplicationGlobalSetSelectedColor	Selected color
seApplicationGlobalSetSketchColor	Sketch color
seApplicationGlobalSetUnderDefinedColor	User Defined Color
seApplicationGlobalSheetColorDraft	Sheet Color
seApplicationGlobalSheetMetalDynamicDisplay	Sheet Metal Dynamic Display
seApplicationGlobalSheetMetalGage	Sheet Metal Gage
seApplicationGlobalSheetNoNameSeprator	Sheet Number and name seprator
seApplicationGlobalSheetTabIColorDraft	Sheet Tab1 Color in Draft
seApplicationGlobalSheetTab2ColorDraft	Sheet Tab2 Color in Draft
seApplicationGlobalSheetTabDisplayInfo	Specifies what information is displayed in the sheet tab : 0 = sheet number; 1 = sheet name; 2 = sheet number and sheet name
seApplicationGlobalSheetTabDisplayInfoSeparator	Specifies the separator string used between the sheet number and sheet name when both are displayed in the sheet tabs.
seApplicationGlobalShowCommandTips	Show command tips
seApplicationGlobalShowDragRectTimeValue	Show drag rectangle when full display time exceeds
seApplicationGlobalShowDrawingViewScale	Show drawing view scale

seApplicationGlobalShowEdgesByCuttingPlane	Show Edges by cutting plane
seApplicationGlobalShowLiveRules	Show Live rules
seApplicationGlobalShowOrientationTriad	Show Orientaion Triad
seApplicationGlobalShowPartReference	Show Part reference
seApplicationGlobalShowPFDocView	Show Path finder in doc view
seApplicationGlobalShowRotationAngle	Show rotation angle
seApplicationGlobalShowSensorIndicator	Show sensor indicator
seApplicationGlobalShowSheetNumberParentAnnotation	Show sheet number if parent annotation
seApplicationGlobalShowStartUpScreen	Show Start up screen
seApplicationGlobalShowtriangulationbendlines	Show triangulation bend lines
seApplicationGlobalShowUnitsInValueFields	Shows units of measure in value fields.
seApplicationGlobalShowUnitsValueField	Show units in value field
seApplicationGlobalSimplifyBSplineEdges	Simplify BSpline edges
seApplicationGlobalSketchRelationshipPeerEdges	Sketch Relationship to peer Edges
seApplicationGlobalSmallAsmDynamicDisplay	Small Asm dynamic display
seApplicationGlobalSmallAsmOccurence	Small Asm Occurance
seApplicationGlobalSoftwareVHL	Enable software VHL/Disable hardware VHL
seApplicationGlobalStartEnvironmentSyncOrOrdered	Start part or sheet metal
seApplicationGlobalStartLastSaveDoc	Start with my last saved document
seApplicationGlobalStartUsingTemplate	Start using template
seApplicationGlobalStatusBarCmdFinder	Status Bar Command Finder
seApplicationGlobalStatusBarCommonViews	Status Bar Common Views
seApplicationGlobalStatusBarFit	Status Bar Fit
seApplicationGlobalStatusBarLookAtFace	Status Bar Look At Face
seApplicationGlobalStatusBarNamedViews	Status Bar Named View
seApplicationGlobalStatusBarPan	Status Bar Pan
seApplicationGlobalStatusBarPreviousView	Status Bar Previous View
seApplicationGlobalStatusBarQuickViewCube	Quick View Cube
seApplicationGlobalStatusBarRecordVideo	
seApplicationGlobalStatusBarRotateView	Status Bar Rotate View
seApplicationGlobalStatusBarSketchView	Status Bar Sketch View
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seApplicationGlobalStatusBarZoomArea	Status Bar Zoom Area
seApplicationGlobalStatusBarZoomSlider	Status Bar Zoom Slider
seApplicationGlobalStatusBarZoomTool	Status Bar Zoom Tool
seApplicationGlobalSteeringwheelSize	Steerinf Wheel Size
seApplicationGlobalStopLifeCycleProcess	Stop life cycle process
seApplicationGlobalStoreGeomPart	Store geometry in part file
seApplicationGlobalStudyType	Study Type
seApplicationGlobalSubAsmSimplificationLargeAsm	Sub asm simplification for large asm
seApplicationGlobalSubAsmSimplificationMedAsm	Sub asm simplification for medium asm
seApplicationGlobalSubAsmSimplificationSmallAsm	Sub asm simplification for small asm
seApplicationGlobalSystemInfo	Retrieve system information [readonly]
seApplicationGlobalTangentEdgeStyle	Tangent edge style
seApplicationGlobalTangentEdgeStyleFromAsm	Tangent edge style from asm
seApplicationGlobalTeamcenterFormula	
seApplicationGlobalTeamCenterMode	TeamCenter Mode
seApplicationGlobalTemplatePathFolder	
seApplicationGlobalTemplateRootFolder	Template Root Folder
seApplicationGlobalThreadDisplayMode	Thread Display mode
seApplicationGlobalUpdateAll	Inter-Part_Genral Update All
seApplicationGlobalUpdateDraft	Update draft files also
seApplicationGlobalUpdateLinkAutomatically	Update links automatically

seApplicationGlobalUploadOnClose	Upload On Close
seApplicationGlobalUseAllSimplifiedParts	Use All Simplified Parts
seApplicationGlobalUseAllSimplifiedSubAssemblies	Use All Simplified SubAssemblies
seApplicationGlobalUseConfigOrModelViewShow	Use configuration or model view show/hide states
seApplicationGlobalUseDefaultPlacementName	Use default placement name during Replace Part
seApplicationGlobalUseDetailEnvelopeDisplayAsCircle	Show Detail Envelope as Circle when TRUE.
seApplicationGlobalUseDetailEnvelopeStandard	Use to set the Detail Envelope Standard Constants. Valid values are defined by enum DetailEnvelopeStandardConstants.
seApplicationGlobalIlseDimensionStyleElementMapAngularCoordinateDim	
seApplicationGlobalUseDimensionStyleElementMapAngularDim	
seApplicationGlobalUseDimensionStyleElementMapBalloon	
seApplicationGlobalUseDimensionStyleElementMapBHC	
seApplicationGlobalUseDimensionStyleElementMapBlockLabels	
seApplicationGlobalUseDimensionStyleElementMapCallouts	
seApplicationGlobalUseDimensionStyleElementMapCenterline	
seApplicationGlobalUseDimensionStyleElementMapCenterMark	
seApplicationGlobalUseDimensionStyleElementMapChamferDim	
seApplicationGlobalUseDimensionStyleElementMapCircularDiameterDim	
seApplicationGlobalUseDimensionStyleElementMapConnectors	
seApplicationGlobalUseDimensionStyleElementMapCoordinateDim	
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seApplicationGlobalUseDimensionStyleElementMapRadialDim	
seApplicationGlobalUseDimensionStyleElementMapSTSymbols	
seApplicationGlobalUseDimensionStyleElementMapSymmetricDiameterDim	
seApplicationGlobalUseDimensionStyleElementMapWeldSymbols	
seApplicationGlobalUseDimensionStyleMapping	Use Dimension Style Mapping
seApplicationGlobalUseDrawingViewCommandBarDRWcommand	Use Drawing View Command Bar when the Drawing View Wizard command is run
seApplicationGlobalUseDrawingViewShowCroppingEdges	Show cropping edges when TRUE.
seApplicationGlobalUseDrawingViewShowEdgesHiddenTangentEdgesHiddenByOtherParts	Show tangent edges that are hidden by other parts when TRUE.
seApplicationGlobalUseDrawingViewShowEdgesHiddenTangentEdgesSelfHidden	Show the hidden tangent edges when TRUE.
seApplicationGlobalUseDrawingViewWizardModelsDragged	Use drawing view wizard when models are dragged onto the drawing sheet
seApplicationGlobalUseformulaPlacementName	Use formula for placement name
seApplicationGlobalUseGesters	Use gesters
seApplicationGlobalUselSOToleranceTable	Use ISO Tolerance Table when TRUE. Use ANSI Tolerance Table when FALSE.
seApplicationGlobalUselevelbaseNumbers	Use level bases numbers
seApplicationGlobalUseOnLineHelp	
seApplicationGlobalUserType	User Type
seApplicationGlobalUseshading2DFence	Use Shading 2D Fence
seApplicationGlobalUseShadingDraft	Use shading in Draft
seApplicationGlobalUseShadingRefplanes	Use Shading Reference Plane
seApplicationGlobalUseShadingSelection	Use Shading on Selection
seApplicationGlobalUseSimplifiedModels	Use simplified models
seApplicationGlobalUseWebBrowser	Use my web browser
seApplicationGlobalUsewhenPlacingAsmPartslib	Use when placing asm from parts lib
seApplicationGlobalVerticalScrollBar	Vertical scroll bar
seApplicationGlobalViewTransitionValue	View Translation
seApplicationGlobalVisibleEdgesStyleFromAsm	Visible edge style from asm
seApplicationGlobalVisibleEdgeStyle	Visible edge style
seApplicationGlobalWeldSymbols	Weld Symbols

seApplicationGlobalWeldSymbolsDraft	Weld symbols
seApplicationGlobalZebraFolderMapping	Set ZebraFolderMapping
seApplicationGlobalZebraFormula	
seApplicationGlobalZone	
seApplicationGlobalZoomToolLeftClick	Zoom Tool left click
seApplicationGlobalZoomToolLeftDrag	Zoom Tool left drag
seApplicationGlobalZoomToolRightClick	Zoom Tool right click
seApplicationGlobalZoomToolRightDrag	Zoom Tool right drag

Specifies the global parameter for which the setting is to be returned. Valid values are members of the ApplicationGlobalConstants for Application objects, AssemblyGlobalConstants for AssemblyDocument objects, PartGlobalConstants for PartDocument objects, SheetMetalGlobalConstants for SheetMetalDocument objects, or WeldmentGlobalConstants for WeldmentDocument objects.

Value

Returns the setting of the specified global parameter. The data type returned depends on the data being queried. For Application objects, the return data type is Long or Variant. For all other objects, the returned data type is Double.

Example

Example (VB.NET)

```
Imports System.10
Imports System.Reflection
Imports System.Runtime.InteropServices

Module Example
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objType As Type = Nothing
        Dim objFieldInfos() As System.Reflection.Fieldinfo
        Dim objFieldInfo As System.Reflection.Fieldinfo
        Dim objValue As Object = Nothing
        Dim constantvalue As SolidEdgeFramework.ApplicationGlobalConstants

        Try
            OleMessageFliter .Register()

            ' Connect to Solid Edge
            objApplication = Marshal.GetActiveObject("SolidEdge.Application")

            ' Demonstrate getting specific global value.
            objApplication.GetGlobalParámeter( SolidEdgeFramework.ApplicationGlobalConstants . seApplicationGlobalSysteminfo, objVali

            '
            ' Now demonstrate getting all global values by dynamically enumerating the ApplicationGlobalConstants via .NET reflec
            '

            ' Get ApplicationGlobalConstants type
            objType = GetType(Solid EdgeFramework.ApplicationGlobalConstants)

            ' Get all ApplicationGlobalConstants constants
            objFieldInfos = objType.GetFields(System.Reflection.BindingFlags.Static Or System.Reflection.BindingFlags.Public)

            ' Loop through all of the constants and demonstrate getting their values.
            For Each objFieldInfo In objFieldInfos
                objValue = Nothing
                If Not objFieldInfo.IsSpecialName Then
                    constantvalue = objFieldInfo.GetRawConstantValue()

                    ' Wrap GetGlobalParameter() in a try\catch because it can throw an exception.
                    Try
                        ' Get global parameter for specified constant.
                        objApplication.GetGlobalParameter(constantvalue, objValue)
                    Catch ex As Exception
                        objValue = ex.Message
                    End Try

                    If objValue Is Nothing Then
                        objValue = String.Empty
                    End If

                    ' Write out constant and value.
                    Console.WriteLine("{0} = {1}", constantvalue, objValue)
                End If
            Next
        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter. Revoke()
        End Try
    End Sub
End Module
```

See Also

Application Object | Application Members

Solid Edge FrameworkSupport Type Library

GetJogPointlDistance Method

Collapse All

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Description

Returns the JogPoint! distance for the jogged line in a Dimension.

Syntax

Visual Basic

```
Public Sub GetJogPointlDistance( _
    ByRef VaLue As Double _
)
```

Parameters

Value
Returns the distance along the dimension line.

Remarks

This method is applicable only for dimensions with jogged capability (for example, all radius, diameter, and coordinate dimensions).

Example

Example (Visual Basic)

```
Private Sub Form_Load()
    Dim objApp As SolidEdgeFrameWork.Application
    Dim objDoc As SolidEdgeDraft.DraftDocument
    Dim objSheet As SolidEdgeDraft.Sheet
    Dim objArc As SolidEdgeFrameworkSupport.Arc2d
    Dim objDimension As SolidEdgeFrameworkSupport.Dimension
    Dim dblIogDistance As Double
    ' Report errors
    Const PI = 3.14159265358979
    ' Create/get the application with specific settings
    On Error Resume Next
    Set objApp = GetObject(, "SolidEdge.Application")
    If Err Then
        Err.Clear
        Set objApp = CreateObject("SolidEdge.Application")
        Set objDoc = objApp.Documents.Add("SolidEdge.DraftDocument")
        objApp.Visible = True
    Else
        Set objDoc = objApp.ActiveDocument
    End If
    'getting the Active Sheet object
    Set objSheet = objDoc.ActiveSheet
    'create an arc2d
    Set objArc = objSheet.Arcs2d.AddByCenterStartEnd(xCenter:=0.2, _
                                                     yCenter:=0.2, xStart:=0.1j yStart:=0.1, xEnd:=0.3, yEnd:=0.3)

    'create a dimension
    Set objDimension = objSheet.Dimensions.AddRadius(Object:=objArc)
    'convert to jogged dimension
    objDimension.logged = True
    'get the jogpointl distance
    Call objDimension.Get!ogPointlDistanee(Value:=dblIogDistance)
    ' USER DISPLAY
    ' Release objects
    Set objApp = Nothing
    Set objDoc = Nothing
    Set objSheet = Nothing
    Set objDimension = Nothing
    Set objArc = Nothing
End Sub
```

See Also

Dimension Object | Dimension Members

Solid Edge FrameworkSupport Type Library

GetJogPoint2 Method

Collapse All

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Description

Returns the JogPoint2 distance for the jogged line in a Dimension.

Syntax

Visual Basic

```
Public Sub GetJogPoint2( _
    ByRef x As Double, _
    ByRef y As Double _
)
```

Parameters

x
The x offset from the Jog Pointl.

y
The y offset from the Jog Pointl.

Remarks

The values are returned as the x and y offset distances. This method is applicable only for dimensions with jogged capability (for example, all radius, diameter, and coordinate dimensions).

Example

Example (Visual Basic)

```
Private Sub Form_Load()
    Dim objApp As SolidEdgeFrameWork.Application
    Dim objDoc As SolidEdgeDraft.DraftDocument
    Dim objSheet As SolidEdgeDraft.Sheet
    Dim objArc As SolidEdgeFrameworkSupport.Arc2d
    Dim objDimension As SolidEdgeFrameworkSupport.Dimension
    Dim dblx As Double, dbly As Double
    ' Report errors
    Const PI = 3.14159265358979
    ' Create/get the application with specific settings
    On Error Resume Next
    Set objApp = GetObject(, "SolidEdge.Application")
    If Err Then
        Err.Clear
        Set objApp = CreateObject("SolidEdge.Application")
        Set objDoc = objApp.Documents.Add("SolidEdge.DraftDocument")
        objApp.Visible = True
    Else
        Set objDoc = objApp.ActiveDocument
    End If
    'getting the Active Sheet object
    Set objSheet = objDoc.ActiveSheet
    'create an arc2d
    Set objArc = objSheet.Arcs2d.AddByCenterStartEnd(xCenter:=0.2, _
                                                     yCenter:=0.2, xStart:=0.1, yStart:=0.1, xEnd:=0.3, yEnd:=0.3)
    'create a dimension
    Set objDimension = objSheet.Dimensions.AddRadius(Object:=objArc)
    'convert to jogged dimension
    objDimension.logged = True
    'get jog point2 offsets
    Call objDimension.GetlogPoint2(x:=dblx, y:=dbly)
    ' USER DISPLAY
    ' Release objects
    Set objApp = Nothing
    Set objDoc = Nothing
    Set objSheet = Nothing
    Set objDimension = Nothing
    Set objArc = Nothing
End Sub
```

See Also

Dimension Object | Dimension Members

Solid Edge FrameworkSupport Type Library

GetJogPoint3Distance Method

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Description

Sets the JogPoint3 distance for the jogged line in a Dimension.

Syntax

Visual Basic

```
Public Sub GetJogPoint3Distance( _
    ByRef VaLue As Double _
)
```

Parameters

Value
Returns the distance along the dimension line from the second jog point.

Remarks

This method is applicable only for dimensions with jogged capability (for example, all radius, diameter, and coordinate dimensions).

Example

Example (Visual Basic)

```
Private Sub Form_Load()
    Dim objApp As SolidEdgeFrameWork.Application
    Dim objDoc As SolidEdgeDraft.DraftDocument
    Dim objSheet As SolidEdgeDraft.Sheet
    Dim objArc As SolidEdgeFrameworkSupport.Arc2d
    Dim objDimension As SolidEdgeFrameworkSupport.Dimension
    Dim DblDistance As Double
    ' Report errors
    Const PI = 3.14159265358979
    ' Create/get the application with specific settings
    On Error Resume Next
    Set objApp = GetObject(, "SolidEdge.Application")
    If Err Then
        Err.Clear
        Set objApp = CreateObject("SolidEdge.Application")
        Set objDoc = objApp.Documents.Add("SolidEdge.DraftDocument")
        objApp.Visible = True
    Else
        Set objDoc = objApp.ActiveDocument
    End If
    'getting the Active Sheet object
    Set objSheet = objDoc.ActiveSheet
    'create an arc2d
    Set objArc = objSheet.Arcs2d.AddByCenterStartEnd(xCenter:=0.2, _
                                                     yCenter:=0.2j xStart:=0.1, yStart:=0.1, xEnd:=0.3, yEnd:=0.3)
    'create a dimension
    Set objDimension = objSheet.Dimensions.AddRadius(Object:=objArc)
    'convert to jogged dimension
    objDimension.logged = True
    'get jog point2 offsets
    Call objDimension.Get!ogPoint3Distance(Value:=DblDistance)
    ' USER DISPLAY
    ' Release objects
    Set objApp = Nothing
    Set objDoc = Nothing
    Set objSheet = Nothing
    Set objDimension = Nothing
    Set objArc = Nothing
End Sub
```

See Also

Dimension Object | Dimension Members

Search

304	SetToFaceOffsetData Method
305	Add Method
306	Distance Property
307	Distance Property
308	GetMinSearchDistValue Method
309	SetMinSearchDistValue Method
310	Distance Property
311	Depth Property
312	Duplicate Method
313	Depth Property
314	Add Method
315	AddExtrudedProtrusion Method
316	AddFiniteExtrudedProtrusion Method
317	AddFiniteRevolvedProtrusion Method
318	AddThickenFeature Method
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Solid Edge Part Type Library

SetMinSearchDistVal Method

Collapse All

Solid Edge Part Type Library > FEAConnector Object : SetMinSearchDistValue Method

Description

Set minimum search distance value for a connector.

Syntax

Visual Basic

```
Public Sub SetMinSearchDistV
    ByVai dConnectorVaLue As
)
```

Parameters

dConnectorValue

Example

Example (VB.NET)

```
Imports System.10
Imports System.Runtime.In

Module Example
    <STAThread()> _
    Sub Main()
        Dim objApplicatio
        Dim objPartDocume
        Dim objStudyOwner
        Dim objStudy As S
        Dim objConnectorO
        Dim objConnector
        Dim dMinSearchDis

        Try
            OleMessageFil

            ' Connect to
            objApplicatio
            objPartDocume
```

Solid Edge Part Type Library

GetMinSearchDistValue Method

Colla pseAH

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Description

Get minimum search distance value for a connector.

Syntax

Visual Basic

```
Public Sub GetMinSearchDistValue( _
    ByRef dConnectorVaLue As Double _
)
```

Parameters

dConnectorValue

See Also

FEAConnector Object | FEAConnector Members | Solid Edge ST3 - What's New

Solid Edge Assembly Type Library

GetOrigin Method

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Description

Returns the origin for the referenced object.

Syntax

Visual Basic

```
Public Sub GetOrigin( _
    ByRef OriginX As Double, _
    ByRef OriginY As Double, _
    ByRef OriginZ As Double _
)
```

Parameters

OriginX
Returns the x coordinate of the Occurrence's point of origin.

OriginY
Returns the y coordinate of the Occurrence's point of origin.

OriginZ
Returns the z coordinate of the Occurrence's point of origin.

Example

Example (Visual Basic)

```
Private Sub Form_Load()
    Dim objApp As SolidEdgeFramework.Application
    Dim objDoc As SolidEdgeAssembly.AssemblyDocument
    Dim objPart As SolidEdgeAssembly.Occurrence
    Dim dblOriginX As Double
    Dim dblOriginY As Double
    Dim dblOriginZ As Double
    Const TESTFILE = "T:\vbtests\testcases\Sample.asm"
    ' Report errors
    Const PI = 3.14159265358979
    ' Create/get the application with specific settings
    On Error Resume Next
    Set objApp = GetObject(, "SolidEdge.Application")
    If Err Then
        Err.Clear
        Set objApp = CreateObject("SolidEdge.Application")
        Set objDoc = objApp.Documents.Add("SolidEdge.AssemblyDocument")
        objApp.Visible = True
    Else
        Set objDoc = objApp.ActiveDocument
    End If
    Call objDoc.Close
    ' opening an existing assembly file
    Set objDoc = objApp.Documents.Open(TESTFILE)
    ' getting a part in the assembly
    Set objPart = objDoc.Occurrences.Item(3)
    ' getting the origin of the placed part
    Call objPart.GetOrigin(OriginX:=dblOriginX, OriginY:=dblOriginY, OriginZ:=dblOriginZ)
```

```
        ' USER DISPLAY
        ' Release objects
        Set objApp = Nothing
        Set objDoc = Nothing
        Set objPart = Nothing
    End Sub
```

See Also

Occurrence Object | Occurrence Members

Solid Edge Geometry Type Library

GetStrokeData Method

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Description

Returns information on the strokes on the referenced Edge.

Syntax

Visual Basic

```
Public Sub GetStrokeData( _
    ByVal ToLerance As Double, _
    ByRef StrokeCount As Long, _
    ByRef Points() As Double, _
    ByRef Params() As Double _
)
```

Parameters

Tolerance
This is the chordal tolerance between a curve and its facet edges, i.e. through upper bound on distance from chord to the curve. This distance is in Meters.

StrokeCount
Returns the number of strokes for the Edge object based on the specified tolerance.

Points
Returns the x, y, and z coordinates of the points on the Edge represented by the strokes. The size of the array is three times the StrokeCount.

Params
Returns the parameter values that are used to calculate the points on the Edge. The size of the array is equal to the StrokeCount.

Example

Example (Visual Basic)

```
Private Sub Form_Load()
    Dim objApp As SolidEdgeFramework.Application
    Dim objDoc As SolidEdgePart.PartDocument
    Dim objBody As SolidEdgeGeometry.Body
    Dim objEdge As SolidEdgeGeometry.Edge
    Dim lngStrokes As Long
    Dim dblPoints() As Double
    Dim dblParams() As Double
    ' Report errors
    Const PI = 3.14159265358979
    ' Create/get the application with specific settings
    On Error Resume Next
    Set objApp = GetObject(, "SolidEdge.Application")
    If Err Then
        Err.Clear
        Set objApp = CreateObject("SolidEdge.Application")
        Set objDoc = objApp.Documents.Add("SolidEdge.PartDocument")
        objApp.Visible = True
    Else
        Set objDoc = objApp.ActiveDocument
    End If
    ' creating the base feature
    If CreateModel(objDoc) <> "" Then
        MsgBox "Error creating the model"
        Exit Sub
    End If
```

```
    ' getting the body object of the model
    Set objBody = objDoc.Models(l).Body
    'getting an edge form the collection of edges of the body object
    Set objEdge = objBody.Edges(EdgeType:=igQueryAll).Item(2)
    ¹ getting the stroke data for an edge
    Call objEdge.GetStrokeData(TOLERANCE:=0.001, StrokeCount:=lngStrokes, Points:=dblPoints, _
                               Params:=dblParams)
    ¹ USER DISPLAY
    ¹ Release objects
    Set objApp = Nothing
    Set objDoc = Nothing
    Set objBody = Nothing
    Set objEdge = Nothing
End Sub
```

See Also

Edge Object | Edge Members

Solid Edge ST7 SDK

Glossary

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A

ActiveX automation
A technology that allows any object to expose a set of commands and functions that some other piece of code can invoke. Automation is intended to allow applications to create system macro programming tools.

API
An acronym for Application Programming Interface. An API is a collection of formal definitions that describe the function calls and other protocols a program uses to communicate with the outside world.

Application Domain
Application Domains are isolated boundaries of memory allocated by the framework for applications to execute inside of. When your programs execute, the framework will create a "Default AppDomain" for your application.

argument
A value passed to a procedure. The value can be a constant, variable, or expression.

assembly
A group of parts specified as a single component. The parts in an assembly are usually individually specified elsewhere, combined according to requirements, and physically connected. When the assembly is a unit to be used in higher level assemblies, it is called a subassembly ;when the assembly is a product, it is a top-level assembly.

Assembly (.NET)
Assemblies are the building blocks of .NET framework applications. When you compile a .NET application, you are building an assembly.

associative
A condition in which an element is related to another element.

available
A condition in which a document can be accessed by a user for review or revision.

B

base feature
A feature that defines the basic part shape. Solid models are constructed by adding material to and removing material from a base feature.

C

chamfer
A corner that has been cut at an angle.

class
The definition of a data structure and the functions that manipulate that structure.

collection
A special type of object whose purpose is to provide methods of creating objects and also providing a way of accessing all the objects of a specific type.

COM
The acronym for the Component Object Model. This model specifies a binary standard for object implementation that is independent of the programming language you decide to use. This binary standard lets two applications communicate through object-oriented interfaces without requiring either to know anything about the other's implementation. It is supported at the operating system level and is the basis of all Microsoft products.

COM Interop
The .NET framework is a natural progression from COM because the two models share many central themes,

including component reuse and language neutrality. For backward compatibility, COM Interop provides access to existing COM components without requiring that the original component be modified. You can incorporate COM components into a .NET Framework application by using COM Interop tools to import the relevant COM types. Once imported, the COM types are ready to use.

When you add a reference to a Solid Edge type library, Visual Studio .NET automatically build a .NET assembly for use in your project. This assembly will contain all of the type library definitions in a format that can be consumed from any .NET programming language. These generated assemblies are referred to as "Interop Assemblies".

command stacking
The process of temporarily deactivating a command without terminating it.

compound document
A document that contains files with various formats. For example, a Word document that has a Solid Edge file embedded in it.

container
A document that contains documents created with other applications. Through ActiveX, you can access the application that created the document and link and embed a document created by another application.

cutout
A feature created by removing material from a part by extrusion, revolution, sweeping, or lofting. A profile defines the feature's shape.

D

dimension
A control that assigns and maintains a dimensional value to an individual element or establishes a dimensional relationship between multiple elements. Dimensions are represented graphically by a label consisting of text, lines, and arrows.

dimension axis
An axis for dimension orientation that you define by selecting a line. You can place linear dimensions that run parallel or perpendicular to the axis. By default, dimensions are placed horizontally or vertically.

dimension group
A series of dimensions. You can place a chained dimension group, a coordinate dimension group, or a stacked dimension group.

driven dimension
A dimension whose value depends on the value of other dimensions or elements.

E

element
A single, selectable unit. You can select geometric elements, dimensions, annotations, objects placed in the drawing through ActiveX, and so forth. The type of element that can be selected is determined by command context.

embed
A method for inserting information from a source document into the active document. Once embedded, the information becomes part of the active document; if changes are made to the source document, the updates are not reflected in the active document.

enumerator
An object that iterates through a sequence of items.

F

feature
A characteristic of a part that is usually created by adding material to or removing material from the basic part shape. Features include holes, cutouts, protrusions, and so forth.

G

Garbage Collection
The .NET framework employs a non-deterministic approach to freeing memory whereas languages like Visual Basic

6.0 or C++ use a deterministic approach. This means that when you free a reference to an object, the memory allocated for the object is not immediately freed but rather is flagged as being available for garbage collection. Garbage collection occurs when the framework identifies an opportune time to perform the actual collection. You have no control of when this occurs nor will you be notified when it happens.

In regards to COM Interop, this can lead to trouble releasing COM objects in a timely manner.

H

host object
Any object in Solid Edge to which you have attached an attribute set.

I

Interop Assembly
Interop Assemblies are generated by the Type Library Importer (Tlbimp.exe) tool. Visual Studio .NET automatically generates Interop assemblies for you when you reference COM type libraries. Once create this reference, your application will depend upon the Interop assembly for execution.

iterator
A statement that sequences through a set of objects for a specified number of times or until a certain condition is met.

See Also: enumerator

K

key point
A recognizable point on an element. Key points include vertices, mid points, center points, and so forth.

L

link
A method for inserting information stored in a source document into the active document. The two documents are connected, and changes made in the source document are reflected in the active document.

M

macro
A sequence of actions or commands that can be named and stored. When you run the macro, the software performs the actions or runs the commands.

method
Any operation that can be performed on an object.

N

native data
All the information that an application requires to edit an object.

O

object
Information that can be linked or embedded into an ActiveX-compliant product.

object browser
A feature of Microsoft Visual Studio that allows you to examine the contents of a specified object library to access information about the objects in that library.

object hierarchy, object model
A diagram that shows the relationships among objects.

object persistance
A method for saving a complex network of objects in a permanent binary form, usually disk storage, that persists

after those objects are deleted from memory.

p

parent feature
A feature upon which another feature is based. For example, a circular pattern feature is created by copying other features in a circular arrangement. The copied feature is the parent of the circular pattern feature.

profile
A 2-D set of variational elements used to construct a feature.

property
A unique characteristic of an element or object in a file. The characteristics can include the name, parent, application, and so forth.

R

reference element
An element that is not included in the part model. Reference elements, such as profile planes and centerline axes, are used for construction only.

reference plane
A flat, rectangular, infinite surface whose position and orientation provide a frame of reference for creating and working with profiles in 3-D space.

routing slip
A list that indicates the users that should receive electronic mail containing an attached document.

RCW, Runtime Callable Wrapper
The .NET runtime exposes COM objects through a proxy called the runtime callable wrapper. Although the RCW appears to be an ordinary object to .NET clients, its primary function is to marshal calls between a .NET client and a COM object. The runtime creates exactly one RCW for each COM object within an application domain.

S

selection set
A single selected object or a group of selected objects.

server
The application that created objects that are linked or embedded in a compound document (container).

share embed
A document copied directly into a drawing. When you embed the same document more than once in the same document, the document elements are copied each time. When you share embed the same document more than once in the same document, the documents are not copied each time. Instead, the other documents reference the initial placement of the document.

SmartFrame
A placeholder for an ActiveX object. The contained ActiveX object can be 2-D or 3-D, but will be mapped to 2-D space. SmartFrames are generally rectangles on a sheet that enclose embedded or linked object(s) and have some intelligence about how to deal with the data in that frame.

symbol
A document placed in a drawing. You can override and edit the properties and style of the symbol. A document can be linked, embedded, shared embedded, or inserted as elements.

T

object library, type library
A standalone file or a component within another file (typically a DLL) containing type information that is used by automation controllers such as Visual Basic to find and invoke the properties and methods on an object.

Solid Edge Draft Type Library

GraphicMembers Collection Members

Collapse All

Sonu cuye Draft Type Library : GraphicMembers Collection

Public Methods

-V Item	The items in a collection.

Public Properties

@ Application	Returns the active application object.
Ü=| Count	Returns the number of objects in the referenced collection.
Parent	Returns the parent object for the referenced object.

See Also

GraphicMembers Collection

Solid Edge Draft Type Library

GraphicMembers Collection

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Description

Represents a collecion of GraphicMember objects.

Remarks

The GraphicMembers object is a collection of all derived graphic Drawing View objects. It is a collection of all DVArc2d, DVBSplineCurve2d, DVCircle2d, DVEllipse2d, DVEllipticalArc2d, DVLine2d, and DVLineString2d objects. This collection can only be obtained from a Drawing View or from a ModelMember object.

See Also

GraphicMembers Members

Solid Edge ST7 SDK

Handling "Application is Busy" and "Call was Rejected By Callee" errors

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If you automate Solid Edge from an external (out-of-proc) application, you may receive one of the following errors:

- Application is busy **(RPC_E_CALL_REJECTED** 0x80010001)
- Call was rejected by callee **(RPC_E_SERVERCALL_RETRYLATER** 0x8001010A)

These errors occur due to threading contention issues between automation clients and Solid Edge. They can be handled by implementing the OleMessageFilter error handlers in your automation application. Once you have the code in place, simply call *OleMessageFilter.RegisterO* before you make any calls to COM clients and *OleMessageFilter.RevokeQ* at the end of your program.

> A In order for OleMessageFilter to work, you must ensure that Thread.GetApartmentState method returns STA. In Windows Forms applications, this is the default. In Console applications, it is not the default. You will need to specify the STAThreadAttribute on your Main() function.

OleMessageFilter for Visual Basic .NET

In Console applications, you must decorate your Sub MainQ function with STAThreadAttribute.

VB.NET example usage — Copy Code

```
<STAThread()> _
Sub Main()
  OleMessageFilter.Register()
  ' Make COM calls.
  OleMessageFilter.Revoke()
End Sub
```

In your Visual Basic .NET project, create a new class and name it OleMessageFilter.vb. Copy this example code and paste it into the new class.

VB.NET implementation of OleMessageFilter — Copy Code

```
Imports System.Runtime.InteropServices
Imports System.Threading

Public Class OleMessageFilter
    Implements IQleMessageFilter

    Public Shared Sub Register()
        Dim newFilter As IQleMessageFilter = New OleMessageFilter()
        Dim oldFilter As IQleMessageFilter = Nothing
        Dim iRetVal As Integer

        If (Thread.CurrentThread.GetApartmentState() = Apartmentstate.STA) Then
            iRetVal = CoRegisterMessageFilter(newFilter, oldFilter)
        Else
            Throw New COMException("Unable to register message filter because the current thread apartment state is not STA.")
        End If

    End Sub

    Public Shared Sub Revoke()
        Dim oldFilter As IQleMessageFilter = Nothing
        CoRegisterMessageFilter (Nothingj oldFilter)
    End Sub

    Private Function HandleInComingCall(ByVal dwCallType As Integer, _
                                        ByVai hTaskCaller As System.IntPtr, _
                                        ByVai dwTickCount As Integer, _
                                        ByVai ipInterfaceInfo As System.IntPtr) _
                                        As Integer Implements IQleMessageFilter.HandleInComingCall

        Return SERVERCALL.SERVERCALL_ISHANDLED
    End Function

    Private Function RetryRejectedCall(ByVal hTaskCallee As System.IntPtr, _
                                       ByVai dwTickCount As Integer, _
                                       ByVai dwRejectType As Integer) _
                                       As Integer Implements IQleMessageFilter.RetryRejectedCall

        If dwRejectType = SERVERCALL. SERVERCALL_RETRYLATER Then
            Return 99
        End If

        Return -1
    End Function

    Private Function MessagePending(ByVal hTaskCallee As System.IntPtr, _
                                    ByVai dwTickCount As Integer, _
                                    ByVai dwPendingType As Integer) _
                                    As Integer Implements IQleMessageFilter.MessagePending

        Return PENDINGMSG.PENDINGMSG_WAITDEFPROCESS
```

```
    End Function

    <DllImport("Ole32.dll")> _
    Private Shared Function CoRegisterMessageFilter(ByVal newFilter As IOleMessageFilter, _
                                                    ByRef oldFilter As IOleMessageFilter) _
                                                    As Integer
    End Function
End Class

Enum SERVERCALL
    SERVERCALL_ISHANDLED = 0
    SERVERCALL_REJECTED = 1
    SERVERCALL_RETRYLATER = 2
End Enum

Enum PENDINGMSG
    PENDINGMSG_CANCELCALL = 0
    PENDINGMSG_WAITNOPROCESS = 1
    PENDINGMSG_WAITDEFPROCESS = 2
End Enum

<ComImport(), _
Guid("00000016-0000-0000-C000-000000000046"), _
InterfaceTypeAttribute(ComInterfaceType.InterfaceIsIUnknown)> _
Interface IOleMessageFilter
    <PreserveSig()> _
    Function HandleInComingCall(ByVal dwCallType As Integer, _
                                ByVal hTaskCaller As IntPtr, _
                                ByVal dwTickCount As Integer, _
                                ByVal lpInterfaceInfo As IntPtr) _
                                As Integer

    <PreserveSig()> _
    Function RetryRejectedCall(ByVal hTaskCallee As IntPtr, _
                               ByVal dwTickCount As Integer, _
                               ByVal dwRejectType As Integer) _
                               As Integer

    <PreserveSig()> _
    Function MessagePending(ByVal hTaskCallee As IntPtr, _
                            ByVal dwTickCount As Integer, _
                            ByVal dwPendingType As Integer) _
                            As Integer
End Interface
```

OleMessageFilter for Visual C#

In Console applications, you must decorate your Main() function with STAThreadAttribute.

C# example usage — Copy Code

```
[STAThread]
static void Main(string[] args)
{
  OleMessageFilter.Register();
  // Make COM calls.
  OleMessageFilter.Revoke();
}
```

In your Visual C# project, create a new class and name it OleMessageFilter.es. Copy this example code and paste it into the new class. You will want to modify the namespace of the example code to match your own namespace.

C# implementation of OleMessageFilter — Copy Code

```
using System;
using System.Runtime.InteropServices;
using System.Threading;

namespace SolidEdge.SDK
{
    class OleMessageFilter : IOleMessageFilter
    {
        public static void Register()
        {
            IOleMessageFilter newFilter = new 01eMessageFilter();
            IOleMessageFilter oldFilter = null;

            if (Thread.CurrentThread.GetApartmentState() == Apartmentstate.STA)
            {
                CoRegisterMessageFilter(newFilter, out oldFilter);
            }
            else
            {
                throw new COMException("Unable to register message filter because the current thread apartment state is not STA.");
            }
```

```
        }

        public static void Revoke()
        {
            IOleMessageFilter oldFilter = null;
            CoRegisterMessageFilter(null, out oldFilter);
        }

        int IOleMessageFilter.HandleInComingCall(
            int dwCallType,
            System.IntPtr hTaskCaller,
            int dwTickCount,
            System.IntPtr lpInterfaceInfo)
        {
            return (int)SERVERCALL.SERVERCALL_ISHANDLED;
        }

        int IOleMessageFilter.RetryRejectedCall(
            System.IntPtr hTaskCallee,
            int dwTickCount,
            int dwRejectType)
        {
            if (dwRejectType == (int)SERVERCALL.SERVERCALL_RETRYLATER)
            {
                return 99;
            }

            return -1;
        }

        int IOleMessageFilter.MessagePending(
            System.IntPtr hTaskCallee,
            int dwTickCount,
            int dwPendingType)
        {
            return (int)PENDINGMSG.PENDINGMSG_WAITDEFPROCESS;
        }

        [DllImport("Ole32.dll")]
        private static extern int CoRegisterMessageFilter(
            IOleMessageFilter newFilter,
            out IOleMessageFilter oldFilter);
    }

    enum SERVERCALL
    {
        SERVERCALL_ISHANDLED = 0,
        SERVERCALL_RE1ECTED = 1,
        SERVERCALL_RETRYLATER = 2
    }

    enum PENDINGMSG
    {
        PENDINGMSG_CANCELCALL = 0,
        PENDINGMSG_WAITNOPROCESS = 1,
        PENDINGMSG_WAITDEFPROCESS = 2
    }

    [ComImport(), Guid("00000016-0000-0000-C000-000000000046"),
    InterfaceTypeAttribute(ComInterfaceType.InterfaceIsIUnknown)]
    interface IOleMessageFilter
    {
        [PreserveSig]
        int HandleInComingCall(
            int dwCallType,
            IntPtr hTaskCaller,
            int dwTickCount,
            IntPtr lpInterfaceInfo);

        [PreserveSig]
        int RetryRejectedCall(
            IntPtr hTaskCallee,
            int dwTickCount,
            int dwRejectType);

        [PreserveSig]
        int MessagePending(
            IntPtr hTaskCallee,
            int dwTickCount,
            int dwPendingType);
    }
}
```

Solid Edge Draft Type Library

Item Method

1

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Description

The items in a collection.

Syntax

Visual Basic

```
Public Function Item( _
    ByVai Index As Variant _
) As DrawingView
```

Parameters

Index
Variant (input) index.

Example

Example (VB.NET)

```
Imports System.10
Imports System.Runtime.InteropServices

Module Example
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objDraftDocument As SolidEdgeDraft.DraftDocument
        Dim objSheet As SolidEdgeDraft.Sheet = Nothing
        Dim objDrawingViews As SolidEdgeDraft.Drawingviews = Nothing
        Dim objDrawingView As SolidEdgeDraft.Drawingview = Nothing

        Try
            OleMessageFilter.Register()

            objApplication = Marshal.GetActiveObject("SolidEdge.Application")
            objDraftDocument = objApplication.ActiveDocument
            objSheet = objDraftDocument.ActiveSheet
            objDrawingViews = objSheet.Drawingviews
            objDrawingView = objDrawingViews.Item(I)

        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter.Revoke()
        End Try
    End Sub
End Module
```

See Also

DrawingViews Collection | DrawingViews Members

Solid Edge FrameworkSupport Type Library

LeaderDistance Property

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Description

Sets and returns the leader distance of the referenced Dimension object.

Property type

Read-write property

Syntax

Visual Basic
`Public Property LeaderDistance As Double`

Remarks

The leader distance defines the leader point at the text position. The leader line is valid only when the break is positioned to left or right of the Dimension object.

See Also

Dimension Object | Dimension Members

Solid Edge FrameworkSupport Type Library

LeaderLineLength Property

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Description

Sets and returns the default leader distance.

Property type

Read-write property

Syntax

Visual Basic
`Public Property LeaderLineLength As Double`

Remarks

The value is a ratio of dimension text height. The leader line is valid only when the break is positioned to the left or right of the associated Dimension object.

See Also

DimensionStyle Object | DimensionStyle Members

Solid Edge FrameworkSupport Type Library

Leftindent Property

Collapse All

Description

Sets and returns the distance from the left margin to the end of the line.

Property type

Read-write property

Syntax

Visual Basic
`Public Property Leftindent As Double`

Remarks

The TextEdit object always has an insert index or an active selection. Setting any property on the TextEdit object applies to the current selection, if one is present, or to the text at the current insert index.

See Also

TextEdit Object | TextEdit Members

Solid Edge FrameworkSupport Type Library

Lines2d Collection Members

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Sunci cuye rrameworkSupport Type Library : Lines2d Collection

Public Methods

AddAsChamfer	Creates a Line2d object that connects two intersecting objects.
AddBy2Points	Creates a text box by specifying the diagonal points of the text box.
AddByPointAngleLength	Creates a Line2d object by specifying a start point, an angle, and a length.
Item	The items in a collection.

Public Properties

Application	Returns the active application object.
Count	Returns the number of objects in the referenced collection.
Parent	Returns the parent object for the referenced object.

See Also

Lines2d Collection

Solid Edge FrameworkSupport Type Library

Relations2d Collection

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Description

A collection of Relation2d objects.

Remarks

This object can be created from a Sheet object, a Profile object, a Sketch object, or a Layout object.

See Also

Relations2d Members

Solid Edge FrameworkSupport Type Library

Margin Property

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Description

Sets and returns the minimum distance from the extents of the OLE object to the extents of the frame boundary.

Property type

Read-write property

Syntax

Visual Basic
`Public Property Margin As Double`

Remarks

The distance is specified in container units.

Example

Example (Visual Basic)

```
Private Sub Form_Load()
    Dim objApp As SolidEdgeFrameWork.Application
    Dim objDoc As SolidEdgeDraft.DraftDocument
    Dim objSheet As SolidEdgeDraft.Sheet
    Dim objSmartFrame2d As SolidEdgeFrameworkSupport.SmartFrame2d
    Dim objSmartFrame2dStyle As SolidEdgeFrameworkSupport.SmartFrame2dStyle
    Dim objFrame As SolidEdgeFrameworkSupport.Frame
    Dim dblMargin As Double
    ' Report errors
    Const PI = 3.14159265358979
    ' Create/get the application with specific settings
    On Error Resume Next
    Set objApp = GetObjectG "SolidEdge.Application")
    If Err Then
        Err.Clear
        Set objApp = CreateObject("SolidEdge.Application")
        Set objDoc = objApp.Documents.Add("SolidEdge.DraftDocument")
        objApp.Visible = True
    Else
        Set objDoc = objApp.ActiveDocument
    End If
    'getting the Active Sheet object
    Set objSheet = objDoc.ActiveSheet
    'Get the SamrtFrames2d object
    Set objSmartFrame2d = objSheet.SmartFrames2d.AddBy2Points(StyleName:="Reference"j xl:=0.1, yl:=0.1, x2:=0.2, y2:=0.2)
    'Get the Frame object
    Set objFrame = objSmartFrame2d.Frame
    objFrame.BorderVisible = True
    'Get the Margin ogf the Frame
    dblMargin = objFrame.Margin
    'Set the Margin to 0.4
    objFrame.Margin = 0.4
    ' USER DISPLAY
    ' Release objects
    Set objApp = Nothing
    Set objDoc = Nothing
    Set objSheet = Nothing
    Set objSmartFrame2d = Nothing
    Set objSmartFrame2dStyle = Nothing
    Set objFrame = Nothing
End Sub
```

See Also

Frame Object | Frame Members

Solid Edge Part Type Library

MeasureDistance Method

Collapse All

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Description

Measures the distance between two elements.

Syntax

Visual Basic

```
Public Sub MeasureDistance( _
    ByVal Element1 As Object, _
    ByVal Element2 As Object, _
    ByVal DistanceType As MeasureDistanceTypeConstants, _
    ByRef Distance As Double, _
    ByRef DX As Double, _
    ByRef DY As Double, _
    ByRef DZ As Double, _
    ByRef Point1X As Double, _
    ByRef Point1Y As Double _
)
```

Parameters

Element1
Specifies the first element from which the minimum distance is to be measured.

Element2
Specifies the second element from which the minimum distance is to be measured.

DistanceType

Value	Description
MeasureDistanceTypeConstantsJVIaximumDistance	
MeasureDistanceTypeConstantsJMinimumDistance	
MeasureDistanceTypeConstants_SmartDistance	

Specify the type of distance to be measured.

Distance
Returns the distance between Element1 and Element2.

DX
Returns dX value. This is the distance calculated along X axis between the point1 on Element1 that is closest to point2 on Element2.

DY
Returns dY value. This is the distance calculated along Y axis between the point1 on Element1 that is closest to point2 on Element2.

DZ
Returns dZ value. This is the distance calculated along Z axis between the point1 on Element1 that is closest to point2 on Element2.

Point1
Returns the point on Element1 that is closest to Element2.

Point2
Returns the point on Element2 that is closest to Element1.

Example

Example (VB.NET)

```
Imports System.10
Imports System.Runtime.InteropServices

Public Class Form1

    Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
        Dim objApp As SolidEdgeFramework.Application
        Dim objPartDoc As SolidEdgePart.PartDocument = Nothing
        Dim objDocuments As SolidEdgeFramework.Documents = Nothing
        Dim objFolder As Directoryinfo
        Dim objFileInfo As Fileinfo
        Dim distance As Double
        Dim dx As Double
        Dim dy As Double
        Dim dz As Double
        Dim dgoem(2) As Double
        Dim featedges As SolidEdgeGeometry.Edges
        Dim featedge1 As SolidEdgeGeometry.Edge
        Dim featedge2 As SolidEdgeGeometry.Edge
        Dim featedge3 As SolidEdgeGeometry.Edge
        Dim objFace1 As SolidEdgeGeometry.Face
        Dim objFace2 As SolidEdgeGeometry.Face
        Dim dblPoint1(0) As Double
        Dim dblPoint2(0) As Double
        Dim dblPoint3(0) As Double
        Dim angle1 As Double
        Dim angle2 As Double
        Dim angle3 As Double
        Dim angle4 As Double
        Dim SurfArea As Double
        Dim Volume As Double
```

```
        Dim Length As Double
        Dim coSyss As SolidEdgePart.CoordinateSystem

        dgoem(0) = 0
        dgoem(l) = 0
        dgoem(2) = 0

        Try
            objApp = Marshal.GetActiveObject("SolidEdge.Application")
            objFolder = GetTrainingFolder()
            objFileInfo = New FileInfo(Path.Combine(objFolder.FullName, "anchor.par"))
            objDocuments = objApp.Documents
            objPartDoc = objDocuments.Open(objFileInfo.FullName)
            objFacel = objPartDoc.Models.Item(l).ExtrudedProtrusions.Item(l).Faces(SolidEdgeGeometry.FeatureTopologyQueryTypeCons
            objFace2 = objPartDoc.Models.Item(l).ExtrudedProtrusions.Item(l).Faces(SolidEdgeGeometry.FeatureTopologyQueryTypeCons

            Call objPartDoc.MeasureDistance(objFacel, objFace2; SolidEdgePart.MeasureDistanceTypeConstants.MeasureDistanceTypeCon:
            MsgBox("Distance between the elements is " + distance.ToString)
            -Featedges = objPartDoc.Models.Item(l).ExtrudedProtrusions.Item(2).Edges(SolidEdgeConstants.FeatureTopologyQueryTypeCot
            featedgel = featedges.Item(18)
            featedge2 = featedges.Item(19)
            featedge3 = featedges.Item(20)
            Call objPartDoc.MeasureAngleEx(featedgel, featedge2; featedge3, anglelj angle2, angle3, angle4)
            MsgBox("Angle between the edges is " + Math.Round(RadiansToDegrees(anglel)).ToString + " degrees")

            Call objPartDoc.InquireElement(featedgel, dgoem Nothing; dblPointl; SurfArea, Volume; Length)

            coSyss = objPartDoc.CoordinateSystems.Item(l)

            Call objPartDoc.InquireElement(featedgel, dblPoint2; coSyss; dblPoint3; SurfArea, Volume; Length)
            MsgBox("Length of Inquired Element is " + Length.ToString)
        Catch ex As Exception
            MessageBox.Show(ex.ToString)
            Clipboard.SetText(ex.ToString)
        End Try
    End Sub
    Pnivate Function RadiansToDegrees(ByVal radians As Double) As Double
        Return radians * (180 / Math.PI)
    End Function
    Function GetTrainingFolder() As Directoryinfo
        Dim objInstallData As SEInstallDataLib.SEInstallData = Nothing
        Dim objInstallFolder As Directoryinfo = Nothing
        Dim objTrainingFolder As Directoryinfo = Nothing

        Try
            objInstallData = New SEInstallDataLib.SEInstallData
            objInstallFolder = New DirectoryInfo(objInstallData.GetInstalledPath())
            objTrainingFolder = New DirectoryInfo(Path,Combine(objInstallFolder.Parent.FullName, "Training"))
        Catch
        Finally
            If Not (objInstallData Is Nothing) Then
                Marshal.FinaiReleaseComObject(objInstallData)
                objInstallData = Nothing
            End If
        End Try

        Return objTrainingFolder
    End Function

End Class
```

See Also

PartDocument Object | PartDocument Members

Solid Edge Frameworksupport Type Library

MeasurementAxisDirection Property

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Solid Edge. Design better.

Description

Specifies the direction in which the linear distance is measured for the referenced Dimension object.

Property type

Read-write property

Syntax

Visual Basic

```
Public Property MeasurementAxisDirection As Boolean
```

Remarks

Dimension objects measure the linear distance along or perpendicular to the measurement axis defined by the default/implicit or explicit axis. If the flag is True, the dimension measures perpendicular to the defined axis.

Example

Example (Visual Basic)

```
Private Sub Form_Load()
    Dim objApp As SolidEdgeFrameWork.Application
    Dim objDoc As SolidEdgeDraft.DraftDocument
    Dim objSheet As SolidEdgeDraft.Sheet
    Dim objDimensions As SolidEdgeFrameworkSupport.Dimensions
    Dim objDim As SolidEdgeFrameworkSupport.Dimension
    Dim objLine As SolidEdgeFrameworkSupport.Line2d
    Dim objLine1 As SolidEdgeFrameworkSupport.Line2d
    Dim bMeasAxiDir As Boolean
    ' Report errors
    Const PI = 3.14159265358979
    ' Create/get the application with specific settings
    On Error Resume Next
    Set objApp = GetObject(, "SolidEdge.Application")
    If Err Then
        Err.Clear
        Set objApp = CreateObject("SolidEdge.Application")
        Set objDoc = objApp.Documents.Add("SolidEdge.DraftDocument")
        objApp.Visible = True
    Else
        Set objDoc = objApp.ActiveDocument
    End If
    'getting the Active Sheet object
    Set objSheet = objDoc.ActiveSheet
    'create a line
    Set objLine = objSheet.Lines2d.AddBy2Points(x1:=0.1, y1:=0.1, x2:=0.3, y2:=0.3)
    'Create another line to be set as DimensionAxis
    Set objLine1 = objSheet.Lines2d.AddBy2Points(x1:=0.05, y1:=0.05, x2:=0.15, y2:=0.1)
    'Set the Dimension axis
    Set objDimensions = objSheet.Dimensions
    objDimensions.AxisMode = igDimAxisModeExplicit
    objDimensions.Axis = objLine1
    'create the dimension
    Set objDim = objDimensions.AddDistanceBetweenObjects(Object1:=objLine, x1:=0.1, y1:=0.1, z1:=0, _
                                                         Keypoint1:=True, Object2:=objLine, x2:=0.3, y2:=0.3, z2:=0, Keypoint2:=Ti
    'get MeasurementAxisDirection
    bMeasAxiDir = objDim.MeasurementAxisDirection
    'set MeasurementAxisDirection
    objDim.MeasurementAxisDirection = True

    ' USER DISPLAY
    ' Release objects
    Set objApp = Nothing
    Set objDoc = Nothing
    Set objSheet = Nothing
    Set objDimensions = Nothing
    Set objDim = Nothing
    Set objLine = Nothing
    Set objLine1 = Nothing
End Sub
```

See Also

Dimension Object | Dimension Members

Solid Edge Part Type Library

MinimumDistance Method

Collapse All

Solid Edge.
Design better.

Description

Measures the minimum distance between two elements.

Syntax

Visual Basic

```
Public Sub MinimumDistance( _
    ByVal Element1 As Object,, _
    ByVal Element2 As Object, _
    ByRef Distance As Double, _
    ByRef Point1() As Double, _
    ByRef Point2(') As Double _
)
```

Parameters

Element!
Specifies the element from which the minimum distance is to be measured.

Element2
Specifies the element to which the minimum distance is to be measured.

Distance
Returns the minimum distance between Element! and Element2.

Point!
Returns the point on Element! that is closest to Element2.

Point2
Returns the point on Element2 that is closest to Element!.

See Also

PartDocument Object | PartDocument Members

Solid Edge Draft Type Library

ModelMembers Collection Members

Collapse All

Sonu Luye Draft Type Library : ModelMembers Collection

Public Methods

-V Item	The items in a collection.

Public Properties

@ Application	Returns the active application object.
Count	Returns the number of objects in the referenced collection.
Parent	Returns the parent object for the referenced object.

See Also

ModelMembers Collection

Solid Edge Draft Type Library

ModelMembers Collection

CollapseAI1

Solid Edge.
Design better.

Description

Represents a collecion of ModelMember objects.

Example

Example (VB.NET)

```
Imports System.10
Imports System.Runtime.InteropServices

Module Example
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objDraftDocument As SolidEdgeDraft.DraftDocument
        Dim objSheet As SolidEdgeDraft.Sheet = Nothing
        Dim objDrawingviews As SolidEdgeDraft.Drawingviews = Nothing
        Dim objDrawingView As SolidEdgeDraft.Drawingview = Nothing
        Dim objModelMembers As SolidEdgeDraft.ModelMembers = Nothing

        Try
            OleMessageFliter .Register()

            objApplication = Marshal.GetActiveObject("SolidEdge.Application")
            objDraftDocument = objApplication.ActiveDocument
            objSheet = objDraftDocument.ActiveSheet
            objDrawingViews = objSheet.Drawingviews
            objDrawingView = objDrawingView.Item(l)
            objModelMembers = objDrawingView.ModelMembers

        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFliter. Revoke()
        End Try
    End Sub
End Module
```

See Also

ModelMembers Members

Solid Edge Draft Type Library

ModelMembers Collection

CollapseAI1

Solid Edge.
Design better.

Description

Represents a collecion of ModelMember objects.

Example

Example (VB.NET)

```
Imports System.10
Imports System.Runtime.InteropServices

Module Example
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objDraftDocument As SolidEdgeDraft.DraftDocument
        Dim objSheet As SolidEdgeDraft.Sheet = Nothing
        Dim objDrawingviews As SolidEdgeDraft.Drawingviews = Nothing
        Dim objDrawingView As SolidEdgeDraft.Drawingview = Nothing
        Dim objModelMembers As SolidEdgeDraft.ModelMembers = Nothing

        Try
            OleMessageFliter .Register()

            objApplication = Marshal.GetActiveObject("SolidEdge.Application")
            objDraftDocument = objApplication.ActiveDocument
            objSheet = objDraftDocument.ActiveSheet
            objDrawingViews = objSheet.Drawingviews
            objDrawingView = objDrawingView.Item(l)
            objModelMembers = objDrawingView.ModelMembers

        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFliter. Revoke()
        End Try
    End Sub
End Module
```

See Also

ModelMembers Members

Solid Edge Part Type Library

NormalDistance Method

Colla pseAH

Solid Edge.
Design better.

Description

Measures the normal distance between lines or planes.

Syntax

Visual Basic

```
Public Sub NormalDistance( _
    ByVai ELementl As Object,, _
    ByVai ELement2 As Object, _
    ByRef TrueDistance As Double, _
    ByRef ApparentDistance As Double, _
    ByRef DeLtaX As Double, _
    ByRef DeLtaY As Double, _
    ByRef DeLtaZ As Double, _
    Optional ByVai CoordinateSystem As Variant _
)
```

Parameters

ElementI
The first element.

Element2
The second element.

TrueDistance
The true distance between ElementI and Element2.

ApparentDistance
The apparent distance between ElementI and Element2.

DeltaX
The X axis component of the true normal distance.

DeltaY
The Y axis component of the true normal distance.

DeltaZ
The Z axis component of the true normal distance.

CoordinateSystem
Specifies the coordinate system that defines one of the elements. If you include CoordinateSystem input, the returned values are relative to the coordinate system.

See Also

PartDocument Object | PartDocument Members

Solid Edge ST7 SDK

Working with Occurrences

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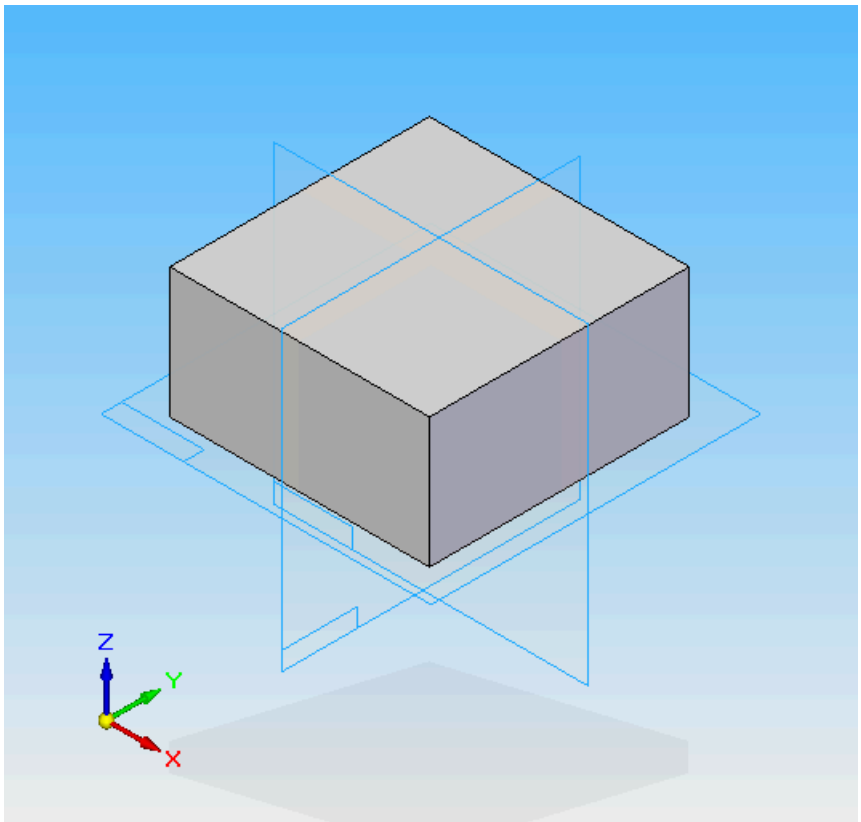
The Occurrence object represents an instance of a part or subassembly within an assembly. It is an OLE link to a part or assembly document. An Occurrence object can only be added to an assembly through an Occurrences collection object.

The automation interface for the assembly environment allows you to place parts and subassemblies into an assembly. This is handled by the AddByFilename method, which is provided on the Occurrences collection object. Parts and subassemblies are differentiated by the Subassembly property on each Occurrence object. The following example shows how to place a part into an assembly.

Parts or subassemblies are initially placed into the assembly at the same location and position they maintain in their original files. The following illustration shows a block and its position relative to the three initial global reference planes. The block is positioned so its corner is at the coordinate (0,0,0). When this part is placed into an assembly using the AddByFilename method, it is placed in the same location and orientation in the assembly file as it existed in the original part file. Subassemblies follow the same rules.



See Handling 'Application is Busy' and 'Call was Rejected By Callee' errors for information regarding the use of OleMessageFilter.

Adding a new Occurrence in Visual Basic.NET — Copy Code

```
Imports SolidEdgeConstants
Imports System.Runtime.interopServices

Module Program
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objDocuments As SolidEdgeFramework.Documents = Nothing
        Dim objAssembly As SolidEdgeAssembly.AssemblyDocument = Nothing
        Dim objOccurrences As SolidEdgeAssembly.Occurrences = Nothing
        Dim objOccurrence As SolidEdgeAssembly.Occurrence = Nothing

        Try
            OleMessageFilter.Register()

            ' Connect to a running instance of Solid Edge
            objApplication = Marshal.GetActiveObject("Solid Edge.Application")

            ' Get a reference to the documents collection
            objDocuments = objApplication.Documents

            ' Create a new assembly document
            objAssembly = objDocuments.Add("SolidEdge.AssemblyDocument")
```

```
            ' Get a reference to the occurrences collection
            objOccunnences = objAssembly.Occurrences

            ' Add a part document to the assembly
            objOccurrence = objOccurrences.AddByFilename("C:\Partl.par")

        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter.Revoke()
        End Try
    End Sub
End Module
```

Adding a new Occurrence in C# Copy Code

```
using System;
using System.Reflection;
using System.Runtime.InteropSenvices;

namespace SolidEdge.SDK
{
    class Program
    {
        [STAThread]
        static void Main(strlng[] args)
        {
            SolidEdgeFramework.Application application = null;
            SolidEdgeFramework.Documents documents = null;
            SolidEdgeAssembly.AssemblyDocument assembly = null;
            SolidEdgeAssembly.Occurrences occurrences = null;
            SolidEdgeAssembly.Occurrence occurrence = null;

            tny
            {
                OleMessageFilter.Register();

                // Connect to a running instance of Solid Edge
                application = (SolidEdgeFramewonk.Application)
                    Marshal.GetActiveObject("SolidEdge.Application");

                // Get a reference to the documents collection
                documents = application.Documents;

                // Create a new assembly document
                assembly = (SolidEdgeAssembly.AssemblyDocument)
                  documents.Add("SolidEdge.AssemblyDocument", Missing.Value);

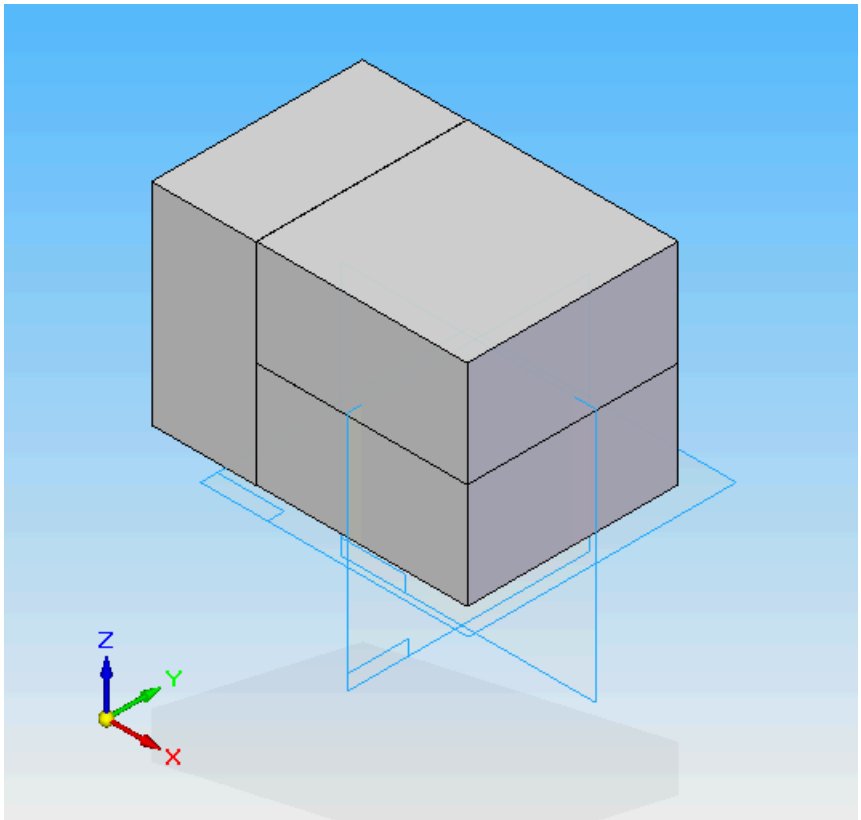
                // Get a reference to the occurrences collection
                occurrences = assembly.Occurrences;

                // Add a part document to the assembly
                occurrence = occurrences.AddByFilename(@"C:\Partl.par"， Missing.Value);

            }
            catch (System.Exception ex)
            {
                Console.WriteLine(ex.Message);
            }
            finally
            {
                OleMessageFilter.Revoke();
            }
        }
    }
}
```

Because occurrences are placed in the same relative location and orientation in which they were initially created, you will typically change the part or subassembly's position and orientation after placement. These methods apply only to grounded occurrences. Occurrences that are placed with relationships to other occurrences have their location and orientation defined by their relationships to the other occurrences.

To show how to use these methods, consider a block with dimensions of 100 mm in the x axis, 100 mm in the y axis, and 50 mm in the z axis. Assume that you need to place three of these parts to result in the following assembly:



A See Handling 'Application is Busy' and 'Call was Rejected By Callee' errors for information regarding the use of OleMessageFilter.

Manipulating Occurrences in Visual Basic.NET Copy Code

```
Imports SolidEdgeConstants
Imports System.Runtime.InteropServices

Module Program
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objDocuments As SolidEdgeFramework.Documents = Nothing
        Dim objAssembly As SolidEdgeAssembly.AssemblyDocument = Nothing
        Dim objOccurrences As SolidEdgeAssembly.Occurrences = Nothing
        Dim objOccurrence As SolidEdgeAssembly.Occurrence = Nothing
        Dim objOccurrence1 As SolidEdgeAssembly.Occurrence = Nothing
        Dim objOccurrence2 As SolidEdgeAssembly.Occurrence = Nothing
        Dim objOccurrence3 As SolidEdgeAssembly.Occurrence = Nothing

        Try
            OleMessageFilter.Register()

            ' Connect to a running instance of Solid Edge
            objApplication = Marshal.GetActiveObject("SolidEdge.Application")

            ' Get a reference to the documents collection
            objDocuments = objApplication.Documents

            ' Create a new assembly document
            objAssembly = objDocuments.Add("SolidEdge.AssemblyDocument")

            ' Get a reference to the occurrences collection
            objOccurrences = objAssembly.Occurrences

            ' Add the first block to the assembly
            objOccurrence1 = objOccurrences.AddByFilename("C:\Part1.par")

            ' Add the second block to the assembly
            objOccurrence2 = objOccurrences.AddByFilename("C:\Part1.par")
```

```
            ' It is currently in the same position and orientation as the first
            ' block, so reposition it
            objOccurnence2.Move(0, 0, 0.05)

            ' Add the third block to the assembly
            objOccurrence3 = objOccurrences.AddByFilename("C:\Partl.par")

            ' Rotate the third block to a vertical position.
            objOccurrence3.Rotate(0, 0, 0, 0, 1, 0, -Math.PI / 2)

            ' Reposition the third block.
            objOccurrence3.Move(-0.049, 0, 0.049)

        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFliter.Revoke()
        End Tny
    End Sub
End Module
```

Manipulating Occurrences in C# Copy Code

```
using System;
using System.Reflection;
using System.Runtime.InteropServices;

namespace SolidEdge.SDK
{
    class Program
    {
        [STAThread]
        static void Main(string[] args)
        {
            SolidEdgeFramework.Application application = null;
            SolidEdgeFramework.Documents documents = null;
            SolidEdgeAssembly.AssemblyDocument assembly = null;
            SolidEdgeAssembly.Occurrences occurrences = null;
            SolidEdgeAssembly.Occurrence occurrencel = null;
            SolidEdgeAssembly.Occurrence occurrence2 = null;
            SolidEdgeAssembly.Occurrence occurrences = null;

            try
            {
                OleMessageFilt er.Register();

                // Connect to a running instance of Solid Edge
                application = (SolidEdgeFramework.Application)
                  Marshal.GetActiveObject("SolidEdge.Application");

                // Get a reference to the documents collection
                documents = application.Documents;

                // Create a new assembly document
                assembly = (SolidEdgeAssembly.AssemblyDocument)
                  documents.Add("SolidEdge.AssemblyDocument", Missing.Value);

                // Get a reference to the occurrences collection
                occurrences = assembly.Occurrences;

                // Add the first block to the assembly
                occurrencel = occurrences.AddByFilename(@"C:\Partl.par", Missing.Value);

                // Add the second block to the assembly
                occurrence2 = occurrences.AddByFilename(@"C:\Partl.par", Missing.Value);

                // It is currently in the same position and orientation as the first
                // block, so reposition it
                occurrence2.Move(0, 0, 0.05);

                // Add the third block to the assembly
```

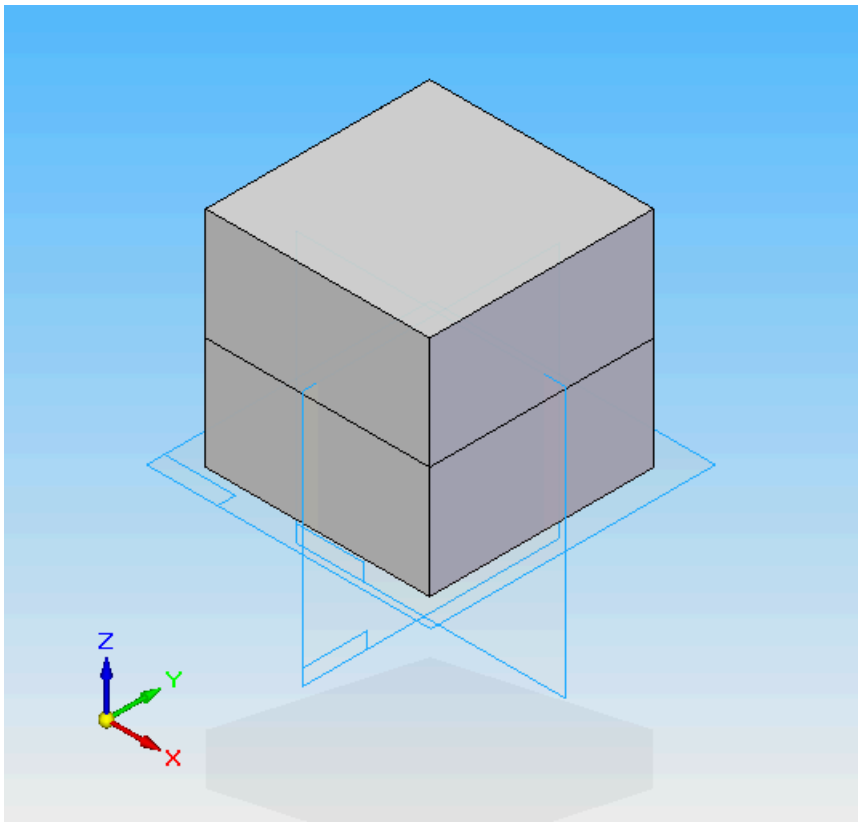
```
                occurrences = occurrences.AddByFilename(@"C:\Part1.par", Missing.Value);

                // Rotate the third block to a vertical position,
                occurrences.Rotate(0, 0, 0, 0, 1, 0, -Math.PI / 2);

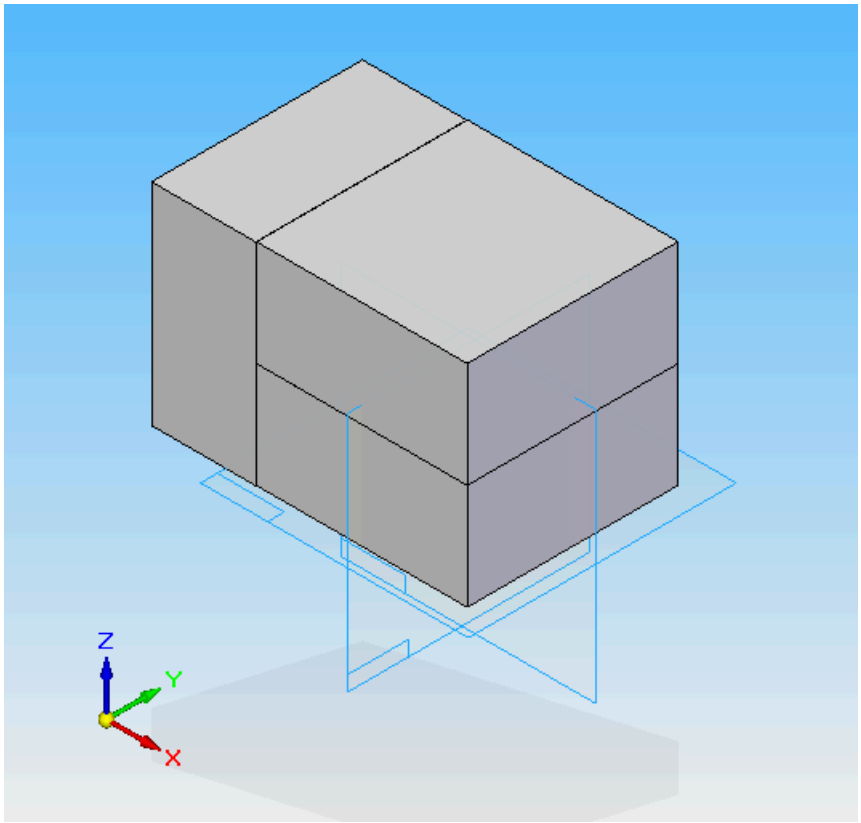
                // Reposition the third block.
                occurrence3.Move(-0.049, 0, 0.049);

            }
            catch (System.Exception ex)
            {
                Console.WriteLine(ex.Message);
            }
            finally
            {
                OleMessageFilter.Revoke();
            }
        }
    }
}
```

The following illustration shows the assembly after placing the second block and moving it into place:



And after placing the third block, rotating it, and moving it into place, the assembly is as follows:



This example positions the blocks using the Move method. You can also use the SetOrigin method, which is available on the Occurrence object, to move occurrences. SetOrigin works together with GetOrigin; GetOrigin returns the coordinates of the occurrence origin with respect to the assembly origin, and SetOrigin positions the occurrence's origin relative to the assembly's origin.

Solid Edge Draft Type Library

Parent Property

Solid Edge.
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Description

Returns the parent object for the referenced object.

Property type

Read-only property

Syntax

Visual Basic
`Public Property Parent As Object`

See Also

DrawingViews Collection | DrawingViews Members

Solid Edge Draft Type Library

PartDrawingViewTypeConstants Enumeration

Collapse All

Solid Edge Draft Type Library : PartDrawingViewTypeConstants Enumeration

Description

Part Drawing View Type Constants

Members

Member	Value	Description
sePartDesignedView	0	Part Designed Drawing View Type - This is a view of a part with no simplification
sePartSimplifiedView	1	Part Simplified Drawing View Type - This is a view of a part with simplification

Solid Edge Draft Type Library

PopulateQuicksheetTemplate Method

Colla pseAH

Solid Edge.
Design better.

Description

Populates the existing Quicksheet template drawing views with the model specified.

Syntax

Visual Basic

```
Public Sub PopulateQuicksheetTemplate( _
    ByVai FiLeName As String _
)
```

Parameters

FlleName

See Also

DraftDocument Object | DraftDocument Members | Solid Edge V18 - What's New

Solid Edge FrameworkSupport Type Library

ProjectionExtensionGap Property

Colla pseAH

Solid EdcjO.
Design better.

Description

Sets and returns the distance by which the projection lines are extended beyond the dimension line.

Property type

Read-write property

Syntax

Visual Basic
`Public Property ProjectionExtensionGap As Double`

Remarks

The extension gap value is a ratio of dimension text height.

See Also

DimStyle Object | DimStyle Members

Solid Edge FrameworkSupport Type Library

ProjectionExtensionGap Property

Colla pseAH

Solid EdcjO.
Design better.

Description

Sets and returns the distance by which the projection lines are extended beyond the dimension line.

Property type

Read-write property

Syntax

Visual Basic
`Public Property ProjectionExtensionGap As Double`

Remarks

The extension gap value is a ratio of dimension text height.

See Also

DimensionStyle Object | Dimensionstyle Members

Solid Edge FrameworkSupport Type Library

RadialExtensionDistance Property

Colla pseAH

Solid EdcjO.
Design better.

Description

Sets and returns the extension distance of radial and half displayed diameter dimensions. Negative is away from the terminator.

Property type

Read-write property

Syntax

Visual Basic
`Public Property RadialExtensionDistance As Double`

See Also

Dimension Object | Dimension Members | Solid Edge ST5 - What's New

Solid Edge Assembly Type Library

Range Method

Collapse All

Solid Edge.
Design better.

Description

Returns the high and low range values for the referenced object.

Syntax

Visual Basic

```
Public Sub Range( _
    ByRef X1 As Double., _
    ByRef Y1 As Double; _
    ByRef Z1 As Double; _
    ByRef X2 As Double; _
    ByRef Y2 As Double; _
    ByRef Z2 As Double _
)
```

Parameters

X1
Specifies the x coordinate of the lower-left corner the referenced object.

Y1
Specifies the y coordinate of the lower-left corner of the referenced object.

Z7
Specifies the z coordinate of the lower-left corner of the referenced object.

X2
Specifies the x coordinate of the upper-right corner of the referenced object.

Y2
Specifies the y coordinate of the upper-right corner of the referenced object.

Z2
Specifies the z coordinate of the upper-right corner of the referenced object.

See Also

AssemblyDrivenPartFeaturesRevolvedCutout Object | AssemblyDrivenPartFeaturesRevolvedCutout Members | Solid Edge V17 - What's New

Solid Edge Framework Type Library

Reference Object Members

Colla pseAl1

Solid Edge. Design better.

Sonu Luyt rramework Type Library : Reference Object

Public Methods

GetMatrix	Returns the transformation matrix for the referenced Occurrence or Reference object.
GetOccurrencesInPath	Method to get occurrences from Reference.

Public Properties

Application	Returns the active application object.
AttributeSets	Returns the AttributeSets collection object for the referenced object.
Hsl ImmediateParent	Returns the immediate parent for the referenced object.
Hsl IsAttributeSetPresent	Returns True if a specified Attribute Set exists on the referenced object.
Object	Returns the dispatch interface associated with the site.
Parent	Returns the parent object for the referenced object.
Style	For collection objects, Body objects, RefPlane objects, Sketch objects, CoordinateSystem objects, CommandBarButton objects, SmartFrame2d objects, View objects, and TextCharStyle objects, Style is read-write. For all other objects, Style is read-only.
Type	Specifies the type of the object being referenced.

See Also

Reference Object

Solid Edge Framework Type Library

Reference Object

CollapseAI1

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Description

Represents an object that references another object (a part or assembly, for example).

Remarks

It consists of an object and a matrix that specifies where in the assembly the referenced object is located. For example, a part in an assembly may be used many times. The data regarding the construction of the part exists only once in the file. All occurrences of the part are references to the original part with a matrix that specifies the location of the specific occurrence.

See Also

Reference Members

Solid Edge Framework Type Library

Reference Object

CollapseAI1

Solid Edge.
Design better.

Description

Represents an object that references another object (a part or assembly, for example).

Remarks

It consists of an object and a matrix that specifies where in the assembly the referenced object is located. For example, a part in an assembly may be used many times. The data regarding the construction of the part exists only once in the file. All occurrences of the part are references to the original part with a matrix that specifies the location of the specific occurrence.

See Also

Reference Members

Solid Edge FrameworkSupport Type Library

Relations2d Collection Members

Solid EdCfO.
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Colla pseAl1

Sonu Luyt rrameworkSupport Type Library : Relations2d Collection

Public Methods

AddAutoConstraints	
AddChamfer	Creates a chamfer relationship between a Line2d object and the two objects it intersects.
AddColinear	Adds a colinear relationship to a Relations2d collection.
AddConcentric	Adds a concentric relationship to a Relations2d collection.
AddEqual	Adds an equal relationship to a Relations2d collection.
AddEqualDirection	This method adds a relationship to the referenced Relations2d collection in which the two specified objects have equal direction at the specified keypoints.
AddEqualDirectionEqualCurvature	This method adds a relationship to the referenced Relations2d collection in which the two specified objects have equal direction and equal curvature at the specified keypoints.
AddFillet	Creates a fillet relationship between an Arc2d object and the two objects it intersects.
AddFix	Adds a fix relationship to a Relations2d collection.
AddHorizontal	Adds a horizontal relationship to a Relations2d collection.
AddHorizontalAlign	Adds a horizontal alignment relationship to a Relations2d collection.
Addintersection	Adds an intersection relationship to a Relations2d collection.
AddKeypoint	Adds a key point relationship to a Relations2d collection.
AddKeypointFix	Adds a Fix relationship (called a Lock relationship in the Solid Edge user interface) to a keypoint of an object. A Fix relationship eliminates two degrees of freedom when placed on a keypoint of a geometric element.
AddLink	Adds a link relationship to a Relations2d collection.
AddParallel	Adds a parallel relationship to a Relations2d collection.
AddPerpendicular	Adds a perpendicular relationship to a Relations2d collection.
AddPointOn	Adds a point-on relationship to a Relations2d collection.
AddPointOnCompoundCurve	This method adds a point-on relationship to the referenced Relations2d collection. The method creates a point-on relationship between the key point index of the specified object to a point on the specified compound curve.
"*** AddSet	Adds a set relationship to a Relations2d collection.
AddSymmetric	Adds a symmetric relationship to a Relations2d collection.
AddTangent	Adds a tangent relationship to a Relations2d collection.

AddTangentBetweenCompoundCurves	This method adds a tangent relationship to the referenced Relations2d collection. The method creates a tangent relationship between two specified compound curves. Each set of curves must be endpoint-connected through keypoint constraints.
AddTangentEqualCurvature	This method adds a relationship to the referenced Relations2d collection in which the two specified objects are tangent and have equal curvature at the specified keypoints.
AddVertical	Adds a vertical relationship to a Relations2d collection.
AddVerticalAlign	Adds a vertical alignment relationship to a Relations2d collection.
Item	The items in a collection.

Public Properties

Count	Returns the number of objects in the referenced collection.

See Also

Relations2d Collection

Solid Edge FrameworkSupport Type Library

Rightindent Property

Colla pseAH

Solid EdcjO.
Design better.

Description

Sets and returns the distance from the right margin to the end of the line.

Property type

Read-write property

Syntax

Visual Basic
`Public Property Rightindent As Double`

Remarks

The TextEdit object always has an insert index or an active selection. Setting any property on the TextEdit object applies to the current selection, if one is present, or to the text at the current insert index.

See Also

TextEdit Object | TextEdit Members

Solid Edge Part Type Library

SaveAsFlatDXFEx Method

Colla pseAH

Solid EdCJO.
Design better.

Description

Creates a flat pattern file from specified geometry.

Syntax

Visual Basic

```
Public Sub SaveAsFlatDXFEx( _
    ByVal FiLeName As String, _
    ByVal Face As Object, _
    ByVal Edge As Object, _
    ByVal Vertex As Object, _
    ByVal UseFLatPatternModeL As Boolean _
)
```

Parameters

FileName
Specifies the name of the exported file. The file extension can be .dxf, .par, or .psm.

Face
Specifies the input face.

Edge
Specifies the input edge. The edge must be part of the input face.

Vertex
Specifies the input vertex. The vertex must be part of the input face.

UseFlatPatternModel
Indicates to use existing flat pattern if there is one and it is up to date otherwise it uses the face, edge and vertex information.

Remarks

The new file can be a Solid Edge part file, a Solid Edge sheet metal file, or an AutoCAD file. Face does not have to be planar and vertex is not mandatory.

See Also

Models Collection | Models Members | Solid Edge ST3 - What's New

Solid Edge FrameworkSupport Type Library

SetJogPointlDistance Method

Collapse All

Solid Edge.
Design better.

Description

Sets the JogPointl distance for the jogged line in a Dimension.

Syntax

Visual Basic

```
Public Sub SetJogPointlDistance( _
    ByVal Value As Double _
)
```

Parameters

Value
Specifies the JogPointl distance along the dimension line.

Remarks

This method is applicable only for dimensions with jogged capability (for example, all radius, diameter, and coordinate dimensions).

Example

Example (Visual Basic)

```
Private Sub Form_Load()
    Dim objApp As SolidEdgeFrameWork.Application
    Dim objDoc As SolidEdgeDraft.DraftDocument
    Dim objSheet As SolidEdgeDraft.Sheet
    Dim objArc As SolidEdgeFrameworkSupport.Arc2d
    Dim objDimension As SolidEdgeFrameworkSupport.Dimension
    ' Report errors
    Const PI = 3.14159265358979
    ' Create/get the application with specific settings
    On Error Resume Next
    Set objApp = GetObject(_, "SolidEdge.Application")
    If Err Then
        Err.Clear
        Set objApp = CreateObject("SolidEdge.Application")
        Set objDoc = objApp.Documents.Add("SolidEdge.DraftDocument")
        objApp.Visible = True
    Else
        Set objDoc = objApp.ActiveDocument
    End If
    'getting the Active Sheet object
    Set objSheet = objDoc.ActiveSheet
    'create an arc2d
    Set objArc = objSheet.Arcs2d.AddByCenterStartEnd(xCenter:=0.2, _
                                                     yCenter:=0.2, xStart:=0.1j yStart:=0.1, xEnd:=0.3, yEnd:=0.3)
    'create a dimension
    Set objDimension = objSheet.Dimensions.AddRadius(Object:=objArc)
    'set the jogged property
    objDimension.logged = True
    'SetJogPointlDistance
    Call objDimension.SetJogPointlDistance(Value:=0.5)
    ' USER DISPLAY
    ' Release objects
    Set objApp = Nothing
    Set objDoc = Nothing
    Set objSheet = Nothing
    Set objDimension = Nothing
    Set objArc = Nothing
End Sub
```

See Also

Dimension Object | Dimension Members

Solid Edge FrameworkSupport Type Library

SetJogPoint2 Method

col| ap seA|1

Solid Edge. Design better.

Description

Sets the JogPoint2 distance for the jogged line in a Dimension.

Syntax

Visual Basic

```
Public Sub SetJogPoint2( _
    ByVal x As Double, _
    ByVal y As Double _
)
```

Parameters

x
Specifies the x offsets from JogPointl.

y
Specifies the y offsets from JogPointl.

Remarks

The values are passed as the x and y offset distances. This method is applicable only for dimensions with Jogged capability (for example, all radius, diameter, and coordinate dimensions).

Example

Example (Visual Basic)

```
Private Sub Form_Load()
    Dim objApp As SolidEdgeFrameWork.Application
    Dim objDoc As SolidEdgeDraft.DraftDocument
    Dim objSheet As SolidEdgeDraft.Sheet
    Dim objArc As SolidEdgeFrameworkSupport.Arc2d
    Dim objDimension As SolidEdgeFrameworkSupport.Dimension
    ' Report errors
    Const PI = 3.14159265358979
    ' Create/get the application with specific settings
    On Error Resume Next
    Set objApp = GetObject(, "SolidEdge.Application")
    If Err Then
        Err.Clear
        Set objApp = CreateObject("SolidEdge.Application")
        Set objDoc = objApp.Documents.Add("SolidEdge.DraftDocument")
        objApp.Visible = True
    Else
        Set objDoc = objApp.ActiveDocument
    End If
    'getting the Active Sheet object
    Set objSheet = objDoc.ActiveSheet
    'create an arc2d
    Set objArc = objSheet.Arcs2d.AddByCenterStartEnd(xCenter:=0.2, _
                                                     yCenter:=0.2, xStart:=0.1, yStart:=0.1, xEnd:=0.3, yEnd:=0.3)
    'create a dimension
    Set objDimension = objSheet.Dimensions.AddRadius(Object:=objArc)
    'set the jogged property
    objDimension.logged = True
    'SetlogPoint2
    Call objDimension.SetJogPoint2(x:=-0.1, y:=-0.1)
    ' USER DISPLAY
    ' Release objects
    Set objApp = Nothing
    Set objDoc = Nothing
    Set objSheet = Nothing
End Sub
```

See Also

Dimension Object | Dimension Members

Solid Edge FrameworkSupport Type Library

SetJogPoint3Distance Method

Collapse All

Solid Edge.
Design better.

Description

Sets the JogPoint3 distance for the jogged line in a Dimension.

Syntax

Visual Basic

```
Public Sub SetJogPoint3Distance( _
    ByVal Value As Double _
)
```

Parameters

Value
Specifies the JogPoint3 distance along the dimension line from JogPoint2.

Remarks

This method is applicable only for dimensions with Jogged capability (for example, all radius, diameter, and coordinate dimensions).

Example

Example (Visual Basic)

```
Private Sub Form_Load()
    Dim objApp As SolidEdgeFrameWork.Application
    Dim objDoc As SolidEdgeDraft.DraftDocument
    Dim objSheet As SolidEdgeDraft.Sheet
    Dim objArc As SolidEdgeFrameworkSupport.Arc2d
    Dim objDimension As SolidEdgeFrameworkSupport.Dimension
    ' Report errors
    Const PI = 3.14159265358979
    ' Create/get the application with specific settings
    On Error Resume Next
    Set objApp = GetObject(_, "SolidEdge.Application")
    If Err Then
        Err.Clear
        Set objApp = CreateObject("SolidEdge.Application")
        Set objDoc = objApp.Documents.Add("SolidEdge.DraftDocument")
        objApp.Visible = True
    Else
        Set objDoc = objApp.ActiveDocument
    End If
    'getting the Active Sheet object
    Set objSheet = objDoc.ActiveSheet
    'create an arc2d
    Set objArc = objSheet.Arcs2d.AddByCenterStartEnd(xCenter:=0.2, _
                                                     yCenter:=0.2, xStart:=0.1j yStart:=0.1, xEnd:=0.3, yEnd:=0.3)
    'create a dimension
    Set objDimension = objSheet.Dimensions.AddRadius(Object:=objArc)
    'set the jogged property
    objDimension.logged = True
    'Set JogPoint3Distance
    Call objDimension.SetlogPoint3Distance(Value:=-0.05)
    ' USER DISPLAY
    ' Release objects
    Set objApp = Nothing
    Set objDoc = Nothing
    Set objSheet = Nothing
    Set objDimension = Nothing
    Set objArc = Nothing
End Sub
```

See Also

Dimension Object | Dimension Members

Solid Edge Assembly Type Library

SetOrigin Method

Collapse All

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Description

Syntax

Visual Basic

```
Public Sub SetOrigin( _
    ByVal x As Double, _
    ByVal y As Double _
)
```

Parameters

x
Specifies the x coordinate for the new origin point.

y
Specifies the y coordinate for the new origin point.

See Also

VirtualComponentOccurrence Object | VirtualComponentOccurrence Members | Solid Edge V17 - What's New

Solid Edge Draft Type Library

Sheet Object Members

Overview Collapse All

Solid Edge Draft Type Library : Sheet Object

Public Methods

Activate	Makes the referenced object the active one.
ArrangeDimensionsInSelectSet	Arranges the dimensions in the select set. API SolidEdgeDraft.DrawingView.AddConnectedDimensionsToSelectSetO can be used to add dimensions to select set which are connected to DV
ChainLocate	Returns the end connected elements starting from the element located at the input point. This function will clear the SelectSet and add the located elements to the SelectSet.
CleanGeometry2d	Cleans the selected 2d geometry elements on the sheet.
ConvertToCurve	Converts the specified element(s) to a b-spline curve.
CopyEMFToClipboard	Copies an enhanced metafile to the operating system clipboard.
CreateVFS	Creates a ViewFilterSet object.
Delete	Deletes the referenced object.
DeleteVFS	Deletes a ViewFilterSet object from the referenced Sheet.
FenceLocate	Clears the SelectSet, then adds all objects located within the specified rectangular fence to the SelectSet.
GetAreaProperties	Returns the properties of an area, based on a point or group of points on the active sheet. The points must be inside areas of existing elements.
MoveSelectSet	Moves the objects in the select set. All of the selected objects must belong to the sheet.
Offset2d	Offsets the end connected elements from the SelectSet in the direction of the 2d point by the given distance. This function will clear the SelectSet and add the offset elements to the SelectSet.
PrintArea	Prints the specified area.
Printsheet	Prints the sheet.
RotateSelectSet	Rotates the objects in the select set about the specified point. All of the selected objects must belong to the sheet.
ScaleSelectSet	Scales the objects in the select set about the specified point. All of the selected objects must belong to the sheet.
SmartSelect	Selects objects with the same specified characteristics as the specified object.

Public Properties

AnnotAlignmentShapes	Returns the Alignment shapes collection object for the referenced object.
0 Application	Returns the active application object.
Arcs2d	Returns the Arcs2d collection object for the referenced object.
AreaPropertiesCollection	Returns the AreaPropertiesCollection collection object for the referenced Sheet.

AttributeSets	Returns the AttributeSets collection object for the referenced object.
Background	Sets and returns the background sheet assigned to the referenced sheet.
0 BackgroundVisible	Specifies whether or not the Background Sheet for this Sheet object is visible. Only valid for sheets in the WorkingSection.
S Balloons	Returns the Balloons collection object for the referenced object.
BlockLabels	Returns the collection of block labels for the selected sheet.
BlockOccurrences	Returns the BlockOccurrences collection.
0 BoltHoleCircles	Returns the BoltHoleCircles collection object for the referenced object.
Boundaries2d	Returns the Boundaries2d collection object for the referenced object.
BsplineCurves2d	Returns the BSplineCurves2d collection object for the referenced object.
Centerlines	Returns the Centerlines collection object for the referenced object.
CenterMarks	Returns the CenterMarks collection object for the referenced object.
Circles2d	Returns the Circles2d collection object for the referenced object.
ComplexStrings2d	Returns the ComplexStrings2d collection object for the referenced object.
Connectors	Returns the Connectors collection.
CornerAnnotations	Returns the CornerAnnotations collection object for the referenced object.
Curves2d	Returns the Curves2d collection object for the referenced object.
DatumFrames	Returns the DatumFrames collection object for the referenced object.
DatumPoints	Returns the DatumPoints collection object for the referenced object.
DatumTargets	Returns the DatumTargets collection object for the referenced object.
DefaultTextScale	Get or set the default text scale for the sheet. The text scale is used by dimension and annotation commands.
Dimensions	Returns the Dimensions collection object for the referenced object.
Document	Returns the document to which the referenced object belongs.
DrawingObjects	Returns a collection of drawing objects.
Drawingviews	Returns the DrawingViews collection object for the referenced Sheet.
0 Ellipses2d	Returns the Ellipses2d collection object for the referenced object.
EllipticalArcs2d	Returns the EllipticalArcs2d collection object for the referenced object.
FeatureControlFrames	Returns the FeatureControlFrames collection object for the referenced object.
0 GostWeldSymbols	Returns the GostWeldSymbols collection object for the referenced object.
Groups	Returns the Groups collection object for the referenced object.
H2I Images2d	Returns the Images2d collection.
Index	Returns the numeric index for the active object.
IsAttributeSetPresent	Returns True if a specified Attribute Set exists on the referenced object.
Key	Returns a key (a string) to find the object relative to its owning parents.
[±2] Layers	Returns the Layers collection object.

0 Leaders	Returns the Leaders collection object for the referenced object.
t] Lines2d	Returns the Lines2d collection object for the referenced object.
0 LineStrings2d	Returns the LineStrings2d collection object for the referenced object.
Name	Returns the name of the referenced object.
Number	Index of sheet within section tab bar. This value is the same as the property text sheet number.
Parent	Returns the parent object for the referenced object.
0 Points2d	Returns the Points2d collection object for the referenced object.
Relations2d	Returns the Relations2d collection.
0 Section	Returns the section to which the sheet belongs. Sheets are grouped into sections: WorkingSection, Backgroundsection, and 2dModelSection.
SectionBoundaries2d	Returns the Boundaries2d collection object for the referenced object.
SectionType	Specifies the type of the referenced Sheet object.
SheetGroup	Returns the SheetGroup to which this working sheet belongs. This property is only valid for working sheets.
0 SheetSetup	Returns a SheetSetup object for setting model-to-paper scale, paper size, and so forth.
SmartFrames2d	Returns the SmartFrames2d collection.
SurfaceFinishSymbols	Returns the SurfaceFinishSymbols collection object for the referenced object.
Symbols	Returns the Symbols collection object for the referenced object.
H2] TextBoxes	Returns the TextBoxes collection object for the referenced object.
TextProfiles	Returns the TextProfiles collection object for the referenced Sheet.
Type	Specifies the type of the object being referenced.
0 VFS	Returns the ViewFilterSet for the referenced Sheet.
Visible	Specifies whether or not the referenced object is displayed to the screen.
WeldSymbols	Returns the WeldSymbols collection object for the referenced object.

See Also

Sheet Object

Solid Edge Draft Type Library

Sheet Object

Members Collapse All

Solid Edge.
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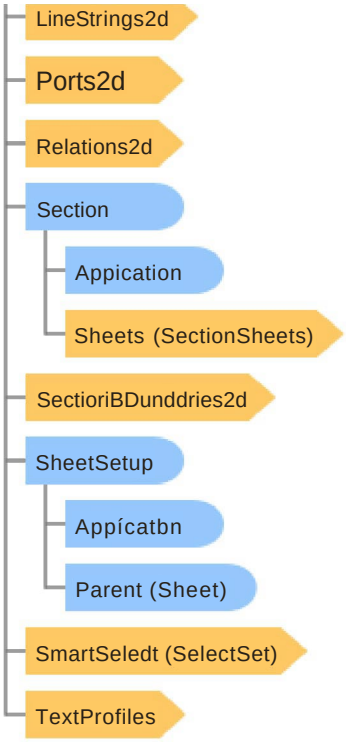
Description

Represents a sheet in a document.

Object Model

- Sheet
  - AppiodlKHi
    - AcLveSelectSet (SelectSet)
    - Addins
    - AppicaLkm
    - AppicdLionEvents
    - AppicatbriVSAfterDocumentOpenEvent (AppficdikjnV8DocurrienlOpenEvent)
    - AppicationWiridowEvents
    - CreateCommand (Command)
    - Customization
    - Documents
    - Environments
    - FeatureLibraryEvents
    - FileUIEvents
    - GetMatenarTable (MatTable)
    - Insight
    - NewFileIII Events
    - Paient (Application)
    - SEECEvents
    - ShortcutMenuEvents
    - SolidEdgeInsightXT
    - SolidEdgeTCE
    - Windows
  - Arcs2d
  - ArreaPropertiesCollecliun
  - Background (Sheet)
    - Appication
    - Arcs2d
    - AreaPropertiesCollectiun
    - Background (Sheet)
    - BbckLabels
    - BkidkOccuii erices
    - Boundaries2d

- BsplineCurves2d (BSplineCurves2d)
- Circles2d
- ComplexStrings2d
- Curves2d
- DrawingObjects
- DrawingViews
- Elipses2d
- ElipticalArcs2d
- Images2d
- Layers
- Lines2d
- LineStrings2d
- Points2d
- Relations2d
- Section
- SectionBoundaries2d
- SheetSetup
- SmartSelect (SelectSet)
- TextProfiles
- BludkLabels
- BkickOccurrences
- Boundaries2d
- BspírieCuives2d (BSpfineCurves2d)
- Cides 2d
- ComplexStrhgs2d
- Curves2d
- DrawingObjects
- DrawingViews
- Elipses2d
- ElípticalAi cs2d
- Images2d
- Layers
- Lines2d



Remarks

A sheet is typically a container that has graphical objects. It includes properties that specify its dimensions and the scale factor to be applied when mapping graphics on the sheet.

Example

Example (VB.NET)

```
Imports System.10
Imports System.Runtime.InteropServices

Module Example
    <STAThread()> _
    Sub Main()

        Dim objApplication As SolidEdgeFramework.Application
        Dim objDraftDocument As SolidEdgeDraft.DraftDocument
        Dim objSheet As SolidEdgeDraft.Sheet

        Try
            OleMessageFilter.Register()

            objApplication = Marshal.GetActiveObject("SolidEdge.Application")
            objDraftDocument = objApplication.Documents.Add("SolidEdge.DraftDocument")
            'Get the active sheet
            objSheet = objDraftDocument.ActiveSheet

        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter.Revoke()
        End Try
    End Sub
End Module
```

See Also

Sheet Members

Solid Edge ST7 SDK

Working with Sections and Sheets

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The structure of Draft documents differs significantly from other Solid Edge document types. From the DraftDocument object, you access the Sheets collection and then the individual Sheet objects. The Sheets collection contains both working sheets and background sheets.

In addition, DraftDocument supports a Sections object. The Sections object is a collection of Section objects that group Sheets by the characteristics of the data they contain. As users create drawings interactively, data from these drawings is automatically placed in one of three Section objects:

- Working Section - Contains Sheet objects (Working Sheets) on which normal 2-D drawing objects (Lines, Arcs, DrawingViews, and so forth) are placed.
- Background Section - Contains background sheets, which hold the sheet borders.

Sections are a part of the graphical interface, although they are not immediately apparent. When the interactive user selects View > Background Sheet, Solid Edge internally changes to the Backgrounds section and displays its sheets. Similarly, the View > Working Sheet command allows you to modify the sheets that are in the Sections! section. When a DrawingView is added, a new sheet is added to the DrawingViews section.

However, it is not possible through the graphical interface to create and manipulate sections directly. Although it is possible through automation to create new Sections, it is not a supported workflow. Although the same information is available on the Sheets collection that is a child of the DraftDocument object, within Sections, the information is separated by its functional characteristics.

| /i\ See Handling 'Application is Busy' and 'Call was Rejected By Callee' errors for information regarding the use of OleMessageFilter.

Sections and Sheets in Visual Basic .NET Copy Code

```
Imports SolidEdgeDraft
Imports System.Runtime.InteropServices

Module Program
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objDocuments As SolidEdgeFramework.Documents = Nothing
        Dim objDraft As SolidEdgeDraft.DraftDocument = Nothing
        Dim objSections As SolidEdgeDraft.Sections = Nothing
        Dim objSection As SolidEdgeDraft.Section = Nothing
        Dim objSectionSheets As SolidEdgeDraft.Sectionsheets = Nothing
        Dim objSheets As SolidEdgeDraft.Sheets = Nothing
        Dim objSheet As SolidEdgeDraft.Sheet = Nothing
        Dim strFormat1 As String = "Section = {0}"
        Dim strFormat2 As String = "Sheet = {0}"

        Try
            OleMessageFilter.Register()

            ' Connect to a running instance of Solid Edge
            objApplication = Marshal.GetActiveObject("SolidEdge.Application")

            ' Get a reference to the documents collection
            objDocuments = objApplication.Documents

            ' Add a Draft document
            objDraft = objDocuments.Add("SolidEdge.DraftDocument")

            ' Get a reference to the sections collection
            objSections = objDraft.Sections

            ' Loop through the sections
            ' igWorkingSection, igBackgroundSection & ig2dModelSection
            For Each objSection In objSections
                ' Output the section type
                Console.WriteLine( _
                  String.Format(strFormat1, objSection.Type.ToString()))

                ' Get a reference to the section sheets collection
                objSectionSheets = objSection.Sheets


                ' Loop through the sheets
                For Each objSheet In objSectionSheets
                    ' Output the sheet name
                    Console.WriteLine(String.Format(strFormat2, objSheet.Name))
                Next
```

```
            Next

            ' Access the igWorkingSection directly
            objSection = objSections.Workingsection

            ' Access the igBackgroundSection directly
            objSection = objSections.Backgroundsection

            ' Get a reference to the sheets collection
            objSheets = objDraft.Sheets

            ' Add a new sheet
            objSheet = objSheets.AddSheet( _
              "Sheet2", SheetSectionTypeConstants.igWorkingSection)

            ' Make the newly added sheet active
            objSheet.Activate()

        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter.Revoke()
        End Try
    End Sub
End Module
```

Sections and Sheets in C# Copy Code

```
using SolidEdgeDraft;
using System;
using System.Reflection;
using System.Runtime.InteropServices;

namespace SolidEdge.SDK
{
    class Program
    {
        [STAThread]
        static void Main(string[] args)
        {
            SolidEdgeFramework.Application application = null;
            SolidEdgeFramework.Documents documents = null;
            SolidEdgeDraft.DraftDocument draft = null;
            SolidEdgeDraft.Sections sections = null;
            SolidEdgeDraft.Section section = null;
            SolidEdgeDraft.Sectionsheets sectionsheets = null;
            SolidEdgeDraft.Sheets sheets = null;
            SolidEdgeDraft.Sheet sheet = null;
            string formatl = "Section = {0}";
            string format? = "Sheet = {0}";

            try
            {
                OleMessageFilter.Register();

                // Connect to a running instance of Solid Edge
                application = (SolidEdgeFramework.Application)
                  Marshal.GetActiveObject("SolidEdge.Application");

                // Get a reference to the documents collection
                documents = application.Documents;

                // Add a Draft document
                draft = (SolidEdgeDraft.DraftDocument)
                  documents.Add("SolidEdge.DraftDocument", Missing.Value);

                // Get a reference to the sections collection
                sections = draft.Sections;

                // Loop through the sections
                // IgWorkingSection, igBackgroundSection & ig2dModelSection
                for (int i = 1; i <= sections.Count; i++)
                {
```

```
                section = sections.Item(i);

                // Get a reference to the section sheets collection
                sectionsheets = section.Sheets;

                // Output the section type
                Console.WriteLine(String.Format(format1, section.Type.ToString()));

                // Loop through the sheets
                for (int j = 1; j <= sectionsheets.Count; j++)
                {
                    sheet = sectionsheets.Item(j);

                    // Output the sheet name
                    Console.WriteLine(String.Format(format2, sheet.Name));
                }
            }

            // Access the igWorkingSection directly
            section = sections.Workingsection;

            // Access the igBackgroundSection directly
            section = sections.Backgroundsection;

            // Get a reference to the sheets collection
            sheets = draft.Sheets;

            // Add a new sheet
            sheet = sheets.AddSheet(
              "Sheet2",
              SheetSectionTypeConstants.igWorkingSection,
              Missing.Value,
              Missing.Value);

            // Make the newly added sheet active
            sheet.Activate();
        }
        catch (System.Exception ex)
        {
            Console.WriteLine(ex.Message);
        }
        finally
        {
            OleMessageFilter.Revoke();
        }
    }
  }
}
```

Solid Edge Draft Type Library

SheetGroup Object

Members Collapse All

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Solid Edge Draft Type Library : SheetGroup Object

Description

A group of working sheets.

Object Model

- SheetGroup
  - Appication
    - ActiveSelectSet (SelectSet)
    - Addins
    - Appication
    - AppicationEvents
    - Appic tiuriV8Aiter DocurnentOpenEvent (ApplicdtionV8DocumentOperiEvent)
    - App¡cationWindo**wEvents**
    - CredteCommarid (Command)
    - Customization
    - Documents
    - Environments
    - FeatureldbraryEvenits
    - FieUIEvents
    - GetMateridlTdble (MdtTable)
    - Insight
    - NewFieUI Events
    - Parent (Application)
    - SEECEvents
    - ShortcutMenuEvents
    - SolidEdgeInsightXT
    - SoidEdgeTCE
    - *—**Widows**
  - Sheets (SheetGroupSheets)

Example

Example (VB.NET)

```
Imports System.10
Imports System.Runtime.InteropServices

Module Example
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objDraftDocument As SolidEdgeDraft.DraftDocument = Nothing
        Dim objSheetGroups As SolidEdgeDraft.SheetGroups = Nothing
        Dim objSheetGroup As SolidEdgeDraft.SheetGroup = Nothing

        Try
            OleMessageFliter .Register()

            objApplication = Marshal.GetActiveObject("SolidEdge.Application")
            objDraftDocument = objApplication.ActiveDocument
            objSheetGroups = objDraftDocument.SheetGroups
            objSheetGroup = objSheetGroups.Item(l)

        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter.Revoke()
        End Try
    End Sub

End Module
```

See Also

SheetGroup Members | Solid Edge ST5 - What's New

Solid Edge Draft Type Library

SheetGroups Collection

Members Collapse All

Solid EdcjO. DPSICJII better.

Solid Edge Draft Type Library : SheetGroups Collection

Description

A collection of sheet groups.

Example

Example (VB.NET)

```
Imports System.10
Imports System.Runtime.InteropServices

Module Example
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objDraftDocument As SolidEdgeDraft.DraftDocument = Nothing
        Dim objSheetGroups As SolidEdgeDraft.SheetGroups = Nothing

        Try
            OleMessageFilter.Register()

            objApplication = Marshal.GetActiveObject("SolidEdge.Application")
            objDraftDocument = objApplication.ActiveDocument
            objSheetGroups = objDraftDocument.SheetGroups

        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter.Revoke()
        End Try
    End Sub

End Module
```

See Also

SheetGroups Members | Solid Edge ST5 - What's New

Solid Edge Part Type Library

SheetMetalDocument Object Members

Overview Collapse All

Solid Edge. Design better.

Solid Edge Part Type Library : SheetMetalDocument Object

Public Methods

Activate	Makes the referenced object the active one.
ActivateReflectivePlane	This method is added to turn ON reflective plane command which is similar to check box at ribbon bar of Inspect Tab in Solid Edge UI.
AddMeasureVariable	Returns the measure variable object.
BindKeyToObject	Binds a specified reference key to a specified Occurrence, SubOccurrence, or TopologyReference object.
Break	
BreakAllInterpartLinks	
Close	Closes the referenced object.
CreateFeatureLibrary	
CreatePreview	
DeActivateReflectivePlane	This method will turn OFF reflective plane command which is similar to unchecked box at ribbon bar of Inspect Tab in Solid Edge UI.
DeleteEntities	
EditProperties	Displays the EditProperties dialog box to allow the user to interactively set and view various properties of the referenced document object.
FreezeAllInterpartLinks	
GetBaseStyle	Returns the base style.
GetCapturedRelationshipInformation	
GetContainerDocumentAndMatrixOfIPADoc	Returns the container document and transform matrix for an in-place-active document.
GetContainerDocumentAndOccurrenceOfIPADoc	
GetDefaultCutSizeValues	
GetGlobalParameter	Returns the setting of a specified global parameter for the referenced Application or document object.
GetInContextAssemblyNameForInterpartLinks	
GetLiveRules	Get Live Rules
GetMultiBodyPublishMembers	
GetNumberOfParentsAndDependents	Returns the number of parents and dependents of an object.
GetParentsAndDependents	Returns the parents and dependents of an object.

GetReflectivePlane	This method is added to get individual reflective plane(i.e. Top\Front\Right) status and distance at Model Reflective Display command.
GetReflectivePlaneTransparency	This method is added to get transparency status and level at Model Reflective Display command.
GetTopDocumentAndSubOccurrenceOfIPADoc	
GetUserPhysicalProperties	Method to get the user defined physical properties
GoalSeek	
HasInterpartLinks	
Hide	This method hides the referenced object.
Importstyles	Imports the styles from the document given by Filename into the referenced document. If Overwrite is true, any duplicate styles will be overwritten.
ImportStyles2	Copies styles of the specified type from the specified document. If bReplace is False, only styles that do not already exist will be copied.
InquireElement	
MeasureAngle	Measures the sharp angle between lines or planes or keypoints.
MeasureAngleEx	
MeasureDistance	
MinimumDistance	Measures the minimum distance between two elements.
MoveToSynchronous	This method moves ordered feature to synchronous feature.
NewWindow	Creates a new window that contains the referenced document.
NormalDistance	Measures the normal distance between lines or planes.
PlaceFeatureLibrary	Places features from an existing document into the current document.
PMI_ByModelState	Get PMI dimensions and annotations by model state. Returns failure for invalid states.
Printout	Prints the contents of the referenced object.
PutUserPhysicalProperties	Method to set the user defined physical properties
QueryByEntity	
QueryByProperty	Returns an AttachedPropertyTables collection that consists of AttachedPropertyTable objects that match all of the query inputs.
Recompute	Recalculates the relationships in the referenced object.
Redo	Reapplies actions reversed via Undo.
RestoreLiveRules	Restore default Live Rules
Save	Saves the referenced object.

SaveAs	Saves the referenced document to a new name, directory, or format.
SaveAsJT	Saves the referenced document to JT format with specified options.
SaveBody	Saves the input bodies of the input models into the given format
SaveCopyAs	Saves a copy of the referenced document to a file without modifying the open document.
SaveMultiBodyPublish	
SaveMultiBodyPublishMembers	
SeekReadOnlyAccess	
SeekWriteAccess	Seeks write access to the document if it is available.
SendMail	Distributes the referenced document using the installed electronic mail system.
Separate	
SetBaseStyle	Sets the base style.
SetDefaultCutSizeValues	
SetGlobalParameter	Sets the value of a specified global parameter for the referenced Application or document object.
SetLiveRules	Set Live Rules
SetReflectivePlane	This method is added to turn ON individual reflective plane(i.e. Top\Front\Right) and apply distance to the same which is similar to checked box at Model Reflective Display Dialogue in Solid Edge UI.
SetReflectivePlaneTransparency	This method is added to turn transparency and applying transparency level of reflective plane which is similar to checked box of transparency control at Model Reflective Display Dialogue in Solid Edge UI.
Show	
ShowOnly	
ShowParentsAndChildren	
SuspendLiveRules	Suspend Live Rules
SuspendPersistedRelationships	Suspend/Relax Persisted Relationships until successful edit
SuspendPMI	Suspend/Relax PMI until successful edit
ThawAllInterpartLinks	
ToggleRefPlanesDisplay	Controls the display of the reference planes.
TransformToSynchronousSheetmetal	
Undo	Reversed a specified number of actions.

Public Properties

0 ActiveSketch	Returns the active Sketch object.
0 AddInsStorage	Returns an IUnknown pointer to the object that implements the IStorage corresponding to the add-in substorage for the referenced object.
AdjustableDefinition	Returns the adjustable definition object.
Application	Returns the active application object.
AttachedPropertyTables	Returns the AttachedPropertyTables object for the referenced document object.
S AttributeQuery	Returns the AttributeQuery object.
AutomaticTransitionToSolutionManager	Determines whether a transition will occur to solution manager automatically after a successful edit.
Hsl BendTable	
BodyCheck	Specifies whether or not the referenced document object contains a valid parasolid body.
0 CapturedRelationshipCount	
0 ComponentName	
Hsl Componentsketches	
0 Constraints	
Hsl Constructions	Returns the Constructions collection object.
Hsl CoordinateSystems	Represents a collection of CoordinateSystem objects.
Hsl CreatedVersion	Returns the version of Solid Edge that was used to create the referenced document.
Hsl DesignBodyVisible	Specifies whether or not the referenced object is displayed to the screen.
Hsl DesignEdgebarFeatures	This read-only property returns the EdgeBarFeatures collection object (ordered features) for the referenced object. The collection is dynamically updated, based on history changes.
Hsl Dimensionstyles	Returns the Dimensionstyles collection object for the referenced document.
Hsl Dirty	Specifies whether or not the referenced document has been modified since the last time it was saved.
Hsl DisableMoveToSynchronous	
Hsl DividedParts	Returns the DividedParts collection object for the referenced Part of Sheet Metal document.
Hsl DividePartEvents	Returns the DividePartEvents object for the referenced document object.
I±sl DocumentEvents	Returns the DocumentEvents object for the referenced document object.
Hsl FaceStyles	Returns the FaceStyles collection object for the referenced document.
Hsl FamilyMembers	Returns the FamilyMembers collection object for the referenced document object.

O FamilyOfPartsEvents	This read-only property returns the FamilyOfPartsEvents collection object for the referenced object.
FamilyOfPartsExEvents	
FillStyles	Returns the FillStyles collection object for the referenced document.
S FlatPatternEdgebarFeatures	This read-only property returns the EdgeBarFeatures collection object (ordered features) for the referenced object. The collection is dynamically updated, based on history changes.
O FlatPatternModels	Returns flat pattern models collecton
Hsl FullName	Returns the name and path of the referenced object.
Hsl GeometricVersion	Returns a value that is incremented each time the geometry of an associated document is changed.
Hsl HardwareFile	
Hsl HasCapturedRelationships	
Hsl HatchPatternStyles	Returns the HatchPatternStyles collection object for the referenced document.
Hsl HighlightSets	Returns the HighlightSets collection object for the referenced document object.
Hsl HoleDataCollection	Returns the HoleDataCollection object for the referenced document object.
Hsl InPlaceActivated	Specifies whether or not the referenced document is in-place activated.
Hsl InterDocumentUpdate	
Hsl InterpartLinks	
Hsl IsAdjustablePart	Returns whether file is adjustable or not.
Hsl IsFeatureLibrary	Specifies whether or not the referenced PartDocument or SheetMetalDocument is a feature library.
Hsl IsGeometricVersionDirty	Retrieves whether the geometric version number of the assembly document is dirty or not.
Hsl IsInsightFile	Returns indication of whether the file is an Insight managed file.
Hsl IsMultiCADDriven	
Hsl IsTemplate	Specifies whether or not the referenced document is a template.
Hsl LastSavedVersion	Returns the version of Solid Edge that was used the last time the referenced document was saved.
Hsl Layers	Returns the Layers collection object.
Hsl LinearStyles	Returns the LinearStyles collection object for the referenced document.
Hsl LockedSketch	
Hsl ModelingMode	The property transitions the document between Synchronous and Ordered as is done in the UI via Pathfinder or a context menu.
Hsl Models	Returns the Models collection object for the referenced document object.

Name	Returns the name of the referenced object.
Parent	Returns the parent object for the referenced object.
0 Path	Returns the directory path name of the referenced document.
Hsl PhysicalPropertiesStatus	This read-only property returns the status of the physical properties of the model for the referenced object.
ProfileSets	Returns the ProfileSets collection object for the referenced document object.
ProfileUndoSteps	This read-write Long property sets and returns the number of profile undo steps for the referenced document object.
Properties	For document objects, returns the PropertiesCollection colletion object. For Variables, sets and returns the variable properties.
PropertyTableDefinitions	Returns the PropertyTableDefinitions object for the referenced document object.
Readonly	Specifies whether or not the referenced document has been opened as read-only.
RefAxes	Returns the RefAxes collection object for the referenced document object.
RefPlanes	Returns the RefPlanes collection object for the referenced document object.
Rootstorage	Returns the IUnknown pointer to the object implementing the root storage of the referenced document.
Hsl RoutingSlip	Returns the RoutingSlip object for the referenced document.
Hsl SelectSet	Returns the current selection set for the referenced document.
Hsl Sensors	Returns the Sensors collection object for the referenced document object.
Hsl SimplifiedModels	Returns the Simplified Models collection object for the referenced document object.
Hsl SimplifyEdgebarFeatures	This read-only property returns the EdgeBarFeatures collection object (ordered features) for the referenced object. The collection is dynamically updated, based on history changes.
Hsl Sketches	Returns the Sketches collection object for the referenced document object.
Hsl Sketches3D	Returns the 3D Sketches collection object for the referenced document object.
Hsl Status	Returns the status of the specified object.
Hsl StudyOwner	Get Study Owner object from the document.
Hsl Summaryinfo	Returns a SheetSummaryInfo object to provide access to descriptive properties for the referenced document.
Hsl Terminals	
Hsl Type	Specifies the type of the object being referenced.
Hsl UndoSteps	This read-write Long property sets and returns the number of undo steps for the referenced document object.

UnitsOfMeasure	Returns the UnitsOfMeasure object for the referenced document.
UserDefinedSets	
0 Variables	Returns the Variables collection object for the referenced document.
ViewStyles	Returns the ViewStyles collection object for the referenced document.
Hzl Windows	Returns the Windows collection object.

See Also

SheetMetalDocument Object

Solid Edge Draft Type Library

SheetSetup Object

Members Collapse All

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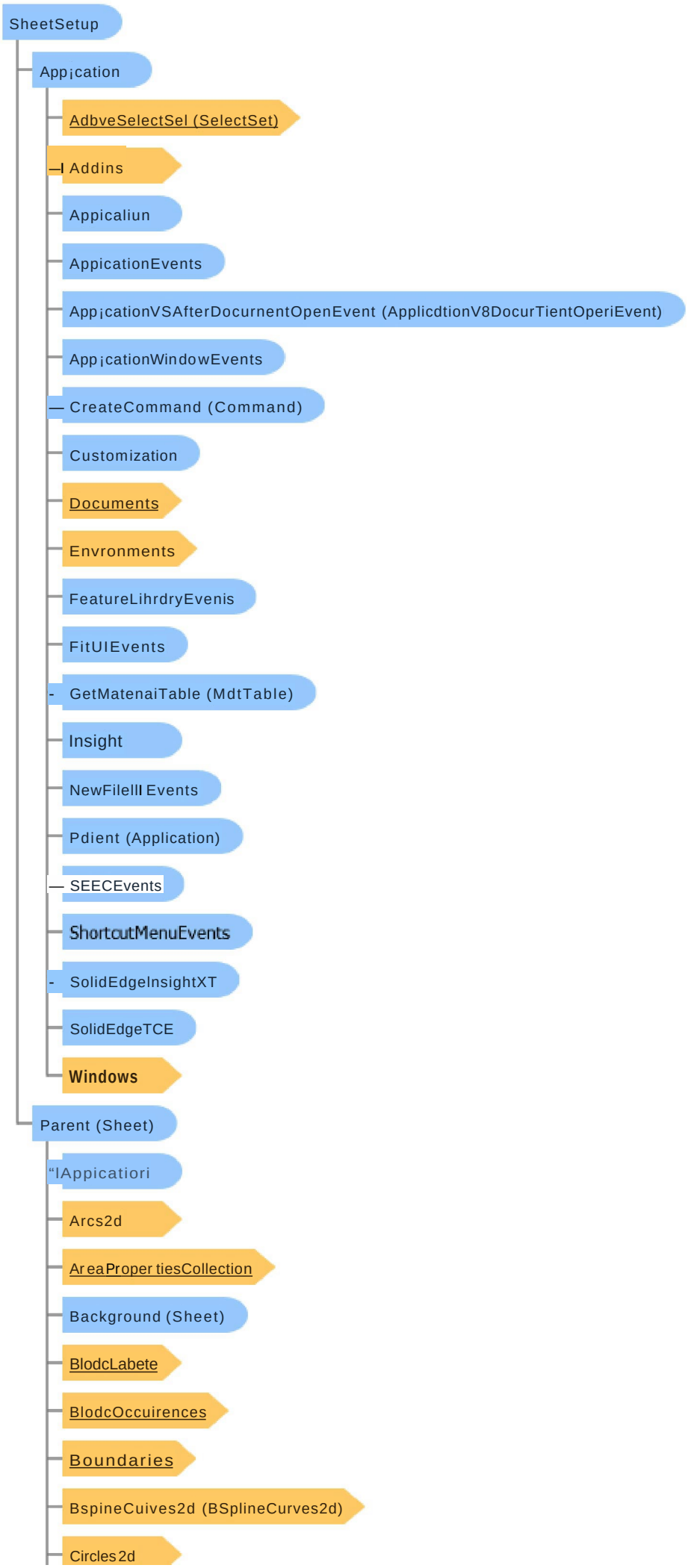
Description

Represents a sheet setup description.

Object Model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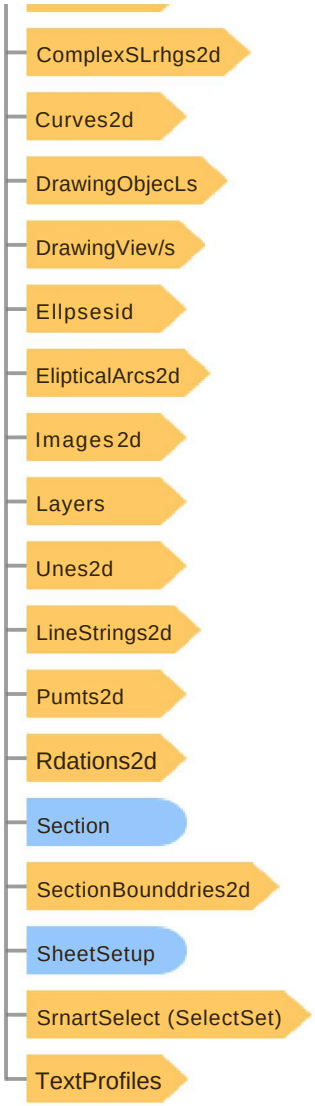


Remarks

This object includes properties that specify all sheet setup attributes such as margins, dimensions, and scale.

See Also

SheetSetup Members

Solid Edge FrameworkSupport Type Library

SmartFrames2d Collection Members

Overview Collapse All

Solid Edge FrameworkSupport Type Library : SmartFrames2d Collection

Public Methods

AddBy2Points	Creates a text box by specifying the diagonal points of the text box.
AddByOrigin	Adds a SmartFrame2d object by specifying the origin and frame cropping distances.
Item	The items in a collection.

Public Properties

Application	Returns the active application object.
Count	Returns the number of objects in the referenced collection.
Parent	Returns the parent object for the referenced object.

See Also

SmartFrames2d Collection

Solid Edge Draft Type Library

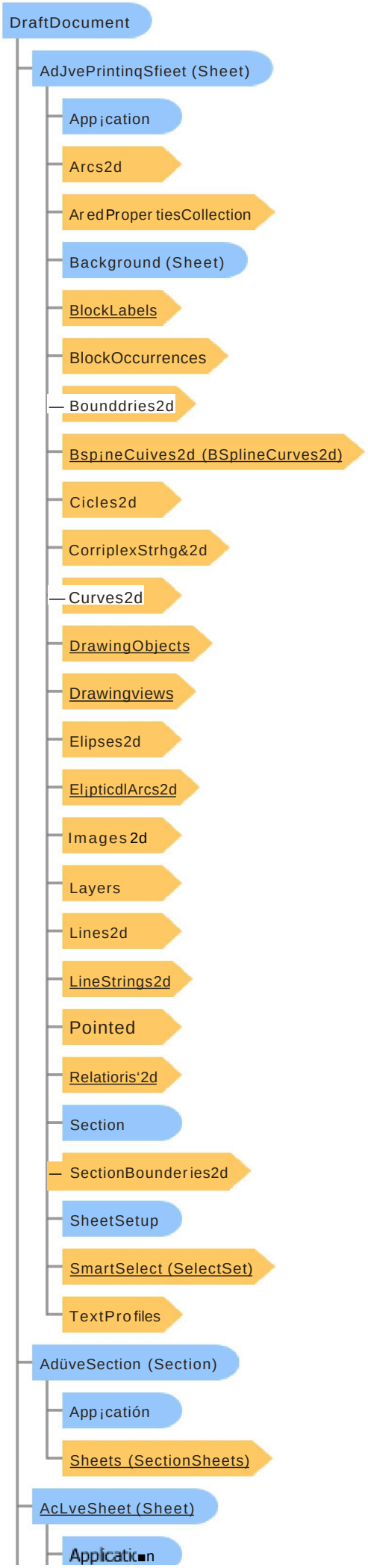
Project Overview

Collapse All

Description

Solid Edge Draft Type Library

Object Model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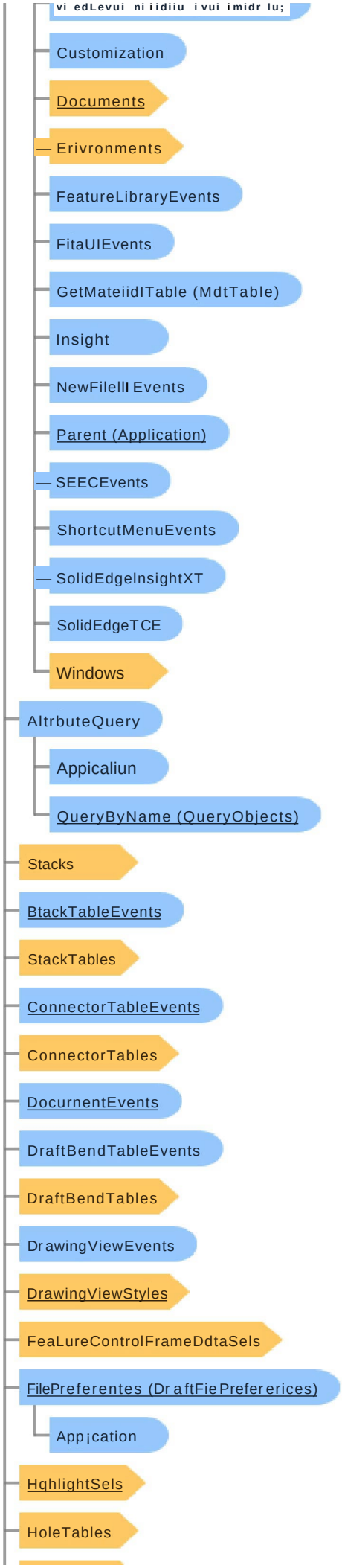
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  - FeatureLibraryEvents
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  - ShortoutMenuEvents
  - SolidEdgeInsightXT
  - SolidEdgeTCE
  - Windows
- PartsListEvents
- PartsLists
- Sections
- SelectSet
- Sheets
- Sui faceFinishSymbolDataSets
- Tables
- TableSLyles
- UnitsOFMeasure
- WeldSymbolDataSets

Whdaws

Classes

Class Module	Description
Block	Represents a block.
BlockLabel	Represents a block label.
BlockLabelOccurrence	Represents an occurrence of a block label in a block occurrence
BlockLabelOccurrences	Returns the BlockLabels collection.
BlockLabels	Returns the BlockLabels collection.
Blockoccurrence	Represents an occurrence of a block view.
Blockoccurrences	Returns the Blockoccurrences collection.
Blocks	Blocks collection.
BlockTable	Represents an occurrence of a BlockTable in a Draft document.
BlockTables	Represents a collection of BlockTable objects.
BlockView	Represents a block view.
BlockViews	Blocks collection.
BrokenOutSectionProfile	Broken Out Section Profile
BrokenOutSectionProfiles	Broken Out Section Profiles Collection
ConnectorTable	Represents an occurrence of a ConnectorTable in a Draft document.
ConnectorTables	Represents a collection of ConnectorTable objects.
CoordinateSystem2d	Represents an occurrence of a user-defined coordinate system from a 3d model in a DrawingView.
CoordinateSystems2d	Represents a collection of CoordinateSystem2d objects.
CuttingPlane	
CuttingPlanes	Cutting Planes Collection
DetailEnvelope	Provides methods and properties of a detail view envelope object.
DetailEnvelopes	Detail Envelopes Collection
DraftBendTable	Represents an occurrence of a DraftBendTable in a Draft document.
DraftBendTables	Represents a collection of DraftBendTable objects.
DraftDocument	Represents the data file that typically contains drafting information, such as part and assembly views, dimensions, annotation, and so forth.
DraftFilePreferences	Get and set file level preferences.
DraftPrintUtility	An object that is sued to print one or more Draft files.
DraftProfile	Draft Profile Object
DrawingView	A 2-D representation of a 3-D part or assembly model.
DrawingViews	Represents a collecion of DrawingView objects.
DVArc2d	A part or assembly derived arc in the DrawingView.
DVArcs2d	A collection of DVArc2d objects.

DVBSplineCurve2d	A part or assembly derived b-spline curve in the DrawingView.
DVBSplineCurves2d	A collection of BSplineCurve2d objects.
DVCircle2d	A part or assembly derived circle in the DrawingView.
DVCircles2d	A collection of DVCircle2d objects.
DVEllipse2d	A part or assembly derived ellipse in the DrawingView.
DVEllipses2d	A collection of DVEllipse2d objects.
DVEllipticalArc2d	A part or assembly derived elliptical arc in the DrawingView.
DVEllipticalArcs2d	A collection of DVEllipticalArc2d objects.
DVLine2d	A part or assembly derived line in the DrawingView.
DVLines2d	A collection of DVLine2d objects.
DVLineString2d	A part or assembly derived linestring in the DrawingView.
DVLineStrings2d	A collection of DVLineString2d objects.
DVPoint2d	A part or assembly derived point in the DrawingView.
DVPoints2d	A collection of DVPoint2d objects.
GraphicMembers	Represents a collecion of GraphicMember objects.
HighlightDrawingViewMembers	Highlight Drawing View Members
HoleTable	Represents an instance of a Hole Table in an associated DraftDocument object.
HoleTables	Represents a collection of HoleTable objects.
HTHole	Represents an occurrence of a hole characteristic in a hole table entry.
HTHoles	Represents a collection of HTHole objects.
ModelLink	Represents a member of the ModelLinks collection object.
ModelLinks	Represents a collection of ModelLink objects.
ModelMember	A part or assembly model member in the DrawingView.
ModelMembers	Represents a collecion of ModelMember objects.
ModeiNode	Represents the relationship between a drawing view and a part, assembly, or weld.
ModeiNodes	Represents a collection of ModeiNode objects.
ModelWeld	Represents an occurrence of a Weld feature.
ModelWelds	Represents a collection of ModelWeld objects.
PartsList	Represents an occurrence of a PartsList in a Draft document.
PartsLists	Represents a collection of PartsList objects.
Section	A partition in a document that contains a list of sheets that belong to the partition.
SectionBoundaries2d	A collection of Boundary2d objects.
Sections	Represents a collection of Section objects.
SectionSheets	Represents a collection of sheets that exist in a section.
SelectedSheets	Represents a collection of currently selected sheets.
Sheet	Represents a sheet in a document.

SheetGroup	A group of working sheets.
SheetGroups	A collection of sheet groups.
SheetGroupSheets	A collection of working sheets belonging to a sheet group.
Sheets	A collection of Sheet objects.
SheetSetup	Represents a sheet setup description.
SheetWindow	A 2-D view type that displays sheets and has a tab user interface.
Table	The Table object provides access to all of the objects and properties associated with a generic table.
TableCell	The TableCell object provides access to all of the objects and properties associated with a generic table cell.
TableColumn	The TableColumn object provides access to all of the objects and properties associated with a generic table column.
TableColumns	The TableColumns collection object supports the standard collection interfaces and includes a method to add new generic table columns to the collection.
TableGroup	An object in the Groups object collection.
TableGroups	A collection of Group objects.
TablePage	
TablePages	Collection of Table Page objects.
TableRow	The TableRow object provides access to all of the objects and properties associated with a generic table row.
TableRows	The TableRows collection object supports the standard collection interfaces and includes a method to add new generic table rows to the collection.
Tables	The Tables collection object supports the standard collection interfaces and includes a method to add new generic tables to the collection.
TableTitle	The TableTitle object provides access to all the objects and properties associated with a generic table title.
TableTitles	The TableTitles collection object supports the standard collection interfaces and includes a method to add new generic table titles to the collection.
ViewPlane	Provides methods and properties that extend to an auxiliary DrawingView object.
ViewPlanes	View Planes Collection

Public Enumerations

Enumeration	Description
AssemblyDrawingViewTypeConstants	Assembly Drawing View Type Constants
BlockLabelOriginLocationconstants	Block Label Origin Location Constants
BlockTableType	
CleanProfileOptions	The options available for cleaning profiles.
CoordinateSystemZdAxisConstants	Coordinate Systems 2d Axis Constants
DetailEnvelopeStandardConstants	Enumerates the standards used to display the detail envelope.
DimensionTrackerReasonCode	

DimWeldBeadWeldImportconstants	Weld Bead Weld Imports
DimWeldBeadWeldmentShapeConstants	Weld Bead Weldment Shapes
DimWeldBeadWeldmentTypeConstants	Weld Bead Weldment Types
DimWeldBeadWeldStandardConstants	Weld Bead Weld Standards
DimWeldBeadWeldTypeConstants	Weld Bead Weld Types
DraftGlobalConstants	Draft Global Constants
DraftPrintOrientationConstants	Orientation of print output - portrait or landscape.
DraftPrintPaperSizeConstants	Common printer paper sizes
DraftPrintScaleTooLargeActionConstants	Defines what action will be taken when printing a sheet that will not fit on the selected paper size.
DraftPrintSheetsPerPageConstants	Number of sheets to be printed per page.
DraftPrintUnitsConstants	Units for Draft Print Utility options.
DraftSectionViewType	
DrawingViewBsplineSimplificationConstants	DrawingView Bspline Simplification Constants
DrawingViewDefaultsConstants	Specifies the defaults for drawing view creation using AddEmptyView.
DrawingViewEdgeStyleMappingEdgeType	Drawing view edge type used to get or set its edge style
DrawingViewIntersectionProcessingConstants	DrawingView Intersection Processing Constants
DrawingViewProjectionAngleConstants	Specifies first or third angle projection drawing view preference
DrawingViewShadingQualityConstants	Drawing View Shading Quality Constants
DrawingViewSimplifiedAssemblyOptionConstants	Specifies how simplified assemblies will be used in the drawing view.
DrawingViewSimplifiedPartOptionConstants	Specifies how simplified parts will be used in the drawing view.
DrawingViewSnapShotQualityConstants	DrawingView SnapShot/Draft Quality Constants
DrawingViewStyleMappingElementType	Drawing view type used to get or set the style name to use when creating drawing views of the specified type.
DrawingViewTypeConstants	Drawing View Type Constants
DrawingViewVHL_ToleranceOverrideQualityConstants	DrawingView VHL Tolerance Constants
DVShowHideEdgeOverrideType	Show/Hide edge override types
DVThreadDisplayModeConstants	Specifies drawing view thread display preference
FoldTypeConstants	Fold Type Constants
GraphicMemberEdgeTypeConstants	GraphicMember Edge Type Constants
GridDisplayOptionsConstants	Options for displaying the grid.
GridSnapOptionsConstants	Options for snapping to the grid.
HTHoleTypeConstants	H T Hole Type Constants
ModelLinkTypeConstants	ModelLink Constants
ModelMemberComponentTypeConstants	ModelMember Type Constants

ModelMemberDisplayTypeConstants	ModelMember Display Type Constants
ModelNodeTypeConstants	Model Node Type Constants
PaperSizeConstants	Predefined Values for Paper Size
PaperToModelScaleConstants	Predefined Values for Paper Scale
PaperUnitConstants	Predefined Values for Paper Units
PartDrawingViewTypeConstants	Part Drawing View Type Constants
PartsListComponentType	
PartsListType	
Precisionconstants	Predefined Values for Paper Unit Precision
ProfileValidationStatus	
SheetFitConstants	Sheet Fit Constants
SheetMetalDrawingViewTypeConstants	Sheet Metal Drawing View Type Constants
SheetSectionTypeConstants	Sheet Section Type Constants
TableAnchorPoint	
TableTextOrientation	
TitlePosition	
ViewOrientationConstants	View orientation constants
WeldmentDrawingViewTypeConstants	Weldment Drawing View Type Constants

Solid Edge FrameworkSupport Type Library

StackPitch Property

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Description

Sets and returns the minimum distance track distance maintained between individual dimensions within a group.

Property type

Read-write property

Syntax

Visual Basic
`Public Property StackPitch As Double`

Remarks

The value is defined as ratio of text height.

See Also

DimensionStyle Object | DimensionStyle Members

Solid Edge FrameworkSupport Type Library

StackPitch Property

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Description

Sets and returns the minimum distance track distance maintained between individual dimensions within a group.

Property type

Read-write property

Syntax

Visual Basic
`Public Property StackPitch As Double`

Remarks

The value is defined as ratio of text height.

See Also

DimStyle Object | DimStyle Members

Solid Edge ST7 SDK

Starting Solid Edge

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You may still use the familiar CreateObjectQ method from the Visual Basic 6 days if you'd like. In Visual Basic .NET, CreateObject() simply calls Activator.CreateInstanee. This is important to understand because CreateObjectQ is a language specific feature of Visual Basic .NET. No other .NET language has a comparable feature. The other .NET languages must rely on Activator.CreateInstance.

A See Handling 'Application is Busy' and 'Call was Rejected By Callee' errors for information regarding the use of OleMessageFilter.

Starting Solid Edge from Visual Basic .NET Copy Code

```
Imports System.Runtime.Interopservices

Module Program
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objType As Type = Nothing

        Try
            'Old VB6 Syntax
            'objApp = CreateObject("SolidEdge.Application")

            OleMessageFilter.Register()

            ' Get the type from the Solid Edge ProgID
            objType = Type.GetTypeFromProgID("SolidEdge.Application")

            ' Start Solid Edge
            objApplication = Activator.CreateInstance(objType)

            ' Make Solid Edge visible
            objApplication.Visible = True

        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter.Revoke()
        End Try
    End Sub
End Module
```

Starting Solid Edge from C# Copy Code

```
using System;
using System.Runtime.InteropServices;

namespace SolidEdge.SDK
{
    class Program
    {
        [STAThread]
        static void Main(string[] args)
        {
            SolidEdgeFramework.Application application = null;
            Type type = null;

            try
            {
                OleMessageFilter.Register();

                // Get the type from the Solid Edge ProgID
                type = Type.GetTypeFromProgIDfSolidEdge.Application");

                // Start Solid Edge
                application = (SolidEdgeFramework.Application)
                  Activator.CreateInstance(type);

                // Make Solid Edge visible
                application.Visible = true;
            }
            catch (System.Exception ex)
            {
                Console.WriteLine(ex.Message);
```

```
            }
            finally
            {
                OleMessageFilter.Revoke();
            }
        }
    }
}
```

See Also

Connecting To Solid Edge
Handling 'Application is Busy' and 'Call was Rejected By Callee' errors

Solid Edge FrameworkSupport Type Library

TextBox Object Members

Overview Collapse All

Solid Edge FrameworkSupport Type Library : TextBox Object

Public Methods

BringForward	Changes the display order of the referenced object by moving it in front of the first object that obscures it.
BringToFront	Changes the display order of the referenced object by moving it in front of all other overlapping objects.
"*** Copy	Places a copy of the referenced object on the clipboard.
Cut	Moves the referenced object to the system clipboard.
Delete	Deletes the referenced object.
Duplicate	Copies the referenced object to a specified location, creating a new instance of the object with the same property values as the original.
GetKeyPoint	Returns information about a specified key point.
GetOrigin	Returns the origin for the referenced object.
Mirror	Copies the referenced object about an axis.
Move	Moves the referenced object.
PromoteToCurrentVersion	Promotes text box to the latest version behavior, the same as if placing a new object.
Range	Returns the high and low range values for the referenced object.
Rotate	Rotates the referenced object about the point specified by the angle in radians.
Scale	Scales the referenced object by a specified factor.
Select	Selects the referenced object in the active window.
SendBackward	Changes the display order of visible objects by moving the referenced object so that it is in back of the first object behind it.
SendToBack	Changes the display order of visible objects by moving the referenced object behind all overlapping objects.
SetKeyPoint	Sets the position for a specified key point.
SetOrigin	Redefines the offset between the lower left corner of the window and the object's origin.

Public Properties

Hsl ActualTextLength	Returns the length of the character string in the longest line of text.
Hsl Application	Returns the active application object.
Hsl AttributeSets	Returns the AttributeSets collection object for the referenced object.
Hsl BorderLineColor	Sets and returns the color used to draw the border of the referenced TextBox object.

0 BorderLineStyle	Sets and returns the line style used to draw the border line of the referenced TextBox object.
BorderLineWidth	Sets and returns the color of the border when BorderVisible is True.
Hsl BorderOffset	Sets and returns the distance between the border and the text box for the referenced TextBox object.
BorderType	Sets and returns the border type for the referenced TextBox object.
Document	Returns the document to which the referenced object belongs.
Edit	Returns the TextEdit object associated with the referenced TextBox object.
0 Fill	Specifies whether or not the background of the referenced TextBox is to be filled.
FillColor	Sets and returns the color with which to fill the background of an object.
FlowOrientation	Sets and returns the orientation of the text flow.
Height	Returns the height of the referenced object.
Hsl HorizontalAlignment	Sets and returns the horizontal alignment of the text within a paragraph.
Index	Returns the numeric index for the active object.
IsAttributeSetPresent	Returns True if a specified Attribute Set exists on the referenced object.
Hsl Justification	Sets and returns the horizontal and vertical justification of the text for the referenced TextBox object.
Key	Returns a key (a string) to find the object relative to its owning parents.
KeyPointCount	Returns the number of keypoints/handles that are supported by the referenced object.
Layer	Sets and returns the named layer on which the referenced object resides.
LineSpacing	Sets and returns the amount of space between two lines in a paragraph for the referenced object.
LineSpacingType	Sets and returns the Line spacing type for the referenced TextBox object.
MinimumAspectRatio	Sets and returns the Minimum Aspect Ratio Textbox can have
Hsl Name	Returns the name of the referenced object.
Parent	Returns the parent object for the referenced object.
PlacementType	Sets and returns the placement type for the referenced TextBox object.
RotationAngle	Sets and returns the angle, in radians, at which the object is rotated along its major axis.
Hsl Shadow	Specifies the status of the shadow feature for the referenced TextBox object.
Hsl ShadowOffset	Sets and returns the offset of the shadow feature from the border for the referenced TextBox object.
Hsl TabStopWidth	Sets and returns the default Tab Stop width for the referenced TextBox object.
Hsl Text	Sets and returns the text within the referenced TextBox object.
Hsl TextControlType	Sets and returns the Text Control type for the referenced TextBox object.
Hsl TextScale	Sets and returns the text scale of the object.
I±sl Type	Specifies the type of the object being referenced.
Hsl VerticalAlignment	Sets and returns the vertical alignment of the text within the text box.

0 Width	Returns the width of the referenced object.
0 ZOrder	Specifies the front-to-back order in which an object is displayed relative to other objects.

See Also

TextBox Object

Solid Edge FrameworkSupport Type Library

TextClearanceGap Property

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Description

Sets and returns the gap distance between the dimension text and the dimension line.

Property type

Read-write property

Syntax

Visual Basic
`Public Property TextClearanceGap As Double`

Remarks

The text clearance gap value is a ratio of dimension text height.

See Also

DimensionStyle Object | Dimensionstyle Members

Solid Edge FrameworkSupport Type Library

TextEdit Object Members

Overview Collapse All

Solid Edge FrameworkSupport Type Library : TextEdit Object

Public Methods

Clear	Removes the currently active text from the editor.
ContinueNumbering	Adds the selcted paragraphs to the last list and continues the numbered sequence
Copy	Places a copy of the referenced object on the clipboard.
Cut	Moves the referenced object to the system clipboard.
Decreaseindent	Decrease the left indent of paragraphs in slection range
GetBulletList	Returns whether the paragraph selection has bulleted list and its data if exists
GetFraction	Returns the fraction data if it is there at insert index
GetNumberList	Returns whether the paragraph selection has numbered list and its data if exists
GetSelect	Returns the current selection range.
Increaseindent	Increase the left indent of paragraphs in slection range
InsertFraction	Inserts the fraction at insert index
Paste	Copies the data from the system clipboard to the referenced object.
PointInsideBox	Returns True if the specified point lies inside the referenced TextBox object.
RestartNumbering	Starts a new numbered sequence in the list
SetBulletList	Assigns the bullet to the paragraph selection
SetNoList	Removes the number or bullet list for the paragraphs in selction range
SetNumberList	Creates the numbered list over the paragraph selection
SetSelect	Assigns the selection range start and end values.

Public Properties

0 AfterSpacing	Sets and returns the amount of space below the first line of the paragraph for the referenced text object.
S AspectRatio	Sets and returns the character width-to-charager height ratio.
BeforeSpacing	Sets and returns the amount of space above the first line of the paragraph for the referenced text object.
Bold	Specifies whether or not the associated text is bold.
Bullet	Specifies whether or not paragraph bulleting is turned on.
Characters	Sets and returns character data for the referenced TextEdit object.
Characterspacing	Sets and returns the spacing between characters for the associated text.
Color	Sets or returns the color of the referenced text.
FirstLineIndent	Sets and returns the distance from the margin to the start of the line.

Font	Sets and returns the name of the font.
HorizontalAlignment	Sets and returns the horizontal alignment of the text within a paragraph.
0 Insertindex	Sets and returns the index at which the next character will be inserted for the referenced TextEdit object.
IsEmpty	Specifies whether or not the referenced object is empty.
Italic	Specifies whether or not the associated text is italic.
Language	Sets and returns the language in which the text is written.
Leftindent	Sets and returns the distance from the left margin to the end of the line.
LineSpacing	Sets and returns the amount of space between two lines in a paragraph for the referenced object.
LineSpacingType	Sets and returns the Line Spacing type for the referenced TextBox object.
Name	Returns the name of the referenced object.
Number	Specifies whether or not paragraph numbering is turned on.
NumberingStartValue	Sets the start value for the referenced numbered list
Rightindent	Sets and returns the distance from the right margin to the end of the line.
SpecialIndentType	Sets and returns the Special Indent type None, First line, or Hanging
SubScript	Specifies whether or not the associated text is subscripted.
@ SuperScript	Specifies whether or not the associated text is superscripted.
TextScale	Sets and returns the text scale of the object.
TextSize	Sets and returns the size of the text for the referenced object.
UnderLine	Specifies whether or not the associated text is underlined.
Verticalposition	Sets and returns the vertical position of superscript and subscript text.

See Also

TextEdit Object

Solid Edge Constants Type Library

TextLineSpacingTypeConstants Enumeration

Collapse All

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Description

Text Line Spacing Types

Members

Member	Value	Description
igAtLeast	3	Text Line Spacing At Least
igDouble	2	Text Line Spacing Double
igExactly	4	Text Line Spacing Exactly
igMultiple	5	Text Line Spacing Multiple
igOneAndHalf	1	Text Line Spacing 1.5
igSingle	0	Text Line Spacing Single

See Also

Solid Edge ST5 - What's New

Solid Edge FrameworkSupport Type Library

TrackDistance Property

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Description

Sets and returns the track distance of the referenced Dimension object.

Property type

Read-write property

Syntax

Visual Basic
`Public Property TrackDistance As Double`

Remarks

The track distance defines the distance that separates the dimension from the object it measures. In linear dimensions, it is the distance between the origin object to the dimension line. In angular dimensions, it is the distance between the center point (the intersection of the origin and the measurement vectors) to the dimension line.

Example

Example (Visual Basic)

```
Private Sub Form_Load()
    Dim objApp As SolidEdgeFrameWork.Application
    Dim objDoc As SolidEdgeDraft.DraftDocument
    Dim objSheet As SolidEdgeDraft.Sheet
    Dim objline As SolidEdgeFrameworkSupport.Line2d
    Dim objDim As SolidEdgeFrameworkSupport.Dimension
    Dim dblTradis As Double
    ' Report errors
    Const PI = 3.14159265358979
    ' Create/get the application with specific settings
    On Error Resume Next
    Set objApp = GetObjectG "SolidEdge.Application")
    If Err Then
        Err.Clear
        Set objApp = CreateObject("SolidEdge.Application")
        Set objDoc = objApp.Documents.Add("SolidEdge.DraftDocument")
        objApp.Visible = True
    Else
        Set objDoc = objApp.ActiveDocument
    End If
    'getting the Active Sheet object
    Set objSheet = objDoc.ActiveSheet
    'create a line
    Set objline = objSheet.Lines2d.AddBy2Points(xl:=0.1, yl:=0.1, x2:=0.3, y2:=0.3)
    'create a dimension object
    Set objDim = objSheet.Dimensions.AddLength(Object:=objline)
    'get TrackDistance
    dblTradis = objDim.TrackDistance
```

```
        'set TrackDistance
        objDim.TrackDistance = 0.05
        ' USER DISPLAY
        ' Release objects
        Set objApp = Nothing
        Set objDoc = Nothing
        Set objSheet = Nothing
        Set objline = Nothing
        Set objDim = Nothing
    End Sub
```

See Also

Dimension Object | Dimension Members

Solid Edge Draft Type Library

Unalign Method

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Description

Removes the alignment relationship between selected principal and/or auxiliary DrawingView objects that are contained in a selection set.

Syntax

Visual Basic
`Public Sub Unalign()`

Remarks

As a result of alignment, views that are unaligned can be moved, rotated, and scaled without affecting other views.

See Also

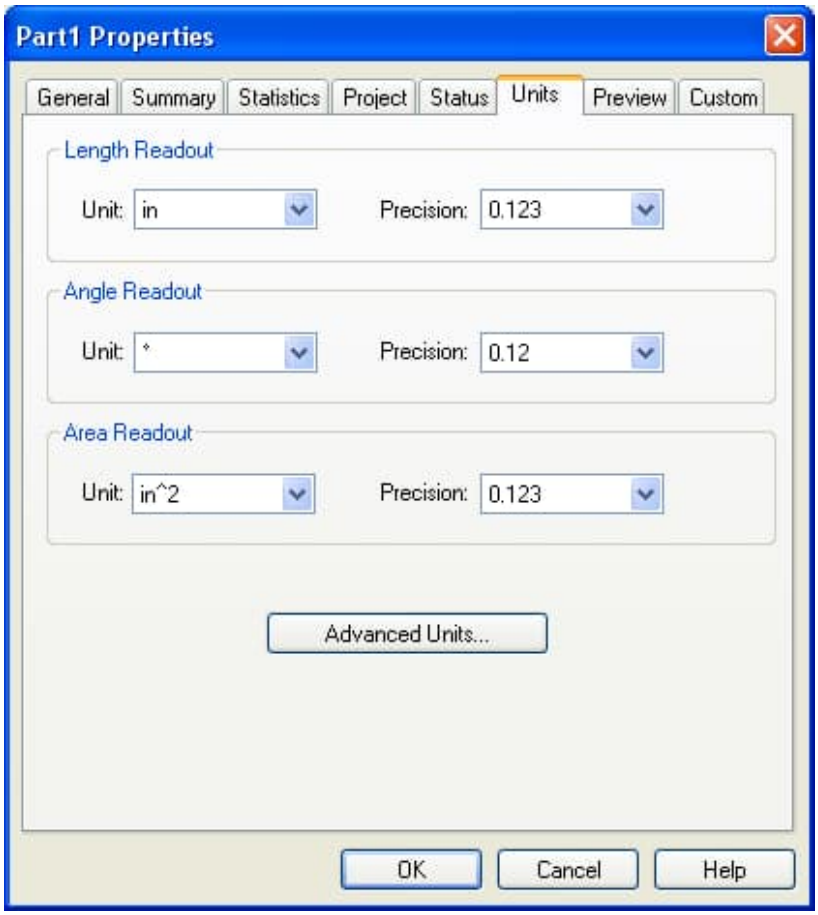
DrawingViews Collection | DrawingViews Members

Solid Edge ST7 SDK

Units of Measure

Overview

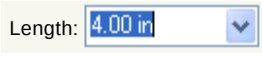
In the interactive environment, Solid Edge allows you to specify the length, angle, and area units to use when placing, modifying, and measuring geometry. For example, you can specify millimeters as the default length unit of measurement; you can also specify the degree of precision of the readout. You specify these properties on the Units and Advanced Units tabs of the Properties dialog box. (On the File menu, click Properties to display the dialog box.)



This is strictly a manipulation of the display of the precision; internally all measurements are stored at their full precision.

With a Length Readout precision of 0.12, the length of any linear measurement is displayed as follows:

Because millimeters are the default units in this example, whenever distance units are entered, they have to be in millimeters. If a user enters a distance value in inches, for example, the units are automatically converted to millimeters.



converts to

The units system in Solid Edge allows users to specify the default units and to control how values are displayed for each of the units. Users can change the default units and their display at any time and as often as necessary.

You can customize Solid Edge so that your commands behave in a similar way. For example, suppose you are creating a program to place hexagons. The program displays a dialog box that allows you to enter the size of the hexagon and then creates the hexagon at a location specified by a mouse click. When users enter the size of the hexagon, they should be able to enter the value in the user-specified default unit. Also, users should be able to override the default unit and specify any linear unit. The program will need to handle any valid unit input.

Internal Units

Unit Type	Internal Units
Distance	Meter
Angle	Radian
Mass	Kilogram
Time	Second
Temperature	Kelvin
Charge	Ampere
Luminous Intensity	Candela
Amount of Substance	Mole
Solid Angle	Steradian

All other units are derived from these. All calculations and geometry placements use these internal units. When values are displayed to the user, the value is converted from the internal unit to the user-specified unit.

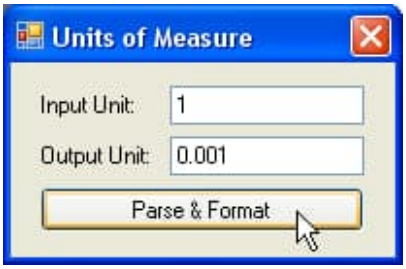
When automating Solid Edge, first convert user input to internal units. Calculations and geometric placements use the internal units. When displaying units, you must convert from internal units to default units. The UnitsOfMeasure object handles these conversions.

Working with Units of Measure

The UnitsofMeasure object provides two methods: ParseUnit and FormatUnit. In addition, a set of constants is provided to use as arguments in the methods. The ParseUnit method uses any valid unit string to return the corresponding database units. The FormatUnit method uses a value in database units to return a string in the user-specified unit type, such as igUnitDistance, igUnitAngle, and so forth. The units (meters, inches, and so forth) and precision are controlled by the active units for the document.

The following programs uses both the ParseUnit and FormatUnit methods to duplicate the behavior of unit fields in Solid Edge. The code also checks whether the input is a valid unit key-in and outputs the correctly formatted string according to the user-specified setting.

Error handling is used to determine if a valid unit has been entered. The Text property from the text field is used as input to the ParseUnit method, and the unit is a distance unit. If the ParseUnit method generates an error, focus is returned to the text field, and an error is displayed, giving the user a chance to correct the input. This cycle continues until the user enters a correct unit value. If the key-in is valid, then the database value is converted into a unit string and displayed in the text field.



Formatting and Displaying Units

Formatting and Displaying Units in Visual Basic .NET C

```
Imports SolidEdgeFramework
Imports System.Runtime.InteropServices

Public Class Forml
  Private m_application As SolidEdgeFramework.Application

  Private Sub Forml_Load(ByVal sender As System.Object, _
    ByVai e As System.EventArgs) Handles MyBase.Load

    Try
```

```
      ' Connect to a running instance of Solid Edge
      m_application = Marshal.GetActiveObject("SolidEdge.Application")
    Catch ex As Exception
      MessageBox.Show(ex.Message, "Error")
    End Try
  End Sub

  Private Sub Buttonl_Click(ByVal sender As System.Object, _
    ByVai e As System.EventArgs) Handles Buttonl.Click
    Dim objDocument As SolidEdgeFramework.SolidEdgeDocument = Nothing
    Dim objUOM As SolidEdgeFramework.UnitsOfMeasure = Nothing
    Dim dHexSize As Double

    Try
      ' Get a reference to the active document
      objDocument = m_application.ActiveDocument

      ' Get a reference to the active document's unit of measure
      objUOM = objDocument.UnitsOfMeasure

      ' Attempt to parse the UOMinput by user
      dHexSize = objUOM.ParseUnit( _
        UnitTypeConstants.igUnitDistance, TextBoxl.Text)

      ' Update the 2nd textbox with the parsed UOM
      TextBox2.Text = dHexSize.ToString()
    Catch ex As Exception
      MessageBox.Show(ex.Message, "Invalid unit")
    Finally
      If Not (objUOM Is Nothing) Then
        Marshal.ReleaseComObject(objUOM)
        objUOM = Nothing
      End If
      If Not (objDocument Is Nothing) Then
        Marshal.ReleaseComObject(objDocument)
        objDocument = Nothing
      End If
    End Try
  End Sub

  Private Sub Forml_FormClosing(ByVal sender As System.Object, _
    ByVai e As System.Windows.Forms.FormClosingEventArgs) _
    Handles MyBase.FormClosing

    Try
      If Not (m_application Is Nothing) Then
        Marshal.ReleaseComObject(m_application)
        m_application = Nothing
      End If
    Catch ex As Exception
      MessageBox.Show(ex.Message, "Error")
    End Try
  End Sub
End Class
```

Formatting and Displaying Units in C# Cc

```
using SolidEdgeFramework;
using System;
```

```
using System.Runtime.InteropServices;
using System.Windows.Forms;

namespace Walkthrough
{
    public partial class Form1 : Form
    {
        private SolidEdgeFramework.Application m_application;

        public Form1()
        {
            InitializeComponent();
        }

        private void Form1_Load(object sender, EventArgs e)
        {
            try
            {
                // Connect to a running instance of Solid Edge
                m_application = (SolidEdgeFramework.Application)
                  Marshal.GetActiveObject("Solid Edge.Application");
            }
            catch (System.Exception ex)
            {
                MessageBox.Show(ex.Message, "Error");
            }
        }

        private void Button1_Click(object sender, EventArgs e)
        {
            SolidEdgeFramework.SolidEdgeDocument document = null;
            SolidEdgeFramework.UnitsOfMeasure uom = null;
            double dHexSize = 0;
            try
            {
                // Get a reference to the active document
                document = (SolidEdgeFramework.SolidEdgeDocument)
                  m_application.ActiveDocument;

                // Get a reference to the active document's unit of measure
                uom = document.UnitsOfMeasure;

                // Attempt to parse the UOM input by user
                dHexSize = (double)uom.ParseUnit(
                  (int)UnitTypeConstants.igUnitDistance, TextBox1.Text);

                // Update the 2nd textbox with the parsed UOM
                TextBox2.Text = dHexSize.ToString();
            }
            catch (System.Exception ex)
            {
                MessageBox.Show(ex.Message, "Invalid unit");
            }
            finally
            {
                if (uom != null)
                {
                    Marshal.ReleaseComObject(uom);
```

```
                    uom = null;
                }
                if (document != null)
                {
                    Marshal.ReleaseComObject(document);
                    document = null;
                }
            }
        }

        private void Forml_FormClosing(object sender, FormClosingEventArgs e)
        {
            try
            {
                if (m_application != null)
                {
                    Marshal.ReleaseComObject(m_application);
                    m_application = null;
                }
            }
            catch (System.Exception ex)
            {
                MessageBox.Show(ex.Message, "Error");
            }
        }
    }
}
```

See Also

UnitsOfMeasure Collection
UnitOfMeasure Object

Solid Edge Framework Type Library

View Object Members

Overview Collapse All

Solid Edge Framework Type Library : View Object

Public Methods

ApplyNamedView	Apply a named view.
AreaZoomCamera	Zooms the camera view to the specified window area.
BeginCameraDynamics	Starts the dynamic effects of the camera view.
ClearRotationFocus	Clearview rotation focus parameters.
CreateIlserRange	Creates a user range identifier.
DeleteIlserRange	Deletes a user range by identifier.
EndCameraDynamics	Ends the dynamic effects of the camera view.
Fit	Changes the scale of the referenced object to fit within a frame.
GetAttribute	Gets in the referenced View object the named attribute value.
GetCamera	Returns the settings of the camera.
GetCameraEx	Returns the settings of the camera.
GetCounter	Returns a counter value chosen by Type from the current view. If bReset is false, the count is cumulative.
GetDisplayDepths	Gets in the referenced View object the definitions of the active display depths.
GetModelRange	Return the model range.
GetRotationFocus	This method returns all the current rotation focus settings including the options flags and can be used to detect manual rotation focus parameters.
GetSectionPlanes	Gets in the referenced View object the definitions of the active section planes.
GetUserRange	Gets a user range limits by identifier.
Locate	Return the closest point at the specified screen location.
ModelToDC	Transforms model coordinates to Device Context coordinates.
ModelToView	Transforms model coordinates to View coordinates.
OrientCamera	Orients the camera. Converts three mouse positions on the screen into one of three motions: rotate when cmdtype = 1, pan when cmdtype - 2, or zoom when cmdtype = 3.
OrientCameraEx	Orients the camera. Converts three mouse positions on the screen into one of three motions: rotate when cmdtype = 1, pan when cmdtype = 2, or zoom when cmdtype = 3.
PanCamera	Pans the camera view (dx, dy) pixels across the screen.
RotateCamera	Rotates the camera Angle degrees about an axis centered at (CenterX, CenterY, CenterZ), with the axis pointing in the direction (AxisX, AxisY, AxisZ).
SaveAsImage	Saves the current view as an image.
SaveCurrentView	Save current view to named view.

SetAttribute	Sets in the referenced View object the named attribute value.
SetCamera	Specifies the settings of the camera.
SetCameraEx	Specifies the settings of the camera.
SetDisplayDepths	Sets in the referenced View object the definitions of the active display depths.
SetRenderMode	Sets the render mode used in this view.
SetRotationAxis	Set the rotation focus about an explicit axis where X, Y, and Z define a point in top level model coordinates and NX, NY, and NZ define the direction of the axis in the top level model.
SetRotationFocus	Set view rotation focus parameters.
SetRotationPoint	Set the rotation focus to be free rotation about an explicit point where X, Y, and Z define the point in top level model coordinates.
SetSectionPlanes	Sets in the referenced View object the definitions of the active section planes [0 will remove all planes].
SetUserRange	Sets a user range limits by identifier.
ShowDrawDC	Displays the Device Context into which the drawing is done.
SwapBuffers	Swap drawing buffers in this view.
TransformDCToModel	Converts device context coordinates to model coordinates.
TransformGLProjectionToView	Converts OpenGL Projection coordinates to View coordinates.
TransformModelToDC	Converts Model coordinates to device context coordinates.
TransformModelToView	Converts Model coordinates to View coordinates.
TransformViewToGLProjection	Converts View coordinates to OpenGL projection coordinates.
TransformViewToModel	Converts View coordinates to Model coordinates.
Update	Updates the referenced object.
ViewToGLProjection	Transforms View coordinates to OpenGL projection coordinates.
ZoomCamera	Zooms the camera view by a scale factor.

Public Properties

Hsl CullingEnabled	Specifies whether or not culling will be used in the display.
DisplayEnabled	Specifies whether or not a referenced view is displayed.
DisplayEvents	Returns the DisplayEvents object.
DrawDC	Returns the Device Context into which the drawing is done.
GDIBufferModified	Specified whether or not the GDI buffer has been modified.
GLDisplayEvents	Returns the GLDisplayEvents object.
RenderEvents	Returns the RenderEvents object.
RenderModeType	Sets and returns the render mode used in this view.
SectionPlanesEnabled	Sets and returns the flag in the referenced View object that determines whether section planes will be active.

SharpnessLevel	Sets or returns the sharpness level of the display.
SharpnessLevelCount	Returns the maximum sharpness level of the display.
0 SilhouettesEnabled	Sets and returns the flag in the referenced View object that determines whether silhouettes will be displayed in Smooth with VHL overlay render mode.
S StereoAngle	Sets or returns the half-angle (specified in degrees) between the two displays of a stereoscopic display.
StereoDeviation	Sets or returns the planar shift away from the center of the view plane necessary to distinctly focus the two displays of a stereoscopic display.
StereoEnabled	Enables and disables the display state of the referenced View.
Style	For collection objects, Body objects, RefPlane objects, Sketch objects, CoordinateSystem objects, CommandBarButton objects, SmartFrame2d objects, View objects, and TextCharStyle objects, Style is read-write. For all other objects, Style is read-only.
StyleFallbackEnabled	Specifies whether or not style fallback will be used in the display.
0 ViewEvents	Returns the ViewEvents object.
ViewStyle	Sets or returns the view style used in this view.
Window	Returns the Window object.

See Also

View Object

Solid Edge Framework Type Library

view object

Members Collapse All

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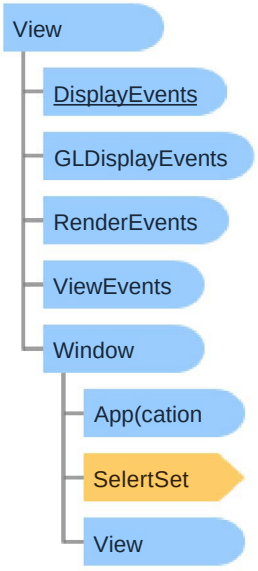
Description

Provides methods and properties that allow you to manipulate Solid Edge window views.

Object Model



See Also

View Members

Solid Edge Constants Type Library

ViewAttributeConstants Enumeration

Collapse All

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Description

View Attributes constants

Members

Member	Value	Description
seApplicationDisplay	5	Application display setting [1=Backing store, 2=Graphics card driven, 3=Software driven]
seAutoSharpenLevel	7	Automatic sharpen level
seDefaultSharpness	6	Default sharpness setting for new views
seDisplaySectionCaps	2	Display section caps
seDisplayStatistics	1	Display view statistics
seDynamicTransition	4	Enable dynamic view transitions
seResetAllDefaults	65535	Reset all view attributes to default values
seSoftwareVHL	3	Enable software VHL

See Also

Solid Edge ST4 - What's New

Solid Edge Draft Type Library

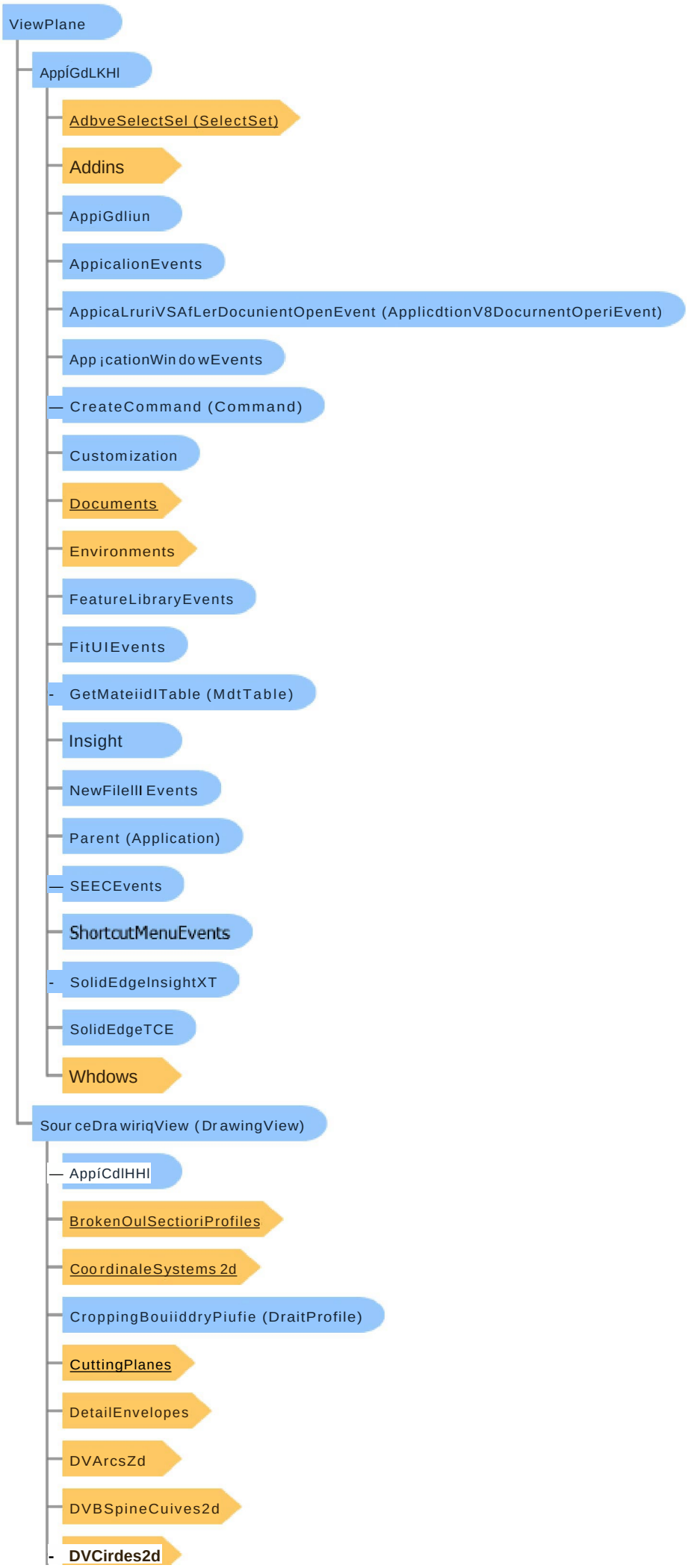
ViewPlane Object

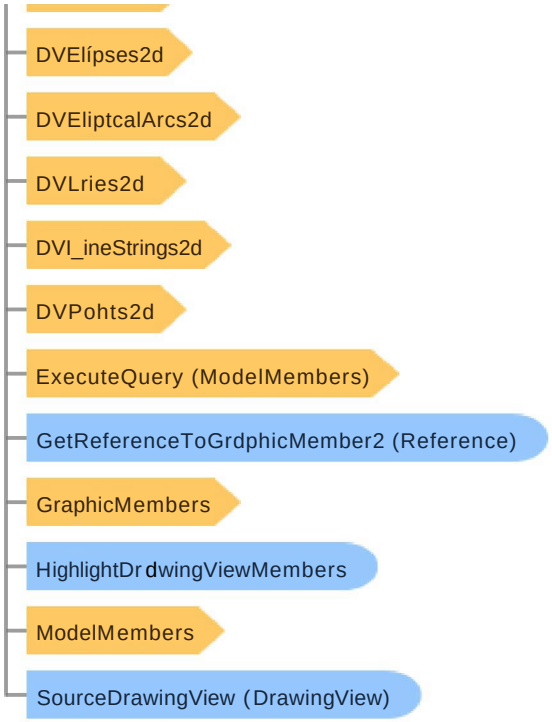
Members Collapse All

Description

Provides methods and properties that extend to an auxiliary DrawingView object.

Object Model


ViewPlane
AppíGdLKHl
AdbveSelectSel (SelectSet)
Addins
AppiGdliun
AppicalionEvents
AppicaLruriVSAfLerDocunientOpenEvent (ApplicdtionV8DocurnentOperiEvent)
App¡cationWindowEvents
CreateCommand (Command)
Customization
Documents
Environments
FeatureLibraryEvents
FitUIEvents
GetMateiidITable (MdtTable)
Insight
NewFileIlI Events
Parent (Application)
SEECEvents
ShortcutMenuEvents
SolidEdgeInsightXT
SolidEdgeTCE
Whdows
SourceDrawiriqView (DrawingView)
AppíCdlHHl
BrokenOulSectioriProfiles
CoordinaleSystems 2d
CroppingBouiiddryPiufie (DraitProfile)
CuttingPlanes
DetailEnvelopes
DVArcsZd
DVBSpineCuives2d
DVCirdes2d



Remarks

These methods and properties provide information and controls for the display and characteristics (such as text color) of the view plane's caption, the view plane line type and line width, the view plane terminator ratio and type, and so forth. The ViewPlane methods and properties extend to a DrawingView object whose type is igAuxiliaryView, through the DrawingView.Annotation property. An auxiliary DrawingView object is created with the AddByAuxiliaryFold method of the Drawingviews collection object.

See Also

ViewPlane Members

Solid Edge Draft Type Library

Views Property

Collapse All

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Description

Block views collection

Property type

Read-only property

Syntax

Visual Basic
Public Property Views As Blockviews

See Also

Block Object | Block Members (Solid Edge V19 - What's New

Solid Edge ST7 SDK

Working with Dimensions

Working with Dimensions

This topic contains these sections:

Solid Edge allows you to place and edit dimensions on elements. In the Profile environment, dimension objects control the size and orientation of geometry.

Dimensions can be linear, radial, or angular. Dimensions supply information about the measurements of elements, such as the angle of a line or the distance between two points. A dimension is related to the element on which it is placed. In the Profile environment, the dimensions control the geometry; if the dimension changes, the geometry updates.

In a Part document, the Dimensions collection is accessed through the Profile object. The Dimensions collection provides the methods for placing dimensions and for iterating through all the dimensions on the entire sheet or profile.

Linear Dimensions
A linear dimension measures the distance between two or more elements, the length of a line, or an arc's length. For a complete description of the properties that define how a linear dimension is placed, see the Programming with Solid Edge on-line Help.

Radial Dimensions
Radial dimensions measure the radius or diameter at a point on the element. These dimensions are similar except that they show the radius or diameter value depending on the type. With the ProjectionArc and TrackAngle properties, you can define the measurement point on the element.

Angular Dimensions
Angular dimensions measure the angle between two lines or three points. An angular dimension defines two intersecting vectors and four minor sectors. These sectors are distinguished according to whether the angle is measured in the sector where the vector direction goes outward from the intersection point or comes inward, and whether the angle is measured in the clockwise or counterclockwise direction.

The angles are always measured in the counterclockwise direction with both vector directions going outward from the intersection point (sector one condition). To measure in any other angle, certain properties are set so that the dimension object modifies the vector direction and computes the angle.

Placing Dimensions

In the Profile environment, driving dimensions control the elements to which they refer. When you edit a driving dimension, the geometry of the element that is related to that dimension is modified.

You can place dimensions only on existing elements. A set of Add methods is provided on the Dimensions collection, one for each type of dimension. The element to which the dimension is attached determines the type of dimension (driving or driven) that will be placed. The Add methods on the Dimensions collection object take minimal input and place the dimensions with specific default values.

When you place dimensions between two elements interactively, the dimensions are measured at a specific location on an element For example, when you place a dimension between the end points of two lines, you select one end of each line. When you place dimensions through automation, you specify a point on the element and a key point flag to define the dimension.

In the following program, four lines are drawn and connected with key point relationships. The lengths of two of the lines and the distance between two lines are dimensioned. The dimension is set to be a driving dimension so it will control the length and position of the geometry. The sample also shows how to modify a dimension style by changing the units of measurement of one of the dimensions to meters.

| A See Handling 'Application is Busy' and 'Call was Rejected By Callee' errors for information regarding the use of OleMessageFilter.

Placing Dimensions in Visual Basic.NET

```
Imports SolidEdgeConstants
Imports System.Runtime.InteropServices

Module Program
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objDocuments As SolidEdgeFramework.Documents = Nothing
        Dim objPart As SolidEdgePart.PartDocument = Nothing
        Dim objProfileSets As SolidEdgePart.ProfileSets = Nothing
        Dim objProfileSet As SolidEdgePart.ProfileSet = Nothing
        Dim objProfiles As SolidEdgePart.Profiles = Nothing
        Dim objProfile As SolidEdgePart.Profile = Nothing
        Dim objRefPlanes As SolidEdgePart.RefPlanes = Nothing
        Dim objRefPlane As SolidEdgePart.RefPlane = Nothing
        Dim objLines2d As SolidEdgeFrameworkSupport.Lines2d = Nothing
        Dim aLine2d(0 To 3) As SolidEdgeFrameworkSupport.Line2d
```

```
Dim objRelations2d As SolidEdgeFrameworkSupport.Relations2d = Nothing
Dim objDimensions As SolidEdgeFrameworkSupport.Dimensions = Nothing
Dim objDimension As SolidEdgeFrameworkSupport.Dimension = Nothing
Dim objDimStyle As SolidEdgeFrameworkSupport.DimStyle = Nothing
Dim i As Integer

Try
    OleMessageFilter.Register()

    ' Connect to a running instance of Solid Edge
    objApplication = Marshal.GetActiveObject("Solid Edge.Application")

    ' Get a reference to the documents collection
    objDocuments = objApplication.Documents

    ' Add a Part document
    objPart = objDocuments.Add("SolidEdge.PartDocument")

    ' Get a reference to the ref planes collection
    objRefPlanes = objPart.RefPlanes

    ' Get a reference to the first reference plane
    objRefPlane = objRefPlanes.Item(1)

    ' Get a reference to the profile sets collection
    objProfileSets = objPart.ProfileSets

    ' Add a new profile set
    objProfileSet = objProfileSets.Add()

    ' Get a reference to the profiles collection
    objProfiles = objProfileSet.Profiles

    ' Add a new profile on the selected reference plane
    objProfile = objProfiles.Add(objRefPlane)

    ' Get a reference to the lines 2d collection
    objLines2d = objProfile.Lines2d

    ' Get a reference to the relations 2d collection
    objRelations2d = objProfile.Relations2d

    ' Get a reference to the dimensions collection
    objDimensions = objProfile.Dimensions

    ' Draw the geometry.
    aLine2d(0) = objLines2d.AddBy2Points(0, 0? 0.1, 0)
    aLine2d(1) = objLines2d.AddBy2Points(0.1, 0, 0.1, 0.1)
    aLine2d(2) = objLines2d.AddBy2Points(0.1, 0.1, 0, 0.05)
    aLine2d(3) = objLines2d.AddBy2Points(0, 0.05, 0, 0)

    ' Add endpoint relationships between the lines
    objRelations2d.AddKeypoint( _
      aLine2d(0), _
      KeypointIndexConstants.igLineEnd, _
      aLine2d(1), _
      KeypointIndexconstants.igLineStart)

    objRelations2d.AddKeypoint( _
      aLine2d(1), _
      KeypointIndexConstants.igLineEnd, _
      aLine2d(2), _
      KeypointIndexConstants.igLineStart)

    objRelations2d.AddKeypoint( _
      aLine2d(2), _
```

```
                KeypointIndexConstants.igLineEnd, _
                aLine2d(3), _
                KeypointIndexconstants.igLineStart)

            objRelations2d.AddKeypoint( _
                            aLine2d(3), _
                KeypointIndexConstants.igLineEnd, _
                aLine2d(0), _
                KeypointIndexconstants.igLineStart)

            ' Add dimensions, and change the dimension units to meters
            objDimension = objDimensions.AddLength(aLine2d(l))

            objDimension.Constraint = True
            objDimStyle = objDimension.Style
            objDimStyle.Primaryunits = _
                DimLinearUnitConstants.igDimStyleLinearMeters

            objDimension = objDimensions.AddLength(aLine2d(3))

            objDimension.Constraint = True
            objDimStyle = objDimension.Style
            objDimStyle.Primaryunits = _
                DimLinearUnitConstants.igDimStyleLinearMeters

            objDimension = objDimensions.AddDistanceBetweenObjects( _
                aLine2d(l), 0.1, 0.1, 0, False, _
                aLine2d(2), 0, 0.05, 0, False)

            objDimension.Constraint = True
            objDimStyle = objDimension.Style
            objDimStyle.Primaryunits = _
                DimLinearUnitConstants.igDimStyleLinearMeters

        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter.Revoke()
        End Try
    End Sub
End Module
```

Placing Dimensions in C#

```
using SolidEdgeConstants;
using System;
using System.Reflection;
using System.Runtime.InteropServices;

namespace SolidEdge.SDK
{
    class Program
    {
        [STAThread]
        static void Main(string[] args)
        {
            SolidEdgeFramework.Application application = null;
            SolidEdgeFramework.Documents documents = null;
            SolidEdgePart.PartDocument part = null;
            SolidEdgePart.ProfileSets profileSets = null;
            SolidEdgePart.ProfileSet profileSet = null;
            SolidEdgePart.Profiles profiles = null;
            SolidEdgePart.Profile profile = null;
            SolidEdgePart.RefPlanes refPlanes = null;
            SolidEdgePart.RefPlane refPlane = null;
            SolidEdgeFrameworkSupport.Lines2d lines2d = null;
            SolidEdgeFrameworkSupport.Line2d[] aLine2d =
```

```
    new SolidEdgeFrameworkSupport.Line2d[4];
SolidEdgeFrameworkSupport.Relations2d relations2d = null;
SolidEdgeFrameworkSupport.Dimensions dimensions = null;
SolidEdgeFrameworkSupport.Dimension dimension = null;
SolidEdgeFrameworkSupport.DimStyle dimStyle = null;

try
{
    OleMessageFilter.Register();

    // Connect to a running instance of Solid Edge
    application = (SolidEdgeFramework.Application)
      Marshal.GetActiveObject("Solid Edge.Application");

    // Get a reference to the documents collection
    documents = application.Documents;

    // Add a Part document
    part = (SolidEdgePart.PartDocument)
      documents.Add("SolidEdge.PartDocument", Missing.Value);

    // Get a reference to the ref planes collection
    refPlanes = part.RefPlanes;

    // Get a reference to the first reference plane
    refPlane = refPlanes.Item(l);

    // Get a reference to the profile sets collection
    profilesets = part.Profilesets;

    // Add a new profile set
    profileSet = profileSets.Add();

    // Get a reference to the profiles collection
    profiles = profileSet.Profiles;

    // Add a new profile on the selected reference plane
    profile = profiles.Add(refPlane);

    // Get a reference to the lines 2d collection
    lines2d = profile.Lines2d;

    // Get a reference to the relations 2d collection
    relations2d = (SolidEdgeFrameworkSupport.Relations2d)
      profile. Relations2d;

    // Get a reference to the dimensions collection
    dimensions = (SolidEdgeFrameworkSupport.Dimensions)
      profile .Dimensions;

    // Draw the geometry.
    aLine2d[0] = lines2d.AddBy2Points(0j 0, 0.1, 0);
    aLine2d[l] = lines2d.AddBy2Points(0.1, 0, 0.1, 0.1);
    aLine2d[2] = lines2d.AddBy2Points(0.1, 0.1, 0, 0.05);
    aLine2d[3] = lines2d.AddBy2Points(0, 0.05., 0, 0);

    // Add endpoint relationships between the lines
    relations2d.AddKeypoint(
      aLine2d[0],
      (int)KeypointIndexConstants.igLineEnd,
      aLine2d[l],
      (int)KeypointIndexConstants.igLineStart,
      Missing.Value);

    relations2d.AddKeypoint(
      aLine2d[l],
```

```
                    (int)KeypointIndexConstants.igLineEnd,
                    aLine2d[2],
                    (int) KeypointIndexconstants.igLineStart,
                    Missing.Value);

                relations2d.AddKeypoint(
                    aLine2d[2],
                    (int) KeypointIndexconstants.igLineEnd,
                    aLine2d[3],
                    (int) KeypointIndexConstants.igLineStart,
                    Missing.Value);

                rel ations 2d.AddKeypoint (
                    aLine2d[3],
                    (int) Keypoint IndexConstants .igLineEnd,
                    aLine2d[0] ,
                    (int) KeypointIndexconstants.igLineStart.,
                    Missing.Value);

                // Add dimensions, and change the dimension units to meters
                dimension = dimensions.AddLength(aLine2d[l]);
                dimension.Constraint = true;
                dimStyle = dimension.Style;
                dimStyle.PrimaryUnits = SolidEdgeFrameworkSupport.DimLinearUnitConstants.igDimStyleLinearMeters;

                dimension = dimensions.AddLength(aLine2d[3]);
                dimension.Constraint = true;
                dimStyle = dimension.Style;
                dimStyle.PrimaryUnits = SolidEdgeFrameworkSupport.DimLinearUnitConstants.igDimStyleLinearMeters;

                dimension = dimensions.AddDistanceBetweenObjects(
                    aLine2d[l], 0.1, 0.1, 0, false, aLine2d[2], 0, 0.05, 0, false);
                dimension.Constraint = true;
                dimStyle = dimension.Style;
                dimStyle.PrimaryUnits = SolidEdgeFrameworkSupport.DimLinearUnitConstants.igDimStyleLinearMeters;

            }
            catch (System.Exception ex)
            {
                Console.WriteLine (ex. Message);
            }
            finally
            {
                OleMessageFilter.Revoke();
            }
        }
    }
}
```

Dimensions via Variables

When working interactively with the Variable Table, both variables and dimensions are displayed in the table. This enables you to create formulas using the values of dimensions and also have formulas that drive the values of dimensions. In this workflow, there is no apparent difference between variables and dimensions. Internally, however, variables and dimensions are two distinct types of objects that have their own unique collections, properties, and methods.

The Variables collection allows you to work with dimensions in the context of variables through several methods on the collection. These methods include Edit, GetFormula, GetName, PutName, Query, and Translate. The following program uses dimensions through the Variables collection. The following programs assume that Solid Edge is running and in the Profile environment.

| A See Handling 'Application is Busy' and 'Call was Rejected By Callee' errors for information regarding the use of OleMessageFilter.

AccessingDimensionsthroughtheVariableTableinVisualBasic.NET

```
Imponts System.Runtime.InteropSenvices

Module Pnognam
    <STAThnead()> _
```

```
Sub Main()
    Dim objApplication As SolidEdgeFramework.Application = Nothing
    Dim objPart As SolidEdgePart.PartDocument = Nothing
    Dim objProfileSets As SolidEdgePart.ProfileSets = Nothing
    Dim objProfileSet As SolidEdgePart.ProfileSet = Nothing
    Dim objProflies As SolidEdgePart.Profiles = Nothing
    Dim objProfile As SolidEdgePart.Profile = Nothing
    Dim objVariables As SolidEdgeFramework.Variables = Nothing
    Dim objVariable As Object = Nothing
    Dim objl_ines2d As SolidEdgeFrameworkSupport.Lines2d = Nothing
    Dim objLine2d As SolidEdgeFrameworkSupport.Line2d = Nothing
    Dim objDimensions As SolidEdgeFrameworkSupport.Dimensions = Nothing
    Dim objDimensionl As SolidEdgeFrameworkSupport.Dimension = Nothing
    Dim objDimension2 As SolidEdgeFrameworkSupport.Dimension = Nothing
    Dim objVariableList As SolidEdgeFramework.VariableList = Nothing
    Dim sName As String
    Dim sFormula As String

    Try
        OleMessageFilter.Register()

        ' Connect to a running instance of Solid Edge
        objApplication = Marshal.GetActiveObject("SolidEdge.Application")

        ' Get a reference to the active document
        objPart = objApplication.ActiveDocument

        ' Get a reference to the active document
        objProfileSets = objPart.ProfileSets

        ' Get a reference to the profile set
        objProfileSet = objProfileSets.Item(objProfileSets.Count)

        ' Get a reference to the profiles collection
        objProfiles = objProfileSet.Profiles

        ' Get a reference to the profile
        objProfile = objProfiles.Item(l)

        ' Get a reference to the variables collection
        objVariables = objPart.Variables

        ' Get a reference to the lines2d collection
        objLines2d = objProfile.Lines2d

        ' Get a reference to the dimensions collection
        objDimensions = objProfile.Dimensions

        ' Create a new line
        objLine2d = objLines2d.AddBy2Points(0, 0, 0.1, 0.1)

        ' Place a dimension on the line to control its length
        objDimensionl = objDimensions.AddLength(objLine2d)

        ' Make the dimension a driving dimension
        objDimensionl.Constraint = True

        ' Create a second line
        objLine2d = objLines2d.AddBy2Points(0, 0.1, 0.1, 0.2)

        ' Place a dimension on the line to control its length
        objDimension2 = objDimensions.AddLength(objLine2d)

        ' Make the dimension a driving dimension
        objDimension2.Constraint = True
```

```
            ' Assign a name to the dimension placed on the first line
            objVariables.PutName(objDimension1, "Dimension1")

            ' Retrieve the system name of the second dimension, and display
            ' it in the debug window


            sName = objVariables.GetName(objDimension2)
            Console.WriteLine(String.Format("Dimension = {0}", sName))

            ' Edit the formula of the second dimension to be half
            ' the value of the first dimension
            objVariables.Edit(sName, "Dimension1/2.0")

            ' Retrieve the formula from the dimension, and print it to the
            ' debug window to verify
            sFormula = objVariables.GetFormula(sName)
            Console.WriteLine(String.Format("Formula = {0}", sFormula))

            ' This demonstrates the ability to reference a dimension object
            ' by its name
            objDimension1 = objVariables.Translate("Dimension1")

            ' To verify a dimension object was returned, change its
            ' TrackDistance property to cause the dimension to change
            objDimension1.TrackDistance = objDimension1.TrackDistance * 2

            ' Use the Query method to list all all user-defined
            ' variables and user-named Dimension objects and
            ' display in the debug window
            objVariableList = objVariables.Query("*")
            For Each objVariable In objVariableList
                Console.WriteLine(objVariables.GetName(objVariable))
            Next

        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter.Revoke()
        End Try
    End Sub
End Module
```

Accessing Dimensions through the Variable Table in C#

```
using SolidEdgeFramework;
using System;
using System.Runtime.InteropServices;

namespace SolidEdge.SDK
{
    class Program
    {
        [STAThread]
        static void Main(string[] args)
        {
            SolidEdgeFramework.Application application = null;
            SolidEdgePart.PartDocument part = null;
            SolidEdgePart.ProfileSets profileSets = null;
            SolidEdgePart.ProfileSet profileSet = null;
            SolidEdgePart.Profiles profiles = null;
            SolidEdgePart.Profile profile = null;
            SolidEdgeFramework.Variables variables = null;
            Object oVariable = null;
            SolidEdgeFrameworkSupport.Lines2d lines2d = null;
            SolidEdgeFrameworkSupport.Line2d line2d = null;
            SolidEdgeFrameworkSupport.Dimensions dimensions = null;
```

```
SolidEdgeFrameworkSupport.Dimension dimension1 = null;
SolidEdgeFrameworkSupport.Dimension dimension2 = null;
SolidEdgeFramework.VariableList variableList = null;
String sName;
String sFormula;

try
{
    OleMessageFilter.Register();

    // Connect to a running instance of Solid Edge
    application = (SolidEdgeFramework.Application)
      Marshal.GetActiveObject("Solid Edge.Application");

    // Get a reference to the active document
    part = (SolidEdgePart.PartDocument)
      application.ActiveDocument;


    // Get a reference to the profile sets collection
    profileSets = part.ProfileSets;

    // Get a reference to the profile set
    profileSet = profileSets.Item(profileSets.Count);

    // Get a reference to the profiles collection
    profiles = profileSet.Profiles;

    // Get a reference to the profile
    profile = profiles.Item(1);

    // Get a reference to the variables collection
    variables = (SolidEdgeFramework.Variables)part.Variables;

    // Get a reference to the lines2d collection
    lines2d = profile.Lines2d;

    // Get a reference to the dimensions collection
    dimensions = (SolidEdgeFrameworkSupport.Dimensions)
      profile.Dimensions;

    // Create a line
    line2d = lines2d.AddBy2Points(0, 0, 0.1, 0.1);

    // Place a dimension on the line to control its length
    dimension1 = dimensions.AddLength(line2d);

    // Make the dimension a driving dimension
    dimension1.Constraint = true;

    // Create a second line
    line2d = lines2d.AddBy2Points(0, 0.1, 0.1, 0.2);

    // Place a dimension on the line to control its length
    dimension2 = dimensions.AddLength(line2d);

    // Make the dimension a driving dimension
    dimension2.Constraint = true;

    // Assign a name to the dimension placed on the first line
    variables.PutName(dimension1, "Dimension1");

    // Retrieve the system name of the second dimension, and display
    // it in the debug window
    sName = variables.GetName(dimension2);
    Console.WriteLine(String.Format("Dimension = {0}", sName));
```

```
                // Edit the formula of the second dimension to be half
                // the value of the first dimension
                variables.Edit(sName, "Dimensionl/2.0");

                // Retrieve the formula from the dimension, and print it to the
                // debug window to verify
                sFormula = variables.GetFormula(sName);
                Console.WriteLine(String.Format("Formula = {0}", sFormula));

                // This demonstrates the ability to reference a dimension object
                //by its name
                dimensionl = (SolidEdgeFrameworkSupport.Dimension)
                  variables.Translate("Dimensionl");

                //To verify a dimension object was returned, change its
                // TrackDistance property to cause the dimension to change
                dimensionl.TrackDistance = dimensionl.TrackDistance * 2;

                // Use the Query method to list all all user-defined
                // variables and user-named Dimension objects and
                // display in the debug window
                variableList = (SolidEdgeFramework.VariableList)
                  variables.Query("*", null, null, false);

                for (int i = 1; i <= variableList.Count; i++)
                {
                    oVariable = variableList.Item(i);
                    Console.WriteLine(variables.GetName(oVariable));
                }
            }
            catch (System.Exception ex)
            {
                Console.WriteLine(ex.Message);
            }
            finally
            {
                OleMessageFilter.Revoke();
            }
        }
    }
}
```

Solid Edge ST7 SDK

Working with DrawingViews

Solid Edge. Design better.

A DrawingView object is a 2-D representation of a 3-D part or assembly model. A drawing view is used to display design space geometry in document space. A view of design space is enclosed by the drawing view border (a handle that allows manipulation of the drawing view). Only one part or assembly document can be used as the basis for drawing views in a draft document.

> /■\ See Handling 'Application is Busy' and 'Call was Rejected By Callee' errors for information regarding the use of OleMessageFilter.

DrawingViewsinVisualBasic.NET ~ /Code

```
Imports SolidEdgeFramework
Imports System.Runtime.InteropServices

Module Program
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objDocuments As SolidEdgeFramework.Documents = Nothing
        Dim objDraft As SolidEdgeDraft.DraftDocument = Nothing
        Dim objSheet As SolidEdgeDraft.Sheet = Nothing
        Dim objModelLinks As SolidEdgeDraft.ModelLinks = Nothing
        Dim objModelLink As SolidEdgeDraft.ModelLink = Nothing
        Dim objDrawingViews As SolidEdgeDraft.DrawingViews = Nothing
        Dim objDrawingView As SolidEdgeDraft.DrawingView = Nothing

        Try
            OleMessageFilter.Register()

            ' Connect to a running instance of Solid Edge
            objApplication = Marshal.GetActiveObject("SolidEdge.Application")

            ' Get a reference to the documents collection
            objDocuments = objApplication.Documents

            ' Add a Draft document
            objDraft = objDocuments.Add("SolidEdge.DraftDocument")

            ' Get a reference to the active sheet
            objSheet = objDraft.ActiveSheet

            ' Get a reference to the model links collection
            objModelLinks = objDraft.ModelLinks

            ' Add a new model link
            objModelLink = objModelLinks.Add("C:\Part1.par")

            ' Get a reference to the drawing views collection
            objDrawingViews = objSheet.DrawingViews

            ' Add a new drawing view
            objDrawingView = objDrawingViews.AddPartView( _
              objModelLinkj _
```

```
                SolidEdgeDraft.ViewOrientationConstants.igFrontView, _
                1, _
                0.1, _
                0.1, _
                SolidEdgeDraft.PartDrawingViewTypeConstants.sePartDesignedView)

            ' Assign a caption
            objDrawingView.Caption = "My New Drawing View"

            ' Ensure caption is displayed
            objDrawingView.DisplayCaption = True

        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter.Revoke()
        End Try
    End Sub
End Module
```

DrawingViews in C# Code

```
using SolidEdgeFramework;
using System;
using System.Reflection;
using System.Runtime.InteropServices;

namespace SolidEdge.SDK
{
    class Program
    {
        [STAThread]
        static void Main(string[] args)
        {
            SolidEdgeFramework.Application application = null;
            SolidEdgeFramework.Documents documents = null;
            SolidEdgeDraft.DraftDocument draft = null;
            SolidEdgeDraft.Sheet sheet = null;
            SolidEdgeDraft.ModelLinks modelLinks = null;
            SolidEdgeDraft.ModelLink modelLink = null;
            SolidEdgeDraft.DrawingViews drawingviews = null;
            SolidEdgeDraft.Drawingview drawingview = null;

            try
            {
                OleMessageFilter.Register();

                // Connect to a running instance of Solid Edge
                application = (SolidEdgeFramework.Application)
                    Marshal.GetActiveObject("SolidEdge.Application");

                // Get a reference to the documents collection
                documents = application.Documents;

                // Add a Draft document
                draft = (SolidEdgeDraft.DraftDocument)
                    documents.Add("SolidEdge.DraftDocument", Missing.Value);
```

```
                // Get a reference to the active sheet
                sheet = draft.ActiveSheet;

                // Get a reference to the model links collection
                modelLinks = draft.ModelLinks;

                // Add a new model link
                modelLink = modelLinks.Add(@"C:\Part1.par");

                // Get a reference to the drawing views collection
                drawingviews = sheet.DrawingViews;

                // Add a new drawing view
                drawingview = drawingviews.AddPartView(
                    modelLink,
                    SolidEdgeDraft.ViewOrientationConstants.igFrontView,
                    1,
                    0.1,
                    0.1,
                    SolidEdgeDraft.PartDrawingViewTypeConstants.sePartDesignedView);

                // Assign a caption
                drawingView.Caption = "My New Drawing View";

                // Ensure caption is displayed
                drawingView.DisplayCaption = true;

            }
            catch (System.Exception ex)
            {
                Console.WriteLine(ex.Message);
            }
            finally
            {
                OleMessageFilter.Revoke();
            }
        }
    }
}
```

Solid Edge ST7 SDK

Working with References

Colla pseAH

Solid Edge. Design better.

When you work with an assembly document interactively, you can work directly with occurrences and part geometry in subassemblies. For example, you can place relationships between occurrences nested in subassemblies, you can measure distances between faces of occurrences in subassemblies, you can in-place-activate an occurrence within a sub-assembly, and you can apply face styles to occurrences within subassemblies.

This topic contains these sections:

Because you can use the Occurrences collection to access occurrences nested in subassemblies, and because you can access the OccurrenceDocument representing a PartDocument and access geometry within the part, it appears simple to use the automation interface to work with occurrences in subassemblies just as you would through the graphical user interface. However, this appearance is deceptive. When you work with occurrences in subassemblies, and when you work with geometry of parts in occurrences (however deeply nested), use the Reference object to create references to part geometry and to nested occurrences from the top-level assembly. Then use the Reference object to place relationships, measure distances, in-placeactivate nested occurrences, apply face styles, and so forth.

You can create Reference objects with the AssemblyDocument.CreateReference method. This method has two input parameters: an occurrence (which must be an Occurrence object), and an entity, which can be one of several different types of objects.

Analyzing Existing Assembly Relationships

When interactively placing parts in an assembly, you define relationships between parts to control their relative positions. Using the automation interface for the Assembly environment, you can access and modify properties of the assembly relationships.

Relationship objects are accessible through two collections: Relations3d on the AssemblyDocument object and Relations3d on each Part object. The Relations3d collection on the AssemblyDocument allows you to iterate through all relationships in the document. The RelationsSd collection on each Part object allows you to iterate through the relationships defined forthat specific part.

There are five types of 3-D relationships: AngularRelationSd, AxialRelation3d, GroundRelation3d, PlanarRelation3d, and PointRelation3d. These do not directly correlate to the interactive commands that place relationships. The relationships are as follows:

- AngularRelationSd - Defines an angular relationship between two objects.
- AxialRelation3d - Defines a relationship between conical faces. This is an axial align in the interactive interface.
- GroundRelation3d - Defines a ground constraint.
- PointRelationSd - Applies a connect relationship between points (vertices) of the points in an assembly.
- PlanarRelation3d - Defines a relationship between two planar faces. This includes both mates and planar aligns.

The following example shows how to use some of these relationship objects. This sample finds all of the PlanarRelation3d objects that define mates and modifies their offset values.

> A See Handling 'Application is Busy' and 'Call was Rejected By Callee' errors for information regarding the use of OleMessageFilter.

Analyzing Assembly Relationships in Visual Basic .NET ~ ~ >de

```
Imports System.Runtime.InteropServices

Module Program
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objAssembly As SolidEdgeAssembly.AssemblyDocument = Nothing
        Dim objRelations3d As SolidEdgeAssembly.RelationsSd = Nothing
```

```
        Dim objRelation3d As Object = Nothing
        Dim objAngularRelation3d As SolidEdgeAssembly.AngularRelation3d = Nothing
        Dim objAxialRelation3d As SolidEdgeAssembly.AxialRelation3d = Nothing
        Dim objGroundRelation3d As SolidEdgeAssembly.GroundRelation3d = Nothing
        Dim objPointRelation3d As SolidEdgeAssembly.PointRelation3d = Nothing
        Dim objPlanarRelation3d As SolidEdgeAssembly.PlanarRelation3d = Nothing

        Try
            OleMessageFilter.Register()

            ' Connect to a running instance of Solid Edge
            objApplication = Marshal.GetActiveObject("SolidEdge.Application")

            ' Get a reference to the active document
            objAssembly = objApplication.ActiveDocument

            ' Get a reference to the relations 3d collection
            objRelations3d = objAssembly.Relations3d

            ' Loop through the relations 3d objects
            For Each objRelation3d In objRelations3d
                ' Determine the relation type
                Select Case objRelation3d.Type
                    Case SolidEdgeFramework.ObjectType.igAngularRelation3d
                        objAngularRelation3d = objRelation3d
                    Case SolidEdgeFramework.ObjectType.igAxialRelation3d
                        objAxialRelation3d = objRelation3d
                    Case SolidEdgeFramework.ObjectType.igGroundRelation3d
                        objGroundRelation3d = objRelation3d
                    Case SolidEdgeFramework.ObjectType.igPointRelation3d
                        objPointRelation3d = objRelation3d
                    Case SolidEdgeFramework.ObjectType.igPlanarRelation3d
                        objPlanarRelation3d = objRelation3d
                End Select
            Next

        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter.Revoke()
        End Try
    End Sub
End Module
```

Analyzing Assembly Relationships in C# ~ ‘ de

```
using SolidEdgeFramework;
using System;
using System.Reflection;
using System.Runtime.InteropServices;

namespace SolidEdge.SDK
{
    class Program
    {
        [STAThread]
        static void Main(string[] args)
```

```
{
    SolidEdgeFramework.Application application = null;
    SolidEdgeAssembly.AssemblyDocument assembly = null;
    SolidEdgeAssembly.Relations3d relations3d = null;
    object relation3d = null;
    SolidEdgeAssembly.AngularRelation3d angularRelation3d = null;
    SolidEdgeAssembly.AxialRelation3d axialRelation3d = null;
    SolidEdgeAssembly.GroundRelation3d groundRelation3d = null;
    SolidEdgeAssembly.PointRelation3d pointRelation3d = null;
    SolidEdgeAssembly.PlanarRelation3d planarRelation3d = null;

    try
    {
        OleMessageFilter.Register();

        // Connect to a running instance of Solid Edge
        application = (SolidEdgeFramework.Application)
            Marshal.GetActiveObject("SolidEdge.Application");

        // Get a reference to the active document
        assembly = (SolidEdgeAssembly.AssemblyDocument)
          application.ActiveDocument;

        // Get a reference to the relations 3d collection
        relations3d = assembly.Relations3d;

        // Loop through the relations 3d objects
        for (int i = 1; i <= relations3d.Count; i++)
        {
            relation3d = relations3d.Item(i);
            Type type = relation3d.GetType();

            // Get the value of the Type proprety via Reflection
            SolidEdgeFramework.ObjectType objectType =
              (SolidEdgeFramework.ObjectType)type.InvokeMember(
                "Type",
                BindingFlags.GetProperty,
                null,
                relation3d,
                null);

            // Determine the relation type
            switch (objectType)
            {
                case SolidEdgeFramework.ObjectType.igAngularRelation3d:
                    angularRelation3d = (SolidEdgeAssembly.AngularRelation3d)
                      relation3d;
                    break;
                case SolidEdgeFramework.ObjectType.igAxialRelation3d:
                    axialRelation3d = (SolidEdgeAssembly.AxialRelation3d)
                      relation3d;
                    break;
                case SolidEdgeFramework.ObjectType.igGroundRelation3d:
                    groundRelation3d = (SolidEdgeAssembly.GroundRelation3d)
                      relation3d;
                    break;
```

```
                        case SolidEdgeFramework.ObjectType.igPointRelation3d:
                            pointRelation3d = (SolidEdgeAssembly.PointRelation3d)
                              relation3d;
                            break;
                        case SolidEdgeFramework.ObjectType.igPlanarRelation3d:
                            planarRelation3d = (SolidEdgeAssembly.PlanarRelation3d)
                              relation3d;
                            break;
                    }
                }
            }
            catch (System.Exception ex)
            {
                Console.WriteLine(ex.Message);
            }
            finally
            {
                OleMessageFilter.Revoke();
            }
        }
    }
}
```

Adding New Assembly Relationships

There are five methods to define assembly relationships through the automation interface: AddAngular, AddAxial, AddGround, AddPlanar, and AddPoint. These do not exactly correspond with the assembly relationship commands that are available interactively. However, they do correspond to the relationships that the interactive commands create.

For example, the AddPlanar method can be used to create either a Mate or an Align. The inputs to the AddPlanar method are two reference objects which are described below (but they correspond to the faces being mated or aligned), a Boolean that specifies whether or not the normals to the faces are aligned (this determines whether the faces are mated or aligned), and constraining points on each face (that correspond to the locations where you would click to locate the faces when you work interactively).

The following sample demonstrates the AddAxial method. This produces the same relationship that the interactive Align command produces when you align cylindrical faces. The inputs to this method are similar to those for the AddPlanar method. The first two inputs are reference objects that represent the cylindrical faces being aligned, and the third input is the Boolean that specifies whether normals to these faces are aligned. This method does not have input parameters for the constraining points the AddPlanar method uses.

To programmatically create the relationships that the Insert interactive command creates, you would use the AddPlanar and AddAxial methods. This would define the two cylindrical faces whose axes are aligned, and it would define the two planar faces that are mated. To remove the final degree of freedom, you would edit the axial relationship and set its FixedRotate property to True.

To create a Connect relationship, use the AddPoint method. The first input parameter is a reference object corresponding to the face or edge on the first part; the second input parameter is a constant that defines which key point from the input geometry defines the connection point (for example, CenterPoint, EndPoint, MidPoint, and so forth); and the third and fourth input parameters describe the same characteristics of the second part.

Within this general description, there are some important refinements. The methods previously described refer to reference objects, which correspond to part geometry. Each Assembly relationship must store a means of retrieving the geometric Part information that defines it. When using the AddPlanar method, for example, you need to pass in references to two planar faces (or reference planes).

The AssemblyDocument object has a CreateReference method whose job is to create the reference objects. The CreateReference method takes as input an Occurrence (an object that represents a member document of the assembly—which in this case would be a part document) and an Entity. The Entity can be an Edge, Face, or RefPlane object from the Occurrence document. The Reference object stores a path to the geometric representations necessary to construct the relationships.

To create assembly relationships via the automation interface, Occurrence objects (the Part and Subassembly models that comprise the assembly) must be placed in the Assembly document. You do this with the AssemblyDocument.Occurrances.AddByFilename method. This places the Occurrence in the assembly with a ground relationship. Therefore, (except for the first Occurrence added to the assembly) before any other relationships can be applied between this Occurrence and others in the assembly, the ground relationship must be deleted.

> ⚠ See Handling 'Application is Busy' and 'Call was Rejected By Callee' errors for information regarding the use of OleMessageFilter.

AddingNewAssemblyRelationshipinVisualBasic.NET)de

```
Imports SolidEdgeGeometry
Imports System.Runtime.InteropServices

Module Program
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objDocuments As SolidEdgeFramework.Documents = Nothing
        Dim objAssembly As SolidEdgeAssembly.AssemblyDocument = Nothing
        Dim objOccurrences As SolidEdgeAssembly.Occurrences = Nothing
        Dim objOccurrencel As SolidEdgeAssembly.Occurrence = Nothing
        Dim objOccurrence2 As SolidEdgeAssembly.Occurrence = Nothing
        Dim objPart As SolidEdgePart.PartDocument = Nothing
        Dim objModels As SolidEdgePart.Models = Nothing
        Dim objModel As SolidEdgePart.Model = Nothing
        Dim objRevolvedProtrusions As SolidEdgePart.RevolvedProtrusions = Nothing
        Dim objRevolvedProtrusion As SolidEdgePart.RevolvedProtrusion = Nothing
        Dim objRevolvedCutouts As SolidEdgePart.RevolvedCutouts = Nothing
        Dim objRevolvedCutout As SolidEdgePart.RevolvedCutout = Nothing
        Dim objFaces As SolidEdgeGeometry.Faces = Nothing
        Dim objFace As SolidEdgeGeometry.Face = Nothing
        Dim objGeometry As Object = Nothing
        Dim objScrewConicalFace As SolidEdgeGeometry.Face = Nothing
        Dim objNutConicalFace As SolidEdgeGeometry.Face = Nothing
        Dim objRefToCylinderlnScrew As SolidEdgeFramework.Reference = Nothing
        Dim objRefToConelnNut As SolidEdgeFramework.Reference = Nothing
        Dim objRelations3d As SolidEdgeAssembly.Relations3d = Nothing
        Dim objGroundRel As SolidEdgeAssembly.GroundRelation3d = Nothing
        Dim objRelNuttoScrew As SolidEdgeAssembly.AxialRelation3d = Nothing

        Try
            OleMessageFilter.Register()

            ' Connect to a running instance of Solid Edge
            objApplication = Marshal.GetActiveObject("SolidEdge.Application")

            ' Get a reference to the documents collection
            objDocuments = objApplication.Documents

            ' Create a new assembly document
            objAssembly = objDocuments.Add("SolidEdge.AssemblyDocument")

            ' Get a reference to the occurrences collection
            objOccurrences = objAssembly.Occurrences

            ' Add the first occurrence
```

```
objOccurrence1 = objOccurrences.AddByFilename("C:\Screw.par")

' Get a reference to the occurrence document
objPart = objOccurrence1.OccurrenceDocument

' Get a reference to the models collection
objModels = objPart.Models

' Get a reference to the one and only model
objModel = objModels.Item(1)

' Get a reference to the revolved protrusions collection
objRevolvedProtrusions = objModel.RevolvedProtrusions

' Get a reference to the first revolved protrusion
objRevolvedProtrusion = objRevolvedProtrusions.Item(1)

' Get a reference to the side faces collection
objFaces = objRevolvedProtrusion.SideFaces

' Loop through the faces
For Each objFace In objFaces
    ' Get a reference to the geometry object
    objGeometry = objFace.Geometry
    If objGeometry.Type = GNTTypePropertyConstants.igCylinder Then
        objScrewConicalFace = objFace
        Exit For
    End If
Next

' Create the first reference
objRefToCylinderInScrew = objAssembly.CreateReference( _
  objOccurrence1, objScrewConicalFace)

' Add the second occurrence
obj0ccurrence2 = objOccurrences.AddByFilename("C:\Nut.par")

' Get a reference to the occurrence document
objPart = obj0ccurrence2.0ccurrenceDocument

' Get a reference to the models collection
objModels = objPart.Models

' Get a reference to the one and only model
objModel = objModels.Item(1)

' Get a reference to the revolved cutouts collection
objRevolvedCutouts = objModel.RevolvedCutouts

' Get a reference to the first revolved cutout
objRevolvedCutout = objRevolvedCutouts.Item(1)

' Get a reference to the side faces collection
objFaces = objRevolvedCutout.SideFaces

' Loop through the faces
```

```
            For Each objFace In objFaces
                objGeometry = objFace.Geometry
                If objGeometry.Type = GNTTypePropertyConstants.igCone Then
                    objNutConicalFace = objFace
                    Exit For
                End If
            Next

            ' Create the second reference
            objRefToConeInNut = objAssembly.CreateReference( _
                        objOccurrence2, objNutConicalFace)

            ' All Occurrences placed through automation are placed "Grounded."
            ' You must delete the ground constraint on the second Occurrence
            ' before you can place other relationships.
            objRelations3d = objAssembly.Relations3d
            objGroundRel = objRelations3d.Item(2)
            objGroundRel.Delete()

            ' Rather than passing literal axes to the AddAxial method, pass
            ' references to conical faces, lust as you select conical faces
            ' when you use the interactive Align command.
            objRelNuttoScrew = objRelations3d.AddAxial( _
              objRefToConeInNut, objRefToCylinderInScrew, False)

        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter.Revoke()
        End Try
    End Sub
End Module
```

Adding New Assembly Relationship in C#

```
using SolidEdgeFramework;
using SolidEdgeGeometry;
using System;
using System.Reflection;
using System.Runtime.InteropServices;

namespace SolidEdge.SDK
{
    class Program
    {
        [STAThread]
        static void Main(string[] args)
        {
            SolidEdgeFramework.Application application = null;
            SolidEdgeFramework.Documents documents = null;
            SolidEdgeAssembly.AssemblyDocument assembly = null;
            SolidEdgeAssembly.Occurrences occurrences = null;
            SolidEdgeAssembly.Occurrence occurrence1 = null;
            SolidEdgeAssembly.Occurrence occurrence2 = null;
            SolidEdgePart.PartDocument part = null;
            SolidEdgePart.Models models = null;
            SolidEdgePart.Model model = null;
```

```
SolidEdgePart.RevolvedProtrusions revolvedProtrusions = null;
SolidEdgePart.RevolvedProtrusion revolvedProtrusion = null;
SolidEdgePart.RevolvedCutouts revolvedCutouts = null;
SolidEdgePart.RevolvedCutout revolvedCutout = null;
SolidEdgeGeometry.Faces faces = null;
SolidEdgeGeometry.Face face = null;
object geometry = null;
SolidEdgeGeometry.Face screwConicalFace = null;
SolidEdgeGeometry.Face nutConicalFace = null;
SolidEdgeFramework.Reference refToCylinderInScrew = null;
SolidEdgeFramework.Reference refToConeInNut = null;
SolidEdgeAssembly.Relations3d relations3d = null;
SolidEdgeAssembly.GroundRelation3d groundRelation3d = null;
SolidEdgeAssembly.AxialRelation3d relNuttoScrew = null;

try
{
    OleMessageFilter.Register();

    // Connect to a running instance of Solid Edge
    application = (SolidEdgeFramework.Application)
      Marshal.GetActiveObject("SolidEdge.Application");

    // Get a reference to the documents collection
    documents = application.Documents;

    // Get a reference to the active document
    assembly = (SolidEdgeAssembly.AssemblyDocument)
      documents.Add("SolidEdge.AssemblyDocument", Missing.Value);

    // Get a reference to the occurrences collection
    occurrences = assembly.Occurrences;

    // Add the first occurrence
    occurrence1 = occurrences.AddByFilename(@"C:\Screw.par", Missing.Value);

    // Connect to a running instance of Solid Edge
    application = (SolidEdgeFramework.Application)
      Marshal.GetActiveObject("SolidEdge.Application");

    // Get a reference to the documents collection
    documents = application.Documents;

    // Get a reference to the active document
    assembly = (SolidEdgeAssembly.AssemblyDocument)
      documents.Add("SolidEdge.AssemblyDocument", Missing.Value);

    // Get a reference to the occurrences collection
    occurrences = assembly.Occurrences;

    // Add the first occurrence
    occurrence1 = occurrences.AddByFilename(@"C:\Screw.par", Missing.Value);

    // Get a reference to the one and only model
    model = models.Item(1);
```

```
            // Get a reference to the revolved cutouts collection
            revolvedCutouts = model.RevolvedCutouts;

            // Get a reference to the first revolved cutout
            revolvedCutout = revolvedCutouts.Item(1);

            // Get a reference to the side faces collection
            faces = (SolidEdgeGeometry.Faces)revolvedCutout.SideFaces;

            // Loop through the faces
            for (int i = 1; i <= faces.Count; i++)
            {
                // Get a reference to the face object
                face = (SolidEdgeGeometry.Face)faces.Item(i);

                // Get a reference to the geometry object
                geometry = face.Geometry;

                // Determine the face type
                GNTTypePropertyConstants typeProperty = (GNTTypePropertyConstants)
                  geometry.GetType().InvokeMember(
                    "Type",
                    BindingFlags.GetProperty>
                    null,
                    geometry,
                    null);

                if (typeProperty == GNTTypePropertyConstants.igCone)
                {
                    nutConicalFace = face;
                    break;
                }
            }

            // Create the second reference
            refToConeInNut = (SolidEdgeFramework.Reference)
              assembly.GreateReference(occurrence2, nutConicalFace);

            // All Occurrences placed through automation are placed "Grounded."
            // You must delete the ground constraint on the second Occurrence
            // before you can place other relationships.
            relations3d = assembly.Relations3d;
            groundRelation3d = (SolidEdgeAssembly.GroundRelation3d)
              relations3d.Item(2);
            groundRelation3d.Delete();

            // Rather than passing literal axes to the AddAxial method, pass
            // references to conical faces, Just as you select conical faces
            // when you use the interactive Align command.
            relNuttoScrew = relations3d.AddAxial(
              refToConeInNut,
              refToCyUnderInScrew,
              false);

        }
        catch (System.Exception ex)
```

```
            {
                Console.WriteLine(ex.Message);
            }
            finally
            {
                OleMessageFilter.Revoke();
            }
        }
    }
}
```

Solid Edge ST7 SDK

Working with Sections and Sheets

Solid Edge. Design better.

The structure of Draft documents differs significantly from other Solid Edge document types. From the DraftDocument object, you access the Sheets collection and then the individual Sheet objects. The Sheets collection contains both working sheets and background sheets.

In addition, DraftDocument supports a Sections object. The Sections object is a collection of Section objects that group Sheets by the characteristics of the data they contain. As users create drawings interactively, data from these drawings is automatically placed in one of three Section objects:

- Working Section - Contains Sheet objects (Working Sheets) on which normal 2-D drawing objects (Lines, Arcs, Drawingviews, and so forth) are placed.
- Background Section - Contains background sheets, which hold the sheet borders.

Sections are a part of the graphical interface, although they are not immediately apparent. When the interactive user selects View > Background Sheet, Solid Edge internally changes to the Backgrounds section and displays its sheets. Similarly, the View > Working Sheet command allows you to modify the sheets that are in the Sections) section. When a DrawingView is added, a new sheet is added to the Drawingviews section.

However, it is not possible through the graphical interface to create and manipulate sections directly. Although it is possible through automation to create new Sections, it is not a supported workflow. Although the same information is available on the Sheets collection that is a child of the DraftDocument object, within Sections, the information is separated by its functional characteristics.

| A See Handling 'Application is Busy' and 'Call was Rejected By Callee' errors for information regarding the use of OleMessageFilter.

Sections and Sheets in Visual Basic .NET Copy Code

```
Imports SolidEdgeDraft
Imports System.Runtime.InteropServices

Module Program
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objDocuments As SolidEdgeFramework.Documents = Nothing
        Dim objDraft As SolidEdgeDraft.DraftDocument = Nothing
        Dim objSections As SolidEdgeDraft.Sections = Nothing
        Dim objSection As SolidEdgeDraft.Section = Nothing
        Dim objSectionSheets As SolidEdgeDraft.Sectionsheets = Nothing
        Dim objSheets As SolidEdgeDraft.Sheets = Nothing
        Dim objSheet As SolidEdgeDraft.Sheet = Nothing
        Dim strFormatl As String = "Section = {0}"
        Dim strFormat2 As String = "Sheet = {0}"

        Try
            OleMessageFilter.Register()

            ' Connect to a running instance of Solid Edge
            objApplication = Marshal.GetActiveObject("SolidEdge.Application")

            ' Get a reference to the documents collection
            objDocuments = objApplication.Documents

            ' Add a Draft document
            objDraft = objDocuments.Add("SolidEdge.DraftDocument")

            ' Get a reference to the sections collection
            objSections = objDraft.Sections

            ' Loop through the sections
            ' igWorkingSection, igBackgroundSection & ig2dModelSection
            For Each objSection In objSections
                ' Output the section type
                Console.WriteLine( _
                  String.Format(strFormatlj objSection.Type.ToString()))

                ' Get a reference to the section sheets collection
                objSectionSheets = objSection.Sheets


                ' Loop through the sheets
                For Each objSheet In objSectionSheets
                    ' Output the sheet name
                    Console.WriteLine(String. Format(str Format2, objSheet.Name))
```

```
            Next
        Next

        ' Access the igWorkingSection directly
        objSection = objSections.Workingsection

        ' Access the igBackgroundSection directly
        objSection = objSections.Backgroundsection

        ' Get a reference to the sheets collection
        objSheets = objDraft.Sheets

        ' Add a new sheet
        objSheet = objSheets.AddSheet( _
          "Sheet2", SheetSectionTypeConstants.igWorkingSection)

        ' Make the newly added sheet active
        objSheet.Activate()

    Catch ex As Exception
        Console.WriteLine(ex.Message)
    Finally
        OleMessageFliter .Revoke()
    End Try
  End Sub
End Module
```

Sections and Sheets in C# Copy Code

```
using SolidEdgeDraft;
using System;
using System.Reflection;
using System.Runtime.InteropServices;

namespace SolidEdge.SDK
{
    class Program
    {
        [STAThread]
        static void Main(string[] args)
        {
            SolidEdgeFramework.Application application = null;
            SolidEdgeFramework.Documents documents = null;
            SolidEdgeDraft.DraftDocument draft = null;
            SolidEdgeDraft.Sections sections = null;
            SolidEdgeDraft.Section section = null;
            SolidEdgeDraft.Sectionsheets sectionsheets = null;
            SolidEdgeDraft.Sheets sheets = null;
            SolidEdgeDraft.Sheet sheet = null;
            string formatl = "Section = {0}";
            string format2 = "Sheet = {0}";

            try
            {
                OleMessageFilter.Register();

                // Connect to a running instance of Solid Edge
                application = (SolidEdgeFramework.Application)
                  Marshal.GetActiveObject("Solid Edge.Application");

                // Get a reference to the documents collection
                documents = application.Documents;

                // Add a Draft document
                draft = (SolidEdgeDraft.DraftDocument)
                  documents.Add("SolidEdge.Draft Document"j Missing.Value);

                // Get a reference to the sections collection
                sections = draft.Sections;

                // Loop through the sections
                // igWorkingSection igBackgnoundSection & ig2dModelSection
```

```
                for (int i = 1; i <= sections.Count; i++)
                {
                    section = sections.Item(i);

                    // Get a reference to the section sheets collection
                    sectionsheets = section.Sheets;

                    // Output the section type
                    Console.WriteLine(String.Format(formatl, section.Type.ToString()));

                    // Loop through the sheets
                    for (int j = 1; j <= sectionsheets.Count; j++)
                    {
                        sheet = sectionsheets.Item(j);

                        // Output the sheet name
                        Console.WriteLine(String.Format(format2, sheet.Name));
                    }
                }

                // Access the igWorkingSection directly
                section = sections.Workingsection;

                // Access the igBackgroundSection directly
                section = sections.Backgroundsection;

                // Get a reference to the sheets collection
                sheets = draft.Sheets;

                // Add a new sheet
                sheet = sheets.AddSheet(
                  "Sheet2",
                  SheetSectionTypeConstants.igWorkingSection,
                  Missing.Value,
                  Missing.Value);

                // Make the newly added sheet active
                sheet.Activate();
            }
            catch (System.Exception ex)
            {
                Console.WriteLine(ex.Message);
            }
            finally
            {
                OleMessageFilter.Revoke();
            }
        }
    }
}
```

Solid Edge ST7 SDK

Working with SmartFrames

Solid Edge. Design better.

SmartFrames are shapes (rectangles or ellipses) on a sheet that enclose embedded or linked object(s) and have some intelligence about how to deal with the data in that frame. SmartFrames provide control over the way automation objects are displayed and manipulated on a Solid Edge sheet. SmartFrames have intelligence about their contained objects that includes the following features:

- A transformation matrix to convert between the framed object's local coordinate system and the containing document's coordinate system.
- Methods to manipulate the contained object, such as scale, crop, move, or rotate.
- Frame symbology that shows the state of the framed object such as linked, embedded, or a link that needs updating.
- Link update rules (such as automatic and manual).
- In-place activation rules.
- Knowledge about favorite commands.
- Knowledge about a preferred file location or extension used in first associating the file to a frame.
- Knowledge for converting between links, embeddings, and native data.

When using Solid Edge, you may sometimes find it useful to reference data that exists in a format other than a Solid Edge file. For example, while in the Solid Edge drawing environment, you might want to link to a portion of a Microsoft Excel spreadsheet. Solid Edge supports this cross-referencing through the implementation of SmartFrames. A SmartFrame is a Solid Edge object that contains a view of an embedded or linked object.

Initially, you can create an empty SmartFrame without specifying an object to be linked or embedded. A SmartFrame style must be specified, or you can use the default style for a sheet. A SmartFrame style has properties that affect how the object within the SmartFrame can be manipulated. For example, a SmartFrame that is based on a reference file style can either align the origin of the external file with the Solid Edge file or provide an option to scale the contents.

When you create a SmartFrame, four solid black lines are drawn to represent the frame. Once you have created the SmartFrame, you can select and manipulate the object as you would other Solid Edge objects.

You can create and manipulate SmartFrame objects through the automation interface using the methods that are associated with the SmartFrames2d collection object. In the following example, the AddBy2Points method creates a SmartFrame. The first argument of AddBy2Points specifies a style to be applied to the SmartFrame. In this case, the style is set to a blank string (""), so the default style is applied.

| See Handling 'Application is Busy' and 'Call was Rejected By Callee' errors for information regarding the use of OleMessageFilter.

Linking and Embedding in Visual Basic .NET Copy Code

```
Imports SolidEdgeDraft
Imports System.Runtime.InteropServices

Module Program
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objDocuments As SolidEdgeFramework.Documents = Nothing
        Dim objDraft As SolidEdgeDraft.DraftDocument = Nothing
        Dim objSheet As SolidEdgeDraft.Sheet = Nothing
        Dim objSmartFrames2d As SolidEdgeFrameworkSupport.SmartFrames2d = Nothing
        Dim objSmartFrame2d As SolidEdgeFrameworkSupport.SmartFrame2d = Nothing

        Try
            OleMessageFilter.Register()

            ' Connect to a running instance of Solid Edge
            objApplication = Marshal.GetActiveObject("Solid Edge.Application")

            ' Get a reference to the documents collection
            objDocuments = objApplication.Documents

            ' Add a Draft document
            objDraft = objDocuments.Add("SolidEdge.DraftDocument")

            ' Get a reference to the active sheet
            objSheet = objDraft.ActiveSheet

            ' Get a reference to the smart frames 2d collection
            objSmartFrames2d = objSheet.SmartFrames2d

            ' Create a SmartFrame2d object by two points.
            objSmartFrame2d = objSmartFrames2d.AddBy2Points( _
              "", 0.05, 0.05, 0.1, 0.1)

            ' Add a description to the SmartFrame
```

```
            objSmartFrame2d.Description = "My SmartFrame2d"

            ' Embed document within the SmartFrame
            objSmartFrame2d.CreateEmbed("C:\MyFile.doc")

            ' or


            ' Link document within the SmartFrame
            'objSmartFrame2d.CreateLink("C:\MyFile.doc")


        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter.Revoke()
        End Try
    End Sub
End Module
```

Linking and Embedding in C# Copy Code

```
using SolidEdgeDraft;
using System;
using System.Reflection;
using System.Runtime.InteropServices;

namespace SolidEdge.SDK
{
    class Program
    {
        [STAThread]
        static void Main(string[] args)
        {
            SolidEdgeFramework.Application application = null;
            SolidEdgeFramework.Documents documents = null;
            SolidEdgeDraft.DraftDocument draft = null;
            SolidEdgeDraft.Sheet sheet = null;
            SolidEdgeFrameworkSupport.SmartFrames2d smartFrames2d = null;
            SolidEdgeFrameworkSupport.SmartFrame2d smartFrame2d = null;

            try
            {
                OleMessageFilter.Register();

                // Connect to a running instance of Solid Edge
                application = (SolidEdgeFramework.Application)
                  Marshal.GetActiveObject("Solid Edge.Application");

                // Get a reference to the documents collection
                documents = application.Documents;

                // Add a Draft document
                draft = (SolidEdgeDraft.DraftDocument)
                  documents.Add("Solid Edge.DraftDocument", Missing.Value);

                // Get a reference to the active sheet
                sheet = draft.ActiveSheet;

                // Get a reference to the smart frames 2d collection
                smartFrames2d = (SolidEdgeFrameworkSupport.SmartFrames2d)sheet.SmartFrames2d;

                // Create a SmartFrame2d object by two points.
                smartFrame2d = smartFrames2d.AddBy2Points("", 0.02, 0.02, 0.07, 0.07);

                // Add a description to the SmartFrame
                smartFrame2d.Description = "My SmartFrame2d";

                // Embed document within the SmartFrame
                smartFrame2d.CreateEmbed(@"C:\MyFile.doc", Missing.Value);
```

```
                11 or

                // Link document within the SmartFrame
                11smartFrame2d.CreateLink(@"C:\MyFile.doc", Missing.Value) j

            }
            catch (System.Exception ex)
            {
                Console.WriteLine(ex.Message);
            }
            finally
            {
                OleMessageFilter.Revoke();
            }
        }
    }
}
```

Solid Edge ST7 SDK

Working with Symbols

Solid Edge. Design better.

Symbols are documents that contain graphic elements. You can place these documents at a specified scale, position, and orientation. The document that contains the graphic elements is the source document; the document into which the source is placed is the container document. A source document is represented in a container document by a symbol. The symbol references the source document as the COM object. Using symbols, you can store a drawing of a nut, bolt, or screw in one document and place it in several documents at a user-defined size. In addition, symbols have the following benefits:

- Save memory when placing multiple instances of the same source document in the same container.
- Automatically update the container document when modified.
- Maintain the properties defined in the source document.

On the Insert menu, click Object to place a symbol in the interactive environment. When using Solid Edge though automation, you can place a symbol using the methods associated with the Symbols collection.

The Symbols collection object provides methods that enable you to place new symbols and to query for information about existing ones. The Symbol2d object provides methods and properties to enable you to review or manipulate the symbol geometry, the attachment between the symbol and the source document, and the user properties. You can also move and copy symbols.

You can place a symbol from any source document that is implemented as an ActiveX object. For example, a source document could be a Microsoft Word file, an Excel spreadsheet, or a Solid Edge document.

When you place a symbol, you must specify an insertion type. The insertion type affects the way the symbol is updated. Three options are available:

- Linked - The symbol and the initial source document are directly connected. The symbol is automatically updated when its source document is edited. The source document is external to the container. It is a separate file that is visible with Explorer.
- Embedded - A copy of the initial source document is stored in the container. The symbol is attached to this copy and is automatically updated when the copy is updated. After placement, the symbol is strictly independent of the initial source document.
- Shared Embedded - When placing a symbol more than one time into the same container, the initial source document is copied only one time. The symbols are attached to that copy and are all updated automatically when the copy of the initial source document is updated. After placement, the symbols are strictly independent of the initial source document.

A See Handling 'Application is Busy' and 'Call was Rejected By Callee' errors for information regarding the use of OleMessageFilter.

SymbolsinVisualBasic.NET Copy Code

```
Imports SolidEdgeConstants
Imports SolidEdgeDraft
Imports System.Runtime.InteropServices

Module Program
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objDocuments As SolidEdgeFramework.Documents = Nothing
        Dim objDraft As SolidEdgeDraft.DraftDocument = Nothing
        Dim objSheet As SolidEdgeDraft.Sheet = Nothing
        Dim objSymbols As SolidEdgeFramework.Symbols = Nothing
        Dim objSymboll As SolidEdgeFramework.Symbol2d = Nothing
        Dim objSymbol2 As SolidEdgeFramework.Symbol2d = Nothing
        Dim objSymbol3 As SolidEdgeFramework.Symbol2d = Nothing
        Dim x As Double
        Dim y As Double
        Dim strSourceDoc As String
        Dim objWordDoc As Object = Nothing
        Dim objWordApp As Object = Nothing

        Try
            OleMessageFilter.Register/)

            ' Connect to a running instance of Solid Edge
            objApplication = Marshal.GetActiveobject("Solid Edge.Application")

            ' Get a reference to the documents collection
            objDocuments = objApplication.Documents

            ' Add a Draft document
            objDraft = objDocuments.Add("SolidEdge.DraftDocument")

            ' Get a reference to the active sheet
            objSheet = objDraft.Activesheet

            ' Get a reference to the symbols collection
            objSymbols = objSheet.Symbols

            ' Create a linked symbol
            objSymboll = objSymbols.Add( _
              OLEInsertionTypeConstant.igOLELinked, _
              "C:\MyFile.doc", 0.1, 0.1)
```

```
            ' Create a embedded symbol
            objSymbol2 = objSymbols.Add( _
              OLEInsertionTypeConstant.igOLEEmbedded, _
              "C:\MyFile.doc", 0.1, 0.2)

            ' Create a shared embedded symbol
            objSymbol3 = objSymbols.Add( _
              OLEInsertionTypeConstant.igOLESharedEmbedded, _
              "C:\MyFile.doc", 0.1, 0.3)

            ' Retrieve the origin of the first symbol
            objSymbol1.GetOrigin(x, y)

            ' Modify the first symbol's origin
            objSymbol1.SetOrigin(x + 0.1, y + 0.1)

            ' Set the angle of rotation of the first symbol to 45 degrees
            ' (in radians)
            objSymbol1.Angle = 45 * (Math.PI / 180)

            ' Find the path to the linked document
            If objSymbol1.OLEType = OLEInsertionTypeConstant.igOLELinked Then
                strSourceDoc = objSymbol1.SourceDoc
            End If

            ' Open the source document to modify it
            objSymbol1.DoVerb( _
              Solid EdgeConstants.StandardOLEVerbConstants.igOLEOpen)

            ' In this case, we know that we're dealing with a word document
            ' Get a reference the source document dispatch interface
            ' At this point, you can use the Word API to manipulate the document
            objWordDoc = objSymbol1.Object

            ' Get a reference to the word application object
            objWordApp = objWordDoc.Application

            ' Save and close the word document
            objWordDoc.Save()
            objWordDoc.Close()

            ' Quit word
            objWordApp.Quit()

        Catch ex As Exception
            Console.WriteLine(ex.Message)
        Finally
            OleMessageFilter.Revoke()
        End Try
    End Sub
End Module
```

Symbols in C# Copy Code

```
using SolidEdgeFramework;
using System;
using System.Reflection;
using System.Runtime.InteropServices;

namespace SolidEdge.SDK
{
    class Program
    {
        [STAThread]
        static void Main(string[] args)
        {
            SolidEdgeFramework.Application application = null;
            SolidEdgeFramework.Documents documents = null;
            SolidEdgeDraft.DraftDocument draft = null;
            SolidEdgeDraft.Sheet sheet = null;
            SolidEdgeFramework.Symbols symbols = null;
            SolidEdgeFramework.Symbol2d symbol1 = null;
            SolidEdgeFramework.Symbol2d symbo!2 = null;
            SolidEdgeFramework.Symbol2d symbols = null;
            double x = 0;
            double y = 0;
            string strSouceDoc;
            object wordDoc = null;
```

```
object wordApp = null;

try
{
    OleMessageFilter.Register();

    // Connect to a running instance of Solid Edge
    application = (SolidEdgeFramework.Application)
      Marshal.GetActiveObject("SolidEdge.Application");

    // Get a reference to the documents collection
    documents = application.Documents;

    // Add a Draft document
    draft = (SolidEdgeDraft.DraftDocument)
      documents.Add("SolidEdge.DraftDocument", Missing.Value);

    // Get a reference to the active sheet
    sheet = draft.ActiveSheet;

    // Get a reference to the symbols collection
    symbols = (SolidEdgeFramework.Symbols)sheet.Symbols;

    // Create a linked symbol
    symbol1 = symbols.Add(
        (int) OLEInsertionTypeConstant.igOLELinked,
        @"C:\MyFile.doc",
        0.1,
        0.1,
        Missing.Value);

    // Create a embedded symbol
    symbol2 = symbols.Add(
        (int)OLEInsertionTypeConstant.igOLEEmbedded,
        @"C:\MyFile.doc",
        0.1,
        0.2,
        Missing.Value);

    // Create a shared embedded symbol
    symbols = symbols.Add(
        (int) OLEInsertionTypeConstant.igOLEShared Embedded,
        @"C:\MyFile.doc"j
        0.1,
        0.3,
        Missing.Value);

    // Retrieve the origin of the first symbol
    symbol1.GetOrigin(out x, out y);

    // Modify the first symbol's origin
    symbol1.SetOrigin(x + 0.2, y + 0.2);

    // Set the angle of rotation of the first symbol to 45 degrees
    // (in radians)
    symbol1.Angle = 45 * (Math.PI / 180);

    // Find the path to the linked document
    if (symbol1.OLEType == OLEInsertionTypeConstant.igOLELinked)
    {
        strSouceDoc = symbol1.SourceDoc;
    }

    // Open the source document to modify it
    symbol1.DoVerb(
      SolidEdgeConstants.StandardOLEVerbConstants.igOLEOpen);

    // In this case., we know that we're dealing with a word document
    // Get a reference the source document dispatch interface
    // At this point, you can use the Word API to manipulate the document
    wordDoc = symbol1.Object;

    // Get a reference to the word application object
    wordApp = wordDoc.GetType().InvokeMember(
      "Application", BindingFlags.GetProperty, null, wordDoc, null);

    // Save and close the word document
```

```
                wordDoc.GetType().InvokeMember(
                  "Save", BindingFlags.InvokeMethod, null., wordDoc, null);
                wordDoc.GetType().InvokeMember(
                  "Close", BindingFlags.InvokeMethod, null, wordDoc, null);

                // Quit word
                wordApp.GetType().InvokeMember(
                  "Quit", BindingFlags.InvokeMethod, null, wordApp, null);

            }
            catch (System.Exception ex)
            {
                Console.WriteLine(ex.Message);
            }
            finally
            {
                OleMessageFilter.Revoke();
            }
        }
    }
}
```

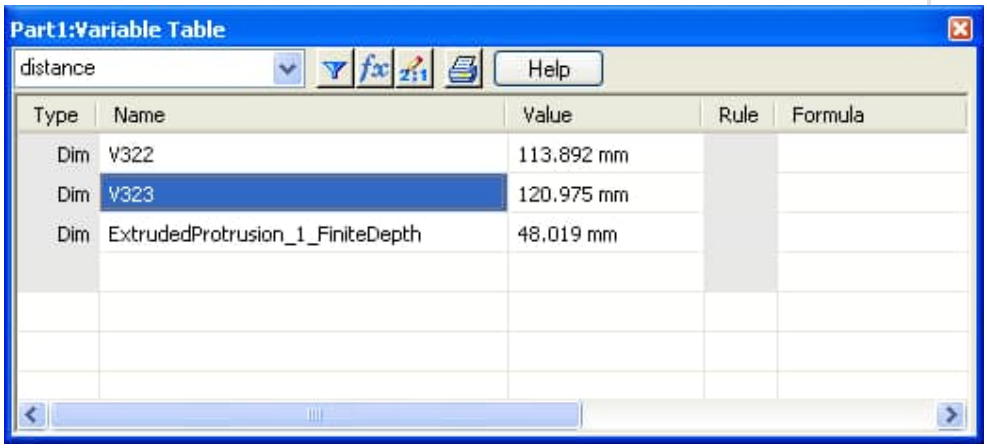
Solid Edge ST7 SDK

Working with the Variable Table

Variables

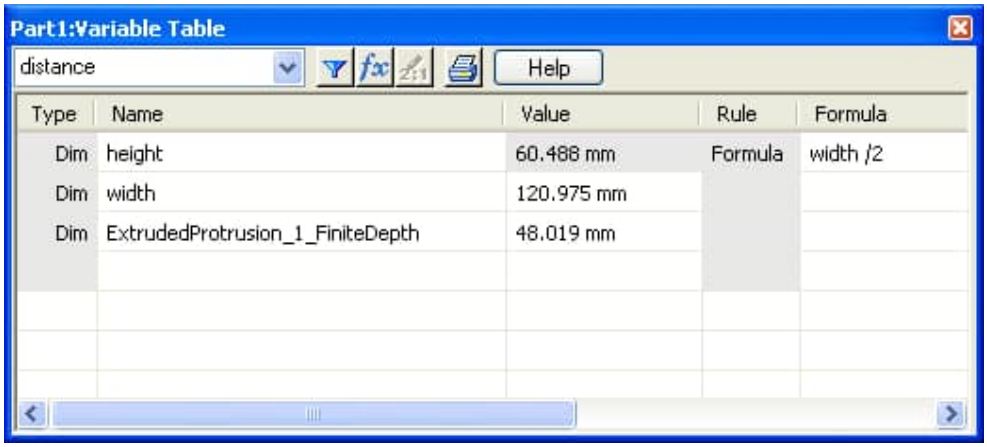
The variable system allows you to define relationships between variables and dimensions using equations, external functions, and spreadsheets. For example, you can construct a rectangular profile using four lines, and place two dimensions to control the height and width. The following illustration shows the Variable Table with these two dimensions displayed. (Every dimension is automatically available as a variable.)

This topic contains these sections:



Using these variables, you can make the height a function of the width by entering a formula. For example, you could specify that the height is always one-half of the width. Once you have defined this formula, the height is automatically updated when the width changes.

All variables have names; it is through these names that you can reference them. In the preceding illustration, the names automatically assigned by the system are V108 and V109. You can rename these dimensions; in the following illustration, V322 has been renamed to "height," and V323 has been renamed to "width."



Every variable has a value. This can be a static value or the result of a formula. Along with the value, a unit type is also stored for each variable. The unit type specifies what type of measurement unit the value represents. In this example, both of the variables use distance units. You can create Variables for other unit types such as area and angle. Values are displayed using this unit type and the unit readout settings.

Variable Table

All variable automation is accessed through the Variables collection and Variable objects. The Variables collection serves two purposes: it allows you to create and access variable objects, and it allows you to work with dimensions as variables.

The Variables collection supports an Add method to create new Variable objects. It also supports the standard methods for iterating through the members of the collection. The following program connects to the Variables collection, creates three

new variables, and lists them.

Units with variables work the same as units in the system. Units are stored using internal values and then are appropriately converted for display. For example, all length units are stored internally as meters. When these units are displayed in the Variable Table, they are converted to the units specified in the Properties dialog box.

In addition to the properties and methods on the Variables collection, properties and methods are also available on the Variable objects. These properties and methods read and set variable names, define formulas, set values, and specify units of measure.

> A See Handling 'Application is Busy' and 'Call was Rejected By Callee' errors for information regarding the use of OleMessageFilter.

VariableTableinVisualBasic.NET Ci

```
Imports System.Runtime.InteropServices

Module Program
    <STAThread()> _
    Sub Main()
        Dim objApplication As SolidEdgeFramework.Application = Nothing
        Dim objPart As SolidEdgePart.PartDocument = Nothing
        Dim objVariables As SolidEdgeFramework.Variables = Nothing
        Dim objVariable As SolidEdgeFramework.variable = Nothing

        Try
            OleMessageFilter.Register()

            ' Connect to a running instance of Solid Edge
            objApplication = Marshal.GetActiveObject("Solid Edge.Application")

            ' Get a reference to the active document
            objPart = objApplication.ActiveDocument

            ' Get a reference to the variables collection
            objVariables = objPart.Variables

            'Add a variable
            objVariable = objVariables.Add("NewVar", "1.5")

            'Change the formula of the variable to a function
            objVariable.Formula = Math.Sin(O.l).ToStringQ

            'Change the name of the variable
            objVariable.Name = "NewName"

            'Change the value of the variable. This will not change
            'the value of the variable
            objVariable.Value = 123

            'Change the formula of the variable to a static value
            'This causes the formula to be removed and sets the value
            objVariable.Formula = "456"

            'Change the value of the variable. It works now
            objVariable.Value = 789

            'Delete the variable
            objVariable.Delete()

        Catch ex As Exception
            Console.WriteLine(ex.Message)
```

```
        Finally
            OleMessageFilter.Revoke()
        End Try
    End Sub
End Module
```

Variable Table in C# Cop

```
using Examples;
using SolidEdgeFramework;
using System;
using System.Runtime.InteropServices;

namespace SolidEdge.SDK
{
    class Program
    {
        [STAThread]
        static void Main(string[] args)
        {
            SolidEdgeFramework.Application application = null;
            SolidEdgePart.PartDocument part = null;
            SolidEdgeFramework.Variables variables = null;
            SolidEdgeFramework.variable variable = null;

            try
            {
                OleMessageFilter.Register();

                // Connect to a running instance of Solid Edge
                application = (SolidEdgeFramework.Application)
                  Marshal.GetActiveObject("SolidEdge.Application");

                // Get a reference to the active document
                part - (SolidEdgePart.PartDocument)
                  application.ActiveDocument;

                // Get a reference to the variables collection
                variables = (SolidEdgeFramework.Variables)part.Variables;

                // Add a variable
                variable = (SolidEdgeFramework.variable)
                  variables.Add("NewVar"j "1.5", UnitTypeConstants.igUnitDistance);

                // Change the formula of the variable to a function
                variable.Formula = Math.Sin(O.l).ToString();

                // Change the name of the variable
                variable.Name= "NewName";

                // Change the value of the variable. This will not change
                // the value of the variable
                variable.Value = 123;

                // Change the formula of the variable to a static value
                // This causes the formula to be removed and sets the value
                variable.Formula = "456";

                // Change the value of the variable. It works now
                variable.Value = 789;
```

```
                //Delete the variable
                variable.DeleteQ;
            }
            catch (System.Exception ex)
            {
                Console.WriteLine(ex.Message);
            }
            finally
            {
                OleMessageFilter.Revoke();
            }
        }
    }
}
```

Solid Edge ST7 SDK

.NET Garbage Collection

Solid Edge.
Design better.

Solid Edge will automatically invoke Garbage Collection for the default application domain when a document is closed and when Solid Edge is exited. Any .NET client that is not running in isolation (that is, it is running in the context of the .NET runtime embedded in Solid Edge), should NOT perform GC when a document close event is encountered.

Solid Edge Draft Type Library

Add2dModelView Method

1

Solid Edcje.
Design better.

Description

This method creates a 2D Model view as a DrawingView object.

Syntax

Visual Basic

```
Public Function Add2dModelView( _
    ByVai ScaLe As Double, _
    ByVai xl As Double, _
    ByVai yl As Double _
) As Drawingview
```

Parameters

Scale
Specifies the scale of the 2D Model view. The scale determines the size of the 2D Model view in that the view sizes automatically to display everything within the view definition box.

xl
Specifies the x coordinate of the 2D Model view origin.

yi
Specifies the y coordinate of the 2D Model view origin.

See Also

DrawingViews Collection | DrawingViews Members | Solid Edge V18 - What's New